UNIVERSIDAD DE GRANADA

Lecture Notes

in

Statistical Physics

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Academic Year 2025 - 2026

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Prologue

Welcome to the course on Statistical Physics!

In this course, we will explore the fascinating world of Statistical Physics and its connection with Thermodynamics. Statistical physics provides a powerful framework for understanding the behaviour of large systems of particles, such as gases, solids, and liquids. It offers a microscopic description of these systems, taking into account the individual interactions between particles. By studying the statistical properties of these systems, we can derive macroscopic quantities and thermodynamic laws, bridging the gap between microscopic and macroscopic descriptions.

Throughout this course, we will investigate the principles of thermodynamics and their statistical foundations. We will delve into topics such as entropy, temperature, and energy, understanding how these quantities emerge from the statistical behaviour of particles. By examining the statistical distributions of particles in different energy states and statistical ensembles, we will establish connections between thermodynamic quantities and microscopic properties.

We hope you find this course intellectually stimulating and rewarding. By the end of the course, you will develop a solid understanding of statistical physics and its connections with thermodynamics, empowering you to unravel the mysteries of matter and energy.

Finally, we would like to acknowledge the invaluable support provided by Prof. Pablo Hurtado (Department of Electromagnetism and Matter Physics, UGR) and Prof. Arturo Moncho (Department of Applied Physics, UGR), who have kindly shared their course material with us, making it possible to arrange the foundation of the present lecture notes. Their excellent work over the years has been especially inspiring to assemble the present document. We stress that mistakes, typos or inconsistencies possibly found in these notes are our responsibility only.

Best wishes for a successful, engaging and enriching learning experience!

Alessandro Patti
Carlos Pérez Espigares

Granada, September 2023

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Course Presentation

This course is designed to introduce you to the fascinating world of *Statistical Physics* and its connections with Thermodynamics. As 3^{rd}-year undergraduate students in Physics, you will learn about the statistical description of physical systems comprising a large number of particles. By studying *Statistical Physics*, you will gain a deeper understanding of the behaviour of matter at the macroscopic level and develop problem-solving skills through the application of statistical methods.

In this course, we will explore fundamental concepts and principles of *Statistical Physics*, starting with an introduction to its basic principles and postulates. We will then focus on the connections between *Statistical Physics* and *Thermodynamics*, understanding how the former provides a microscopic foundation for the laws of the latter. Through a combination of theoretical discussions, mathematical derivations, and problem-solving exercises, we will cover topics such as equilibrium statistical mechanics, quantum statistical physics, collective behaviour, and phase transitions.

The present lecture notes intend to cover the basic principles of *Statistical Physics*. They will serve as a comprehensive resource, containing detailed explanations, examples, and additional readings to enhance your understanding of the subject. Nevertheless, you are highly recommended to extend the concepts here presented by attending lectures and by further reading.

Students who have enrolled in this course should know that it will be entirely delivered in English. Interaction with lecturers, which include in-class discussions and tutorials, as well as coursework and exams, will take place in English.

Course Information

- Course Code: 2671134
- Semester: 2
- ECTS Credits: 6
- Prerequisites: Thermodynamics, Mathematical Methods of Physics, Mechanics and Waves, Quantum Physics. Prior knowledge of Analytical Mechanics is advisable
- Lecturers: Dr. Alessandro Patti (apatti@ugr.es) and Dr. Carlos Pérez Espigares (carlosperez@ugr.es)

Course Description

This course provides an introduction to Statistical Physics, which is the study of the behaviour of large systems of particles using statistical methods. The course provides the necessary tools to understand the statistical behaviour of physical systems. Topics covered include microstates and macrostates, statistical ensembles, Boltzmann statistics, quantum statistics, and thermodynamic properties of gases, among others.

Learning Outcomes

By the end of this course, students should be able to:

- Understand the fundamental concepts and principles of Statistical Physics.
- Apply statistical methods to describe the behaviour of physical systems.
- Derive and analyse statistical distributions relevant to Statistical Physics.
- Relate statistical properties to thermodynamic quantities.
- Apply statistical physics concepts to solve problems in various areas of physics.

Course Topics

- PART I
  - **Introduction to Statistical Physics**
  - **Microstates and Macrostates**
  - **Classical and Quantum Microscopic Descriptions**
  - **Ensembles**
  - **Fluctuations, Equivalence of Ensembles**
  - **Thermodynamic Limit**
- PART II
  - **Ideal Systems**
  - **Solids**
  - **Introduction to the Quantum Ideal Gas**
  - **Degenerate Fermionic Systems**
  - **Degenerate Bosonic Systems**

Textbooks

1. Theory
   - P. Pathria, *Statistical Mechanics* (2nd edition), Butterworth-Heinemann, Oxford (1996).
   - R. Balescu, *Equilibrium and Non-equilibrium Statistical Mechanics*, Wiley & Sons, New York (1975).
   - D. A. McQuarrie, *Statistical Mechanics*, Harper & Row, New York (1976).
   - M. Le Bellac, F. Mortessagne, and G. G. Batrouni, *Equilibrium and non-equilibrium statistical thermodynamics*, Cambridge University Press, Cambridge (2004).
   - J. J. Brey Abalo, J. de la Rubia Pacheco, and J. de la Rubia Sánchez, *Mecánica Estadística*, UNED, Madrid (2001).
   - D. Chandler, *Introduction to Modern Statistical Mechanics*, Oxford University Press (1987).
   - J. J. Sakurai *Modern Quantum Mechanics*, San Fu Tuan, editor (1989).
2. Problems

- Dalvit, D. A. R., Frastai, J., & Lawrie, I. D. *Problems on Statistical Mechanics.* Institute of Physics Publishing, Bristol, 1999.
- Yung-Kuo, L. *Problems and Solutions on Thermodynamics and Statistical Mechanics.* World Scientific Publishing, Singapore, 1990.
- Fernández Tejero, C., & Rodríguez Parrondo, J. M. *100 problemas de Física Estadística.* Alianza, Madrid, 1996.
- Kubo, R. *Statistical Mechanics: an Advanced Course with Problems and Solutions.* 2nd edition, North-Holland, Amsterdam, 1999.

Assessment

- ORDINARY EXAMINATION (continuous assessment mode).
  Students will take one or more exams, which may include theoretical, quantitative and problem-solving questions as well as demonstrations. The weighting corresponding to this exam-style assessment is 70%. The remaining 30%, corresponding to the student's autonomous work, may consist of resolution and submission of proposed problems, practical tests, development of individual or group projects, submission of reports/memos, or, if applicable, personal interviews with students, delivering short seminars, etc. **A minimum grade of 3.5 out of 10 in the exam-style assessment (theoretical-practical exams) is required**. The overall grade will therefore correspond to the weighted sum of the grades obtained in the two separate pieces of assessment (exams and homework) as long as the minimum grade is achieved.

- RE-SIT EXAMINATION
  The re-sit examination will consist of a single piece of assessment. The exam may consist of theoretical, quantitative and problem-solving questions as well as demonstrations. The final grade will be equally split between fundamentals (50%) and applications (50%).

- SINGLE FINAL ASSESSMENT
  Following the protocol and within the strict deadline set by the University of Granada, students who wish to do so may choose the single evaluation model. This evaluation will consist of a theoretical-practical exam, which may include theoretical questions, demonstrations, and problem-solving, with a 100% grade (50% for the fundamentals part and 50% for the applications part). Please note that there is a **deadline** for selecting this option, typically set no later than two weeks from your registration for this course. For further information, please refer to the additional details provided at this link.

  All aspects related to evaluation will be governed by the regulations on teaching planning and examination organisation in force at the University of Granada. The grading system will be expressed using numerical grades in accordance with the provisions of Article 5 of Royal Decree 1125/2003, of September 5, which establishes the European credit system and the grading system for official university degrees valid throughout the national territory.

Course Policies

- Attendance: Regular attendance is expected.
- Late Submissions: Late assignments will incur a penalty of 10% per day unless an extension is granted.
- Academic Integrity: plagiarism or any form of cheating will not be tolerated. Violations will be dealt with according to university policies.

Disclaimer

PART I

Alessandro Patti

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Chapter 1

Introduction to Statistical Physics

1.1 Introduction

Statistical physics is the branch of physics that studies systems composed of a large number of constituents (or *degrees of freedom*). Its objective is to establish a connection between the essential properties of the constituents (*e.g.* atoms, molecules, particles) of a given system and its macroscopic or collective properties. Theoretically, one might determine the complete set of relevant properties of a system if positions and velocities of its constituents were completely known, that is if their dynamics and a model describing their mutual interactions were available. However, this approach would be immediately rejected for two main reasons: (1) describing the time evolution of, say, one mole (roughly 10^{23}) of particles is technically impossible and (2) the time needed to observe a system and measure its properties in a given configuration is much longer than the characteristic time over which that system changes on a microscopic scale due to the collisions between its constituents. In other words, by the time we measure position and velocity of each atom, molecule or particle, these positions and velocities have already changed many times and the system has explored a huge number of *microstates* [1]. What one actually measures in experiments is an average of individual microstates. We therefore conclude that mapping the transition dynamics between two subsequent microstates is not only computationally prohibitive, but also uninteresting. Therefore, it makes more sense to seek a statistical approach.

Considering the system's constituents as microscopic entities, we can say that statistical physics aims to explain the macroscopic properties of matter in terms of its microscopic properties. Nature has a hierarchical structure, composed of various scales: ..., quarks, elementary particles (leptons, bosons), atoms, molecules, macromolecules, gases/liquids/solids, ..., planets, stars and planetary systems, galaxies, clusters and superclusters of galaxies. For example, in a gas, we refer to the microscopic level as defined by the characteristic length and time scales of the atoms that compose it, and the macroscopic level as defined by the length and time scales at which measurements are made in a laboratory. The physical laws that govern them can have different appearances. For example, at microscopic scales, temporal reversibility can exist, but is lost at large scales. By contrast, spontaneous symmetry breaking, which does not happen microscopically, can occur in macroscopic systems. In particular, a hierarchical description appears when the microscopic level has many degrees of freedom, and the typical length and time scales are very different at both levels. The definition of micro-constituents and macro-constituents depend on the type of description in which we are interested:

- In the case of a gas, liquid, or solid, the micro-constituents are the molecules, while the macro-constituents are represented by the collective behaviour of the molecules as a whole.
- In a laser, the micro-constituents are the individual photons, and the macro-constituents are the overall properties of the large number of photons.

[1] We still have to clarify what a microstate is, so don't worry if the term sounds new to you. Imagine, for the time being, that it is a set of different configurations, images or pictures of the same system.

- For a galaxy, the micro-constituents are the stars, and the macro-constituents are the gas and the collective behaviour of the stars.
- In a neural network, the micro-constituents are the individual neurons, and the macro-constituents are the overall behaviour and functionality of the network.
- Proteins are composed of aligned sequences of amino acids, where the micro-constituents are the amino acids, and the macro-constituents are the folded protein structure and its functions.
- In ecological populations, the micro-constituents are the individual organisms and species, while the macro-constituents are the interactions and dynamics of the entire population.
- In economics, the micro-constituents are the individual agents engaging in capital exchange, and the macro constituents are the overall economic behaviour and trends.
- The evolution of norms and linguistic conventions arises from the interaction between speakers, where the micro-constituents are the individual speakers, and the macro-constituents are the collective linguistic norms and conventions that emerge from their interactions.

At each level, there are magnitudes that fully characterise the properties of the system at that level. Furthermore, there are certain autonomous relationships (laws) between these magnitudes.

1.1.1 Microscopic and Macroscopic Perspectives

Statistical physics is a branch of Theoretical Physics that aims to provide an explanation for the macroscopic properties of matter (mechanical, thermal, electrical, magnetic, etc.) based on the knowledge of the microscopic properties of its constituents. It employs concepts of **probability** to establish the connection between the microscopic and macroscopic universe. It involves analysing systems of macroscopic dimensions with a very large number of particles, on the order of Avogadro's number, $N_A \approx 6.023 \times 10^{23}$.

To understand the behaviour of a system at the **microscopic** level, we need an interpretative model that describes the particles and their interactions. This model can be based on either *Classical Mechanics* or *Quantum Mechanics*, depending on the nature of the system. In either case, we rely on the system's Hamiltonian, which represents its energy and governs its dynamics. One striking characteristic of the microscopic realm is the vast number of accessible microscopic states. The system's particles exhibit fluctuating and chaotic behaviour, with their positions and velocities rapidly fluctuating over time. This dynamic nature arises from various factors, such as thermal motion, quantum effects, and the intricate interactions among particles. As already noticed above, mathematically resolving the precise motion of particles at this scale is infeasible due to the enormous number of particles involved. Even if we had complete knowledge of their individual motions, it would not be practical for predicting the macroscopic behaviour of the system. This is because the macroscopic properties of the system emerge from the collective behaviour of an enormous number of particles, making it challenging to directly deduce macroscopic phenomena from microscopic dynamics. Furthermore, the experimental observation of the microscopic state of a system is impossible in practice. Even if it were hypothetically feasible, the sheer volume of data would be overwhelming. Handling N_A number of particles is prohibitively demanding, even for the most powerful supercomputers available today. Hence, **statistical physics offers an alternative approach**. Instead of focusing on individual particle details, it seeks to describe and understand the behaviour of macroscopic systems through statistical analysis and probability. By examining the statistical properties of ensembles of particles, statistical physics allows us to derive macroscopic laws and make predictions about observable phenomena.

By contrast, when we examine a system from a **macroscopic** viewpoint, we focus on global quantities that describe the system as a whole. These quantities provide a comprehensive understanding of the system's macroscopic state. In this context, we typically consider only a handful of variables that capture essential aspects of the system's behaviour. For instance, variables like pressure (p), temperature (T), volume (V), internal energy (U), and entropy (S) are commonly used to characterise and analyse macroscopic systems. Unlike the microscopic level, where particles exhibit rapid and chaotic fluctuations in their positions and velocities, macroscopic properties tend to be more stable and consistent. For example, when we repeatedly measure the pressure of a gas in equilibrium, we expect to obtain the same result each time, indicating a predictable and non-fluctuating macroscopic behaviour. Importantly, macroscopic properties are directly accessible through experimental observations. We can measure and quantify these properties in real-world experiments, enabling us to gain insights into the behaviour and properties of macroscopic systems. It is essential to recognise that, when we perform measurements on macroscopic systems, the process takes time. Unlike the idealised concept of an instant measurement, obtaining an accurate measurement of a macroscopic property requires a finite interval of time, τ. During this time interval, the system may undergo **dynamic changes or fluctuations**, which can arise from various sources, such as the inherent thermal motion of particles, external influences, or the system transitioning between different states. As a result, the measured value of the property A in Eq. 1.1 may exhibit some degree of variability or uncertainty. It is crucial to account for this time interval when interpreting experimental results or conducting observations in statistical physics. By considering the duration of the measurement process, researchers can account for the dynamics and fluctuations that might impact the observed macroscopic properties.

$$A = \frac{1}{\tau} \int_0^{\tau} A(t) dt \tag{1.1}$$

As a general rule, even if the measurement time is very short, during that interval, the microscopic state of the system undergoes numerous changes and perturbations, transitioning through a vast number of microscopic configurations. **There exists an enormous number of microscopic states that are compatible with a given macrostate**. Regardless of the brief duration of the measurement, the microscopic constituents of the system are in constant motion and interaction. These interactions lead to continuous fluctuations and rearrangements of the particles, resulting in a multitude of possible configurations. Each configuration represents a unique arrangement of particles with specific positions, velocities, and energy states. These fluctuations and changes occur rapidly at the microscopic level, driven by the inherent dynamics of the system. While the macroscopic properties may appear stable and predictable on average, the underlying microscopic reality is highly dynamic and constantly evolving. It is important to consider this vast number of microscopic states when analysing macroscopic systems using statistical physics. The ability to account for the multitude of microscopic configurations allows us to develop statistical frameworks that capture the collective behaviour of the system and derive macroscopic properties and trends.

The macroscopic behaviour that we observe is a consequence of what occurs at the microscopic level. The properties and interactions of individual particles or constituents at the microscopic scale give rise to the collective behaviour and emergent properties at the macroscopic scale. The macroscopic properties, such as temperature, pressure, and electrical conductivity, are the result of the combined effects and interactions of a large number of microscopic entities. For example, the macroscopic behaviour of a gas, such as its pressure and volume, arises from the motion and collisions of countless individual gas molecules. The macroscopic behaviour of a solid material, such as its strength and elasticity, is determined by the arrangement and interactions of its constituent atoms or molecules. Understanding the microscopic behaviour allows us to explain and predict the macroscopic properties and phenomena. By studying the statistical distributions and average behaviour of the microscopic constituents, we can **establish relationships and laws that connect the microscopic and macroscopic worlds**. Statistical physics provides the theoretical framework to bridge this gap and establish connections between the microscopic and macroscopic scales. It enables us to understand how the properties and interactions at the microscopic level give rise to the macroscopic behaviour that we observe and measure.

To this end, it is necessary to employ **statistics** within a theoretical framework that should be as general as possible. In particular, it should be applicable to various types of particles (atoms, molecules, electrons, photons, etc.) and different states of matter (solids, liquids, gases, plasmas, etc.). It should also be relevant to many physical systems (planetary atmospheres, metallic and dielectric solids, etc.). By employing statistical methods, we can study the collective behaviour and properties of these systems, regardless of their specific composition or state. The principles and techniques of statistical physics can be applied to, *e.g.* understanding the behaviour of gases, the dynamics of liquids, the phase transitions of solids, the behaviour of plasmas, and even complex systems like planetary atmospheres. This generality allows us to uncover universal laws and principles that govern the macroscopic behaviour of diverse physical systems. By using statistical approaches, we can extract valuable insights and make predictions about the behaviour of particles and systems at the microscopic and macroscopic levels, providing a broad understanding of the physical world around us. For the application of statistics to yield good results, the following conditions are necessary:

1. *The ensemble of individual states must be sufficiently large.* Statistical methods rely on the principle of averaging over a large number of entities. The accuracy and reliability of statistical predictions improve as the number of individual states in the system increases.

2. *The individual states in the ensemble should share similar features.* Statistical methods assume some level of similarity or homogeneity among the individual states of the system. This assumption allows us to make generalisations and derive meaningful statistical relationships.

3. *The individuals in the ensemble should be independent.* Statistical methods assume that individual states of the system are independent of each other. This independence ensures that the statistical analysis accurately reflects the behaviour of the system as a whole.

To obtain macroscopic observables as an average of corresponding microscopic quantities, we need to perform an appropriate averaging process. The specific method of averaging depends on the system and the properties under investigation. In statistical physics, the averaging process often involves calculating **ensemble averages** or taking **time averages** over a large number of measurements. While we aim to match the average obtained through statistical analysis with experimental measurements, it is essential to recognise the role of **fluctuations** in the microscopic world. Fluctuations occur due to the inherent probabilistic nature of microscopic phenomena. While the average behaviour can be accurately described by statistical methods, individual measurements may deviate due to these fluctuations. Statistical physics provides a framework to understand and quantify the impact of fluctuations. By examining statistical properties, such as variances and probability distributions, we can gain insights into the magnitude and significance of fluctuations and determine the range of expected deviations from the average behaviour.

1.1.2 Key Learning Objectives

Let's explore a few examples that will help us better understand the general objective of statistical physics.

- *A container with helium gas.*

  At the **microscopic** level, helium atoms interact very weakly, and it is a good approximation to consider them as classical free particles. The variables that describe the system at this level are the positions and linear momenta of all helium atoms. Classically, their temporal evolution is straightforward: the atoms move in straight lines (following Newton's laws) until they collide with the walls of the container and rebound. At the **macroscopic** level, helium gas in thermodynamic equilibrium follows the equation of state of an ideal gas: $PV = Nk_BT$, where N represents the number of atoms, k_B is the Boltzmann constant, and V is the volume of the container, regardless of the kinetic state of each individual atom. From the microscopic perspective, what are T and P? Why are they related? From the microscopic perspective, T represents the temperature of the gas, which is a measure of the average kinetic energy of the helium atoms. It quantifies how fast the atoms are moving on average. P represents the pressure exerted by the gas on the walls of the container. It arises due to the collisions of the helium atoms with the container walls, as they bounce off and change their momentum. The pressure is related to the rate of change of momentum due to these collisions. The relationship between T and P emerges from statistical physics. The average kinetic energy of the helium atoms is proportional to T. Meanwhile, the pressure exerted by the gas is proportional to the average momentum change during collisions, which is directly related to the kinetic energy and, therefore, to temperature. This relationship is encapsulated in the above-mentioned ideal gas law. It demonstrates that, at a macroscopic level, temperature and pressure are connected and can be described by the ideal-gas equation of state.

- *Water.*

  At the **microscopic** level, water is simply a vast collection of H_2O molecules. However, at the **macroscopic** level, it can exhibit various distinct structures or phases, each with different and well-known properties, despite the fact that the molecules and their interactions remain the same. These different phases of water demonstrate how macroscopic properties can emerge from the same microscopic constituents (H_2O molecules) but with varying arrangements and degrees of freedom.

- *A piece of copper.*

  At the **microscopic** level, a piece of copper is composed of atoms interacting with an effective potential. The variable that describes the system at this level is the wave function. Its temporal evolution is determined by solving the Schrödinger equation. By contrast, at the **macroscopic** level, a piece of copper is a solid at room temperature with well-defined properties. For example, when it is in thermodynamic equilibrium, it has its own equation of state, specific heat capacity as a function of temperature, Young's modulus, hardness, speed of sound propagation, and so on. When it conducts electricity, it is in a non-equilibrium state, and Ohm's law $V = IR$, which is directly measurable, applies.

- *Brownian Motion*

  Brownian motion, first observed by the botanist Robert Brown in 1827, refers to the random and erratic motion exhibited by microscopic particles suspended in a fluid. It is caused by the continuous and random collisions of fluid molecules with the suspended particles. Brownian motion serves as compelling evidence for the existence of atoms and molecules and provides insights into the fundamental nature of matter. This phenomenon has significant implications in various scientific disciplines, including physics, chemistry, and biology. By studying Brownian motion, scientists can uncover important information about diffusion processes, molecular interactions, and the properties of fluids. Additionally, Brownian motion plays a crucial role in fields such as nanotechnology, where it influences particle transport and self-assembly. Statistical physics plays a crucial role in explaining the random motion of particles suspended in a fluid, known as Brownian motion. By considering the statistical distribution of particle velocities and collisions, we can accurately describe and predict the macroscopic diffusion behaviour observed in Brownian motion.

- *Neural Systems.* Neural systems are incredibly complex, consisting of a vast number of neurons interconnected through intricate networks. At the microscopic level, each neuron communicates with others through electrical and chemical signals, forming a complex web of connections and interactions. While we have knowledge of the individual behaviour of neurons and their mechanisms, understanding how these individual elements come together to perform complex cognitive functions remains a significant challenge.

These examples illustrate how statistical physics allows one to bridge the gap between microscopic and macroscopic phenomena. By applying statistical methods and analysing ensembles of particles, we can derive laws, principles, and predictive models that help us understand and describe a wide range of physical systems. An important observation is that, in general, the whole is greater than the sum of its parts. This means that the overall behaviour of a system composed of many bodies does NOT usually coincide with the sum of the individual behaviours of its elementary constituents. Thus, statistical physics is essentially a nonlinear theory. It is also relevant to note that there are many microscopic states that give rise to the same macroscopic state. As we will see, Boltzmann correctly identified this degeneracy with the thermodynamic concept of *entropy*. Many details of the interactions between particles at the microscopic level do not qualitatively influence their macroscopic behaviour. Therefore, it is possible to formulate *highly simplified models* that capture the essential ingredients and provide excellent results. In light of these considerations, the general objective of Statistical Physics is to relate the microscopic level to the macroscopic level of a given system. Some specific objectives include:

- *Determining the range of applicability of macroscopic descriptions.* For example, the time and spatial scales in which they operate, the validity range of macroscopic parameter values, etc.

- *Characterising and understanding cooperative phenomena.* In general, there are macroscopic properties that are not a result of linearly superposing microscopic behaviours. Examples include phase transitions.

- *Deducing new levels of description.* There are systems where we do not have "macroscopic laws" but suspect their existence, such as economics, neuroscience, non-equilibrium systems, etc. There are also mesoscopic (intermediate) levels that exhibit their own regularities (e.g. fluctuating hydrodynamics).

In this course, we will focus on studying **equilibrium** Statistical Physics. That is, we will develop the theory that connects the microscopic and macroscopic levels of systems in thermodynamic equilibrium. Within this context, systems show a *microscopic description* provided by Classical or Quantum Mechanics (depending on the situation), and a *macroscopic description* provided by Thermodynamics, which is based on a few principles deduced from a set of experiments. Our aim is to establish a relationship between these two levels of description and explore the connections between the microscopic behaviour of individual constituents and the macroscopic properties of the system as a whole.

1.1.3 Statistical Physics and Thermodynamics

Another fundamental objective of statistical physics is to deduce the laws or principles on which thermodynamics is based. These principles form the foundation of thermodynamics and provide important insights into the behaviour of systems at the macroscopic level. Statistical physics aims to understand the statistical nature of these principles, studying the probabilistic behaviour of microscopic particles and their collective effects. They can be summarised as follows:

1. *Zeroth Principle.* If two bodies are in equilibrium with a third body, they are also in equilibrium with each other. The Zeroth Principle establishes the concept of thermal equilibrium, where systems that are in equilibrium with a common third system are also in equilibrium with each other. This principle allows us to define temperature and establish the basis for the measurement and comparison of thermal states.

2. *First Principle.* Although energy can take many forms, the total amount of energy remains constant. The First Principle, also known as the law of *conservation of energy*, states that the total energy of an isolated system remains constant. This principle is of utmost importance in understanding the interplay between different forms of energy, such as heat, work, and internal energy.

3. *Second Principle.* There is no cyclic device that can absorb heat from a single source and convert it entirely into work. The Second Principle, often associated with the concept of entropy, highlights the irreversibility of certain processes. It states that in any natural process, the entropy of an isolated system tends to increase, leading to a more disordered state. This principle provides valuable insights into the directionality of physical phenomena and the limitations on energy conversion.

4. *Third Principle.* The Third Principle, known as Nernst's theorem, sets a limit on cooling processes by stating that it is impossible to reach absolute zero temperature through a finite number of processes. This principle elucidates the unattainability of absolute zero and its implications for the behaviour of matter at extremely low temperatures.

From these experimentally deduced principles, a self-consistent and comprehensive theory is developed that describes the macroscopic physics of a wide range of equilibrium systems. However, there are numerous questions that cannot be answered within the framework of thermodynamics alone. Some fundamental questions include:

- Why do the laws of thermodynamics hold perfectly regardless of the type of atoms and/or molecules that compose substances (reduction of variables and universality)?
- Do these laws apply to any system with any type of interaction? What are the limits of applicability?
- What role does the microscopic dynamics play in the macroscopic behaviour of a system?
- Do the laws of thermodynamics depend on the quantum or classical nature of the microscopic system?
- Why are processes irreversible if the microscopic laws of motion are temporally reversible?

Addressing these questions requires entering the realm of statistical physics and **bridging the gap between the microscopic and macroscopic levels of description**. While thermodynamics provides powerful insights into the equilibrium behaviour of macroscopic systems, statistical physics explores the statistical nature and microscopic underpinnings of these macroscopic laws. The *reductionist* approach, where the macroscopic behaviour emerges from the collective behaviour of a large number of microscopic constituents, is a key concept in understanding the *universality* of thermodynamic laws. By studying the statistical distribution of microscopic states and the probabilistic behaviour of particles, statistical physics provides a framework to answer these fundamental questions. The interplay between microscopic dynamics and macroscopic behaviour is a central focus of statistical physics. Understanding how the collective behaviour of particles gives rise to macroscopic observables, such as temperature, pressure, and entropy, requires analysing the statistical properties of the system. Additionally, the question of irreversibility in the face of reversible microscopic laws of motion is a topic of ongoing research and debate. The study of non-equilibrium and far-from-equilibrium systems explores the origins of irreversibility and the emergence of complex behaviour. In summary, while thermodynamics provides powerful principles and laws for describing equilibrium systems, statistical physics goes beyond and addresses fundamental questions related to the **universality**, **applicability**, **microscopic dynamics**, and **irreversibility** of thermodynamic behaviour. By integrating statistical methods, probabilistic reasoning, and a deep understanding of microscopic interactions, statistical physics offers insights into the foundations of thermodynamics and sheds light on the behaviour of systems at both the microscopic and macroscopic levels.

An interesting example in the early stages of Statistical Physics is given by the specific heat, which is defined as the amount of heat required to raise the temperature of a substance or thermodynamic system by one unit per unit mass. In general, the value of specific heat depends on temperature:

- In an ideal gas, the specific heat is constant at any temperature.
- For a crystalline solid, the specific heat tends to approach zero as the temperature decreases and tends towards a constant value at high temperatures.
- In a magnetic solid, the specific heat exhibits a maximum at a certain temperature.
- During phase transitions, the specific heat shows a pronounced peak or divergence.

The variations in specific heat depend on several factors, such as energy levels, vibrational modes, magnetic interactions and phase transitions, and understanding them requires a link to microscopic physics. Statistical physics provides tools and theoretical frameworks to connect the macroscopic behaviour, as reflected in specific heat variations, with the underlying microscopic physics. Through statistical ensembles, probabilistic reasoning, and the analysis of collective behaviour, statistical physics allows us to elucidate the connections between specific heat and the microscopic constituents of a system[2].

More practical questions arise:

- How can we obtain the equation of state or the thermodynamic potentials from the microscopic properties of the system?
- Water is a liquid between 0°C and 100°C. Why does it solidify at -0.00000001°C or vaporize at 100.00000001°C? The interactions between water molecules do not change with temperature. What is happening?

[2]For an **ideal gas**, the constant specific heat is related to the translational degrees of freedom of the gas particles, such as their kinetic energy. The energy levels of the particles are evenly spaced, resulting in a uniform increase in energy with temperature. In a **crystalline solid**, the specific heat is influenced by the vibrational modes of the lattice. At low temperatures, the lattice vibrations decrease, leading to a decrease in specific heat. At higher temperatures, additional degrees of freedom, such as phonons, contribute to the specific heat, causing it to approach a constant value. In a **magnetic solid**, the specific heat is affected by the presence of magnetic moments and their interactions. The maximum in specific heat occurs at the Curie temperature, where the magnetic ordering undergoes a transition. The specific heat reflects the energy required to disrupt or reorient the magnetic moments in the material. During **phase transitions**, the specific heat exhibits pronounced features due to the energy involved in the change of phase. The peak or divergence in specific heat is associated with the absorption or release of latent heat during the transition between different phases

- How do the thermodynamic potentials change during a phase transition? Are there mathematical singularities? What type?
- Under what conditions does a first- or second-order phase transition occur?

Equilibrium Statistical Physics, also known as Statistical Mechanics, is capable of answering all of the above questions and many others. It allows us to understand how the **microscopic and macroscopic descriptions of a system are related**, enabling us to derive the values of macroscopic quantities from their microscopic interactions. The formal mathematical framework that establishes the micro-macro connection in equilibrium systems has a much broader range of applicability, with applications in *neuroscience*, *ecology*, *sociology* and other fields. In this sense, it is clear that Statistical Physics is an intrinsically interdisciplinary science, and that is where its success and popularity lie in recent decades, leading to what is now known as the science of complex systems. The interdisciplinary nature of Statistical Physics allows researchers to apply its concepts and methodologies to a wide range of complex systems, fostering collaborations between physicists, biologists, economists, sociologists, and other scientists. This interdisciplinary approach leads to new insights, connections, and a deeper understanding of the fundamental principles governing complex systems in nature and society. In summary, Statistical Physics, or Statistical Mechanics, is a powerful and versatile framework that not only provides a comprehensive understanding of equilibrium systems, but also serves as a unifying language for investigating complex systems in various disciplines. Its success lies in its ability to bridge the gap between the microscopic and macroscopic realms, paving the way for a deeper understanding of the complexities of nature.

1.1.4 A historical perspective

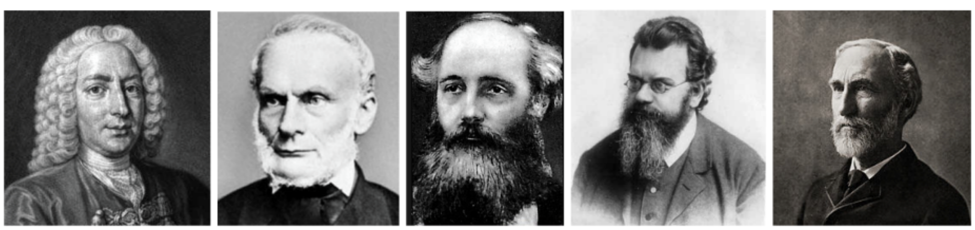


Figure 1.1: From left to right: Daniel Bernoulli (1700-1782), Rudolf Clausius (1822-1888), James Clerk Maxwell (1831-1879), Ludwig Boltzmann (1844-1906), and Josiah Willard Gibbs (1839-1903).

- Bernoulli (1738) and Herapath (1821) established that pressure is a consequence of the motion of atoms. They deduced the ideal gas law ($PV = Nk_BT$) using kinetic theory.
- Clausius (1859) and Maxwell (1860) rigorously developed the arguments of Bernoulli, working on the molecular distribution of velocities, transport coefficients (conductivities, viscosities, etc.), and further advancing the kinetic theory.
- Boltzmann (1868-1871), who is considered the father of Statistical Physics, made significant contributions to the field. Among many other things:
    1. He derived his famous formula (Boltzmann factor) that relates the energy of a state to its probability of occurrence. This established a connection between probability and entropy.
    2. He formulated the Equipartition Theorem, which states that in thermal equilibrium, each degree of freedom of a system contributes equally to its total energy.
    3. Boltzmann culminated the kinetic theory with his development of the Boltzmann Equation, which describes the temporal evolution of the particle density in phase space (positions and velocities). This equation is a powerful tool for studying the macroscopic properties of systems out of equilibrium.

4. He formulated the H-theorem, providing a microscopic justification for the second law of thermodynamics (increase of entropy). It explains on a molecular level the tendency of a physical system to relax towards states of thermodynamic equilibrium.

Ludwig Boltzmann's contributions laid the foundation for understanding the statistical behaviour of systems and their connection to macroscopic observables. His work revolutionized the field of Statistical Physics and significantly advanced our understanding of equilibrium and non-equilibrium systems.

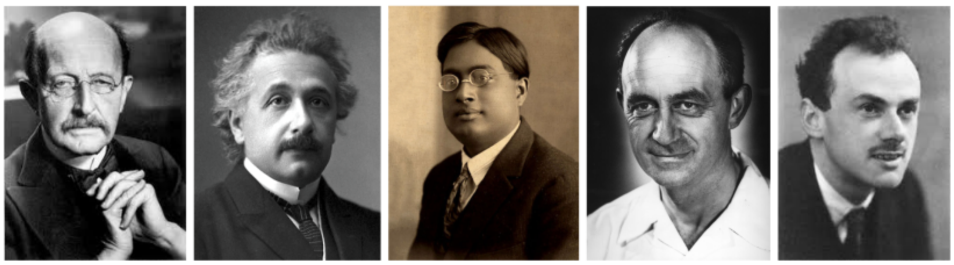


Figure 1.2: From left to right: Max Planck (1858-1947), Albert Einstein (1879-1955), Satyendra Nath Bose (1894-1974), Enrico Fermi (1901-1954), and Paul Dirac (1902-1984).

- The H-theorem and the irreversibility it implied were heavily criticized during Boltzmann's time, mainly by Lochsmith (conflict with microscopic reversibility) and Zermelo (conflict with the quasi-periodicity of dynamics). Mach and Ostwald also strongly criticised the molecular basis of Boltzmann's theory. These attacks pushed Boltzmann to the brink of despair. Unfortunately, he took his own life in 1906 during a vacation in Duino (Italy, near Trieste).
- From 1908 onwards, various discoveries began to confirm Boltzmann's theories and the acceptance of the existence of atoms.
- Gibbs (1902) further developed and formalised the theory of ensembles, consolidating the statistical formulation of Boltzmann's work. This laid the foundation for modern Statistical Mechanics.
- Planck (1901) formulated the law of black-body radiation.
- Einstein (1905) applied stochastic techniques to explain Brownian motion, providing evidence for the atomic theory. He also explained the photoelectric effect and introduced the idea of quantized light or photons, which became a cornerstone of the quantum revolution.
- Bose and Einstein (1924) made significant contributions. Bose derived Planck's radiation law by considering black-body radiation as a gas of photons. Einstein emphasised the crucial role of particle indistinguishability. They defined the Bose-Einstein statistics, and Einstein discovered the phenomenon of Bose-Einstein condensation.
- Fermi and Dirac (1926): Following the formulation of Pauli's exclusion principle, Fermi demonstrated that certain physical systems obey a different statistics, known as Fermi-Dirac statistics.
- Landau and von Neumann (1927): Introduced the concept of density matrix in quantum mechanics and generalised the theory of ensembles to the quantum case.
- Belifante (1939) and Pauli (1940): Discovered the spin-statistics connection and established which type of particles obey Bose-Einstein statistics and which obey Fermi-Dirac statistics.
- van Hove (1949), Yang and Lee (1952), Ruelle (1963), Fisher (1964), Dyson and Lenard (1967), Lebowitz and Lieb (1969): Theorems of the existence of the thermodynamic limit.

- Wilson, Fisher, Kadanoff, and Widom (1970-1975): Precise theoretical description of phase transitions using renormalisation group as a tool.
- Gallavotti, Cohen, Lebowitz, Spohn, Jona-Lasinio, Jarzinsky, Crooks, and others (¿ 1990): Significant advancements in non-equilibrium statistical physics, formulation of fluctuation theorems.

Figure 1.3: Lev Landau (1908-1968), John von Neumann (1903-1957), and Wolfgang Pauli (1900-1958)

1.2 Equilibrium and Fluctuations

In statistical physics, the concepts of equilibrium and fluctuations play crucial roles in understanding the behaviour of physical systems. Let's explore these concepts. A system is said to be in equilibrium when its macroscopic properties remain constant over time and its microscopic constituents are in a balanced state. In equilibrium, the system reaches a stable configuration where there is no net flow of energy or particles. This state allows for the characterisation of the system's macroscopic properties using thermodynamic quantities such as temperature, pressure, and entropy. The study of equilibrium systems provides insights into their stability, phase transitions, and the relationships between different thermodynamic variables. Fluctuations refer to the spontaneous and random variations observed in a system, even when it is in equilibrium. These variations arise from the inherent probabilistic nature of microscopic processes and can affect different properties of the system, such as energy, density, or particle number. Fluctuations are essential as they provide information about the underlying dynamics and interactions of the system's constituents. By studying fluctuations, one can gain insights into the statistical properties of the system and uncover hidden patterns and correlations.

The relationship between equilibrium and fluctuations is intricately connected. While equilibrium represents a state of balance and stability, fluctuations remind us that even in equilibrium, there are inherent fluctuations at the microscopic level. These fluctuations can give rise to small deviations from the average behaviour and contribute to the richness and complexity of physical systems. Statistical physics provides a framework to quantitatively describe and analyse both equilibrium and fluctuation phenomena. Through statistical methods, such as probability distributions, correlation functions, and ensemble averages, researchers can quantify the extent of fluctuations, study their statistical properties, and derive relationships between fluctuations and macroscopic observables. Understanding equilibrium and fluctuations is essential for various fields, including condensed matter physics, thermodynamics, statistical mechanics, and many branches of natural sciences. It allows us to comprehend the behaviour of systems ranging from simple gases to complex materials, biological systems, and even the dynamics of the universe itself.

Let's consider the particular case of a system in equilibrium. For this system, it is known that all its macroscopic properties remain unchanged over time. When measuring a macroscopic observable, $X = \langle x \rangle$, with x the microscopic associated magnitude, we always obtain the same fixed value. This measurement is carried out over a time interval that is very large from a microscopic perspective. In particular, the absolute fluctuation around the mean reads

$$\delta x = x - \langle x \rangle = x - X \tag{1.2}$$

and the associated mean-squared deviation is

$$\langle \delta x^2 \rangle = \langle (x - X)^2 \rangle = \langle x^2 \rangle - X^2 = \langle x^2 \rangle - \langle x \rangle^2 \tag{1.3}$$

It follows that the relative fluctuation around the mean can be estimated as follows

$$\frac{\sqrt{\langle \delta x^2 \rangle}}{\langle x \rangle} = \frac{\sqrt{\langle x^2 \rangle - \langle x \rangle^2}}{\langle x \rangle} \tag{1.4}$$

A question that now arises is the following. How is it possible to interpret the behaviour of a macroscopic system in equilibrium through the fluctuating microscopic world? The key to this interpretation lies in the concept of statistical averages. Although microscopic properties fluctuate, macroscopic properties emerge as statistical averages or averages of the microscopic properties. Due to the **large number of particles in a macroscopic system**, these fluctuations tend to average out, and the collective behaviour of the system becomes more predictable and stable. Equivalently:

$$\lim_{N \to \infty} \frac{\sqrt{\langle \delta x^2 \rangle}}{\langle x \rangle} = 0 \tag{1.5}$$

As the number of particles, N, increases, it becomes more challenging to treat the system mechanically (as the number of degrees of freedom and chaos increase), but it becomes easier to treat it statistically. The inherent chaos in the system plays a beneficial role by allowing us to study the system at a macroscopic level, leveraging statistical methods to extract meaningful information and understand its behaviour.

Example 1.1

Consider a system consisting of N particles within a container divided into two compartments by a permeable wall, as reported in Fig. 1.4. The particles are identical and distinguishable, meaning that each particle has its own unique identity. The compartments have equal volumes, and the permeable wall enables the particles to move freely between the two regions.

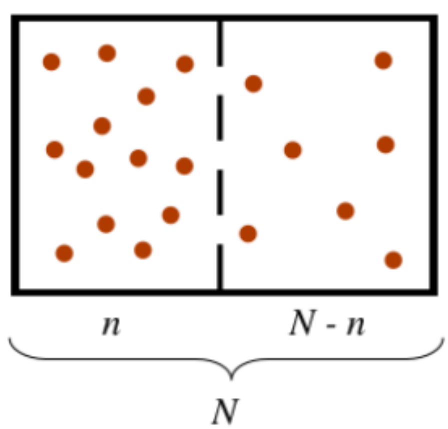


Figure 1.4: N particles randomly distributed in a container divided into two compartments by a permeable wall.

Since the particles do not interact with each other, their behaviour is independent. Each particle follows its own trajectory within its compartment, undergoing random motion due to collisions with the container walls. The particles are free to cross the permeable wall, moving from one compartment to the other. At the microscopic level, the positions and velocities of individual particles fluctuate due to their independent motion. However, at the macroscopic level, we can analyse the system statistically. More specifically, the total number of possible distributions of N particles in the two compartments is $N_T = 2^N$, as also reported in Fig. 1.5. Since the volumes are equal, each of these configurations has an equal probability of occurring. These different possibilities are referred to as **microstates** or microscopic states. There are groups of microstates that give rise to the same macroscopic properties, which are referred to as **macrostates**.

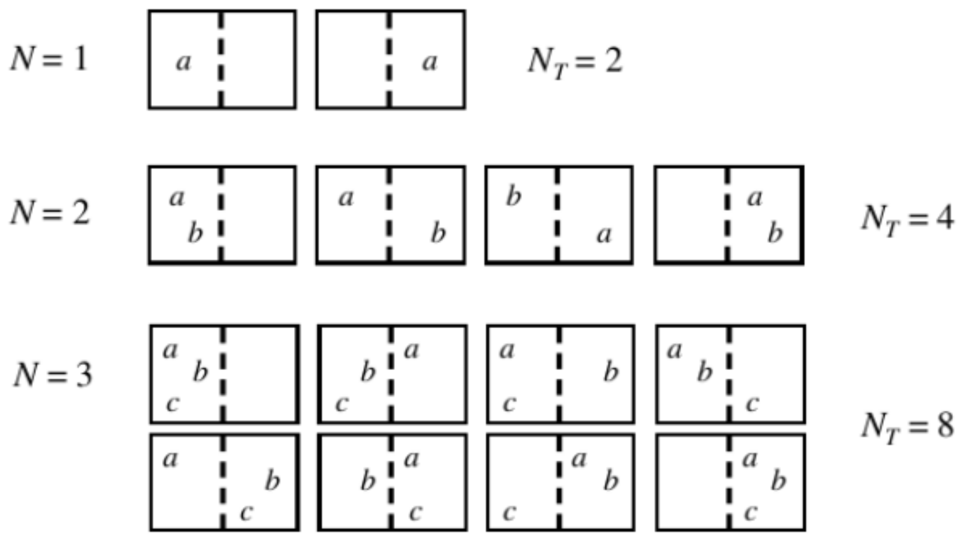


Figure 1.5: Possible distributions of $N = 1, 2, 3$ particles in the two compartments of a container.

In the context of this example system, a microstate would involve providing a detailed description of which particles are present in each compartment (the particles are distinguishable!). For instance, a microstate could be particle a and particle b in compartment A, and particle c in compartment B. Each microstate corresponds to a specific arrangement of particles within the compartments. In contrast, a macrostate is characterized by the overall distribution of particles, without considering their specific identities. For example, a macrostate could be 2 particles in compartment A, 1 particle in compartment B, which indicates the total number of particles in each compartment. The macrostate describes the macroscopic properties of the system, such as the particle density or the average number of particles in each compartment. For example, the macrostate associated to the system of Fig. 1.5 may be defined by specifying the total energy, volume, and number of particles. Within this macrostate, there can be numerous microstates that correspond to different arrangements of individual particles with the same macroscopic properties. It is important to note that while all microstates are equally probable, the same cannot be said for macrostates. Each macrostate corresponds to a different number of compatible microstates. Some macrostates may have a larger number of compatible microstates, resulting in higher probabilities, while others may have fewer compatible microstates, leading to lower probabilities.

1.2.1 Some insight into combinations and probabilities

In this section, we expand the ideas introduced in the above example and suggest the interested student to read Appendix A for further details on theory of probability. Here, we would like to estimate the number of microstates, $W_N(n)$, that are compatible with a given macrostate consisting of n particles in the left compartment and $N-n$ particles in the right compartment. Because both compartments are supposed identical, left and right are interchangeable. To this end, we need to apply *combinatorial principles*. More specifically, we have a total of N particles, and we want to determine the number of ways we can distribute n particles in the left compartment and $N-n$ particles in the right compartment. This can be visualised by arranging N particles in a line and choosing n of them to be in the left compartment, with the remaining $N-n$ particles automatically assigned to the right compartment. The number of microstates can be calculated using the binomial coefficient $\binom{N}{n}$. Mathematically, it can be expressed as:

$$W_N(n) = \binom{N}{n} = \frac{N!}{n!(N-n)!} \tag{1.6}$$

The binomial coefficient accounts for all possible ways to choose n particles from the total N particles, considering that the order of selection does not matter and each arrangement is considered a distinct microstate. By evaluating the binomial coefficient, we can determine the number of microstates that correspond to the given macrostate of having n particles in the left compartment and $N-n$ particles in the right compartment. For instance, if $N=5$ and we choose to have $n=2$ particles in the left compartment, then we have $W_5(2) = \frac{5!}{2!3!} = \frac{120}{12} = 10$ possible combinations of arranging our 5 particles in the two compartments.

Let's take a step further. What is the probability, $P_N(n)$, of observing a macrostate with n particles on a side and $N-n$ particles on the other side? This probability is given by the ratio of the number of microstates corresponding to that macrostate to the total number of microstates in the system. We have already mentioned that the total number of microstates or possible distributions of N particles across the whole system is $N_T = 2^N$. Therefore,

$$P_N(n) = \frac{\text{favourable configurations}}{\text{total configurations}} = \frac{W_N(n)}{N_T} = \frac{N!}{n!(N-n)!2^N} \tag{1.7}$$

It is important to note that in some cases, this probability may also depend on other factors such as energy considerations, constraints, or specific interactions present in the system. These additional factors can influence the likelihood of a macrostate occurring and may require additional considerations in the probability calculation.

We first notice that the probability of finding all particles in a single compartment, that mathematically implies that $n = N$, would result in $P_N(n = N) = 1/2^N \to 0$ for large N as can be easily proved by applying Eq. 1.7. This result indicates that it is highly unlikely to find particles occupying only a single compartment. That said, it makes sense to ask the questions: what is the most probable state, $n = n^*$, and what is the probability $P_N(n^*)$ of observing that state? In other words, what is the most probable number of particles that one would find in a compartment? The most probable state is the one that maximises the probability $P_N(n)$:

$$\frac{\partial P_N(n)}{\partial n}|_{n^*} = 0 \tag{1.8}$$

To differentiate the probability in Eq. 1.7 with respect to n, we first notice that

$$\frac{\partial P_N(n)}{\partial n}|_{n^*} = 0 \Leftrightarrow \frac{\partial \ln(P_N(n))}{\partial n}|_{n^*} = 0 \tag{1.9}$$

and

$$\frac{\partial \ln(P_N(n))}{\partial n}|_{n^*} = \frac{\partial[\ln N! - \ln n! - \ln(N-n)! - N\ln 2]}{\partial n}|_{n^*} \tag{1.10}$$

Then, we apply the Stirling's approximation, which reads

$$\ln N! = N \ln N - N + \mathcal{O}(\ln N), \tag{1.11}$$

where the big O notation indicates that, for sufficiently large N, the difference between $\ln N!$ and $N \ln N - N$ will be at most proportional to the logarithm[3]. Substituting this result into Eq. 1.9, one gets:

$$\frac{\partial \ln P_N(n)}{\partial n}|_{n^*} = \frac{\partial [N \ln N - N - n \ln n + n - (N-n)\ln(N-n) + (N-n) - N \ln 2]}{\partial n}|_{n^*} = 0 \tag{1.12}$$

Finally, we obtain

$$\frac{\partial \ln P_N(n)}{\partial n}|_{n^*} = \frac{\partial [N \ln N - n \ln n - (N-n)\ln(N-n) - N \ln 2]}{\partial n}|_{n^*} = \ln \frac{N-n^*}{n^*} = 0 \tag{1.13}$$

It follows that

$$\frac{N-n^*}{n^*} = 1 \tag{1.14}$$

and

$$n^* = \frac{N}{2} \tag{1.15}$$

The most probable state is therefore the one that favours a homogeneous distribution of particles across the two compartments. if there were s compartments, the most probable state would be $n_1 = n_2 = n_3 = ... = N/s$.

In light of these results, we know turn our attention to the average value of the distribution, $\langle n \rangle$, given by:

$$\langle n \rangle = \sum_{n=0}^{N} n P_N(n) \tag{1.16}$$

Our aim is obtaining $\langle n \rangle$. To this end, we will show that $P_N(n)$ is actually a Gaussian (normal) distribution with average $\langle n \rangle$. Let's expand the function $\ln P_N(n)$ in Taylor series around $n^* = N/2$ up to the second

[3]Stirling's approximation is a mathematical approximation used in various areas of mathematics and physics. It provides an approximation for the factorial of large numbers N. The approximation states that for large values of N, $N!$ can be approximated by $N! \approx \sqrt{2\pi N}(N/e)^N$. As will shall see later on, Stirling's approximation is particularly useful in statistical physics when dealing with large systems and calculating the entropy or the number of microstates associated with a macrostate. By using Stirling's approximation, we can simplify calculations and obtain approximate values for factorial terms that would otherwise be computationally demanding. It is important to note that Stirling's approximation becomes more accurate as N gets larger, and the relative error decreases significantly. However, for small values of N, it may not provide accurate results and should be used with caution.

order:

$$\ln P_N(n) \approx \ln P_N(n^*) + \left.\frac{\partial \ln P_N(n)}{\partial n}\right|_{n^*}(n-n^*) + \frac{1}{2}\left.\frac{\partial^2 \ln P_N(n)}{\partial n^2}\right|_{n^*}(n-n^*)^2 + ... \tag{1.17}$$

The second order term can be simplified according to Eq. 1.13:

$$\frac{\partial^2 \ln P_N(n)}{\partial n^2} = \frac{\partial}{\partial n}[\ln(N-n) - \ln n] = -\frac{1}{N-n} - \frac{1}{n} \tag{1.18}$$

In particular at $n^* = N/2$ we get

$$\frac{\partial^2 \ln P_N(n)}{\partial n^2}|_{n^*=N/2} = -\frac{2}{N} - \frac{2}{N} = -\frac{4}{N} \tag{1.19}$$

Substituting the above results into Eq. 1.17, we obtain:

$$\ln P_N(n) \approx \ln P_N(n^*) + \frac{1}{2}\left(-\frac{4}{N}\right)\left(n - \frac{N}{2}\right)^2 + ... = \ln P_N(N/2) - \frac{2}{N}\left(n - \frac{N}{2}\right)^2 + ... \tag{1.20}$$

Let's apply the exponential function to left and right hand sides:

$$P_N(n) \approx P_N(N/2)e^{\left(-\frac{2}{N}\left(n-\frac{N}{2}\right)^2\right)} = P_N(N/2)e^{-\left(n-\frac{N}{2}\right)^2/(N/2)} \tag{1.21}$$

Eq. 1.21 highlights that $P_N(n)$ is a Gaussian distribution of the type $f(n) = (1/(\sqrt{2\pi}\sigma)e^{-\frac{(n-\langle n\rangle)^2}{2\sigma^2}}$ with $\langle n\rangle = N/2$ and $\sigma = \sqrt{N}/2$. In particular, $\sigma = \sqrt{\langle n^2\rangle - \langle n\rangle^2}$ is the standard deviation of the distribution, whereas σ^2 is the variance, which quantifies the spread or width of a distribution. Out of all the microscopic configurations, approximately 68.3% of the microscopic configurations are located within the interval $[N/2 - \sigma/2, N/2 + \sigma/2]$, as schematically reported in Fig. 1.6.

1.2.2 Introducing fluctuations

Let's use the above example to introduce the concept of **fluctuations**. The fluctuation of a variable is a measure of how much it deviates from its average value. It is calculated by subtracting the average (also known as the mean) of the variable from its actual value. This deviation, here denoted as δn, represents the fluctuation of the variable n:

$$\delta n = n - \langle n\rangle \tag{1.22}$$

It is important to note that fluctuations are often random in nature, meaning they can vary from one measurement to another. This can be appreciated by observing that its average over the same statistical

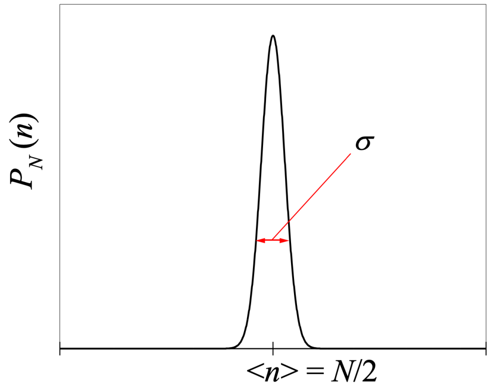


Figure 1.6: Normal distribution of the probability $P_N(n)$ of observing a macrostate with n particles on a side and $N-n$ particles on the other side of the container depicted in Fig. 1.4.

ensemble equals zero:

$$\langle \delta n \rangle = \langle n - \langle n \rangle \rangle = \langle n \rangle - \langle \langle n \rangle \rangle = \langle n \rangle - \langle n \rangle = 0 \tag{1.23}$$

Consequently, the above average cannot be applied to characterise fluctuations' *intensity*. In statistical physics, two commonly used measures to characterise fluctuations are the **root mean squared fluctuation** (r.m.s.) and the **relative fluctuation**:

1. *Root Mean-Squared Fluctuation.* It provides a measure of the average magnitude of the fluctuations. It is calculated by taking the square root of the average of the squared deviations from the mean. Mathematically, it can be expressed as

$$\sqrt{\langle \delta n^2 \rangle} = \sqrt{\langle (n - \langle n \rangle)^2 \rangle} = \sqrt{\langle n^2 - 2n\langle n \rangle + \langle n \rangle^2 \rangle} = \sqrt{\langle n^2 \rangle - 2\langle n \rangle^2 + \langle n \rangle^2} = \sqrt{\langle n^2 \rangle - \langle n \rangle^2} \tag{1.24}$$

   An advantage of the r.m.s. fluctuation is that it has the same dimensionality as the variable itself, so that the ratio between the r.m.s. fluctuation and the average value of the variable is dimensionless and can be employed to characterise the relative intensity of fluctuations. This ratio is defined as the *relative fluctuation.*

2. *Relative Fluctuation.* Also known as the coefficient of variation, the relative fluctuation compares the magnitude of the fluctuation to the average value of the variable. Mathematically, it can be expressed as:

$$\frac{\sqrt{\langle \delta n^2 \rangle}}{\langle n \rangle} = \frac{\sqrt{\langle n^2 \rangle - \langle n \rangle^2}}{\langle n \rangle} \tag{1.25}$$

Using these definitions, we can obtain the values of the root mean squared fluctuation and relative fluctuation for the system of Example 1.1:

$$\sqrt{\langle \delta n^2 \rangle} = \sqrt{\langle n^2 \rangle - \langle n \rangle^2} = \frac{\sqrt{N}}{2}, \tag{1.26}$$

which is the standard deviation of the probability distribution $P_N(n)$ and

$$\frac{\sqrt{\langle \delta n^2 \rangle}}{\langle n \rangle} = \frac{\sqrt{N}/2}{N/2} = \frac{1}{\sqrt{N}}. \tag{1.27}$$

In a macroscopic sample, where the number of particles is on the order of Avogadro's number, the fluctuations are on the order of 10^{-11} and therefore **negligible**. As the width of the distribution tends to zero, the maximum value and the mean always coincide in macroscopic systems. The probability of a macrostate that deviates from the mean value is practically zero, even for macrostates very close to the mean. When we measure the macrostate of a system in equilibrium, the value we will always observe is the mean value. Macroscopically speaking, the system behaves as if it is always in the state $n = N/2$. The same conclusions can be applied to any other macroscopic property $A \equiv \langle a \rangle$ that has a corresponding microscopic counterpart associated with the particles:

$$\frac{\sqrt{\langle a^2 \rangle - A^2}}{A} \propto \frac{1}{\sqrt{N}} \to 0 \tag{1.28}$$

1.3 Conclusions

In this chapter, we have discussed the main goals of Statistical Physics and provided a historical overview of the most relevant advances in the discipline. We have also discussed the concept of Thermodynamic Equilibrium. Equilibrium corresponds to the most probable state of the system (or the mean). The fluctuations in all macroscopic quantities are negligible when the thermodynamic system is large enough. This guarantees that the observables of a macroscopic system in equilibrium take fixed values. The fixed values of macroscopic observables coincide with the average over all possible microscopic configurations. **Fortunately, although the microscopic world is chaotic and fluctuating, the macroscopic observables are quantities that do not fluctuate, and we can consider them as constants**. This property allows us to treat macroscopic observables as stable and predictable quantities, even in the presence of microscopic fluctuations. It is the collective behaviour of a large number of microscopic constituents that gives rise to the stability and predictability of macroscopic systems in equilibrium. By studying the statistical properties and averaging over all possible microscopic configurations, we can effectively describe and understand the macroscopic behaviour of systems in equilibrium. This approach provides a powerful framework for analysing and predicting the properties of complex systems, bridging the gap between the microscopic and macroscopic scales of the physical world.

1.4 Problems

1.1 In a chemical reaction, molecule X decomposes into two different products, Y and Z. The reaction yields a total of n molecules, and the ratio of the number of Y molecules to Z molecules is $2:1$. Determine the number of ways in which the reaction can occur, expressing your answer in terms of combinations and binomial coefficients.

1.2 Obtain the probability of adding up six points if we toss three distinct dice.

1.3 A system consists of three compartments where N identical and distinguishable particles are free to move. The three compartments are divided by a permeable wall. The compartments have equal volumes, and the permeable walls enable the particles to move freely between the three regions. Calculate the most likely number of particles that can be found in each compartment.

1.4 A random variable x is associated with the probability density

$$p(x) = e^{-x}, \quad 0 < x < \infty$$

(a) Find the mean value $\langle x \rangle$.

(b) Two values x_1 and x_2 are chosen independently. Find $\langle x_1 + x_2 \rangle$ and $\langle x_1 x_2 \rangle$.

1.5 The Poisson distribution is a probability distribution that expresses the likelihood of a given number of events occurring in a fixed interval of time or space, given a known average rate of occurrence. It's often used to model events that occur independently of one another in a continuous manner, such as the number of phone calls received at a call center in a given hour, the number of emails received per day, or the number of accidents at a particular intersection in a week. It is useful for situations where the events are rare and random, but the average rate of occurrence is known. It reads

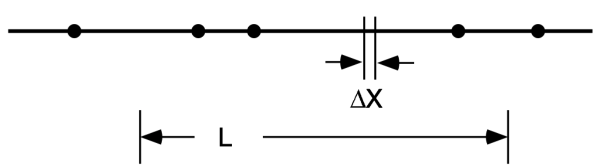
$$P(n) = \frac{e^{-\lambda}\lambda^n}{n!}, \tag{1.29}$$

where λ is the average rate of occurrence of the event. and n is the number of events observed. Show that $P(n)$ is normalised and calculate $\langle n \rangle$ and $\langle (\delta n)^2 \rangle$.

1.6 Probability densities of the Gaussian form are frequently encountered in science and engineering. They occur in thermal equilibrium when the variable makes a quadratic contribution to the energy of the system (the x component of the velocity of an atom in a gas for example). They also occur when the variable is the sum of a large number of small, independent contributions (such as the final displacement after a random walk along a line). Find the mean, variance, and standard deviation for a random variable with a Gaussian probability density. Sketch the cumulative distribution function[4].

1.7 Imagine a situation in which events occur at random along a line X and are governed by the two conditions:

1. In the limit $\Delta X \to 0$ the probability that one and only one event occurs between X and $X + \Delta X$ is given by $r\Delta X$, where r is a given constant independent of X.
2. The probability of an event occurring in some interval ΔX is statistically independent of events in all other portions of the line.



Under these circumstances the probability that exactly n events will occur in an interval of length L is given by the Poisson probability:

$$p(n) = \frac{1}{n!}(rL)^n \mathrm{e}^{-rL}, \quad n = 0, 1, 2, \ldots \tag{1.30}$$

The Poisson probability can be cast in terms of a continuous random variable y as follows:

[4]Problem from *Lecture Notes on Probability for Statistical Physics I* by Thomas J. Greytak, Physics Department, Massachusetts Institute of Technology, 2004

$$p(y) = \sum_{n=0}^{\infty} p(n)\delta(y-n) \tag{1.31}$$

Check the normalisation, find the mean and variance, and show that the density is specified by a single parameter, its mean[5].

[5]Problem from *Lecture Notes on Probability for Statistical Physics I* by Thomas J. Greytak, Physics Department, Massachusetts Institute of Technology, 2004

Chapter 2

Microscopic Descriptions

2.1 Introduction

Statistical physics serves as the bridge that connects the microscopic world of atoms and molecules to the macroscopic phenomena observed in our everyday lives. At its core, this field is concerned with unraveling the mysteries of nature by understanding how countless microscopic particles, when aggregated, give rise to the familiar and predictable behaviour of macroscopic systems. This chapter provides us with the tools and knowledge necessary to comprehend the underlying dynamics of particles at the smallest scales. Here, we shift our focus from macroscopic observables like pressure and temperature to study individual particles – their motions, interactions, and the probabilistic nature of their behaviour. As we walk through the microscopic world, we will uncover the principles governing the behaviour of molecules and atoms. We will decipher the significance of concepts such as phase space, Hamiltonian mechanics, and quantum states, all of which play crucial roles in painting a vivid portrait of the microscopic universe. By the end of this chapter, you will possess a profound understanding of how microscopic details pave the way for macroscopic phenomena.

2.2 Classical Description

In classical physics, the description of a system often involves a fundamental and powerful concept known as **phase space**, here indicated as Γ. Phase space provides a comprehensive mathematical framework for understanding and predicting the behaviour of physical systems. It is particularly essential in classical mechanics for analysing systems with complex configurations and interactions. Phase space provides physicists and engineers with a powerful tool for modelling and analysing complex systems. It allows us to visualise and study the system's evolution over time by tracing its path through the multidimensional space. This approach simplifies the prediction of a system's future behaviour, making it a fundamental concept in classical mechanics and other branches of physics. Understanding phase space is essential for tackling a wide range of problems, from the motion of celestial bodies to the behaviour of particles in a gas. It forms the basis for classical mechanics and serves as a foundation for more advanced topics in physics. At its core, phase space is a multidimensional mathematical space that offers a complete representation of the state of a physical system. Each point within phase space corresponds to a **unique state of the system**. The number of dimensions in phase space equals the number of generalised **coordinates** and generalised **momenta** required to define the system's configuration. For example, in a system with s degrees of freedom, the corresponding phase space would be a $2s$-dimensional space. This concept allows us to track the system's evolution through all possible states by examining its trajectory in phase space.

Let's apply the idea of phase space to mechanical systems, which are of great interest in classical physics. Imagine a mechanical system composed of a specific number of particles, and let's denote the total number of degrees of freedom as s. These degrees of freedom represent the different ways in which the system can move or change. To precisely describe the dynamic state of this mechanical system, we need to know two crucial pieces of information for each degree of freedom: the position and the velocity of every particle in the system. This leads us to a requirement of s position coordinates and s associated momenta, as each degree of freedom has its corresponding position and momentum values. In particular, we can employ generalised coordinates and generalised momenta:

$$\{q_1, q_2, \ldots, q_s, p_1, p_2, \ldots, p_s\} \equiv \{\boldsymbol{q}, \boldsymbol{p}\} \tag{2.1}$$

These generalised coordinates and momenta can correspond to the positions and momenta of the particles within the system. However, they can also represent more abstract quantities such as angles, wave amplitudes, or the various internal degrees of freedom of a molecule.

Example 2.1: A Monoatomic Gas in a Box

Consider a monoatomic gas composed of N molecules confined within a box. In this case, the degrees of freedom are given by $s = 3N$. Its generalized coordinates for positions and momenta can be written as follows:

$$\begin{cases} q_1 = x_1 \quad q_2 = y_1 \quad q_3 = z_1 \\ q_4 = x_2 \quad q_5 = y_2 \quad q_6 = z_2 \\ \ldots \\ q_{3i-2} = x_i \quad q_{3i-1} = y_i \quad q_{3i} = z_i \end{cases}$$

$$\begin{cases} p_1 = m_1\dot{x}_1 \quad p_2 = m_1\dot{y}_1 \quad p_3 = m_1\dot{z}_1 \\ p_4 = m_2\dot{x}_2 \quad p_5 = m_2\dot{y}_2 \quad p_6 = m_2\dot{z}_2 \\ \ldots \\ p_{3i-2} = m_i\dot{x}_i \quad p_{3i-1} = m_i\dot{y}_i \quad p_{3i} = m_i\dot{z}_i \end{cases}$$

In this representation, q_i denotes the position coordinates, while p_i corresponds to the momenta of the i-th particle within the system. These variables, combined with time, allow us to describe the complete state of the system dynamically.

Microstate Definition

A microstate is defined as the set of coordinates and momenta that completely characterise the dynamic state of a system.

$$\textit{microstate} \equiv \{\boldsymbol{q}, \boldsymbol{p}\}$$

2.2.1 Phase Space

The space of phases (or phase space), denoted as Γ, is defined as a $2s$-dimensional metric space, where s is the number of degrees of freedom in the system. The axes of this phase space correspond to the variables $\{q_1, \ldots, q_s, p_1, \ldots, p_s\}$. In the phase space, the microstate of the system is represented by a point. Furthermore, the temporal evolution of the system results in a trajectory within the phase space, traversing different microstates.

2.2.2 Dynamic Functions

In the field of statistical physics, numerous global properties are used to describe a system. Some of these properties are either directly measurable or possess straightforward physical interpretations. For instance, properties like energy, momentum, or angular momentum fall into this category. These properties assume specific values for each microstate $(\boldsymbol{q}, \boldsymbol{p})$, indicating their dependence on the system's dynamic state. These types of properties are known as dynamic functions, typically denoted as $a(\boldsymbol{q}, \boldsymbol{p})$.

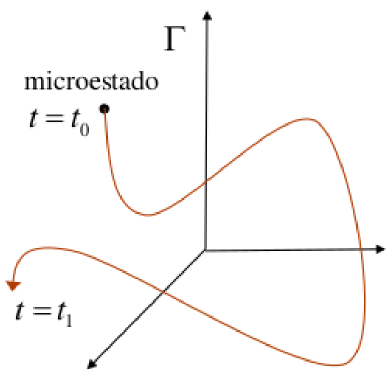


Figure 2.1: Temporal evolution (trajectory) of a system in the phase space Γ, traversing different microstates contained within the red curve. Each microstate is represented by a point, and the system's temporal evolution gives rise to a trajectory as it traverses different microstates.

Dynamic Function	**Macroscopic Observable**	**Physical Interpretation**
Mean Square Displacement	Diffusion coefficient D	Measures how far particles move over time.
Velocity Autocorrelation Function	Momentum transport	Describes how velocity fluctuations persist over time.
Density-Density Correlation Function	Structure factor $S(k,t)$	Tracks density fluctuations in space and time.
Density fluctuations	Dynamic Structure Factor $S(k,\omega)$	Measured in scattering exp.; describes space & time correlations.
Current Autocorrelation Function	Electrical/thermal conductivity	Related to transport properties via Green-Kubo relations.
Magnetisation Autocorrelation Function	Magnetic susceptibility $\chi(\omega)$	Measures how spin/magnetic moments fluctuate over time.

Table 2.1: Dynamic functions and their associated macroscopic observables.

One significant example is the **Hamiltonian** of the system, often represented as $H(\boldsymbol{q},\boldsymbol{p})$. In the case of an isolated and conservative system, $H(\boldsymbol{q},\boldsymbol{p})$ remains independent of time and corresponds to the total energy of the system, denoted as E:

$$H(\boldsymbol{q},\boldsymbol{p}) = E \tag{2.2}$$

In an isolated system, only microstates situated on the hypersurface characterised by the constant energy $H = E$ are accessible. In a classical mechanical system, which is both isolated and conservative, the Hamiltonian function $H(\boldsymbol{q},\boldsymbol{p})$ represents a crucial dynamic quantity. In this context, $H(\boldsymbol{q},\boldsymbol{p})$ signifies the total energy of the system and remains independent of time. It encapsulates the intricate interplay of positions ($\boldsymbol{q}$) and momenta ($\boldsymbol{p}$) of all particles within the system.

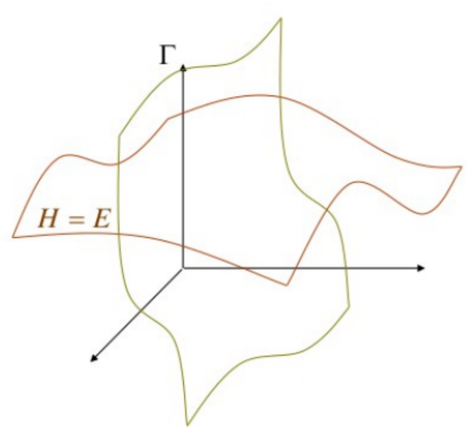


Figure 2.2: Hypersurfaces, characterised by the constant energy $H = E$, in the phase space Γ.

Now, consider the entirety of these dynamic functions $a(\boldsymbol{q}, \boldsymbol{p})$ characterising the system. These functions offer a detailed description of the system's behaviour. For instance, they may represent quantities such as momentum, angular momentum, or energy, all of which have distinct values for each microstate $(\boldsymbol{q}, \boldsymbol{p})$ and thus depend on the dynamic state of the system. As we explore the dynamic evolution of such a system, the trajectory it follows in the phase space reveals a continuous transition between various microstates. This phase space constitutes a metric space with a dimensionality of $2s$, where 's' represents the number of degrees of freedom of the system. Its orthogonal axes align with the variables $\{q_1, \ldots, q_s, p_1, \ldots, p_s\}$.

2.2.3 Temporal Evolution of a Microstate

To understand the concept of the temporal evolution of a microstate within a dynamic system, we need to explore how the state of a system changes over time as it transitions between different microstates. Consider a classical mechanical system composed of a multitude of particles, each with its unique position and momentum. The complete description of the system at any given moment is provided by specifying the positions q and momenta p of all these particles. This set of $2s$ variables defines a microstate of the system at that particular instant. Now, let's focus on how this microstate evolves over time. To do this, we need to introduce the concept of **equations of motion**. These equations, often derived from fundamental principles like Newton's laws of motion, govern how the positions and momenta of particles change as a function of time. In classical mechanics, Hamilton's equations of motion play a central role. These equations are a set of first-order differential equations that describe how the positions q and momenta p evolve over time:

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}, \quad \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}, \tag{2.3}$$

The solution to these equations of motion provides the trajectory of a microstate in the phase space Γ, which was introduced above. This trajectory represents how the microstate's positions and momenta change as time progresses. Understanding the temporal evolution of microstates is fundamental for predicting the macroscopic behaviour of a system. By considering the ensemble of all possible microstates and their trajectories, statistical physics enables us to make probabilistic predictions about macroscopic observables, such as temperature, pressure, and entropy. These predictions are essential for understanding the behaviour of gases, liquids, solids, and many other complex systems in the realm of physics.

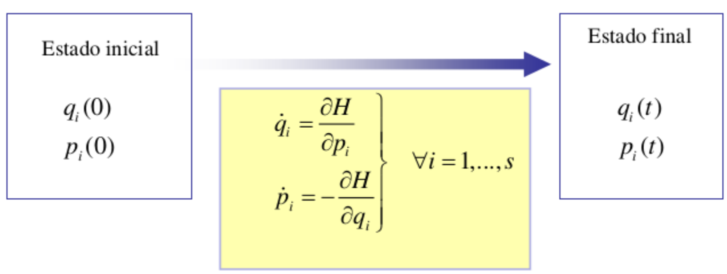


Figure 2.3: Temporal evolution of the degrees of freedom of a generic system of particles from time $t = 0$ to t.

We also stress that the Hamiltonian is nothing more than the Legendre transform of the Lagrangian $L(q, \dot{q})$ of the system[1].

A few comments that help us fix ideas.

[1] The relationship between the Hamiltonian $H(q, p)$ and the Lagrangian $L(q, \dot{q})$ is a fundamental connection in classical mechanics and reads

$$H(q, p) = \dot{q}p - L(q, \dot{q}). \tag{2.4}$$

It can be understood as a Legendre transformation. The Lagrangian, denoted as $L(q, \dot{q})$, represents the difference between the kinetic and potential energies of a system. It is defined as:

$$L(q, \dot{q}) = T(q, \dot{q}) - U(q), \tag{2.5}$$

where $T(q, \dot{q})$ is the kinetic energy and $U(q)$ is the potential energy. The Legendre transformation is essentially a mathematical tool that allows us to switch between these two different descriptions of a mechanical system. It plays a crucial role in the formulation of Hamilton's equations of motion and is a cornerstone in classical mechanics. In summary, the Hamiltonian $H(q, p)$

- Trajectories cannot leave the accessible phase space. Two or more trajectories never intersect each other: the laws of Mechanics are deterministic.
- If two systems are characterised by the same Hamiltonian H and are in the same initial microstate (q_0, p_0), after a certain time t, both will be in the same final microstate.

$$(q_0, p_0) = (q_0', p_0') \Rightarrow (q_t, p_t) = (q_t', p_t') \tag{2.6}$$

- If at $t = 0$, one system is in a microstate (q_0, p_0), and another system with the same Hamiltonian is in a different one (q_0', p_0'), then after a time t, the two will be in different microstates.

$$(q_0, p_0) \neq (q_0', p_0') \Rightarrow (q_t, p_t) \neq (q_t', p_t') \tag{2.7}$$

- Chaos: two systems with the same Hamiltonian prepared in two very similar microstates evolve microscopically in very different ways: high sensitivity to initial conditions.

2.2.4 Temporal Evolution of a Dynamic Function

Let's consider a dynamic function $a(\mathbf{q}, \mathbf{p})$ that doesn't explicitly depend on time. Despite its lack of explicit time dependence, this function can change over time as a result of the system's motion, given that $q_i = q_i(t)$ and $p_i = p_i(t)$.

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^{s} \left(\frac{\partial a}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial a}{\partial p_i} \frac{dp_i}{dt} \right) = \sum_{i=1}^{s} \left(\frac{\partial a}{\partial q_i} \dot{q}_i + \frac{\partial a}{\partial p_i} \dot{p}_i \right) \tag{2.8}$$

and if we use the Hamiltonian equations:

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^{s} \left(\frac{\partial a}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial a}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \equiv \{a, H\} \tag{2.9}$$

where the brackets $\{\quad\}$ in the right-hand-side of the above equation are referred to as **Poisson brackets**[2]. If the dynamic function depends explicitly on time, then

$$\frac{d}{dt} a(\mathbf{q}, \mathbf{p}; t) = \{a, H\} + \frac{\partial a}{\partial t} \tag{2.11}$$

The above equation $\dot{a}(\mathbf{q}, \mathbf{p}) = \{a, H\}$ can be considered the most important equation in Classical Mechanics and includes the canonical equations of Hamilton as a particular case.

$$\dot{q}_j = \{q_j, H\} = \sum_{i=1}^{s} \left(\frac{\partial q_j}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial q_j}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = \frac{\partial H}{\partial p_j} \tag{2.12}$$

---

encapsulates information about a system's dynamics in terms of generalised coordinates and momenta, while the Lagrangian $L(q, \dot{q})$ characterises the same dynamics in terms of generalised coordinates and velocities. This duality provides us with valuable insights into the behaviour of physical systems and is a central concept in theoretical physics. **Starting from the principle of least action, derive the Euler-Lagrange equation. Then, using the definition of the Hamiltonian as the Legendre transform of the Lagrangian, deduce Hamilton's equations. See a possible solution in Appendix B.**

[2] In classical mechanics, Poisson brackets, $\{A, B\}$, are a mathematical notation and concept used to describe the time evolution of physical quantities in a Hamiltonian system. They are named after the French mathematician Siméon Denis Poisson (1781 - 1840), who introduced them. Given two dynamic variables or functions $A(q, p)$ and $B(q, p)$ of the generalised coordinates q_i and their conjugate momenta p_i, the Poisson bracket is defined as:

$$\{A, B\} = \sum_{i=1}^{s} \left(\frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i} \right) \tag{2.10}$$

The Poisson bracket represents the rate of change of A with respect to B (and vice versa) due to the dynamics of the system described by the Hamiltonian $H(q, p)$. In other words, it quantifies how two observables evolve with respect to each other as the system evolves in time.

$$\dot{p}_j = \{p_j, H\} = \sum_{i=1}^{s} \left(\frac{\partial p_j}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial p_j}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = -\frac{\partial H}{\partial q_j} \tag{2.13}$$

$$\dot{H} = \{H, H\} = 0 \tag{2.14}$$

H is basically a constant of motion. In general, if a certain dynamic function a satisfies the condition $\{a, H\} = 0$, then $a(q, p)$ is a constant of motion of the system.

Some points of discussions:

- It is important to note that classical mechanics provides a well-defined and detailed microscopic description based on Newton's laws. However, in general, we can only obtain a formal solution for the microscopic behaviour of the system due to:
  - The **impossibility of experimentally determining** the initial conditions of $\sim 10^{23}$ particles.
  - The **mathematical and numerical impossibility of solving** $\sim 6 \times 10^{23}$ differential equations.
- An interesting question arises: Even if we could solve the equations, how is it possible for thermodynamics, for example, to emerge from such dynamic complexity? Would such a solution provide true understanding? Sometimes, knowing the analytical solution to a complex problem does not necessarily mean understanding what is happening.
- Microscopic Time Reversibility. The microscopic dynamic equations (Hamilton's equations) are temporally reversible. This is demonstrated by reversing the sign of all momenta and the sign of time in the Hamiltonian equations, and observing that they remain invariant[3]. How could this process generate irreversible macroscopic laws?

2.3 Quantum Description

One fundamental distinction between Classical Mechanics and Quantum Mechanics (QM) lies in the *Uncertainty Principle*, a concept introduced by Werner Heisenberg (1901 - 1976). This principle asserts that the precision with which we can simultaneously measure the position and momentum of a particle is fundamentally limited. In mathematical terms, it can be expressed as:

$$\Delta q_i \cdot \Delta p_i \geq \frac{\hbar}{2} \quad \text{for all } i = 1, \ldots, s \tag{2.15}$$

where Δq_i represents the uncertainty in position, Δp_i signifies the uncertainty in momentum, and $\hbar = h/2\pi = 1.054571817... \times 10^{-34}$ J s (or, equivalently, J Hz^{-1}) denotes the reduced Planck constant, being h the **Planck constant**. This principle has profound implications for our understanding of the behaviour of particles at the quantum level. It implies that we cannot precisely know both the position and momentum of a particle simultaneously, no matter how advanced our measurement tools become. In a semiclassical view, this would correspond to having "fuzzy points" in phase space, with a volume proportional to $\hbar$.

Let's discuss some key features of this principle:

- The Uncertainty Principle is a cornerstone of quantum theory and has led to a host of fascinating and counterintuitive phenomena, such as **wave-particle duality** and the **quantisation** of energy levels. It fundamentally challenges our classical intuition and underlines the inherent probabilistic nature of quantum systems.
- The principle implies that there is an **inherent limit** to how precisely both the position and momentum of a particle can be known simultaneously. In other words, the more accurately we know the position of a particle, the less accurately we can know its momentum, and vice versa.

[3]Verify the time reversibility of the Hamiltonian equations of motion. A possible solutions is available in Appendix C

- It fundamentally challenges classical physics, which assumes that it is possible to precisely determine both position and momentum. In QM, measurements are **inherently probabilistic**. Knowing the position of a particle precisely means that its momentum becomes highly uncertain, and vice versa.
- The Uncertainty Principle is intimately connected to the concept of **wave-particle duality**. It implies that particles, such as electrons and photons, exhibit both particle-like and wave-like behaviour. The more precisely you try to determine their position, the more you disturb their momentum, and vice versa.
- The Uncertainty Principle has wide-ranging implications in QM and plays a crucial role in the behaviour of subatomic particles. It has practical consequences in various quantum technologies, including quantum computing and quantum cryptography.
- Beyond its mathematical and physical significance, the Uncertainty Principle has profound philosophical implications. It challenges our classical intuitions about the determinism of physical systems and raises questions about the nature of reality at the quantum level.

In summary, the Uncertainty Principle is a foundational concept in QM, highlighting the fundamental limits on our ability to precisely know certain properties of particles. It underscores the probabilistic nature of quantum systems and has revolutionised our understanding of the behaviour of matter and energy at the smallest scales. An important consequence of the Uncertainty Principle is that a quantum microstate cannot be projected as a point in phase space. The classical description uses the concept of phase space, where each point represents a unique microstate of the system, specified by the generalised coordinates (q) and momenta (p). Instead, we must depict a quantum microstate as **cells** with dimensions $\Delta q_i \Delta p_i$ that satisfy Heisenberg's Uncertainty Principle. Consequently, within a cell in phase space, all points are **quantum mechanically indistinguishable** and correspond to the same quantum state. These cells, which are regions in phase space rather than single points, provide a probabilistic description of a quantum microstate. This quantum description leads to a more **probabilistic and statistical understanding of physical systems**, fundamentally different from classical mechanics. It is essential for understanding the behaviour of particles at the atomic and subatomic levels, where classical mechanics fails to provide an accurate description of their dynamics.

2.3.1 When should a problem be considered quantum or classical?

The distinction between a problem that should be treated quantum mechanically or classically depends on the scale and characteristics of the system studied. QM is fundamental when dealing with microscopic systems, such as atoms, molecules, and subatomic particles. At these scales, quantum effects dominate, and classical mechanics becomes inadequate to describe their behaviour accurately. Classical mechanics, on the other hand, works well for macroscopic systems, where quantum effects are negligible and the classical approximation is valid. More quantitatively, a problem must inevitably be addressed using QM when the action of the system is on the order of Planck's constant, $h = 6.62 \times 10^{-34}$ J s.

Ehrenfest's Theorem

Ehrenfest's Theorem states that a wave packet can be treated using Hamilton's equations (i.e., as a classical particle) if:

- $\Delta x \ll d$, where d is the typical distance traveled;
- $\Delta p_x \ll \overline{p}$, average momentum of the particle (wave packet);
- $\hbar/2 \leq \Delta x \Delta p_x \ll d \cdot \overline{p} \Rightarrow d \gg \hbar/2p$

At high temperatures, thermal agitation causes the particles of a system (molecules, atoms, etc.) to move over long distances ($d \uparrow\uparrow$) and possess very large linear momenta ($p \uparrow\uparrow$). In other words, $d \gg \frac{\hbar}{2p}$, and we can apply the classical treatment. At low temperatures, just the opposite happens: $d < \frac{\hbar}{2p}$, and it's necessary to apply QM. When can we assume that temperature is high enough to apply classical formalism? There is no single answer to this question; it depends on the type of system. For a gas of atoms/molecules, room temperature (300 K) is already high enough to consider the system as classical. For a gas of electrons in a metal at room temperature, the system still exhibits strong quantum behaviour. For most solids at room temperature, classical behaviour holds true; at low temperatures (below 100 K), quantum effects become relevant. For liquid Helium II (which is achieved at temperatures very close to absolute zero), a quantum treatment is necessary.

2.3.2 State Representation in QM

In the formalism of QM, the state of a system is represented by a vector $|\Psi\rangle$ that encodes the complete quantum state of a physical system, including details about the positions, momenta, and other properties of its particles or components. It serves as a mathematical representation that enables predictions about the probabilities of various outcomes when measurements are performed on the system. At a particular instant of time, the components of this vector represent the values of the corresponding **wave function**

$$\Psi = \Psi(q_1, ..., q_s, t) \tag{2.16}$$

where s is the number of degrees of freedom. The link between state vector and wave function, due to the extremely large number of represented values, is better expressed as an integral. In *bra–ket* (Dirac) notation, this vector is written as

$$|\Psi(t)\rangle = \int \Psi(x,t)|x\rangle\, dx \tag{2.17}$$

By manipulating and evolving the wave function using quantum operators, one can make predictions about the behaviour of quantum systems, including particles like electrons, atoms, and molecules. The set of all states or wave functions possesses the structure of a **Hilbert space** $\hat{H}$, which is a complex vector space with an inner product that allows for the mathematical treatment of quantum states, operations, and measurements. This Hilbert space structure provides the mathematical framework for the description of quantum states and their evolution over time.

The main difference between phase space in classical mechanics and configuration space in QM lies in how they represent the state of a physical system and how variables are handled. Let's look at this difference in more detail.

Phase Space in Classical Mechanics:

- **State Representation:** In classical mechanics, phase space represents the state of a physical system. Each point in phase space corresponds to a unique configuration of the system, defined by the coordinates of position and momentum of all particles in the system.

- **Classical Variables:** In classical mechanics, the variables that describe the state are real numbers, i.e., position and momentum coordinates are continuous and can take any real value.

- **Determinism:** In classical phase space, the system follows deterministic trajectories. Given an initial set of initial conditions, the system evolves uniquely along a specific trajectory in phase space, according to Hamilton's equations or Newton's equations of motion.

Configuration Space in QM:

- **State Representation:** In QM, configuration space represents the state of the system. A quantum state is described by a wave function or a state vector that contains information about the probability of finding a particle in a particular configuration.

- **Quantum Variables:** In QM, the variables that describe the state are operators that act on wave functions. These operators generally do not commute, leading to properties such as non-commutativity and Heisenberg's uncertainty.

- **Probabilistic Nature:** In quantum configuration space, the behaviour of particles is described through wave functions that assign probabilities of finding a particle in a specific configuration. Quantum predictions are related to probabilities and superpositions of states.

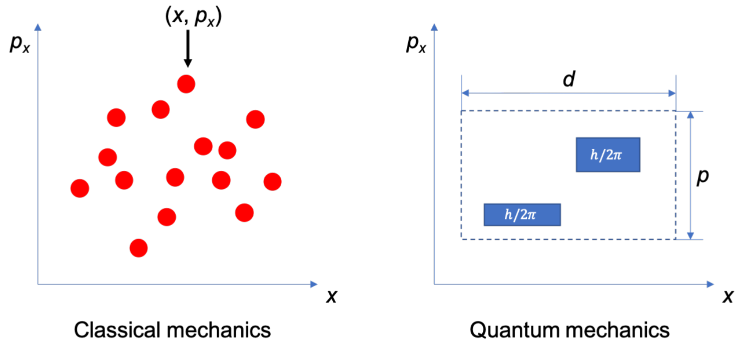


Figure 2.4: Comparison between cllassical (left) and quantum (right) mechanical one-dimensional phase space. In the classical case, a given microstate is represented by a point, whereas in QM it is is described by a phase space volume $p_x \cdot d = \hbar$.

In summary, while phase space in classical mechanics is based on continuous variables and allows for **deterministic evolution** of the system, configuration space in QM is based on wave functions and non-commutative operators, and predictions are related to **probabilities** and the principle of superposition. These fundamental differences reflect the essential distinction between classical physics and quantum physics.

2.3.3 Probability of finding a system in a configuration space.

In QM, one of the fundamental concepts is the probability of finding a physical system in a particular state within its configuration space. This probability is described by the wave function $\Psi(\mathbf{q}, t)$, where $\mathbf{q}$ represents the configuration coordinates of the system and t time. The probability $P(\mathbf{q}, t)$ of finding the system within the region of the configuration space defined by $[q_1, q_1 + dq_1][q_2, q_2 + dq_2]...$ at a specific time t reads

$$P(\mathbf{q}, t) = \int_q^{q+dq} |\Psi(\mathbf{q}, t)|^2 d\mathbf{q} \tag{2.18}$$

where $\Psi(\mathbf{q}, t)$ is the wave function of the system in the configuration space $\mathbf{q}$ at time t; $|\Psi(\mathbf{q}, t)|^2$ is the square modulus of the wave function, providing the probability density; and $d\mathbf{q} = dq_1...dq_s$ represents the infinitesimal volume element in the configuration space.

Example 2.2: Probability of Finding a Particle in a Specific Range

Consider a one-dimensional quantum system with the wave function $\Psi(x, t)$. We want to calculate the probability of finding the particle between $x = 2$ and $x = 4$ at $t = 0$. According to QM, the probability $P(2 \leq x \leq 4; 0)$ of finding the particle in the range $[2, 4]$ at time $t = 0$ is given by:

$$P(2 \leq x \leq 4; 0) = \int_2^4 |\Psi(x, 0)|^2 dx \tag{2.19}$$

Suppose the wave function is given by $\Psi(x, 0) = 2\sqrt{2}e^{-x^2}$. Plugging this into the formula, we can calculate the probability:

$$P(2 \leq x \leq 4; 0) = \int_2^4 |2\sqrt{2}e^{-x^2}|^2 dx \tag{2.20}$$

Now, compute the integral:

$$P(2 \leq x \leq 4; 0) = \int_2^4 8e^{-2x^2} dx \tag{2.21}$$

This integral can be evaluated numerically to find the probability of locating the particle between $x = 2$ and $x = 4$ at $t = 0$. The result will be a numerical value[4].

2.3.4 Wave function and the Schrödinger equation

For a system with constant volume V and a fixed number of particles N, the Schrödinger equation must be satisfied:

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r_1}, \mathbf{r_2}, \ldots, \mathbf{r_N}, t) = \hat{H} \Psi(\mathbf{r_1}, \mathbf{r_2}, \ldots, \mathbf{r_N}, t) \tag{2.22}$$

where $\hat{H}$ represents the Hamiltonian operator describing the total energy of the system; $\Psi(\mathbf{r_1}, \mathbf{r_2}, \ldots, \mathbf{r_N}, t)$ is the wave function of the system, dependent on the coordinates $\mathbf{r_i}$ of the particles and time t; i is the imaginary unit; and $\hbar$ is the reduced Planck constant. This equation is the fundamental form of the Schrödinger equation in QM, governing the evolution of the wave function for a quantum system with N particles in a volume V. Additionally, if the energy E of the system is constant (an isolated system), then the above equation simplifies, and the time dependence disappears:

$$\hat{H} \psi(\mathbf{r_1}, \mathbf{r_2}, \ldots, \mathbf{r_N}) = E \psi(\mathbf{r_1}, \mathbf{r_2}, \ldots, \mathbf{r_N}) \tag{2.23}$$

The above equation is employed only when the Hamiltonian itself does not explicitly depend on time. Nevertheless, even in this scenario, the overall wave function remains time-dependent. In particular, the solutions of this equation are the vectors ψ_{E_n}, in the basis of energy eigenstates, whose linear combination gives the following *time-dependent* state vector

$$|\Psi(t)\rangle = \sum_n A_n e^{-iE_n t/\hbar} |\psi_{E_n}\rangle \tag{2.24}$$

where A_n are complex numbers and $e^{-iE_n t/\hbar}$ represents the time evolution operator, also known as the *propagator*, which describes the time evolution of a quantum state under a time-independent Hamiltonian.

Due to the principle of uncertainty, even if the wave function is known, it is not possible to uniquely determine all the values of q_i and p_i: the wave function Ψ determines the probability distribution of physical quantities but not their exact values. More specifically, the square modulus of the wave function $|\Psi|^2$ provides information about the probability density of finding a particle in a particular state or with specific values of physical quantities such as position or momentum. However, it does not provide the exact values of these quantities. This probabilistic nature of QM is a fundamental feature, and it means that we can predict the likelihood of different outcomes, but we cannot precisely determine individual values. The wave function encodes the probabilities associated with various outcomes, reflecting the inherent uncertainty and indeterminacy at the quantum level.

2.3.5 Pure States and Mixed States

In QM, there are two fundamental types of quantum states.

- Pure states are represented by vectors $|\Psi\rangle$ in a Hilbert space $\hat{H}$. These states are characterised by having a well-defined and definite quantum state. When a system is in a pure state, all information about its quantum properties is contained in a single state vector. Pure states are described by wave functions that are not statistical mixtures of other states.

[4] To solve this integral, substitute $u = \sqrt{2}x \rightarrow du = \sqrt{2}dx$. Then notice that $\int 8e^{-2x^2} dx = 2^{3/2}\sqrt{\pi} \int 2e^{-u^2}/\sqrt{\pi} du = 2^{3/2}\sqrt{\pi} \int \text{erf}(u) du$, where erf(u) is the error function. Note also that P should be normalised by the value of the integral calculated between $-\infty$ and ∞. Show that this value is $8\sqrt{\pi/2}$ and normalise the probability accordingly.

- Mixed states, on the other hand, are represented by density operators or density matrices ρ. These states result from a statistical mixture of pure states. In a mixed state, the system has uncertainty or lack of knowledge about its quantum properties. The density operator ρ contains information about the probabilities of finding the system in different pure states. Mixed states arise in situations where there is incomplete information or when a system is prepared probabilistically.

Understanding and distinguishing between pure states and mixed states is essential in QM as they have different mathematical descriptions and physical interpretations. Pure states correspond to definite quantum states, while mixed states represent statistical ensembles of possible pure states.

Scalar Product Between Two States ψ_1 and ψ_2

The scalar product (inner product) between two quantum states $|\psi_1\rangle$ and $|\psi_2\rangle$ is given by:

$$\langle\psi_1|\psi_2\rangle \equiv \int \psi_1^* \psi_2 dq_1 ... dq_s \tag{2.25}$$

where $\psi_1^*(x)$ is the complex conjugate of the wave function $\psi_1(x)$ associated with $|\psi_1\rangle$ (see Appendix D for additional details). In QM, this mathematical operation allows one to calculate the degree of overlap or similarity between the two states. The result of this scalar product can be a complex number and provides information about the relationship between the states $|\psi_1\rangle$ and $|\psi_2\rangle$.

2.4 Association of physical quantities with quantum operators

In the context of QM, each physical quantity B is associated with a self-adjoint linear operator $\hat{B}$ that operates on the states within the Hilbert space.

1. **Physical Quantity** B: In the context of QM, B represents a physical quantity or observable, such as position, momentum, energy, or any other property that we want to measure or describe.

2. **Self-adjoint Linear Operator** $\hat{B}$: This refers to a mathematical operator denoted as $\hat{B}$ that acts on the quantum states within the Hilbert space. The term "self-adjoint" means that the operator $\hat{B}$ is Hermitian, which implies that it is equal to its own adjoint (or conjugate transpose). Self-adjoint operators are crucial in QM because they correspond to observable physical quantities, and they have real eigenvalues.

3. **Hilbert Space**: The Hilbert space is a mathematical space used in QM to describe the possible states of a quantum system. It is a complex vector space where quantum states, represented as kets (e.g., $|\psi\rangle$, reside. These states can be operated upon by quantum operators like $\hat{B}$ to obtain measurements or predictions of physical quantities.

2.4.1 Properties of the self-adjoint operator

A self-adjoint operator $\hat{B}$ in QM possesses several key properties:

1. **Hermitian Nature**: A self-adjoint operator $\hat{B}$ is Hermitian, which means it is equal to its own adjoint or conjugate transpose:

   $$\hat{B} = \hat{B}^\dagger \tag{2.26}$$

   This property ensures that the inner product (expected value) of $\hat{B}$ acting on a state vector is the same whether one takes the inner product of ψ_1 and $\hat{B}\psi_2$ or ψ_2 and $\hat{B}\psi_1$ (symmetry of the inner product). Mathematically, $\langle\psi_1|\hat{B}|\psi_2\rangle = \langle\psi_2|\hat{B}|\psi_1\rangle^*$.

2. **Real Eigenvalues**: The eigenvalues of a self-adjoint operator $\hat{B}$ are always real numbers. This property is crucial because it ensures that measurements of physical quantities associated with $\hat{B}$ yield real values.

3. **Orthogonal Eigenvectors**: The eigenvectors corresponding to distinct eigenvalues of $\hat{B}$ are orthogonal to each other. In other words, if two eigenvalues λ_i and λ_j are different, then the eigenvectors $|i\rangle$ and $|j\rangle$ associated with these eigenvalues are orthogonal:

$$\langle i|j\rangle = 0 \quad \text{for } \lambda_i \neq \lambda_j \tag{2.27}$$

4. **Completeness**: The set of eigenvectors of $\hat{B}$ forms a complete orthonormal basis for the Hilbert space. This property allows any quantum state to be expressed as a linear combination of these eigenvectors.

2.4.2 Results of measurements

In QM, the numerical value observed in a measurement of a certain physical quantity after preparing the system in a specific microstate ψ is not defined. The only quantity we can define is its average value. This average value is what we would obtain after performing a large number of measurements under identical conditions (the same microstate ψ). This is usually referred to as **quantum average**. The result of a measurement (observable) for a quantum system in the state ψ is equal to the expected value (average) of the operator $\hat{B}$ for that state ψ. In mathematical terms, this can be expressed as:

$$\overline{B} = \langle\psi|\hat{B}|\psi\rangle = \langle\psi|(\hat{B}|\psi\rangle) = \int \psi^*(\hat{B}\psi)dq_1 \dots dq_n \tag{2.28}$$

This equation reflects the fundamental concept that in QM, the result of a measurement is probabilistic, and the expectation value of the operator $\hat{B}$ provides the average or expected outcome for that measurement when the system is in the state ψ. We also notice that $|\psi\rangle$ is normalised to unity, that is $\langle\psi|\psi\rangle = 1$. This condition guarantees that the probability of finding the system in any possible state is conserved and equal to 1. Finally, one should note that the fact that $\hat{B}$ is self-adjoint ensures that its expected value is a real number.

When our quantum system is in a mixed state, meaning it can exist in multiple quantum states with certain probabilities, the outcome of a measurement is probabilistic. In this scenario, *we cannot predict a single definite value for the measurement.* Instead, we can calculate the expected or average value of the measurement, which is known as the "expectation value." The expectation value represents the average outcome we would obtain if we were to repeat the same measurement on identically prepared systems in their respective mixed states many times. It is a fundamental concept in QM and is calculated using the principles of quantum probability.

2.5 Decomposition of $|\psi\rangle$ into an orthonormal basis

In QM, eigenfunctions are special functions associated with a physical observable (represented by the operator $\hat{B}$) that have the property that when the operator $\hat{B}$ acts on them, it returns a scaled version of the same function. Mathematically, this can be expressed as:

$$\hat{B}|\phi_n\rangle = b_n|\phi_n\rangle \tag{2.29}$$

where $\hat{B}$ is the operator corresponding to the physical observable and ϕ_n represents the *eigenfunction* associated with the *eigenvalue* b_n. This equation is usually referred to as the **eigenvalue equation**. It describes how certain special states (called *eigenstates*) of an operator remain the same except for a scaling factor (the eigenvalue) when acted upon by that operator. Eigenfunctions are crucial in QM because they provide a basis in which any quantum state can be expressed through a linear combination. This means that if we know the eigenfunctions of an operator, we can represent any quantum state using those eigenfunctions and coefficients. Decomposition of a state vector into an orthonormal basis $\{\phi_n\}_{n=1}^{\infty}$, is a fundamental concept in vector space theory and also applies in QM. In this context, an orthonormal basis is a set of linearly independent vectors in a vector space that are normalised, meaning they have a norm (length) equal to 1 and are mutually orthogonal (the inner product between two different vectors is 0). In QM, the states of a system are represented as vectors in a Hilbert space, and these states can be decomposed in terms of an orthonormal basis. There are two options here: either the orthonormal basis $\{\phi_n\}_{n=1}^{\infty}$ is an eigenfunction of $\hat{B}$ or it is not.

2.5.1 Case I. The orthonormal basis $\{\phi_n\}_{n=1}^{\infty}$ is an eigenfunction of $\hat{B}$.

The decomposition of a quantum state $|\psi\rangle$ in terms of the eigenfunctions of the operator $\hat{B}$ can be written as:

$$|\psi\rangle = \sum_{n=1}^{\infty} c_n |\phi_n\rangle \tag{2.30}$$

where $|\psi\rangle$ is the quantum state we want to decompose; $|\phi_n\rangle$ are the elements of the orthonormal basis (eigenfunctions of operator $\hat{B}$); and c_n are complex numbers describing how much each element of the basis contributes to the state $|\psi\rangle$. To find the coefficients c_n, inner products and properties of the orthonormal basis are used. In particular, c_n is calculated as the inner product between the state $|\psi\rangle$ and the basis element $|\phi_n\rangle$:

$$c_n = \langle \phi_n | \psi \rangle = \int \phi_n^* \psi dq_1 \dots dq_N \tag{2.31}$$

Additionally

$$\langle \psi | = \sum_{n=1}^{\infty} \langle \phi_n | c_n^* \tag{2.32}$$

where $|c_n|^2$ is the probability that, given a system in the pure state $|\psi\rangle$, after a measurement it will be found in the state $|\phi_n\rangle$. It must hold that

$$\sum_{n=1}^{\infty} |c_n|^2 = 1 \tag{2.33}$$

It follows that the expected value of the observable $\hat{B}$ reads

$$\overline{B} = \langle \psi | \hat{B} | \psi \rangle = \left(\sum_{m=1}^{\infty} \langle \phi_m | c_m^* \right) \hat{B} \left(\sum_{n=1}^{\infty} c_n |\phi_n\rangle \right) = \sum_{m=1}^{\infty} c_m^* \sum_{n=1}^{\infty} c_n \langle \phi_m | \underbrace{\hat{B} |\phi_n\rangle}_{b_n |\phi_n\rangle} \tag{2.34}$$

where $|\phi_n\rangle$ are eigenvalues of $\hat{B}$. This equation reduces to

$$\overline{B} = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} c_m^* c_n b_n \langle \phi_m | \phi_n \rangle = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} c_m^* c_n b_n \delta_{nm} = \sum_{n=1}^{\infty} |c_n|^2 b_n \tag{2.35}$$

where $\delta_{nm} \equiv \langle \phi_m | \phi_n \rangle$ is an orthonormal basis and b_n is the value of B in state n (real number). In other words, $\overline{B}$ is an average, where each b_n should be weighted by its corresponding probability $|c_n|^2$.

2.5.2 Case II. The orthonormal basis $\{\phi_n\}_{n=1}^{\infty}$ is not an eigenfunction of $\hat{B}$.

In this case,

$$\hat{B} |\phi_n\rangle \neq b_n |\phi_n\rangle \tag{2.36}$$

and

$$\overline{B} = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} c_m^* c_n \underbrace{\langle \phi_m | \hat{B} | \phi_n \rangle}_{b_{mn}} \tag{2.37}$$

where b_{mn} is an element of matrix (m, n) and can be a complex number. We conclude that $\overline{B}$ is a real number, but its expression is not that typical of an average: it is a quantum average and its contributions b_{mn} are consequence of quantum interferences: they do not have a classical analogue.

Generally speaking, decomposition into an orthonormal basis is fundamental for understanding and working with quantum systems, as it allows us to express any state in terms of well-defined basic states and calculate probabilities and expectations of observables. Additionally, it ensures that the norm of the state $|\psi\rangle$ is preserved in the decomposition.

2.6 Temporal evolution of an operator

The temporal evolution of states $|\psi_t\rangle \in \hat{H}$ follows the Schrödinger equation:

$$i\hbar\frac{\partial}{\partial t}|\psi_t\rangle = \hat{H}|\psi_t\rangle \tag{2.38}$$

This equation has a formal solution. In the case where the Hamiltonian is independent of time:

$$|\psi_t\rangle = \mathrm{e}^{-\frac{i\hat{H}t}{\hbar}}|\psi(0)\rangle \tag{2.39}$$

where $|\psi(0)\rangle$ is the wave function at $t = 0$ (independent from time). Additionally

$$\langle\psi_t| = \langle\psi(0)|\mathrm{e}^{\frac{i\hat{H}t}{\hbar}} \tag{2.40}$$

The expected value of an operator $\hat{B}$ at time t reads

$$\langle\psi_t|\hat{B}|\psi_t\rangle = \langle\psi(0)|\underbrace{\mathrm{e}^{\frac{i\hat{H}t}{\hbar}}\hat{B}\mathrm{e}^{-\frac{i\hat{H}t}{\hbar}}}_{\hat{B}_t}|\psi(0)\rangle = \langle\psi(0)|\hat{B}_t|\psi(0)\rangle \tag{2.41}$$

where we have transferred the time dependence to the operator $\hat{B}_t$. It follows that we can calculate the derivative of $\hat{B}_t$ with respect to time:

$$\frac{d\hat{B}_t}{dt} = \frac{d}{dt}\left(\mathrm{e}^{\frac{i\hat{H}t}{\hbar}}\hat{B}\mathrm{e}^{-\frac{i\hat{H}t}{\hbar}}\right) = \frac{i}{\hbar}\hat{H}\mathrm{e}^{\frac{i\hat{H}t}{\hbar}}\hat{B}\mathrm{e}^{-\frac{i\hat{H}t}{\hbar}} - \frac{i}{\hbar}\mathrm{e}^{\frac{i\hat{H}t}{\hbar}}\hat{B}\hat{H}\mathrm{e}^{-\frac{i\hat{H}t}{\hbar}} = \frac{i}{\hbar}[\hat{H}\hat{B}_t - \hat{B}_t\hat{H}] = \frac{1}{i\hbar}[\hat{B}_t\hat{H} - \hat{H}\hat{B}_t] \tag{2.42}$$

Equivalently:

$$\frac{d\hat{B}_t}{dt} = \frac{1}{i\hbar}[\hat{B}_t, \hat{H}] \tag{2.43}$$

If $[\hat{B}_t, \hat{H}] = 0$, then $\hat{B}_t$ is a constant of motion of the system[5] and there exists a basis that diagonalises both operators simultaneously.

2.7 Systems of Identical Particles: Bosons and Fermions

In the field of statistical physics, it is common to encounter systems composed of a large number of identical particles, and these particles are *quantum mechanically indistinguishable*. This concept plays a significant role in understanding the behaviour of matter in various physical systems.

In gases and liquids, particles are quantum mechanically indistinguishable, meaning it is impossible to tell one particle apart from another based on their individual properties. In solids, however, particles are typically considered distinguishable. This distinction arises because, in the classical viewpoint, it is always possible to determine the coordinates and momenta of a particle and track its evolution over time. Therefore,

[5]A motion integral or constant of motion in a mechanical problem is a function of position and velocities (or equivalently, generalised coordinates and their conjugate momenta) that remains constant along a trajectory of the system throughout its phases.

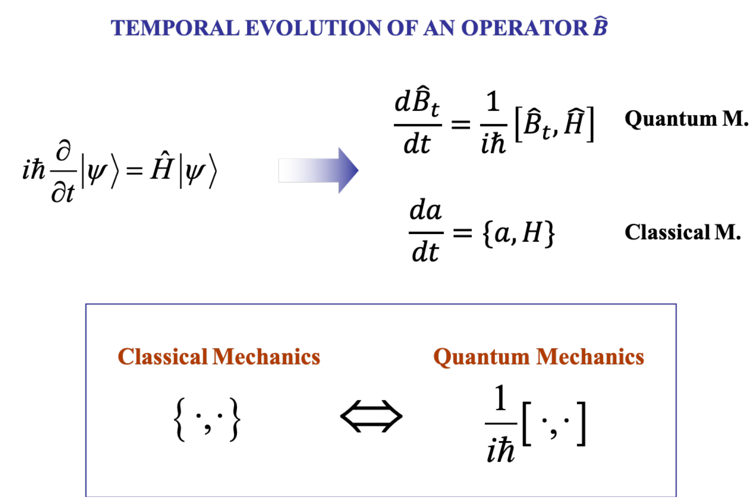


Figure 2.5: Summary of the key equations describing the temporal evolution of an operator $\hat{B}$ quantum-mechanically and classically.

in classical mechanics, particles can be labeled or numbered, making them distinguishable. The concept of indistinguishability is a fundamental property that **cannot be explained within classical physics**. In classical mechanics, particles are distinguishable because they have well-defined trajectories and properties.

By contrast, in QM, particles are fundamentally indistinguishable. This indistinguishability arises from the fact that **particles cannot be localised in configuration space as individual points** but rather as regions of uncertainty, known as cells of indeterminacy. The indistinguishability of identical particles is closely connected to the **invariance of the Hamiltonian under the exchange of labels or enumerations of two particles**. In QM, the particles' properties and behaviour are described by wave functions, and it is not meaningful to label or enumerate individual particles, as they exist in a state of quantum superposition.

2.7.1 Permutation operator in systems of identical indistinguishable particles

Consider a system composed of N identical and indistinguishable particles. In the context of QM, we introduce the **permutation operator**, denoted as $\hat{P}_{ij}$, which operates on the wave function ψ of the system as follows:

$$\hat{P}_{ij}\psi(\vec{r}_1, \ldots, \vec{r}_i, \ldots, \vec{r}_j, \ldots, \vec{r}_N) = \psi(\vec{r}_1, \ldots, \vec{r}_j, \ldots, \vec{r}_i, \ldots, \vec{r}_N) \tag{2.44}$$

The permutation operator $\hat{P}_{ij}$ essentially exchanges the positions of particle i and particle j in the wave function. Now, if the system's Hamiltonian operator $\hat{H}$ is invariant under the interchange of labels or enumeration of any two particles, then the following commutation relation holds:

$$[\hat{H}, \hat{P}_{ij}] = 0 \tag{2.45}$$

This means that the Hamiltonian operator commutes with the permutation operator $\hat{P}_{ij}$. Consequently, the eigenfunctions (wave functions) of both $\hat{H}$ and $\hat{P}_{ij}$ are the same. This property is of significant importance in QM, particularly when dealing with systems of identical particles. **It reflects the symmetry of the system with respect to the exchange of two indistinguishable particles**. When such symmetry exists, the wave functions corresponding to different permutations of particle positions have the same energy, and this leads to the concept of **degeneracy** in the energy levels of the system. The degeneracy of a particle state is the number of different quantum states that have the same energy. It is represented mathematically by the Hamiltonian for the system having more than one linearly independent eigenstate with the same energy eigenvalue.

Eigenvalues of the Permutation Operator $\hat{P}_{ij}$

Let's explore the eigenvalues of the permutation operator $\hat{P}_{ij}$ in more detail. The permutation operator that operates on a wave function $\psi(\vec{r}_1, \dots, \vec{r}_i, \dots, \vec{r}_j, \dots, \vec{r}_N)$, results in:

$$\hat{P}_{ij}\psi(\vec{r}_1, \dots, \vec{r}_i, \dots, \vec{r}_j, \dots, \vec{r}_N) = \lambda\psi(\vec{r}_1, \dots, \vec{r}_i, \dots, \vec{r}_j, \dots, \vec{r}_N) = \psi(\vec{r}_1, \dots, \vec{r}_j, \dots, \vec{r}_i, \dots, \vec{r}_N) \tag{2.46}$$

This equation indicates that applying the permutation operator $\hat{P}_{ij}$ to the wave function ψ yields another wave function ψ with a possible scaling factor λ. Let's examine the square of the permutation operator $\hat{P}_{ij}^2$:

$$\hat{P}_{ij}^2\psi(\vec{r}_1, \dots, \vec{r}_i, \dots, \vec{r}_j, \dots, \vec{r}_N) = \lambda^2\psi(\vec{r}_1, \dots, \vec{r}_i, \dots, \vec{r}_j, \dots, \vec{r}_N) = \psi(\vec{r}_1, \dots, \vec{r}_i, \dots, \vec{r}_j, \dots, \vec{r}_N) \tag{2.47}$$

Now, since $\hat{P}_{ij}^2$ is equivalent to applying the permutation operator twice, we obtain the same wave function ψ with a scaling factor λ^2. Therefore:

$$\lambda^2 = 1 \implies \lambda = \pm 1 \tag{2.48}$$

This means that the eigenvalues of the permutation operator $\hat{P}_{ij}$ are $\lambda = \pm 1$. These eigenvalues are fundamental in QM when dealing with systems of identical particles. They indicate that applying the permutation operator to a wave function either leaves it unchanged (for $\lambda = 1$) or changes its sign (for $\lambda = -1$), reflecting the indistinguishable nature of the particles and the symmetry of the quantum states.

Symmetric and Antisymmetric Wave Functions

When we examine the eigenvalues of the permutation operator $\hat{P}_{ij}$, we find that if $\lambda = +1$, the wave function is symmetric under the exchange of two particles, indicating the particles are **bosons**:

$$\psi(\vec{r}_1, \dots, \vec{r}_i, \dots, \vec{r}_j, \dots, \vec{r}_N) = \psi(\vec{r}_1, \dots, \vec{r}_j, \dots, \vec{r}_i, \dots, \vec{r}_N) \quad \text{(for bosons)} \tag{2.49}$$

Bosons are particles that follow Bose-Einstein statistics. They can occupy the same quantum state simultaneously. This property leads to phenomena like Bose-Einstein condensation, where a large number of bosons collapse into the lowest quantum state, resulting in exotic states of matter. On the other hand, if $\lambda = -1$, the wave function becomes antisymmetric under the exchange of two particles, revealing that the particles are **fermions**:

$$\psi(\vec{r}_1, \dots, \vec{r}_i, \dots, \vec{r}_j, \dots, \vec{r}_N) = -\psi(\vec{r}_1, \dots, \vec{r}_j, \dots, \vec{r}_i, \dots, \vec{r}_N) \quad \text{(for fermions)} \tag{2.50}$$

Fermions, on the other hand, obey Fermi-Dirac statistics. The Pauli exclusion principle dictates that no two fermions can occupy the same quantum state simultaneously. This principle underlies the stability of matter, preventing electrons in atoms from all falling into the lowest energy state, which would result in the collapse of all matter. Symmetric wave functions are associated with bosons, which can occupy the same quantum state, while antisymmetric wave functions are associated with fermions, which obey the Pauli exclusion principle and cannot occupy the same quantum state simultaneously.

Invariance of Observables

In a system of identical particles, all observables must be invariant under the exchange of particle labels. This is represented by the commutation of the observable $\hat{B}$ with the permutation operator $\hat{P}_{ij}$ for all particles i and j:

$$[\hat{B}, \hat{P}_{ij}] = 0 \quad \text{for all } i, j, \text{ and } \hat{B}. \tag{2.51}$$

In QM, two quantum states that differ only by the exchange of two particles are considered the same quantum state and are indistinguishable. They are not counted as two separate states, but as a single state.

Ideal System of Identical Particles

Let's consider an ideal system of identical particles i where these particles do not interact with each other. In such a system, the total energy can be expressed as the sum of individual energies:

$$E = \sum_{i=1}^{N} \varepsilon^{(i)} \tag{2.52}$$

The Hamiltonian operator for this system can be written as the sum of individual single-particle Hamiltonian operators:

$$\hat{H} = \sum_{i=1}^{N} \hat{h}^{(i)} \tag{2.53}$$

where $\hat{h}^{(i)}$ operates on the state $\phi_k^{(i)}$, resulting in the energy $\varepsilon_k^{(i)}$, and position $\vec{r}_i$. In other words:

$$\hat{h}^{(i)} \phi_k^{(i)}(\vec{r}_i) = \varepsilon_k^{(i)} \phi_k^{(i)}(\vec{r}_i) \tag{2.54}$$

If we want to describe the quantum state ψ of the entire system as the product of individual states, $\phi_{k_1}^{(1)} \phi_{k_2}^{(2)} \dots \phi_{k_N}^{(N)}$, we need to take into account the **indistinguishability** of particles. However, simply taking the product of individual states is incorrect, because swapping two particles should not lead to a physically distinct state. In particular, in case we deal with a system of N bosons, the **symmetric** wave function reads:

$$\psi = \frac{1}{\sqrt{\mathcal{N}!}} \sum_P \hat{P} \phi_{k_1}^{(1)} \phi_{k_2}^{(2)} \dots \phi_{k_N}^{(N)} \tag{2.55}$$

where $\hat{P}$ is the permutation operator, the sum $\sum_P$ goes over all permutations of the particles and the factor $\sqrt{\mathcal{N}!}$ normalises the wave function ($\mathcal{N}!$ is basically the number of permutations). By contrast, if we deal with a system of N fermions, the **antisymmetric** wave function reads:

$$\psi = \frac{1}{\sqrt{\mathcal{N}!}} \sum_P (-1)^P \hat{P} \phi_{k_1}^{(1)} \phi_{k_2}^{(2)} \dots \phi_{k_N}^{(N)} \tag{2.56}$$

Generally speaking, in ideal systems, the spectrum of energies is the same for all identical particles. However, it's crucial to correctly account for the symmetrisation of the quantum state when dealing with such systems of identical particles.

Example 2.3: Three Particles in Three Distinct Quantum States

Let's consider a system of three identical particles, each occupying only one of three distinct quantum states: a, b, and c. We'll describe the quantum state for bosons and fermions separately.

Bosons are particles that follow Bose-Einstein statistics, which means they are indistinguishable and obey **symmetric exchange properties under particle exchange**. For bosons, the quantum state must be symmetric under the exchange of any two particles. Since there are three particles and three states, we need to construct symmetric combinations of the single-particle states. Therefore, the symmetric combination for three bosons reads

$$\begin{aligned}\psi_{\text{bosons}} = \frac{1}{\sqrt{6}} \Big[&\phi_a(1)\phi_b(2)\phi_c(3) + \phi_a(1)\phi_b(3)\phi_c(2) + \phi_a(2)\phi_b(1)\phi_c(3) \\ &+ \phi_a(2)\phi_b(3)\phi_c(1) + \phi_a(3)\phi_b(2)\phi_c(1) + \phi_a(3)\phi_b(1)\phi_c(2) \Big]\end{aligned} \tag{2.57}$$

Note that when, for example, we exchange the positions of particle 2 and 3 keeping the position of particle 1, the sign stays positive. Here, $\phi_a(1)$ represents the state a of the first particle, $\phi_b(2)$ represents the state b

of the second particle, and so on.

Fermions, on the other hand, follow Fermi-Dirac statistics, which means they are indistinguishable but obey **antisymmetric exchange properties under particle exchange**. For fermions, the quantum state must be antisymmetric under the exchange of any two particles. Similar to bosons, we need to construct antisymmetric combinations of the single-particle states, which read

$$\begin{aligned}\psi_{\text{fermions}} = \frac{1}{\sqrt{6}}\Big[&\phi_a(1)\phi_b(2)\phi_c(3) - \phi_a(1)\phi_b(3)\phi_c(2) - \phi_a(2)\phi_b(1)\phi_c(3) \\ &+ \phi_a(2)\phi_b(3)\phi_c(1) - \phi_a(3)\phi_b(2)\phi_c(1) + \phi_a(3)\phi_b(1)\phi_c(2)\Big]\end{aligned} \tag{2.58}$$

Note that when, for example, we exchange the positions of particle 2 and 3 keeping the position of particle 1, the sign changes. Again, $\phi_a(1)$ represents the state a of the first particle, $\phi_b(2)$ represents the state b of the second particle, and so on.

Example 2.4: Three Particles in Two Distinct Quantum States

Now, let's consider a system of three identical particles, but this time two of them can occupy the quantum state a and one of them the quantum state b. Let's describe the quantum state for bosons and fermions.

For bosons, the three-particle state reads

$$\psi_{\text{bosons}} = \frac{1}{\sqrt{3}}\left[\phi_a(1)\phi_a(2)\phi_b(3) + \phi_a(1)\phi_a(3)\phi_b(2) + \phi_a(2)\phi_a(3)\phi_b(1)\right]. \tag{2.59}$$

For fermions, we already know that two particles cannot be in the same quantum state. Let's see what we find then:

$$\begin{aligned}\psi_{\text{fermions}} = \frac{1}{\sqrt{6}}\Big[&\phi_a(1)\phi_a(2)\phi_b(3) - \phi_a(1)\phi_a(3)\phi_b(2) - \phi_a(2)\phi_a(1)\phi_b(3) \\ &+ \phi_a(2)\phi_a(3)\phi_b(1) - \phi_a(3)\phi_a(2)\phi_b(1) + \phi_a(3)\phi_a(1)\phi_b(2)\Big] = 0\end{aligned} \tag{2.60}$$

It is possible to find two or more bosons in the same quantum state. This is because bosons follow Bose-Einstein statistics and have no restrictions on the occupation of identical quantum states. In contrast, for fermions, the antisymmetry of the wave function prevents two particles from simultaneously occupying the same quantum state. This result reflects the **Pauli exclusion principle**, which prohibits multiple fermions from occupying the same quantum state. In this case, all states that would involve two fermions in the same state are canceled out, resulting in a net value of zero for the quantum state. This exclusion forms the basis of the electronic structure of atoms and the formation of chemical bonds.

2.7.2 Rephrasing the Problem in Terms of Occupation Numbers

Rather than jumping straight to the question, "*In which state can we find particle 1, particle 2, and so on?*" it's crucial to begin by asking, "*How many particles are currently occupying state 1? How many are residing in state 2, and so forth?*" This subtle shift in perspective sets the stage for a more profound understanding of quantum systems' intricate behaviour. By focusing on the occupancy of each state, we lay the groundwork for exploring the statistical characteristics of particles in different quantum states. To deal with ideal systems of identical particles, it is highly advisable to employ the occupation numbers of each state. This approach provides a more effective means of characterising the distribution and behaviour of particles within such systems. In particular, we can represent the conservation of particles and energy in the context of ideal systems of identical particles as follows:

$$\sum_k n_k = N \quad \text{(Conservation of Particles)} \tag{2.61}$$

$$\sum_k \epsilon_k n_k = E \quad \text{(Conservation of Energy)} \tag{2.62}$$

where n_k is the number of particles in the energy state ϵ_k.

The conditions of symmetry or antisymmetry for a particle are not deduced from theoretical proofs but rather from experimental observations. These observations have allowed us to establish a rule for distinguishing between these types of particles based on their spin. More specifically, **bosons** are particles with an integer or zero spin. Examples of bosons include photons, phonons, and more. The number of particles in a boson state can vary from 0 to infinity, represented as $n_j = \{0, 1, 2, \ldots, \infty\}$. By contrast, **fermions** are particles with half-integer spin. Examples of fermions include electrons, protons, and neutrons. In the case of fermions, the number of particles in a given state is either 0 or 1, denoted as $n_j = \{0, 1\}$.

Ehrenfest and Oppenheimer's Rule

When dealing with atoms and molecules, there is a rule formulated by Ehrenfest and Oppenheimer that involves summing the spins of all the protons, neutrons, and electrons present. This total spin, denoted as S, is calculated as:

$$S = s_p + s_n + s_e \tag{2.63}$$

where s_p represents the spin of protons, s_n is the spin of neutrons, and s_e is the spin of electrons. If the total spin S is an integer, then the particle in question is classified as a boson. Examples of such particles include molecular hydrogen (H_2), deuterium (D_2), molecular nitrogen (N_2), molecular oxygen (O_2), and helium-4 (4He). On the other hand, if the total spin S is a half-integer, then the particle falls into the category of fermions. Examples of fermionic molecules include heavy hydrogen (HD), ammonia (NH_3), nitric oxide (NO), and helium-3 (3He).

2.8 Some Examples of Microstate Counting

Example 2.5: Collection of Spins in a Magnetic Field

Consider a system composed of N atoms with a spin of 1/2 each, which do not interact with each other; this represents an ideal system of identical particles. The total energy E of this system can be expressed as:

$$E = \epsilon^{(1)} + \epsilon^{(2)} + \ldots + \epsilon^{(N)} \tag{2.64}$$

In this system, the atoms are arranged in a crystalline structure, and they are **distinguishable** from one another. Therefore, there is no need to concern ourselves with particle permutations. Additionally, all the atoms are subjected to an external magnetic field denoted as $\vec{B}$. Consequently, the spin orientation of each atom can take one of two values: it can either align with the external magnetic field, corresponding to a spin of 1/2, or align inversely to the field, corresponding to a spin of -1/2.

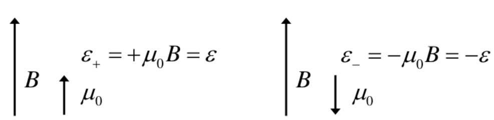


Figure 2.6: The orientation of the spin can take two values: aligned with the external field (spin 1/2) or aligned opposite to the field (spin -1/2).

For each particle, there are only two energy values, $-\epsilon$ and ϵ, forming a bounded energy spectrum. We denote these values as $\epsilon_+ = \epsilon$ with n_+ particles and $\epsilon_- = -\epsilon$ with n_- particles. It follows that the total number of microstates is given by:

$$N_T = 2^N \tag{2.65}$$

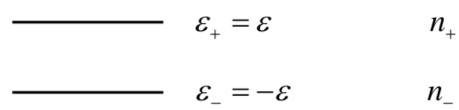


Figure 2.7: Energy states of not-interacting atoms with a spin of 1/2 each.

1 2 3 4 5	E
$-----$	-5ε
$----+$	
$---+-$	
$--+--$	-3ε
$-+---$	
$+----$	
$---++$	
$--+-+$	
$-+--+$	
$+---+$	
$--++-$	$-\varepsilon$
$-+-+-$	
$+--+-$	
$-++--$	
$+-+--$	
$++---$	

1 2 3 4 5	E
$--+++$	
$-+-++$	
$+--++$	
$-++-+$	
$+-+-+$	
$++--+$	$+\varepsilon$
$-+++-$	
$+-++-$	
$++-+-$	
$+++--$	
$-++++$	
$+-+++$	
$++-++$	$+3\varepsilon$
$+++-+$	
$++++-$	
$+++++$	$+5\varepsilon$

Figure 2.8: Number of microstates compatible with each macrostate.

The macrostate of the system is defined by fixing the total number of particles, N, and the total energy of the system, E. **Not all macrostates are equally probable**: some have more compatible microstates, and others have fewer. Let's assume the system contains only 5 particles ($N = 5$). From Fig. 5.8, we can calculate the number of microstates compatible with each macrostate: $\Omega(-5\epsilon) = 1$; $\Omega(-3\epsilon) = 5$; $\Omega(-\epsilon) = 10$; $\Omega(\epsilon) = 10$; $\Omega(3\epsilon) = 5$; and $\Omega(5\epsilon) = 1$. It follows that the total energy E of the system can be expressed in terms of the number of particles n_+ with energy ϵ and the number of particles n_- with energy $-\epsilon$:

$$E = n_+\epsilon_+ + n_-\epsilon_- = (n_+ - n_-)\epsilon \tag{2.66}$$

Similarly, the total number of particles N is given by the sum of n_+ and n_-:

$$N = n_+ + n_- \tag{2.67}$$

Example 2.6: Quantum gas (Bosons and Fermions)

Let's consider an ideal gas of identical particles. As it is a gas, the particles are indistinguishable, and we must take their indistinguishability into account when counting. Failing to do so would lead to an overestimate of the quantum states. In this scenario, each particle has only two accessible energy levels ϵ_1 and $\epsilon_2 = 2\epsilon_1$ that are reported in Fig. 2.9 with degeneracies $g_1 = 2$ and $g_2 = 3$, respectively. What is the number of compatible microstates for the macrostate characterized by a number of particles $N = 3$ and a total energy $E = 5\epsilon$, both for bosons and fermions?

For bosons, the number of microstates compatible with a macrostate characterised by $N = 3$ particles and a total energy $E = 5\epsilon$ can be calculated by considering Fig. 2.10. We stress that bosons can show the same quantum state within a given energy level. In the top left frame of Fig. 2.10, we see for instance three particles with the same quantum state in ϵ_1 and a total energy of 3ϵ. In the next one, the configuration has slightly changed, with one particle moving to a new state, but no changes in the macrostate (still $E = 3\epsilon$). When a particle jumps to the upper level, where $\epsilon_2 = 2\epsilon$, then this new microstate, compatible with the macrostate

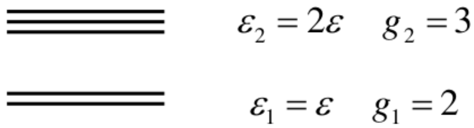


Figure 2.9: Accessible energy levels with respective degeneracies for the system discussed in Example 2.

$E = 4\epsilon$, is observed. If more particle rearrange, then one can finally observe microstates compatible with the desired macrostate of total energy $E = 5\epsilon$. The sum of these compatible microstates results to be $\Omega = 12$. We now turn our attention to fermions and refer to Fig. 2.11. We should remember that, for fermions, the

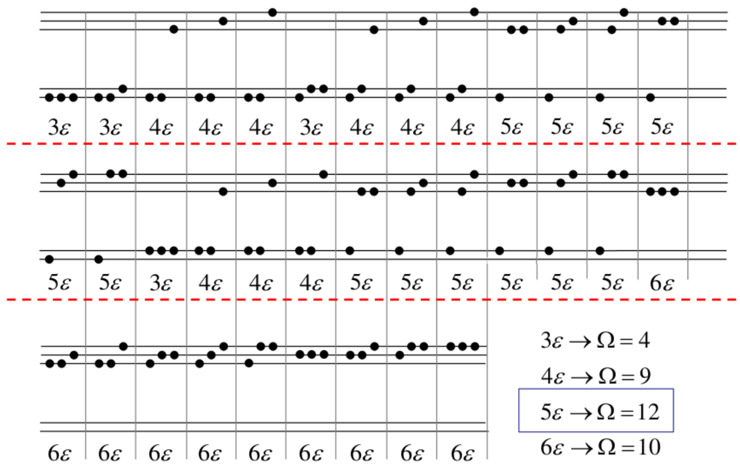


Figure 2.10: Number of compatible microstates for the macrostate characterised by a number of particles $N = 3$ and a total energy $E = 5\epsilon$. Case of bosons.

antisymmetry of the wave function prevents two of these family particles (which exhibit half-integer spin, like electrons) from being simultaneously in the same quantum state (Pauli's Exclusion Principle). It follows that the possible combinations are much less than those reported in Fig. 2.10. More specifically, in this case the number of microstates compatible with the macrostate characterised by a total energy equal to $E = 5\epsilon$ is $\Omega = 6$. We also notice that the macrostate characterised by a total energy $E = 3\epsilon$ is not observed in a gas of fermions, but it is observed in a gas of bosons. While this graphical approach is intuitive, it can be rather

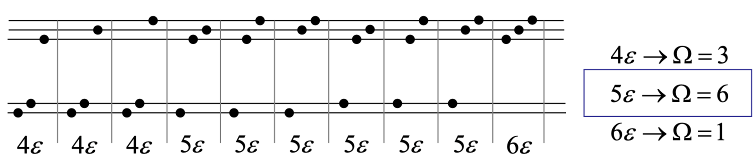


Figure 2.11: Number of compatible microstates for the macrostate characterised by a number of particles $N = 3$ and a total energy $E = 5\epsilon$. Case of fermions.

inefficient as soon as the number of particles or degeneracies increases. In this case, it is more convenient applying the following formula for bosons:

$$W_b = \Pi_i \frac{(n_i + g_i - 1)!}{n_i!(g_i - 1)!} \tag{2.68}$$

and for fermions:

$$W_f = \Pi_i \frac{g_i!}{n_i!(g_i - n_i)!} \tag{2.69}$$

where i indicates energy levels. For the sake of completeness, we also add the formula that can be applied to classical (distinguishable) particles:

$$W_c = N!\,\Pi_i \frac{g_i^{n_i}}{n_i!} \tag{2.70}$$

2.9 Problems

2.1 Consider a two-dimensional quantum system with wave function $\Psi(x, y; t)$ defined in $x = [1, 10]$ and $y = [1, 10]$. Calculate the probability of finding the particle in $x = [2, 4]$ and $y = [3, 6]$ at time $t = 0$, if $\Psi(x, y; 0) = 4e^{-(x+y)}$.

2.2 Determine the total number of compatible microstates for a fixed value of n_+ and n_- and express it in terms of E and N. In this counting, temperature is not considered. What might be expected to happen at very low temperatures? And at very high temperatures?

2.3 Calculate the number of microstates compatible with the macrostate $N = 2$ and $E = 3\epsilon$, considering that the particles are bosons, fermions, or classical particles, and hence, distinguishable from each other. There exist two energy levels with respective degeneracies: ϵ with $g_1 = 2$ and 2ϵ with $g_2 = 3$. Show that, for classical particles, the total number of possible microstates (at any energy) is indeed $\Omega = 25$. Is it possible to find a general expression for the number of microstates (Ω) for bosons and fermions with an arbitrarily large number of particles N?

2.4 Consider a particle of mass $m = 1/2$ subject to the potential $V(x) = Ax^2 + Bx^4$, where $A = -1$ and $B = 1$. Plot the trajectories in the phase space.

2.5 Consider the operator $\exp(-\beta\hat{H})$, where $\beta = \frac{1}{kT}$.

(a) Demonstrate the expression

$$\left\langle \mathbf{r}' \left| \mathrm{e}^{-\beta\hat{H}} \right| \mathbf{r} \right\rangle = \sum_k \phi_k^*(\mathbf{r})\phi_k(\mathbf{r}')\mathrm{e}^{-\beta E_k} \tag{2.71}$$

where $|\phi_k\rangle$ are the eigenstates of $\hat{H}$, and $\phi_k(\mathbf{r})$ is the wave function in position representation.

(b) Calculate this quantity for the specific case of a free particle confined in a cubic box of side L;

(c) Using the result from part (b), obtain the partition function, defined as $Z = \mathrm{Tr}\left[e^{-\beta\hat{H}}\right] = \int \left\langle \mathbf{r} \left| e^{-\beta\hat{H}} \right| \mathbf{r} \right\rangle d\mathbf{r}$.

Chapter 3

Postulates of Statistical Physics. Ensembles

3.1 Introduction

The postulates of Statistical Physics serve as fundamental axioms that clarify the behaviour of systems with numerous particles, bridging the gap between the microcosmic world described by quantum mechanics and the macroscopic domain governed by classical thermodynamics. In this chapter, we'll begin by introducing these postulates and their connection to thermodynamics and then offer a base for understanding the ensemble theory. Ensemble theory will allow us to explore the macroscopic characteristics of matter and connect them to their underlying microscopic dynamics. By doing so, we can make predictions, model complex systems, and gain deeper insights into the intricate world of statistical physics.

3.2 Definition of Ensemble. Postulate I

3.2.1 Systems with Many Particles ($N \sim 10^{23}$)

In the study of systems with an enormous number of particles, typically on the order of 10^{23} or more, we adopt a distinct approach to understand their behaviour. This approach allows us to focus on the macroscopic aspects of these systems, where only a few key variables are needed to define the state of the system. These essential macroscopic variables, such as the internal energy U, volume V, and the number of particles N, encapsulate the system's conditions. How these variables are interconnected for a system in equilibrium is meticulously described by the principles of thermodynamics. In this ensemble view, we consider an ensemble of microstates, each characterised by a specific set of positions and momenta for all particles. The collection of these microstates represents a microcanonical ensemble, encompassing all conceivable microstates that correspond to the given macrostate. These microstates are determined by the positions $\{q_1, q_2, \ldots, q_{3N}\}$ and momenta $\{p_1, p_2, \ldots, p_{3N}\}$ of all particles, totalling $3N$ variables.

The system's evolution over time follows the dynamical laws of either quantum mechanics or classical mechanics, depending on the scale and nature of the system. While thermodynamics guides us in understanding the macroscopic equilibrium properties, the microscopic details are shaped by the principles of mechanics, whether they are classical or quantum in nature. This interplay between macroscopic and microscopic descriptions forms the foundation of statistical physics for systems with a vast number of particles. Microscopically interpreting the macroscopic behaviour of a system involves several challenges when using Mechanics, whether in the quantum or classical context. Tracking the temporal evolution of particles within the system becomes virtually impossible due to two key factors:

- Enormous particle count: the system typically consists of a vast number of particles, making individual tracking impractical;

- Chaotic motion: the motion of these particles is often chaotic, further complicating any attempts at precise tracking.

For each macroscopic configuration or macrostate, there exists a vast number of compatible microscopic configurations or microstates. This multitude of microstates is a fundamental aspect of the microscopic description of the system. To facilitate the study of the macroscopic system's behaviour, a more straightforward approach is adopted. This approach assigns a particular **statistical weight** to each of the microstates that are consistent with a given macrostate.

3.2.2 Satisfactory Application of Statistical Laws to an Ensemble of Individuals

A successful application of statistical laws to number of individuals depends on specific conditions:

1. The number of individuals must be **sufficiently large**.

2. It is essential that the individuals exhibit a certain degree of **similarity**.

3. The individuals within the ensemble must be **independent of each other**.

One approach to addressing this challenge is considering physical systems as composed of a large number, N, of particles, such as atoms or molecules. These particles serve as the constituents of the ensemble under consideration. Can one use the **kinetic theory of gases** to this end? The assumption underlying this approach is that, when dealing with a substantial number of particles, the collective behaviour of these particles, while still following the principles of statistical physics, can be analysed effectively without explicitly tracking each individual particle. The application of statistical laws to such systems, when the conditions mentioned above are met, provides valuable insights into the macroscopic properties and behaviour of these systems. In the context of the kinetic theory of gases, each particle is considered a constituent of the ensemble. Whether this application of statistical physics is satisfactory depends on the following criteria:

1. Is there a sufficiently large number of individuals?

2. Are the individuals somewhat similar to each other?

3. Are the individuals independent of each other?

When addressing these questions in the context of the kinetic theory of gases, we find the following:

1. Indeed, the number of particles in a macroscopic system is extremely large, typically on the order of $N \sim 10^{23}$.

2. The similarity of particles is a requirement for successful statistical analysis. However, in the case of **gases composed of more than one component**, the particles are no longer equivalent from a statistical perspective.

3. The independence of particles is a fundamental consideration. Unless dealing with a diluted gas, **interactions** among particles are common, affecting gases, liquids, solids, and more.

Therefore, in the context of statistical physics, the kinetic theory of gases falls short in providing a comprehensive general theory. An alternative approach, known as the **Gibbs Ensemble Method**, introduced by Gibbs in 1902, offers a more suitable perspective for the development of a broader theoretical framework. This method transcends the limitations of the kinetic theory of gases and holds significant promise for addressing complex systems in the realm of statistical physics.

3.2.3 Gibbs Ensemble Method: What Is It?

A **Gibbs Ensemble** is a concept that involves the coupling of multiple systems, each of which can exhibit various behaviours. These systems can be thermally, mechanically, or materially linked in various ways. When considering a Gibbs Ensemble, the focus is on an isolated collection of systems that share identical macrostates but possess different microstates. In essence, within a Gibbs Ensemble, multiple macroscopically identical systems are analysed, each characterised by its microstates. The **probability** (P_n) of a system residing in a specific microstate n, which is consistent with the macrostate, is a key parameter in this method. These probabilities provide valuable insights into the behaviour of systems within the ensemble.

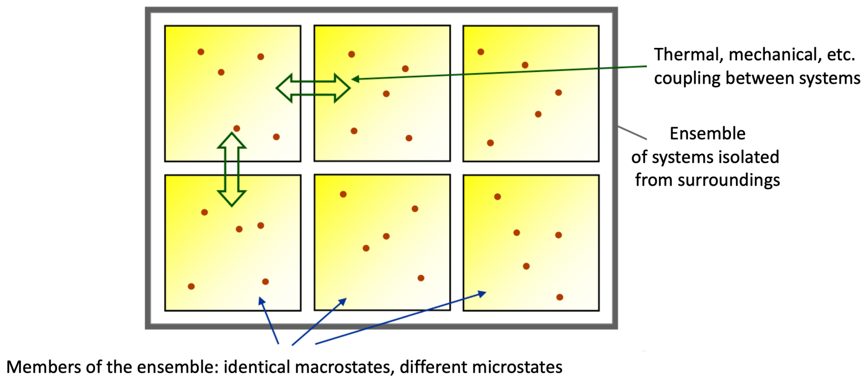


Figure 3.1: Gibbs ensemble: a collection of identical macrostates characterised by different microstates.

Now, with reference to Fig. 3.1 let's assess whether the three conditions are met:

1. **Are there a sufficiently *large number* of individuals?**
   The number of systems in the ensemble can be made arbitrarily large: $N_c \to \infty$.
2. **Are the individuals *similar* to each other?**
   Yes, they are. The systems in the ensemble are macroscopically identical.
3. **Are the individuals *independent* of each other?**
   If a system interacts with others solely through its surface, the interaction energy is negligible compared to the system's energy. Therefore, each system is independent:

$$E = E_1 + E_2 + E_{12} \approx E_1 + E_2 \tag{3.1}$$

3.2.4 First Postulate: Postulate of Averages

In the context of the ensemble method, the conventional portrayal of a system's state as a point in phase space (or a specific quantum state) evolving over time undergoes a substantial change of perspective. In this framework, each point within the phase space (or quantum state) now serves as a representation of a **potential** state of the system, which occurs with a given probability. This change in perspective introduces the concept of a **density function** or **density operator**. This operator, pivotal in the statistical description of a system, reads

$$\hat{\rho}(t) \equiv \rho(q_1, \dots, q_N, p_1, \dots, p_N, t) \tag{3.2}$$

It provides a way to mathematically characterise the probabilities associated with different microstates. If any microstate is incompatible with the macrostate, this function is equal to zero. The idea of a density operator is central to this framework. The density operator is a mathematical construct used to connect the macroscopic observables with the myriad of microstates a system can access. Through the density operator, we can calculate averages of various physical properties, enabling us to describe the system's behaviour in statistical terms. In other words, the macroscopic observables are obtained as averages over all compatible microstates, calculated using the density function (operator). This approach relies on the assumption that the systems forming the collectivity of microstates collectively represent the thermodynamic state and the interactions with the external environment of the original system. This ensures that the statistical properties of the microstates adequately reflect the macroscopic observables. In summary, the transition from macroscopic thermodynamics to the microscopic world of statistical physics is facilitated by the density function ρ, connecting the statistical behaviour of a system with its fundamental microstate descriptions (see Fig. 3.2 for details).

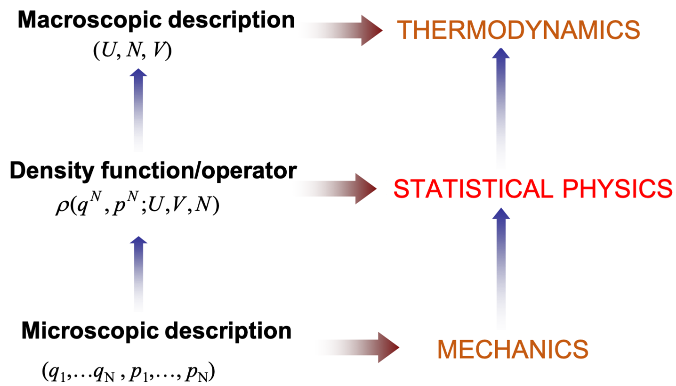


Figure 3.2: In the realm of thermodynamics, a macroscopic description of a system is characterised by internal energy (U), number of particles (N), and volume (V). These thermodynamic variables provide a framework for understanding and analysing the behaviour of macroscopic systems. To delve into the statistical aspects of physical systems, we shift our perspective to consider the density function (or operator) denoted by $\rho(q^N, p^N; U, V, N)$. This function connects the macroscopic parameters (U, N, V) with the microscopic descriptions, allowing us to explore the statistical behaviour of the system. At the level of microscopic descriptions, we analyse the system in terms of individual positions and momenta: $(q_1, \ldots, q_N, p_1, \ldots, p_N)$. This microscopic perspective aligns with the principles of mechanics, enabling us to scrutinise the detailed dynamics and interactions at the particle level.

3.3 Classical Formulation of the First Postulate

Macroscopic Physical Quantities. These are represented by functions of space and time coordinates, often taking the form of fields: $A(\mathbf{r}, t)$. Thermodynamics, in particular, focuses on equilibrium states: $A(\mathbf{r}, t) \equiv A(\mathbf{r})$.

Microscopic Physical Quantities. In contrast, these are described by dynamic functions in phase space coordinates $\{\mathbf{q}, \mathbf{p}\} \equiv \{q_1, \ldots, q_N, p_1, \ldots, p_N\}$. Macroscopic quantities can be simplified as functions of their spatial coordinates only: $a(\mathbf{q}, \mathbf{p}; \mathbf{r})$

3.3.1 Classical Formulation of the First Postulate

The First Postulate (Classical Formulation): "The values of macroscopic physical quantities at time t are given as the averages of microscopic dynamic functions."

$$A(\mathbf{r}, t) \equiv \langle a(\mathbf{q}, \mathbf{p}; \mathbf{r}) \rangle = \int_\Gamma a(\mathbf{q}, \mathbf{p}; \mathbf{r}) \rho(\mathbf{q}, \mathbf{p}; t)\, d\mathbf{q}\, d\mathbf{p} \tag{3.3}$$

$$\int_\Gamma \rho(\mathbf{q}, \mathbf{p}; t)\, d\mathbf{q}\, d\mathbf{p} = 1 \quad \text{Normalization} \quad \rho(\mathbf{q}, \mathbf{p}; t) \geq 0 \tag{3.4}$$

Here, $\rho(\mathbf{q}, \mathbf{p}; t)$ represents the density function and $\rho(\mathbf{q}, \mathbf{p}; t)\, d\mathbf{q}\, d\mathbf{p}$ is the probability of finding a microstate within the hypervolume $d\mathbf{q}\, d\mathbf{p}$ located at the point $(\mathbf{q}, \mathbf{p})$ at time t. A good example is offered by the relation between energy and Hamiltonian of a system:

$$E = \langle H(\mathbf{q}, \mathbf{p}) \rangle = \int_\Gamma H(\mathbf{q}, \mathbf{p}) \rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p}, \tag{3.5}$$

where E represents the average energy of a system; $\langle H(\mathbf{q}, \mathbf{p}) \rangle$ denotes the average value of the system's Hamiltonian, which describes the total energy of the system; $H(\mathbf{q}, \mathbf{p})$ is the Hamiltonian function, which depends on the positions $\mathbf{q}$ and momenta $\mathbf{p}$ of the system's microscopic constituents (*e.g.*, particles) and captures the energy of the system in terms of these microscopic variables; and $\rho(\mathbf{q}, \mathbf{p})$ is the probability density function for the system, which characterises the likelihood of finding the system in a specific microstate defined by positions $\mathbf{q}$ and momenta $\mathbf{p}$. This equation expresses that the average energy E of the system is determined by calculating the expected value of the Hamiltonian using the probability density function $\rho(\mathbf{q}, \mathbf{p})$. In other words, it relates the macroscopic property (average energy) to the microscopic properties (positions and momenta of the particles) and their associated probabilities.

3.3.2 Mechanical and Thermal Magnitudes

Postulate of Averages. Every dynamic function $a(\mathbf{q}, \mathbf{p})$ corresponds to a macroscopic observable A, calculated as an average weighted by the density function.

$$a(\mathbf{q}, \mathbf{p}) \xrightarrow{\rho} A \tag{3.6}$$

Is it always possible to find a dynamic function corresponding to a given macroscopic observable through the averaging process within a given ensemble? No, this is not always the case. For mechanical observables, like energy, pressure, and linear momentum, it's feasible to obtain them as averages of dynamic functions. However, the situation is different for thermal observables, such as entropy, temperature, Helmholtz or Gibbs free energy, and chemical potential. In these cases, there is no straightforward dynamic function at the microscopic level that corresponds to the macroscopic observable, as they lack microscopic counterparts. Entropy, for instance, can be expressed as:

$$S = -\int_\Gamma \ln\left[\rho(\mathbf{q}, \mathbf{p})\right] \rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p} \tag{3.7}$$

In this equation, the natural logarithm of the density function, $\ln\left[\rho(\mathbf{q}, \mathbf{p})\right]$, is averaged by multiplying it with the density function itself and integrating over all possible particle positions and momenta $(\mathbf{q}, \mathbf{p})$. This shows that entropy involves the averaging of the logarithm of the density function and, as a result, it doesn't have a direct microscopic equivalent. Instead, entropy is a global property of the system and is considered a thermal variable.

Temperature, another example of a thermal variable, is also related to the overall state of the system. In fact, it's only well-defined in thermodynamic equilibrium. Temperature is a property that pertains to the entire system as a whole and is not associated with any specific microscopic property. (Remember how Thermodynamics emphasises that temperature doesn't exist for individual atoms and molecules!)

3.3.3 Ensemble Average and Time Average. Ergodicity

Ensemble averaging is a statistical concept used in physics and engineering to describe the average behaviour of a system by considering all possible states or configurations within a certain macrostate. It's like taking a snapshot of a system in time and observing the different microstates it can exist in. The ensemble average of a physical quantity, denoted as $\langle A \rangle$, is calculated by summing the values of that quantity over all microstates, each weighted by the probability of the system being in that microstate. Mathematically, it is expressed as:

$$\langle A \rangle = \sum_i A_i P_i \tag{3.8}$$

where P_i is the probability of observing a given microstate i. For a sufficiently large number of microstates, we can integrate the above equation and employ the density operator:

$$\langle A \rangle = \int_\Gamma a(\mathbf{q}, \mathbf{p})\rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p} \tag{3.9}$$

Ensemble averaging is different from the one obtained through experimental measurements. In an experiment, it's always necessary to measure a certain property of the system over a time τ that is significantly longer than the timescale of the microscopic dynamics. The equation below describes averaging over time by observing the system's properties over a long time interval τ.

$$\bar{A} = \lim_{\tau \to \infty} \frac{1}{\tau} \int_0^\tau a(\mathbf{q}_t, \mathbf{p}_t)\, dt \tag{3.10}$$

Therefore, time averaging focuses on the behaviour of a **single system over time**. It involves measuring the value of a physical quantity repeatedly at different points in time and calculating the average over these measurements. In other words, measuring an observable A involves a **time average** taken along a trajectory in the phase space. In a good measurement, an *enormous* number of microscopic configurations are traversed. For example, if you want to calculate the time-averaged velocity of a particle, you measure its velocity at different times and then calculate the average over those measurements. Fig. 3.3 illustrates this idea. In summary, ensemble averaging considers all possible states in a macrostate and weighs them by their probabilities, while time averaging focuses on a single system's behaviour over time.

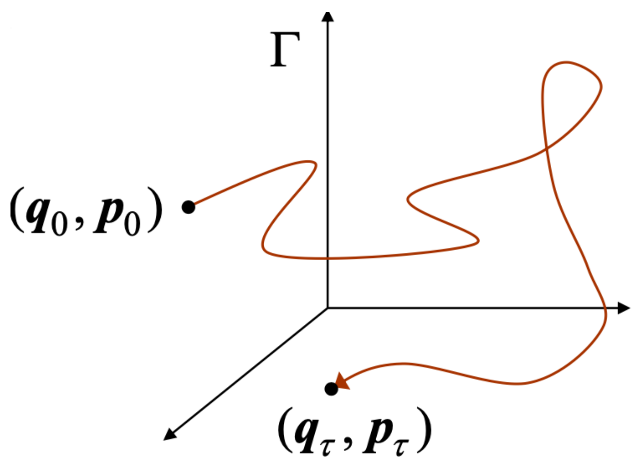


Figure 3.3: Time trajectory of a system in the phase space Γ.

The first postulate leads us to the **ergodic hypothesis**, which states that the time average (measured) equals the ensemble average:

$$\bar{A} = \langle A \rangle = \int_\Gamma a(\mathbf{q}, \mathbf{p})\rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p} \tag{3.11}$$

In simpler terms, the long-term time average of an observable A for a system (measured over a sufficiently long time) is equal to the ensemble average of that observable calculated over all possible microstates, each weighted by their probability of occurrence. In other words, the ergodicity of a system signifies that the statistical behaviour of the system can be adequately represented by time averages over a sufficiently long period. This is essential for linking the microscopic properties (captured in ensemble averages) to macroscopic observations (represented by time averages). The notion of ergodicity simplifies the analysis of complex systems since it allows us to replace the temporal evolution of a physical quantity with an ensemble average over all possible microstates. It relieves us from the need to track the real-time evolution of the physical quantity $a(\mathbf{q}_t, \mathbf{p}_t)$. This significantly eases the theoretical and practical aspects of statistical physics, especially when dealing with a vast number of interacting particles. A system that satisfies the ergodic hypothesis is referred to as an **ergodic system**. For the field of statistical physics to be applicable, it's essential that the system is ergodic. However, it's important to remember that not all systems are ergodic, and this assumption's validity depends on the specific system being studied. When is a system non-ergodic? It's non-ergodic when there are regions in the phase space that are not accessible from others (disconnected regions)[1].

Properties of Ergodic Systems

1. In an ergodic system, given an initial point in phase space, the system's trajectory returns near this point after a sufficiently long time (see Fig. 3.4).
2. In an ergodic system, starting from an initial microstate, one can approach any other microstate in the system as closely as desired.

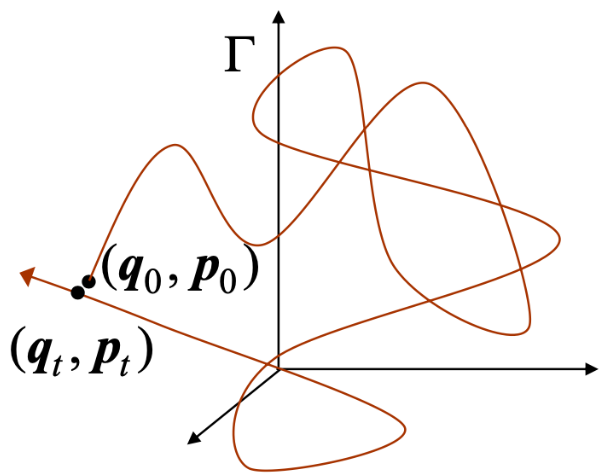


Figure 3.4: In an ergodic system, given an initial point in phase space, the system's trajectory returns near this point after a sufficiently long time.

3.4 Density Function. Liouville's Theorem

Consider a small portion of phase space, denoted by $\Delta\mathbf{q}\Delta\mathbf{p}$. If the system is ergodic, over a sufficiently long time, the system's trajectory will pass through this portion many times. Let's define Δt as the time during which the system's microstate is in this portion, while T is the total time of evolution. In an ergodic system, as T tends to infinity, the ratio between these times approaches a finite value that coincides with the probability of the system being in $\Delta\mathbf{q}\Delta\mathbf{p}$:

$$P = \lim_{T\to\infty} \frac{\Delta t}{T}. \tag{3.12}$$

The density function can then be written as follows

$$\rho(\mathbf{q}, \mathbf{p}) = \lim_{\Delta\mathbf{q}\to 0} \lim_{\Delta\mathbf{p}\to 0} \frac{1}{\Delta\mathbf{q}\Delta\mathbf{p}} \left[\lim_{T\to\infty} \frac{\Delta t}{T} \right] \tag{3.13}$$

[1]Amorphous solids like glasses exhibit properties of non-ergodicity. They have a complex energy landscape with multiple local minima. The system gets "stuck" in one of these minima and doesn't explore all the possible configurations during practical time scales. Biological systems, such as proteins or living cells, often involve a large number of particles with intricate interactions. These systems can be non-ergodic because they have specific functional states and may not have the time to explore all possible configurations due to constraints imposed by their biological functions. Systems like foams, gels, or certain emulsions can be non-ergodic due to their complex and dynamic structure. They might have a large number of metastable states, making them challenging to analyze using traditional statistical physics.

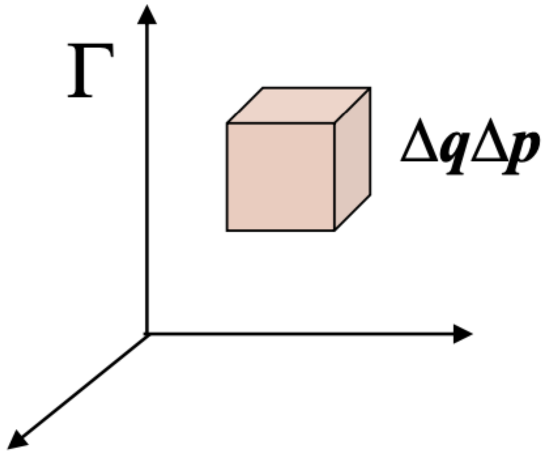


Figure 3.5: Small portion of the phase space Γ. The trajectory of an ergodic system will pass through this portion many times over a sufficiently long time.

where $\rho(\mathbf{q}, \mathbf{p}) \geq 0$ and satisfies normalisation:

$$\int_\Gamma \rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p} = 1. \tag{3.14}$$

Alternative Definition of the Density Function:

The density function can be defined as the probability density of a specific microstate occurring within an infinite ensemble of microstates compatible with a given macrostate. Each element i of the ensemble corresponds to a point $\alpha_i = (\mathbf{q}_i, \mathbf{p}_i)$ in phase space. The ensemble is a set of η points in phase space:

$$\{\alpha_1, \alpha_2, \ldots, \alpha_\eta\} \tag{3.15}$$

We can define $\eta_{\Delta\mathbf{q}\Delta\mathbf{p}}$ as the number of microstates in the ensemble that fall within the volume $\Delta\mathbf{q}\Delta\mathbf{p}$. As η approaches infinity, the cloud of points transforms into a continuous medium in phase space. The density of this continuous medium is precisely the density function:

$$\rho(\mathbf{q}, \mathbf{p}) = \lim_{\Delta\mathbf{q}\to 0} \lim_{\Delta\mathbf{p}\to 0} \frac{1}{\Delta\mathbf{q}\Delta\mathbf{p}} \left[\lim_{\eta\to\infty} \frac{\eta_{\Delta\mathbf{q}\Delta\mathbf{p}}}{\eta} \right] \tag{3.16}$$

This density function characterises the distribution of microstates within the ensemble in the macrostate, allowing us to analyse the system's behaviour statistically.

If the system is ergodic, the definitions of Eq. 3.13 and Eq. 3.16 are entirely equivalent.

Given a macrostate, there is an immense number of compatible microstates for the system. The density function provides the statistical weight (probability) of each microstate $(\mathbf{q}, \mathbf{p})$ in the ensemble. If our system is in **thermodynamic equilibrium**, the macrostate is independent of time, and therefore, the density function does not depend on time either:

$$\rho(\mathbf{q}, \mathbf{p}; t) = \rho(\mathbf{q}, \mathbf{p}) \tag{3.17}$$

Outside of equilibrium, the density function will also evolve in time. What is the time evolution of the density function? This is described by **Liouville's Theorem**.

3.4.1 Liouville's Theorem

In statistical physics, understanding the time evolution of the density function $\rho(\mathbf{q}, \mathbf{p}; t)$ is crucial. This function represents the probability density of finding the system in a specific region of phase space at time t. Liouville's theorem provides crucial insights into this evolution. It states that in phase space, the density of an ensemble of systems evolving under Hamilton's equations remains constant along phase-space trajectories:

$$\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + \{\rho, H\} = 0 \tag{3.18}$$

where $\{\cdot,\cdot\}$ indicates Poisson brackets. In other words, the phase-space density is **conserved** along trajectories of the system. We notice that $\rho(\mathbf{q}, \mathbf{p}; t)$ is defined in Γ with the structure of the dynamic functions. However, while it evolves dynamically, it is not a physical observable in the usual sense, as it does not correspond to a directly measurable quantity. It should be noticed that the **total derivative** of $\rho(\mathbf{q}, \mathbf{p}; t)$ is zero, while the **partial time derivative** is zero only if the system is at the thermodynamic equilibrium (no change over time). In non-equilibrium systems, $\rho(\mathbf{q}, \mathbf{p}; t)$ evolves explicitly with time, but this evolution is exactly balanced by the system's motion in phase space, ensuring that the total derivative remains zero:

$$\frac{\partial \rho}{\partial t} + \sum_i \left(\frac{\partial \rho}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial \rho}{\partial p_i}\frac{\partial H}{\partial q_i} \right) = 0 \tag{3.19}$$

This occurs in systems such as a gas expanding into a vacuum, a system relaxing after an external perturbation, or a system driven by a time-dependent external force. By contrast, if the system is at thermodynamic equilibrium, then all the averages are time-independent and

$$\frac{\partial \rho}{\partial t} = 0 \tag{3.20}$$

which leads to

$$\{\rho, H\} = 0 \tag{3.21}$$

A particular solution of $\{\rho, H\} = 0$ is $\rho =$ const. This implies that all the available states in the phase space are equally probable for the systems in that particular ensemble. Such an ensemble, where number of particles, volume and energy are constant, is referred to as *microcanonical ensemble* (it will be discussed in the following chapter). In other ensembles, ρ is stationary, but not necessarily constant. For instance, in the *canonical ensemble*, where temperature, rather than energy, is constant, $\rho \sim e^{-E/kT}$ and, in the *macrocanonical ensemble*, where the number of particles fluctuates and the chemical potential is constant along with volume and temperature, $\rho \sim e^{-E/kT + \mu N/kT}$.

Note that, in the most general case, the above result does not necessarily means that the *shape* of a given region of phase space remains unchanged, but only that the volume this portion occupies is constant over time. In simpler terms, Liouville's theorem tells us that if we know the initial distribution of microstates in phase space (described by the density function), this distribution will remain constant as the system evolves. This theorem is particularly useful for systems obeying Hamiltonian dynamics, such as ideal gases, and plays a central role in classical statistical physics. For a detailed proof of Liouville's theorem, the interested reader is referred to Appendix E.

Example 3.1

A free particle of mass m is constrained to move in one dimension, with a Hamiltonian given by $H = p^2/2m$. Let's verify that the area of the rectangular enclosure shown in Fig. 3.6 remains invariant during the natural evolution of the system.

In order to demonstrate that the area of the rectangular enclosure remains invariant during the natural evolution of the system, let's begin by defining the rectangular enclosure in the phase space and then make use of Liouville's theorem. The phase space, in this case, is two-dimensional, represented by the coordinates q and momenta p. At time $t = 0$, the rectangular enclosure can be defined in terms of its four corners:

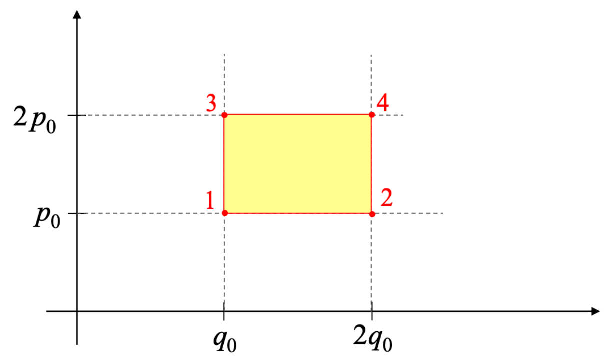


Figure 3.6: Region of the phase space where a free particle of mass m is constrained to move.

(q_0, p_0), $(2q_0, p_0)$, $(q_0, 2p_0)$ and $(2q_0, 2p_0)$, where q_0, $2q_0$, p_0, and $2p_0$ are given values. The total area is then $A(t = 0) = A_0 = q_0 p_0$. Let's write the equations of motion:

$$\dot{p} = -\frac{\partial H}{\partial q} = -\frac{\partial(p^2/2m)}{\partial q} = 0 \longrightarrow p = p_0 = \text{const.} \tag{3.22}$$

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{\partial(p^2/2m)}{\partial p} = \frac{p}{m} \tag{3.23}$$

Integrating with respect to time, we obtain:

$$\int_0^t dq = \int_0^t \frac{p}{m}\, dt \longrightarrow q = q_0 + \frac{p}{m}t = q_0 + \frac{p_0}{m}t \tag{3.24}$$

The new coordinates are obtained with a linear transformation of the old coordinates. This implies that the straight lines of the rectangle are still straight. The locations of the four points at time t are

1. $a_1 = (q_0, p_0) \longrightarrow a_1' = (q_0 + \frac{p_0}{m}t, p_0)$
2. $a_2 = (2q_0, p_0) \longrightarrow a_2' = (2q_0 + \frac{p_0}{m}t, p_0)$
3. $a_3 = (q_0, 2p_0) \longrightarrow a_3' = (q_0 + \frac{2p_0}{m}t, 2p_0)$
4. $a_4 = (2q_0, 2p_0) \longrightarrow a_4' = (2q_0 + \frac{2p_0}{m}t, 2p_0)$

The area at time t can be computed as the cross product of $\vec{u} \equiv \overline{a_2' a_1'}$ and $\vec{v} \equiv \overline{a_3' a_1'}$. That is $A(t) = \vec{u} \times \vec{v} = q_0 p_0$. This result is a consequence of Liouville's theorem and demonstrates that the area of the rectangular enclosure, whose shape changes to rhombus, remains invariant during the natural evolution of the system. A quick test to corroborate the invariance of the volume (area in 2D case) of the phase space over time is showing that the Jacobian associated to the transformation $\mathcal{T} : (q_0, p_0) \to (q, p)$ is equal to 1. The Jacobian of the transformation $\mathcal{T}$ reads

$$J = \frac{\partial(q_t, p_t)}{\partial(q_0, p_0)} = \begin{vmatrix} \frac{\partial q_t}{\partial q_0} & \frac{\partial q_t}{\partial p_0} \\ \frac{\partial p_t}{\partial q_0} & \frac{\partial p_t}{\partial p_0} \end{vmatrix} = \begin{vmatrix} 1 & \frac{t}{m} \\ 0 & 1 \end{vmatrix} = 1 \tag{3.25}$$

We conclude that the Jacobian is indeed equal to 1 and consequently the transformation $\mathcal{T}$ satisfies Liouville's theorem. However, it should be noticed that calculating the Jacobian says nothing on whether the shape of the phase space changes or not.

Some questions that arise from the above discussions. What is the quantum equivalent of the density operator and Liouville's Theorem? How does ρ behave for an isolated system in equilibrium?

3.5 Quantum Formulation of the First Postulate

As we mentioned in the previous section, the first postulate of classical statistical mechanics states that the state of a system is described by a probability distribution $\rho(\mathbf{q},\mathbf{p})$ over its microstates, where $\mathbf{q}$ and $\mathbf{p}$ are the generalized coordinates and momenta of the system. The macroscopic properties of the system, such as the expected value of an observable $B(\mathbf{q},\mathbf{p})$, are calculated as ensemble averages, which are integrals over phase space:

$$\langle B\rangle = \int_\Gamma B(\mathbf{q},\mathbf{p})\rho(\mathbf{q},\mathbf{p})\, d\mathbf{q}\, d\mathbf{p}. \tag{3.26}$$

Now, consider a quantum system prepared in the pure state $|\psi\rangle$. The expected value of an operator $\hat{B}$ in this state is given by

$$\bar{B} = \langle\psi|\hat{B}|\psi\rangle = \int \Psi^*(q_1,\ldots,q_s)(\hat{B}\Psi(q_1,\ldots,q_s))\, dq_1\ldots dq_s \tag{3.27}$$

where $\bar{B}$ is the quantum expectation value of the observable B, $|\psi\rangle$ represents the quantum state of the system (a pure state), $\hat{B}$ is the operator associated with the observable B, $\Psi(q_1,\ldots,q_s)$ is the wavefunction of the system in the chosen representation and $\Psi^*(q_1,\ldots,q_s)$ is the complex conjugate of the wavefunction. This expectation value represents the mean outcome obtained from repeated measurements of B on identically prepared systems in the state $|\psi\rangle$. It is essential to distinguish this **quantum average** from an ensemble average, which involves a second average.

In Quantum Mechanics, the expectation value is a first-level average, determined solely by the quantum state ψ (a pure state). However, in practical scenarios, we often lack complete knowledge of the system's exact microstate. Instead, we only have information about its macrostate, which may correspond to a statistical mixture of pure states, that is a mixed state.

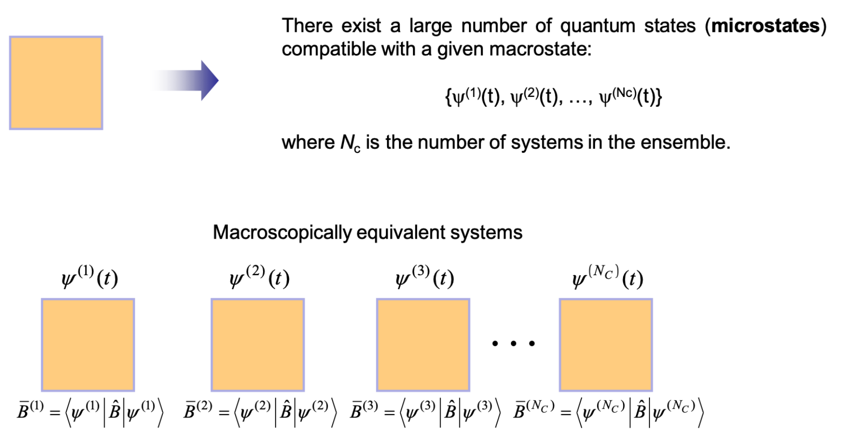


Figure 3.7: Quantum microstates and macroscopically equivalent systems.

In the case of an ensemble consisting of a large number of systems, the ensemble average of an operator $\hat{B}$ at time t is given by:

$$\langle B(t)\rangle = \lim_{N_c\to\infty}\frac{1}{N_c}\sum_{i=1}^{N_c}\bar{B}^{(i)}(t) = \lim_{N_c\to\infty}\frac{1}{N_c}\sum_{i=1}^{N_c}\langle\psi^{(i)}(t)|\hat{B}|\psi^{(i)}(t)\rangle \tag{3.28}$$

This equation states that, in an ensemble of N_c quantum systems, the macroscopic physical quantity B at time t is obtained by averaging the expectation values of the corresponding operator $\hat{B}$ over all individual systems. In the limit as $N_c\to\infty$, this ensemble average provides a **statistical description** of the system's macroscopic behaviour. A more compact and general way to express this ensemble average is through the **density operator** (or **density matrix**) $\hat{\rho}$, which describes the statistical mixture of quantum states within an ensemble. We will see how in the following section. Thus, Eq. (3.28) emphasises that, when dealing with a large number of systems, the expectation value of an operator (representing a macroscopic physical quantity) can be equivalently obtained as an ensemble average.

In summary, the first postulate in quantum mechanics involves (*i*) a quantum average that describes the behaviour of individual quantum systems (specific pure states), and (*ii*) a statistical average that describes the overall system when it is in a mixed state or an ensemble of quantum systems.

3.6 Density Matrix. von Neumann's Equation

The introduction of the density matrix provides a powerful framework for describing quantum systems statistically, bridging the gap between quantum mechanics and thermodynamic ensembles. When we describe a system using a density matrix, it means that the system is not necessarily in a single, well-defined quantum state (a pure state). Instead, it could be in one of several pure states $|\psi^{(i)}(t)\rangle$, each with a probability P_i. Basically, the density matrix encodes both the quantum mechanical superposition (within each $|\psi^{(i)}(t)\rangle$) and the classical uncertainty (the probability P_i) over which state the system is actually in. This is what we are discussing in this section.

In the average of Eq. 3.28, certain systems might be in the same quantum state $|\psi^{(i)}\rangle$. For instance, $N^{(1)}$ systems in $|\psi^{(1)}\rangle$, $N^{(2)}$ systems in $|\psi^{(2)}\rangle$, etc. It follows that

$$\langle B(t)\rangle = \lim_{N_c\to\infty}\left\{\frac{N^{(1)}}{N_c}\bar{B}^{(1)}(t) + \frac{N^{(2)}}{N_c}\bar{B}^{(2)}(t) + \cdots + \frac{N^{(i)}}{N_c}\bar{B}^{(i)}(t) + \dots\right\} \tag{3.29}$$

Let's define the probability to find the quantum state $|\psi^{(i)}\rangle$ in the ensemble as

$$P_i = \lim_{N_c\to\infty}\frac{N^{(i)}}{N_c} \tag{3.30}$$

Then Eq. 3.29 can be rewritten as

$$\langle B(t)\rangle = \sum_i P_i\bar{B}^{(i)}(t) = \sum_i P_i\langle\psi^{(i)}(t)|\hat{B}|\psi^{(i)}(t)\rangle, \tag{3.31}$$

with

$$\sum_i P_i = 1 \tag{3.32}$$

and

$$P_i \geq 0. \tag{3.33}$$

The state vectors $|\psi^{(i)}(t)\rangle$ are normalized to unity, although they may not necessarily be orthogonal to one another. To obtain the matrix representation of the operator, it is necessary to express each state vector $|\psi^{(i)}(t)\rangle$ in an orthonormal basis $\{|\phi_n\rangle\}$ of the Hilbert space:

$$|\psi^{(i)}(t)\rangle = \sum_n c_n^{(i)}|\phi_n\rangle \tag{3.34}$$

$$\langle\psi^{(i)}(t)| = \sum_n c_n^{(i),*}\langle\phi_n| \tag{3.35}$$

where $c_n^{(i)} = c_n^{(i)}(t)$ and $c_n^{(i),*} = c_n^{(i),*}(t)$ are complex numbers. It follows that[2]

[2]Note that the indices m and n are used for clarity here, but they essentially refer to a unique counting of the basis vectors in the same orthonormal basis $\{|\phi_n\rangle\}$.

$$\begin{aligned}\bar{B}^{(i)}(t) &= \langle\psi^{(i)}(t)|\hat{B}|\psi^{(i)}(t)\rangle = \\ &= \left\{\sum_n c_n^{(i),*}\langle\phi_n|\right\}\hat{B}\left\{\sum_m c_m^{(i)}|\phi_m\rangle\right\} = \\ &= \sum_{n,m} c_n^{(i),*}\, c_m^{(i)}\, \underbrace{\langle\phi_n|\hat{B}|\phi_m\rangle}_{B_{nm}}\end{aligned} \tag{3.36}$$

In particular, B_{nm} is the matrix element of $\hat{B}$ in the $\{|\phi_n\rangle\}$ basis, which gives the inner product between the state $|\phi_n\rangle$ and the result of applying the operator $\hat{B}$ to the state $|\phi_m\rangle$. It follows that the ensemble average reads

$$\langle B(t)\rangle = \sum_i P_i \bar{B}^{(i)}(t) = \sum_i P_i \sum_{n,m} c_n^{(i),*}\, c_m^{(i)}\, B_{nm} \tag{3.37}$$

where the first sum on the right-hand side is over different elements of the ensemble, whereas the second sum takes place in the orthonormal basis $\{|\phi_n\rangle\}$. Let's rearrange

$$\langle B(t)\rangle = \sum_{n,m} \underbrace{\left[\sum_i P_i\, c_m^{(i)}\, c_n^{(i),*}\right]}_{\rho_{mn}(t)} B_{nm}, \tag{3.38}$$

where $\rho_{mn}(t)$ is the **density matrix**. Now, the density matrix $\rho_{mn}(t)$ is not generally diagonal. However, there always exists a basis in which it is diagonal. We can assume that the orthonormal basis $\{|\phi_n\rangle\}$ we have chosen is one such basis (see also observations below). It follows that

$$\langle B(t)\rangle = \sum_{n,m} \rho_{mn}(t)\, B_{nm} = \sum_n \underbrace{\left[\sum_m B_{nm}\rho_{mn}(t)\right]}_{(\hat{B}\cdot\hat{\rho}(t))_{nn}} \tag{3.39}$$

The last equation can be further simplified

$$\langle B(t)\rangle = \sum_n (\hat{B}\cdot\hat{\rho}(t))_{nn} = \mathrm{Tr}[\hat{B}\cdot\hat{\rho}(t)] \tag{3.40}$$

Some observations:

- The density operator is self-adjoint: $\rho_{nm}^* = \rho_{mn}$. Proof[3]:

$$\rho_{nm}^* = \left(\sum_i P_i c_n^{(i)}(t) c_m^{(i),*}(t)\right)^* = \sum_i P_i c_m^{(i)}(t) c_n^{(i),*}(t) = \rho_{mn}(t) \tag{3.41}$$

- The elements on the diagonal of the density matrix $\rho_{mn}(t)$ represent the probability P_n of finding a system in the ensemble in the state $|\psi^{(n)}(t)\rangle$:

$$\rho_{nn}(t) = \sum_i P_i\, |c_{nn}^{(i)}(t)|^2 = P_n \tag{3.42}$$

  with $0 \le \rho_{nn}(t) \le 1$. The quantity P_i represents the probability that the system is in the pure quantum state $|\psi^{(i)}(t)\rangle$ within a statistical ensemble. This is a classical (statistical) probability, indicating that the system is in one of the states $|\psi^{(i)}(t)\rangle$ with probability P_i, but without specifying which particular state. On the other hand, $|c_{nn}^{(i)}(t)|^2$ represents the probability that a system, already in the pure state

[3]For simplicity, we are not explicitly showing the dependence on time in the following equations.

$|\psi^{(i)}(t)\rangle$, is found in the basis state $|\phi_n\rangle$ upon measurement. This is a quantum probability, arising from the superposition principle and the probabilistic nature of quantum mechanics[4]. It follows that $\sum_n |c_{nn}^{(i)}(t)|^2 = 1$, and thus,

$$\sum_n \rho_{nn}(t) = \sum_n \sum_i P_i |c_{nn}^{(i)}(t)|^2 = \sum_i P_i \sum_n |c_{nn}^{(i)}(t)|^2 = \sum_i P_i = 1 \longrightarrow \mathrm{Tr}[\hat{\rho}(t)] = 1. \tag{3.43}$$

Thus, while P_i describes the statistical uncertainty regarding which pure state the system occupies, $|c_{nn}^{(i)}(t)|^2$ describes the intrinsic quantum uncertainty in the measurement of a system already in a specific pure state. Combining these two probabilities, the density matrix formalism provides a complete description of both classical and quantum uncertainties in statistical quantum mechanics.

- In general, the density matrix is not diagonal. The off-diagonal elements, usually complex numbers, cannot be interpreted as probabilities. They represent correlations between the basis states and are related to phenomena of **quantum interference** that do not exhibit classical counterparts. However, there always exists an orthonormal basis that makes the density operator diagonal: $\rho_{mn}(t) = P_n \delta_{mn}$.
- The density matrix can be expressed as an operator, that is $\hat{\rho}(t) = \sum_i P_i |\psi^{(i)}(t)\rangle\langle\psi^{(i)}(t)|$. This definition accounts for both the quantum state and the statistical probabilities. Let's prove this result by projecting onto an orthonormal basis $\{\phi_n\}$ and finding the matrix elements of $\hat{\rho}$ in this basis:

$$\langle\phi_m|\,\hat{\rho}(t)\,|\phi_n\rangle = \langle\phi_m| \left\{\sum_i P_i |\psi^{(i)}(t)\rangle\langle\psi^{(i)}(t)|\right\} |\phi_n\rangle = \sum_i P_i \underbrace{\langle\phi_m|\psi^{(i)}(t)\rangle}_{c_m^{(i)}(t)} \underbrace{\langle\psi^{(i)}(t)|\phi_n\rangle}_{c_n^{(i),*}(t)} = \rho_{mn}(t) \tag{3.44}$$

  We then recover the definition of the density matrix $\rho_{mn}(t)$ as introduced in Eq. 3.38.
- If $\{\phi_n\}$ is a basis that diagonalizes the density operator, then the expression for the average is simpler:

$$\langle B(t)\rangle = \mathrm{Tr}[\hat{\rho}(t)\cdot\hat{B}] = \sum_{nm} \rho_{mn}(t) B_{nm} = \sum_{nm} P_n \delta_{nm} B_{nm} = \sum_n P_n B_{nn} \tag{3.45}$$

  It follows that

$$\langle B(t)\rangle = \sum_n P_n(t) \underbrace{\langle\phi_n|\hat{B}|\phi_n\rangle}_{B_{nn}}, \tag{3.46}$$

  where B_{nn} is the expected value of $\hat{B}$ in the basis $\{\phi_n\}$.

3.6.1 Temporal Evolution of the Density Operator

If $|\psi^{(i)}(t)\rangle$ is a pure state corresponding to the state i of the ensemble, then it must follow the Schrödinger equation:

$$i\hbar\frac{\partial}{\partial t}|\psi^{(i)}(t)\rangle = \hat{H}|\psi^{(i)}(t)\rangle \tag{3.47}$$

Equivalently

$$i\hbar\frac{\partial}{\partial t}\langle\psi^{(i)}(t)| = -\langle\psi^{(i)}(t)|\hat{H} \tag{3.48}$$

[4]In quantum mechanics, the system's state $|\psi^{(i)}(t)\rangle$ can generally be expressed as a linear combination of basis states $|\phi_n\rangle$, such that: $|\psi^{(i)}(t)\rangle = \sum_n c_{nn}^{(i)}(t)|\phi_n\rangle$. The coefficients $c_{nn}^{(i)}(t)$ determine the contribution of each basis state $|\phi_n\rangle$ to the overall state. When a measurement is performed in the basis $\{|\phi_n\rangle\}$, the system collapses to one of the basis states $|\phi_n\rangle$ with probability $|c_{nn}^{(i)}(t)|^2$, according to the Born rule. Therefore, $|c_{nn}^{(i)}(t)|^2$ quantifies the likelihood that a system, known to be in the state $|\psi^{(i)}(t)\rangle$, will be observed in the specific basis state $|\phi_n\rangle$ after a measurement.

See the note below for the proof[5]. It follows

$$i\hbar\frac{\partial\hat{\rho}(t)}{\partial t} = i\hbar\frac{\partial}{\partial t}\left\{\sum_i P_i|\psi^{(i)}(t)\rangle\langle\psi^{(i)}(t)|\right\} = i\hbar\sum_i P_i\left\{\frac{\partial|\psi^{(i)}(t)\rangle}{\partial t}\langle\psi^{(i)}(t)| + |\psi^{(i)}(t)\rangle\frac{\partial\langle\psi^{(i)}(t)|}{\partial t}\right\} \tag{3.53}$$

which gives

$$i\hbar\frac{\partial\hat{\rho}(t)}{\partial t} = \sum_i P_i\left\{\hat{H}|\psi^{(i)}(t)\rangle\langle\psi^{(i)}(t)| - |\psi^{(i)}(t)\rangle\langle\psi^{(i)}(t)|\hat{H}\right\} = \hat{H}\hat{\rho} - \hat{\rho}\hat{H} = [\hat{H},\hat{\rho}] \tag{3.54}$$

In its final form:

$$\frac{\partial\hat{\rho}(t)}{\partial t} = \frac{1}{i\hbar}[\hat{H},\hat{\rho}] \tag{3.55}$$

where $[\hat{H},\hat{\rho}] = \hat{H}\hat{\rho} - \hat{\rho}\hat{H}$ is the **commutator**. Eq. 3.55 is known as the **von Neumann equation**. Just as the Schrödinger equation describes the time evolution of pure states, the von Neumann equation, also known as the Liouville-von Neumann equation or the quantum master equation, describes the time evolution of the density matrix for a quantum system. **It is equivalent to the Liouville equation in the classical formalism**: $\partial\rho/\partial t = \{H,\rho\}$ (see Appendix E).

3.7 Postulate II: Equal Prior Probability

We now know that there exists a density operator or matrix that evolves over time, linking the microscopic and macroscopic descriptions of a physical system. This operator plays a crucial role in our understanding of the system's behaviour. Now, let's explore the specific characteristics of this operator when applied to an **isolated system in equilibrium**. This brings us to the *Postulate of Equal A Priori Probability*, shedding light on how probabilities are distributed in such systems. This postulate applies exclusively to macroscopic systems that are both **isolated** and in **thermodynamic equilibrium**. An isolated system is one that remains entirely independent of its surroundings, with no external interactions. This means that the number of particles (or molecules, atoms, etc.) (N), volume (V) and the Hamiltonian (H) of the system, which describes its total energy, remain constant. More specifically:

- No mechanical coupling through rigid walls: V constant.
- No material exchange through impermeable walls: N constant.
- No thermal interchange through adiabatic walls: $H(\mathbf{q},\mathbf{p}) = E$ constant.

In the classical description, thermodynamic equilibrium is characterised by the absence of any time dependence in any macroscopic property, whether it be mechanical or thermal $(A(\vec{r},t) = A(\vec{r}))$. Specifically,

[5]If we project the bra over an orthonormal basis $|\phi_n\rangle$ that does not depend on time, we get

$$i\hbar\frac{\partial}{\partial t}\langle\psi^{(i)}(t)|\phi_n\rangle = i\hbar\frac{\partial}{\partial t}\int\psi^{(i)}(t)^*\phi_n\,dq_1\ldots dq_s = -\int\left[i\hbar\frac{\partial}{\partial t}\psi^{(i)}(t)\right]^*\phi_n\,dq_1\ldots dq_s. \tag{3.49}$$

The term within $[\ldots]$ equals $\hat{H}\psi^{(i)}(t)$ as stated in the Schrödinger equation. Therefore:

$$i\hbar\frac{\partial}{\partial t}\langle\psi^{(i)}(t)|\phi_n\rangle = -\int[\hat{H}\psi^{(i)}(t)]^*\phi_n\,dq_1\ldots dq_s. \tag{3.50}$$

Because $\hat{H}$ is a self-adjoint operator, we can re-write the equation as

$$i\hbar\frac{\partial}{\partial t}\langle\psi^{(i)}(t)|\phi_n\rangle = -\int\psi^{(i)}(t)^*\hat{H}\phi_n\,dq_1\ldots dq_s = -\langle\psi^{(i)}(t)|\hat{H}|\phi_n\rangle,\ \forall\phi_n. \tag{3.51}$$

It then follows that

$$i\hbar\frac{\partial}{\partial t}\langle\psi^{(i)}(t)| = -\langle\psi^{(i)}(t)|\hat{H} \tag{3.52}$$

even the density function (or density matrix) remains independent of time. This property is rigorously expressed through the Liouville equation:

$$\frac{\partial \rho}{\partial t} = 0 \Longleftrightarrow \{\rho, H\} = 0. \tag{3.56}$$

The density matrix (ρ) should be a function of the system's constants of motion. If there are a total of s degrees of freedom, we have a set of $2s$ constants of motion for the system denoted as $\{\alpha_1, \ldots, \alpha_{2s}\}$ such that

$$\rho = \rho(\alpha_1, \alpha_2, \ldots, \alpha_{2s}) \tag{3.57}$$

In particular, it should be a function of the 7 independent additive integrals of motion: the system's energy, the three components of the total linear momentum, and the three components of the total angular momentum:

$$\rho = \rho(E, p_x, p_y, p_z, L_x, L_y, L_z, \alpha_1, \ldots, \alpha_{2s-7}) \tag{3.58}$$

The total linear and angular momenta are associated with the translation and rotation of the system as a whole, which is usually irrelevant for most thermodynamic systems of interest that are typically at rest. On the other hand, $\alpha_1, \alpha_2, \ldots, \alpha_{2s-7}$ constitute the remaining non-trivial constants of motion for the dynamic system, which are practically impossible to compute. Let's distinguish between two types of constants of motion, $\alpha_1, \alpha_2, \ldots, \alpha_{2s-7}$: *controlable* and *non-controlable*.

- *Controlable*. If α_1 is a controlable constant of motion, it is a quantity known to the experimenter and can be set at will. In other words, its value can be controlled externally, and for each chosen value, one can expect the system's behaviour to be different. Therefore, we must explicitly include the dependence of ρ on α_1:

$$\rho = \rho(E, \alpha_1). \tag{3.59}$$

- *Non-controlable*. If α_1 is a non-controlable constant of motion, then $\alpha_1(\mathbf{q}, \mathbf{p})$ is entirely unknown to the experimenter. In this case, there's no need to include it in the expression for the density function:

$$\rho = \rho(E). \tag{3.60}$$

  By excluding it from ρ, we are effectively performing a different averaging in phase space over points where the condition $\alpha_1(\mathbf{q}, \mathbf{p}) = \text{const}$ may be violated. Is this a significant issue?[6]

If all constants of motion are non-controlable (except for energy), then the density function depends exclusively on energy (E). In practice, this is often the case.

$$\rho(\mathbf{q}, \mathbf{p}) = \rho(E) = \rho(H(\mathbf{q}, \mathbf{p})) \tag{3.61}$$

3.7.1 Quantum Description

Thermodynamic equilibrium implies that the density operator does not depend on time:

$$\frac{\partial \hat{\rho}}{\partial t} = 0, \tag{3.62}$$

By applying the von Neumann equation:

[6] It's not a significant concern. Even if we are averaging over regions that should technically be forbidden, it won't have a substantial impact. This is due to our knowledge that relative fluctuations around the mean values are always negligible: $\sqrt{\alpha_1^2 - \langle\alpha_1\rangle^2}/\langle\alpha_1\rangle \to 0$. Consequently, while α_1 may fluctuate, potentially violating the condition $\alpha_1(\mathbf{q}, \mathbf{p}) = \text{const}$, these fluctuations are so small that we can safely disregard them.

$$[\hat{H}, \hat{\rho}(t)] = 0 \tag{3.63}$$

Since both operators commute, it is possible to find an orthonormal basis that simultaneously diagonalises both operators. This means that both operators share a common set of eigenstates, and the action of the Hamiltonian on these eigenstates results in specific energy eigenvalues (E_n).

$$\hat{H}|\phi_n\rangle = E_n|\phi_n\rangle, \tag{3.64}$$

$$\hat{\rho}|\phi_n\rangle = P_n|\phi_n\rangle, \tag{3.65}$$

On one hand, when the Hamiltonian operator operates on the quantum state $|\phi_n\rangle$, it results in the energy eigenvalue E_n times the same state. This equation represents the time-independent Schrödinger equation, signifying that $|\phi_n\rangle$ is an eigenstate of the Hamiltonian associated with the energy E_n. On the other hand, when the density operator acts on the quantum state $|\phi_n\rangle$, it results in another quantum state, represented as $P_n|\phi_n\rangle$, where $P_n = P_n(E_n)$ is a coefficient (probability amplitude). Because they share the same set of eigenstates, the density operator can be expressed as a function of the Hamiltonian operator, $\hat{\rho} = \hat{\rho}(\hat{H})$. In other words, $\hat{\rho}$ depends on the energy eigenvalues (E_n) determined by the Hamiltonian. So, the statement $\hat{\rho} = \hat{\rho}(\hat{H})$ expresses that in thermodynamic equilibrium, the statistical properties of the quantum system are determined by the energy eigenstates and eigenvalues associated with the Hamiltonian operator.

In a macrostate with constant energy, volume, and number of particles, there are numerous compatible microstates. What probability (statistical weight) do we assign to each of these microstates?

The probability, or statistical weight, assigned to each of the microstates in a macrostate with constant energy, volume, and number of particles is based on the fundamental principle of statistical mechanics and is called the "principle of equal a priori probability" or "principle of equiprobability." This principle states that, in the absence of additional information, the same probability is assigned to each microstate compatible with the macrostate. According to the Liouville (von Neumann) equation, we know that for isolated systems in thermodynamic equilibrium, the density function/operator depends exclusively on the Hamiltonian: $\rho = \rho(H)$. With mechanics alone, we can only conclude that ρ is a function of H. Unfortunately, there exists an infinity of functions ρ that satisfy this condition. In 1938, Tolman asserted that it was not possible to arrive at a macroscopic description using only Mechanics, but that some probabilistic hypotheses had to be introduced.We do not have more information regarding which microstates are more likely to occur in an isolated system in equilibrium with energy E. The most we can do is assuming that all microstates with $H = E$ are **equally probable**. Why? Because in the absence of information, there is no a priori reason for one microstate to be more probable than another, provided it is compatible with the condition $H = E$. This is the essence of the **principle of equal a priori probability**, which states:

in the absence of additional information, the same probability is assigned to each microstate compatible with the macrostate

In other words, all microstates compatible with an equilibrium macrostate of an isolated system have the same probability of occurring. When we lack additional information that would allow us to distinguish between microstates, we assume that all compatible microstates are equally likely. This is because we are dealing with systems in thermal equilibrium, and in that context, we have no reason to favor one specific microstate over another[7].

3.8 Problems

3.1 Solve the same problem as that presented in Example 3.1, but this time with $H = p^2/2 + q^2/2$.

[7] We stress that this is a statistical hypothesis, not a mechanical one. The idea that all microstates compatible with an equilibrium macrostate have the same probability of occurring is a concept rooted in statistical reasoning and probability theory, which reflects our limited knowledge about the specific microstate of a system in thermal equilibrium, rather than classical mechanics or deterministic principles.

3.2 Let's consider the harmonic oscillator characterised by the Hamiltonian

$$H = \frac{1}{2}(q^2 + p^2) \tag{3.66}$$

with $m = \omega = 1$ the particle mass and angular frequency, respectively, and equations of motion given by

$$\begin{aligned} q(t) &= q_0 \cos t + p_0 \sin t \\ p(t) &= -q_0 \sin t + p_0 \cos t \end{aligned} \tag{3.67}$$

(a) Plot the trajectories in the phase space.

(b) Be $\rho(q, p)$ the density function defined at $t = 0$ as

$$\begin{aligned} &\rho(q_0, p_0) = \text{const.} \quad \text{if} \quad q_0 > 0, \quad p_0 < 0 \quad \text{and} \quad q_0^2 + p_0^2 < 1 \\ &\rho(q_0, p_0) = 0 \quad \text{in any other case,} \end{aligned} \tag{3.68}$$

calculate the value of the above constant.

(c) Determine ρ at time $t = \frac{\pi}{2}$.

3.3 Consider a harmonic oscillator in one dimension, whose Hamiltonian reads

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 \tag{3.69}$$

(a) Obtain the equation of motion.

(b) Show that the Jacobian of the transformation $\mathcal{T} : (q(t), p(t)) \to (q_0, p_0)$ equals 1.

(c) Demonstrate that the probability density of finding the particle in position q is given by

$$\rho(q) = \frac{1}{\pi\sqrt{a^2 - q^2}}, \tag{3.70}$$

with $a^2 = 2E/m\omega^2$.

3.4 Consider an ideal system in which each particle can occupy one of 5 possible quantum states. The energy of each state increases linearly with the index of the level (equally spaced levels), $E = (i-1)\varepsilon$, with $i = 1, 2, 3, 4$, and 5. The degeneracy of each level is $g_1 = g_2 = g_3 = 1$, $g_4 = 2$, $g_5 = 2$, respectively.

(a) Characterise the ground state of a system composed of $N = 4$ particles in each of the following cases: distinguishable particles, and indistinguishable fermions and bosons. Determine, in each case, the total energy and its degeneracy.

(b) Suppose the system of $N = 4$ particles has a total energy given by $E = 3\varepsilon$. Count the number of allowed states for each of the following cases: distinguishable particles, and indistinguishable fermions and bosons.

3.5 A light source generates photons traveling in the $+z$ direction, such that each of them can be polarised along the x or y axis. Consider $|1\rangle$ as the state with x polarisation, and $|2\rangle$ as the state with y polarisation, both normalised to unity and orthogonal to each other. Obtain the density matrix in the following cases:

(a) The source only produces photons polarised in the x direction.

(b) The source is incoherent and produces photons with both polarisations randomly, with equal probability in either orientation.

(c) The source produces photons coherently with a phase difference α, corresponding to the pure state

$$|\phi\rangle = \frac{1}{\sqrt{2}}\left[|1\rangle + \mathrm{e}^{i\alpha}|2\rangle\right] \tag{3.71}$$

(d) Verify in all cases that the density matrix is self-adjoint, the diagonal elements are always positive, and their sum is equal to unity.

Chapter 4

Microcanonical Ensemble

4.1 Introduction

We have stressed in the *Introduction* that a macroscopic system consists of approximately $N = 10^{23}$ particles and, correspondingly, exhibits an energy spectrum that varies within $\Delta E \sim e^{-N}$. Any attempt to find a solution of the equations of motion at the microscopic level is hopeless and also superfluous as during the measurement process (which takes a finite amount of time) both macroscopic and microscopic properties of the system change. We have also learnt that a many-body system cannot be characterised by a single microstate, but rather by an **ensemble of microstates** which is determined by macroscopic variables $(E, V, N, \dots)$. Experience shows that, at sufficiently long time scales, any microscopic system evolves towards an equilibrium state, in which

$$\frac{\partial \hat{\rho}}{\partial t} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}] = 0 \tag{4.1}$$

must be satisfied. Since, according to Eq. 4.1, the density operator commuted with the Hamiltonian operator, in an equilibrium ensemble ρ can depend only on conserved quantities, the microscopic properties of the system change continually even at equilibrium, but the distribution of microstates within the ensemble does not depend on time. In this chapter, to fix these ideas, we will introduce the microcanonical ensemble, where number of particles (N), system volume (V) and energy (E) are constant. To this end, we will make use of the *Postulate of Averages* and *Postulate of Equal A Priori Probability.* These postulates are the building blocks with which we will construct the thermodynamic ensembles in equilibrium, starting with the simplest: the isolated system.

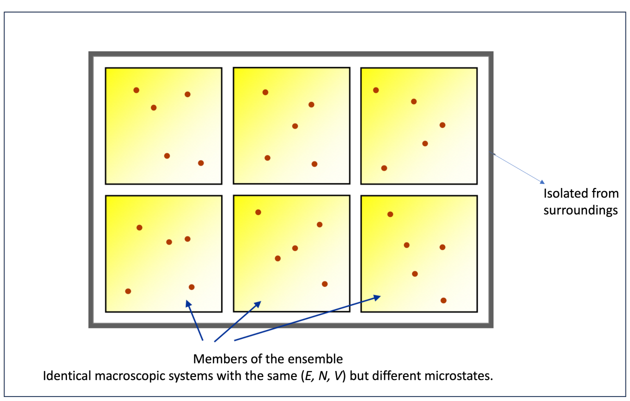


Figure 4.1: Schematic representation of a microcanonical ensemble characterised by different microstates.

The members of the microcanonical ensemble are macroscopic systems identical to the original, **isolated**, and **in equilibrium**. An isolated system does not interact thermally, mechanically, or materially with the outside; it has adiabatic, rigid, and impermeable walls. As such, it exhibits constant volume, constant number of particles. Additionally $H = E$. Systems in equilibrium have no temporal dependencies on any variable, and their macrostate can be described using thermodynamics. The density matrix/function also does not depend on time.

4.2 Classical Microcanonical Ensemble

In the microcanonical ensemble, the constituents of the ensemble represent macroscopic systems that are exact copies of the original system. These systems are kept in isolation, do not interact with the external environment, and are maintained in a state of thermodynamic equilibrium. In the classical context, a microstate of the system is distinctly characterised by a specific point within the system's phase space, denoted as Γ. This phase space **encompasses all possible combinations of positions and momenta for each particle in the system, and each point within this space represents a unique microstate**:

$$(\mathbf{q}, \mathbf{p}) \equiv (q_1, p_1, \ldots, q_s, p_s) \longrightarrow 2\text{s dimensions} \tag{4.2}$$

The probability density associated with the likelihood of the system being in a given microstate, $\rho(q, p)$, characterises the relative probability of finding the system in that specific microstate defined by its generalised coordinates q and momenta p within the phase space. In a sense, it quantifies the "crowding" of microstates within the phase space, indicating which microstates are more or less probable within the constraints of the given macrostate.

What probability (statistical weight) do we assign to each of these microstates? We know that our system is isolated and in equilibrium. This implies that N, V and E are constant and the density function does not change with time. Now, what is the density function for the microcanonical ensemble? With the help of the postulate of equal a priori probability, it is possible to construct this density function clearly and directly. In the microcanonical ensemble, all microstates compatible with a particular macrostate have the same probability of occurring, simplifying the construction of the density function. Generally speaking:

$$\rho(\mathbf{q}, \mathbf{p}) = \begin{cases} \text{constant} & \text{if } (\mathbf{q}, \mathbf{p}) \text{ is compatible with the macrostate} \\ 0 & \text{otherwise} \end{cases}$$

The constant in this function is chosen in such a way that the **sum or integral of the density function over the entire phase space** (q, p) **equals 1**, ensuring that the total probability of finding the system in some microstate within the macrostate is 1. Let's now add some specific details:

1. In any average within this ensemble, it must be taken into account that all mechanical states (q, p) for which $H \neq E$ cannot contribute to the mean value, as they do not conserve energy (see Fig. 4.4).

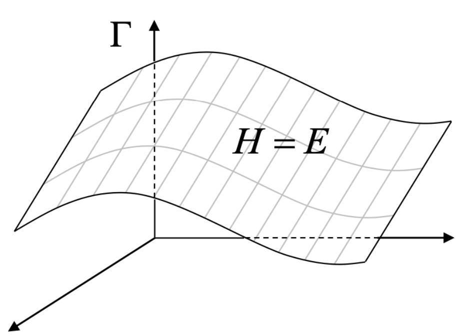


Figure 4.2: Compatible microstates in the phase space Γ.

2. ρ is exclusively a function of H.

3. Equal a priori probability for all microstates with $H = E$.
4. Normalization condition: $\int \rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p} = 1$.

4.2.1 Density of States

An isolated system with a precisely defined energy E is an idealisation. In reality, we can never completely eliminate interactions with the external world. Any small error in determining the energy covers an extremely wide spectrum of microstates. According to the *Heisenberg Uncertainty Principle*, obtaining the energy exactly would require a measurement of infinite duration. As such, instead of specifying the energy precisely, we define an isolated system as one whose energies fall within the interval $[E, E + \Delta E]$ with $\Delta E \ll E$. In particular, we consider a small but finite shell $[E, E + \Delta E]$ close to the energy surface. With reference to Fig. 4.3, $\Omega(N, V, E)$ is the volume[1] of the shell bounded by the two energy surfaces with energies E and $E + \Delta E$ or, equivalently, the volume of phase space accessible to a system with N particles, volume V, and total energy E. In other words, $\Omega(N, V, E)$ quantifies the number of microstates available to the system under those conditions:

$$\Omega(N, V, E) = \int_{E \leq H \leq E+\Delta E} d\mathbf{q}\, d\mathbf{p} \tag{4.3}$$

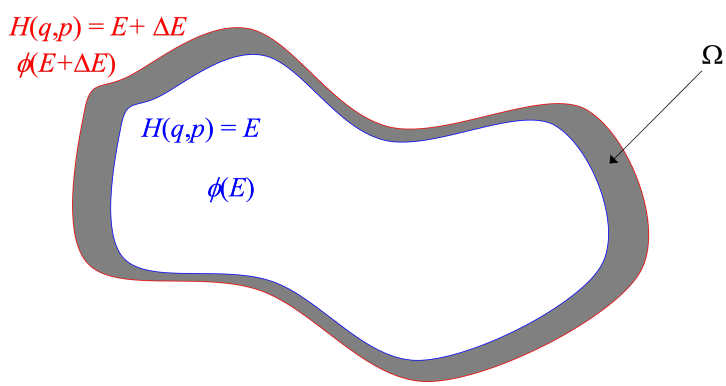


Figure 4.3: Microstates with energy that falls in the interval $[E, E + \Delta E]$ with $\Delta E \ll E$.

Let now ϕ be the total volume of phase space enclosed by the energy surface E. We then have that

$$\Omega(N, V, E) = \phi(E + \Delta E) - \phi(E) \tag{4.4}$$

Taking the limit for $\Delta E \to 0$, corresponding to an infinitesimal shell, one obtains

$$\Omega(N, V, E) \approx \frac{\partial \phi(E)}{\partial E} \Delta E \equiv \omega \Delta E, \quad \Delta E \ll E \tag{4.5}$$

$\Omega(N, V, E)$ refers to the *microcanonical ensemble multiplicity*. It represents the number of microstates that correspond to a specific energy E in a microcanonical ensemble. In other words, $\Omega(N, V, E)$ quantifies the **number of distinct ways in which the energy E can be distributed among the particles in the system** while satisfying the fixed constraints of N and V. From Ω and its dependence on the variables N, V, and E, the entire thermodynamics of the microcanonical ensemble can be deduced. Additionally, the normalisation constant

$$\omega(N, V, E) = \frac{\partial \phi(E)}{\partial E} = \int \delta(E - H(\mathbf{q}, \mathbf{p})) d\mathbf{q}\, d\mathbf{p} \tag{4.6}$$

[1] This volume is not a physical volume like the volume of a container in three-dimensional space. Instead, it is a measure of the "space" occupied by the system's possible configurations in the multi-dimensional phase space spanned by the generalised coordinates and momenta.

represents the **density of states**, indicating the number of compatible microstates per unit energy interval. By contrast, Ω gives the total volume of phase space accessible to the system, quantifying the number of microstates available under specified conditions.

Let's see how the density of states is related to the phase-space density $\rho(\mathbf{q},\mathbf{p})$, which describes how microstates are distributed in phase space (remember this is a probability!). In the microcanonical ensemble, all accessible states are equally probable, meaning:

$$\rho(\mathbf{q},\mathbf{p}) \propto \delta(H(\mathbf{q},\mathbf{p}) - E) \tag{4.7}$$

Since the system is equally likely to be in any of these microstates, the normalisation factor must ensure that:

$$\int \rho(\mathbf{q},\mathbf{p}) d\mathbf{q}\, d\mathbf{p} = 1 \tag{4.8}$$

Using the definition of $\omega(E)$, we then set:

$$\rho(\mathbf{q},\mathbf{p}) = \frac{\delta(H(\mathbf{q},\mathbf{p}) - E)}{\omega(E)} \tag{4.9}$$

where the Dirac delta excludes non-compatible microstates. Equivalently,

$$\rho(\mathbf{q},\mathbf{p}) = \begin{cases} \frac{1}{\omega(E)} & E \leq H(\mathbf{q},\mathbf{p}) \leq E + \Delta E \\ 0 & \text{otherwise} \end{cases} \tag{4.10}$$

The expressions given above for the classical density function provide a useful framework for understanding the statistical behaviour of macroscopic systems in the classical realm. However, it's important to note that these classical expressions may not hold true when we transition to the quantum realm. The behaviour of particles at the quantum level is inherently probabilistic and governed by the Heisenberg uncertainty principle, which is encapsulated in h. Boltzmann realised that the particles in the system are actually indistinguishable. This implies dividing the number of states $\omega(E)$ by an additional $N!$, thereby accounting for the number of different ways to label the particles. Additionally, to satisfy dimensional requirements, Boltzmann suggested to use a constant that has units of momentum times length, which we now know to be h^s, with h the Planck constant and s the number of degrees of freedom of the system. The interested reader is referred to McQuarrie, p. 114 for further details [3]. In particular,

$$\Delta q_i \Delta p_i \sim h \tag{4.11}$$

$$\Delta q^{(s)} \Delta p^{(s)} \sim h^{(s)} \tag{4.12}$$

We therefore end up with the following final expressions:

$$\rho(\mathbf{q},\mathbf{p}) = \frac{1}{h^s N!} \frac{1}{\omega(N,V,E)} \delta(E - H) \tag{4.13}$$

$$\omega(N,V,E) = \frac{1}{h^s N!} \int \delta(E - H(\mathbf{q},\mathbf{p}))\, d\mathbf{q}\, d\mathbf{p} \tag{4.14}$$

The mean value of the dynamic function is given by:

$$\langle a(\mathbf{q},\mathbf{p}) \rangle = \frac{1}{h^s N!} \frac{1}{\omega(N,V,E)} \int a(\mathbf{q},\mathbf{p}) \delta(E - H(\mathbf{q},\mathbf{p}))\, d\mathbf{q}\, d\mathbf{p} \tag{4.15}$$

and the number of states with energy in the interval $[0, E]$ reads

$$\phi(N,V,E) \equiv \phi(E) = \frac{1}{h^s N!} \int \theta(E - H)\, d\mathbf{q}\, d\mathbf{p}, \quad \text{with}\, \theta(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases} \tag{4.16}$$

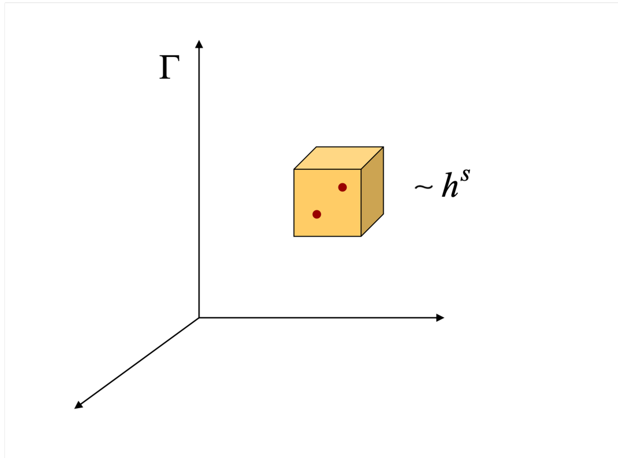


Figure 4.4: Quantum representation of a microstate in the phase space Γ.

where θ is the Heaviside step function[2].

4.3 Connection with Thermodynamics

Let's consider two systems, A_1 and A_2, which are initially separated and in equilibrium, as shown in Fig. 4.5. At a certain moment, we bring both systems into thermal contact, allowing an exchange of energy between them. The separating wall remains rigid and impermeable: N_i = const, V_i = const. However, the walls allows for heat exchange: $E_i \neq$ const. The global system $A = A_1 + A_2$ is also isolated. Its total energy reads $E = E_1 + E_2 + E_{12}$, with E_{12} being negligible compared to E_1 and E_2.

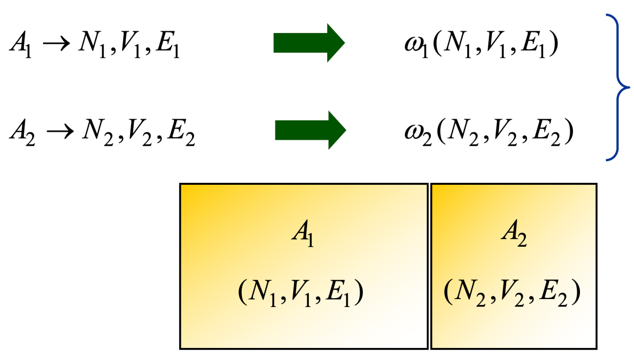


Figure 4.5: Isolated systems that, at a given time, are put in contact to each other.

The coordinates and moment of A_1 and A_2 are, respectively, $(\mathbf{q}_1, \mathbf{p}_1)$ and $(\mathbf{q}_2, \mathbf{p}_2)$, whereas those of the global system A are $(\mathbf{q}, \mathbf{p})$. It follows that the global Hamiltonian can be expressed as

$$H(\mathbf{q}, \mathbf{p}) = H_1(\mathbf{q}_1, \mathbf{p}_1) + H_2(\mathbf{q}_2, \mathbf{p}_2) \tag{4.19}$$

[2]We remember that

$$\frac{d\theta(x-a)}{dx} = \delta(x-a) \tag{4.17}$$

and therefore

$$\omega = \frac{\partial \phi}{\partial E} = \frac{1}{h^s N!} \int \frac{\partial \theta(E-H)}{\partial E} d\mathbf{q}\, d\mathbf{p} = \frac{1}{h^s N!} \int \delta(E-H) d\mathbf{q}\, d\mathbf{p} \tag{4.18}$$

.

Now, let's calculate the number of accessible states of the isolated system $A = A_1 + A_2$. The number of microstates that correspond to energy E, which is the multiplicity introduced in Eq. 4.5, reads

$$\Omega(N, V, E) = \omega(N, V, E)\Delta E, \tag{4.20}$$

with

$$\omega(N, V, E) = \frac{1}{h^{s_1+s_2}N_1!N_2!}\int \delta(E - H(q, p))\, dq\, dp \tag{4.21}$$

This equation can be re-written by incorporating the linearity of the Hamiltonian:

$$\omega(E) = \frac{1}{h^{s_1}N_1!}\int dq_1\, dp_1\, \frac{1}{h^{s_2}N_2!}\int dq_2\, dp_2\, \delta(E - \underbrace{H_1(q_1, p_1)}_{E_1} - H_2(q_2, p_2)) \underbrace{\int_0^E \delta(E_1 - H_1(q_1, p_1))\, dE_1}_{=1} \tag{4.22}$$

the last integral in the right hand side of the above equation has just been added for reasons that become clear in the following equation

$$\omega(E) = \int_0^E dE_1\, \underbrace{\frac{1}{h^{s_1}N_1!}\int dq_1\, dp_1\, \delta(E_1 - H_1(q_1, p_1))}_{\omega_1(E_1)}\, \underbrace{\frac{1}{h^{s_2}N_2!}\int dq_2\, dp_2\, \delta(E - E_1 - H_2(q_2, p_2))}_{\omega_2(E-E_1)} \tag{4.23}$$

It follows that

$$\omega(E) = \int_0^E \omega_1(E_1)\omega_2(E - E_1)\, dE_1 \tag{4.24}$$

The energy E of the global system is fixed, but E_1 can vary. Only the values of E_1 that are sufficiently close to E_1^* contribute to the integral. If $|E_1 - E_1^*| >> 0$, then the number of microstates decreases abruptly. What is the value of E_1 that is compatible with equilibrium? If we refer to it as E_1^*, it follows that the corresponding equilibrium value for A_2 will be $E_2^* = E - E_1^*$. The above integral $\omega(E)$ (see red curve in Fig. 4.6) tends to a distribution that is especially peaked at E_1^*. The maximum of this distribution indicates the equilibrium macrostate with largest probability of being observed. Let's approximate $\omega(E)$ to a Gaussian distribution:

$$\omega_1(E_1)\, \omega_2(E - E_1) \simeq \omega_1(E_1^*)\, \omega_2(E - E_1^*)\mathrm{e}^{-(E_1 - E_1^*)^2/2\sigma_E^2} \tag{4.25}$$

The idea behind this approximation is to assume that around the point E_1^*, the densities $\omega_1(E_1)$ and $\omega_2(E - E_1)$ can be approximated by their values at E_1^*, with a Gaussian "correction" term to account for variations away from E_1^*. This is often used in statistical mechanics when combining two densities of states that are expected to behave similarly near their maximum values (see Fig. 4.6).

It follows that

$$\omega(E) = \int_0^E \omega_1(E_1)\, \omega_2(E - E_1)\, dE_1 \simeq \omega_1(E_1^*)\, \omega_2(E - E_1^*)\int_0^E \mathrm{e}^{-(E_1 - E_1^*)^2/2\sigma_E^2}\, dE_1 \tag{4.26}$$

The integral between 0 and E can be approximated to the integral between $-\infty$ and $+\infty$, that is $\int_0^E \simeq \int_{-\infty}^{+\infty}$. We then note that

$$\int_{-\infty}^{+\infty} \mathrm{e}^{-(E_1 - E_1^*)^2/2\sigma_E^2}\, dE_1 = \sqrt{2\pi}\, \sigma_E \tag{4.27}$$

Therefore

$$\omega(E) \simeq \omega_1(E_1^*)\, \omega_2(\underbrace{E - E_1^*}_{E_2^*})\sqrt{2\pi}\, \sigma_E \tag{4.28}$$

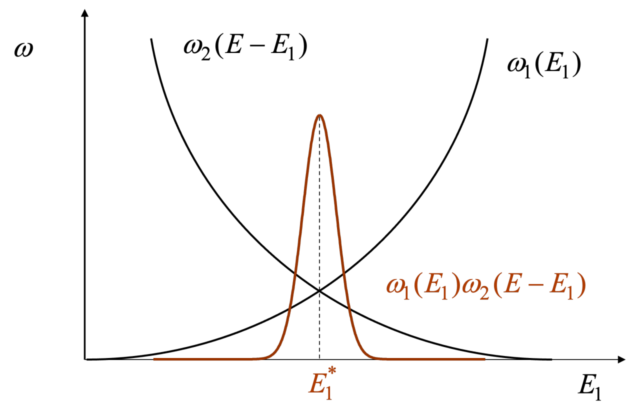


Figure 4.6: Number of compatible microstates as a function of the energy of the isolated system A_1.

If we now multiply both sides by ΔE and take the logarithm, we obtain

$$\omega(E)\Delta E = \Omega(E) \simeq \omega_1(E_1^*)\Delta E\, \omega_2(E_2^*)(\Delta E)^2 \,\frac{\sqrt{2\pi}\ \sigma_E}{\Delta E} = \Omega_1(E_1^*)\,\Omega_2(E_2^*)\,\frac{\sqrt{2\pi}\ \sigma_E}{\Delta E} \tag{4.29}$$

$$\ln\Omega(E) \simeq \ln\Omega_1(E_1^*) + \ln\Omega_2(E_2^*) + \ln\left(\frac{\sqrt{2\pi}\ \sigma_E}{\Delta E}\right) \tag{4.30}$$

Some observations:

- $\Omega(E)$ increases exponentially with N in *normal* systems; $\ln(\Omega(E))$ increases linearly with N; $\sqrt{\sigma_E}$ grows as $\sqrt{N}$; and ΔE does not change.
- $\ln(\sigma_E/\Delta E)$ grows as $\ln(N)$ and in the limit of macroscopic systems $\ln(N) \ll N$ and $N \gg 1$.

We therefore conclude that

$$\ln\Omega(E) \simeq \ln\Omega_1(E_1^*) + \ln\Omega_2(E_2^*) \tag{4.31}$$

This result is especially important as it implies that $\ln\Omega(E)$ is an additive or **extensive** property.

4.3.1 Condition of Thermal Equilibrium

What are the conditions observed when equilibrium is achieved? In other words, what do we observe when $A_1 \to E_1^*$ and $A_2 \to E_2^*$? Let's locate the maximum of the distribution by differentiating the distribution with respect to E_1:

$$\frac{d}{dE_1}\left[\omega_1(E_1)\omega_2(E-E_1)\right]_{E_1^*} = 0 \tag{4.32}$$

It follows that

$$\left.\frac{d\omega_1(E_1)}{dE_1}\right|_{E_1^*} \omega_2(E-E_1^*) + \omega_1(E_1^*)\left.\frac{d\omega_2(E-E_1)}{dE_1}\right|_{E_1^*} = 0 \tag{4.33}$$

Since $E_2 = E - E_1$, then $d/dE_2 = -d/dE_1$ and the above equation can be rewritten as

$$\left.\frac{d\omega_1(E_1)}{dE_1}\right|_{E_1^*} \omega_2(E_2^*) - \omega_1(E_1^*)\left.\frac{d\omega_2(E_2)}{dE_2}\right|_{E_2^*} = 0 \tag{4.34}$$

Equivalently:

$$\left.\frac{1}{\omega_1}\frac{d\omega_1}{dE_1}\right|_{E_1^*} = \left.\frac{1}{\omega_2}\frac{d\omega_2}{dE_2}\right|_{E_2^*} \tag{4.35}$$

which gives

$$\left.\frac{1}{\Omega_1}\frac{d\Omega_1}{dE_1}\right|_{E_1^*} = \left.\frac{1}{\Omega_2}\frac{d\Omega_2}{dE_2}\right|_{E_2^*} \tag{4.36}$$

and equivalently

$$\underbrace{\left.\frac{d\ln\Omega_1}{dE_1}\right|_{E_1^*}}_{\text{depends on } A_1} = \underbrace{\left.\frac{d\ln\Omega_2}{dE_2}\right|_{E_2^*}}_{\text{depends on } A_2} \tag{4.37}$$

We have therefore proved that there exists a property, β that is identical in both systems when equilibrium is achieved:

$$\beta \equiv \left(\frac{\partial \ln\Omega}{\partial E}\right)_{N,V} \tag{4.38}$$

We can finally obtain the *condition of thermal equilibrium*:

$$\beta_1 = \beta_2 \tag{4.39}$$

This property is necessarily a function of intensive variables only, that is $\beta = \beta(T, \rho, \mu, \dots)$, as it does not depend on the system size. Nevertheless, we still have to understand what $\ln\Omega$ and β actually are and to what thermodynamic properties they are associated.

4.3.2 Entropy

Let's consider an isolated system at equilibrium as the one depicted in the yellow box of Fig. 4.7. This system is modified in such a way that it can receive a given amount of energy (heat) over a time interval. It is then isolated again and allowed to achieve the new equilibrium state. We note that number of particles and volume are constant. How does the property $X \equiv \ln\Omega(N, V, E)$ change over the process $(N, V, E) \to (N, V, E + dE)$?

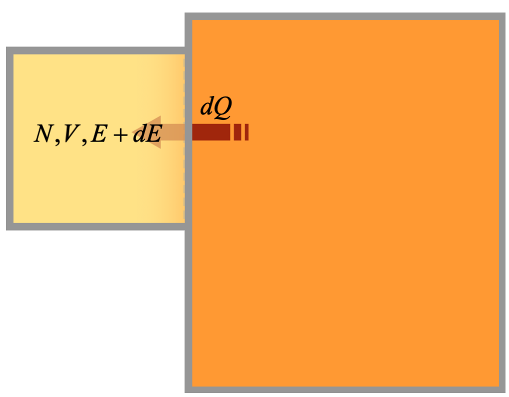


Figure 4.7: Isolated system at equilibrium that receives an amount of heat from a bath and recover isolation soon after.

Let's write its total derivative as follows

$$dX = \left(\frac{\partial \ln\Omega}{\partial E}\right)_{N,V} dE \tag{4.40}$$

Therefore

$$dE = \frac{dX}{\left(\frac{\partial \ln \Omega}{\partial E}\right)_{N,V}} \tag{4.41}$$

From the First and Second Law of Thermodynamics, we know that

$$đQ = TdS = dE + pdV \tag{4.42}$$

Because volume is constant

$$dE = \frac{dX}{\left(\frac{\partial \ln \Omega}{\partial E}\right)_{N,V}} = TdS \tag{4.43}$$

We can then write dS as the total differential of X - it only depends on X

$$dS = \frac{dX}{T\left(\frac{\partial \ln \Omega}{\partial E}\right)_{N,V}} \tag{4.44}$$

Such an exclusive dependence implies that the factor accompanying dX can only depend on X. Mathematically:

$$dS = \frac{dX}{f(X)} \tag{4.45}$$

By integrating the above equation, we obtain

$$S = \underbrace{\int \frac{dX}{f(X)}}_{g(X)} + S_0 \tag{4.46}$$

We know that entropy is an extensive property:

$$S_{1+2} = S_1 + S_2 \tag{4.47}$$

and so is $X = \ln \Omega$:

$$X_{1+2} = X_1 + X_2 \tag{4.48}$$

Therefore

$$S_{1+2} = g(X_{1+2}) + S_{0,1+2} = g(X_{1+2}) + S_{0,1} + S_{0,2} = g(X_1 + X_2) + S_{0,1} + S_{0,2} \tag{4.49}$$

It is also true that

$$S_{1+2} = S_1 + S_2 = g(X_1) + S_{0,1} + g(X_2) + S_{0,2} \tag{4.50}$$

It follows that

$$g(X_1 + X_2) = g(X_1) + g(X_2) \tag{4.51}$$

The only function that satisfies the above equation is a linear function of X:

$$g(X) = kX \tag{4.52}$$

where k is a constant. Therefore

$$S = kX + S_0 \tag{4.53}$$

By substituting the X function with its original value, we get:

$$S = k \ln \Omega + S_0 \tag{4.54}$$

In the limit of a temperature approaching zero ($T \to 0$ K), there exist only one accessible microstate, which is the one with the lowest possible energy. It then follows that, for this particular situation, $\Omega = 1$:

$$\lim_{T \to 0} S = k \ln 1 + S_0 = S_0 \tag{4.55}$$

The Third Law of Thermodynamics states that the entropy of a closed system at equilibrium approaches a constant value when its temperature approaches absolute zero. This constant value, here denoted as S_0, is usually set to zero by convention. It then follows that

$$S = k \ln \Omega. \tag{4.56}$$

This is a formula of fundamental importance in physics. It provides a bridge between the macroscopic and microscopic worlds. Now we know that $\ln \Omega$ refers to. What about β?

$$\beta \equiv \left(\frac{\partial \ln \Omega}{\partial E} \right)_{N,V} = \frac{1}{k} \left(\frac{\partial S}{\partial E} \right)_{N,V} = \frac{1}{kT} \tag{4.57}$$

It is then clear that β is the inverse temperature, whereas k is a *universal constant* that we don't know yet. It is also clear why Eq. 4.39 gives $\beta_1 = \beta_2$ for two systems in contact that achieve equilibrium: their temperature must be the same. Now, how do we calculate the constant k? Since k is a constant, its value is expected to be the same regardless the system. To calculate it, we can then make use of the simplest thermodynamic system, that is the classical monotonic ideal gas. The equation of state of this system is known:

$$pV = nRT \tag{4.58}$$

where $n = N/N_A$ is the number of moles, T the absolute temperature, p pressure, V volume and $R \simeq 8.314$ J Kmol^{-1} the ideal-gas constant. In the next section, we will use this gas to derive an expression of entropy that is functional to understand what k is and calculate its numerical value.

4.4 Classical Monoatomic Ideal Gas

Let's write the Hamiltonian of a monoatomic ideal gas enclosed in a volume V of radius r:

$$H = \sum_{j=1}^{N} \left[\frac{p_j^2}{2m} + U_r(\bar{r}_j) \right] \tag{4.59}$$

To calculate the entropy S of this gas, we need to know the number of microstates:

$$\Omega(N, V, E; \Delta E) \longrightarrow S = k \ln \Omega \tag{4.60}$$

Let's introduce $\phi(E)$, which is the number of microstates with energy in the interval $[0, E]$. Calculating $\phi(E)$ is easier than calculating the total number of compatible microstates with a macrostate of energy E, that is $\Omega(E)$. In particular:

$$\phi(N, V, E) = \frac{1}{h^{3N} N!} \int \theta(E - H) \, dq \, dp = \frac{1}{h^{3N} N!} \int \theta(E - H) \, d\bar{r}_1 \dots d\bar{r}_N d\bar{p}_1 \dots d\bar{p}_N \tag{4.61}$$

Therefore

$$\phi(N, V, E) = \frac{1}{h^{3N} N!} \int d\bar{r}_1 \dots d\bar{r}_N \int d\bar{p}_1 \dots d\bar{p}_N \, \theta \left[E - \sum_{j=1}^{N} \left(\frac{p_j^2}{2m} + U_r(\bar{r}_j) \right) \right] \tag{4.62}$$

If we pay attention to the above integral, we see that the function $\theta(E - H)$ is zero for all points behind radius r and, for all the remaining points $U_r(\bar{r}_j) = 0$. Consequently, Eq. 4.62 can be simplified and re-written as follows

$$\phi(N,V,E) = \frac{1}{h^{3N}N!}\underbrace{\int d\bar{r}_1\dots d\bar{r}_N}_{\text{accessible volume}}\int d\bar{p}_1\dots d\bar{p}_N\;\underbrace{\theta\left[E-\sum_{j=1}^{N}\left(\frac{p_j^2}{2m}\right)\right]}_{\text{independent of }\{\bar{r}_j\}} \tag{4.63}$$

We then obtain

$$\phi(N,V,E) = \frac{V^N}{h^{3N}N!}\int dp_1\dots dp_{3N}\,\theta\left[E-\sum_{j=1}^{3N}\left(\frac{p_j^2}{2m}\right)\right] \tag{4.64}$$

Because $\theta(x-a) = \theta(b(x-a))$, with b positive constant, then

$$\phi(N,V,E) = \frac{V^N}{h^{3N}N!}\;\underbrace{\int dp_1\dots dp_{3N}\,\theta\left[2mE-\sum_{j=1}^{3N}p_j^2\right]}_{\text{volume of a 3N-D hypersphere of } r=\sqrt{2mE}} \tag{4.65}$$

Only momenta that satisfy $\sum_{j=1}^{3N} p_j^2 \le 2mE$ contribute to the above integral. For the volume of a hypersphere see the note below[3]. In particular

$$\phi(N,V,E) = \frac{V^N}{h^{3N}N!}(2mE)^{3N/2}\frac{\pi^{3N/2}}{\Gamma\left(\frac{3N}{2}+1\right)} = \frac{V^N}{h^{3N}N!}\frac{(2\pi mE)^{3N/2}}{\Gamma\left(\frac{3N}{2}+1\right)} \tag{4.70}$$

We can finally estimate the number of states in the interval $[E, E+\Delta E]$:

$$\Omega(E;\Delta E) = \frac{\partial\phi}{\partial E}\Delta E = \frac{V^N}{h^{3N}N!}\frac{(2\pi mE)^{3N/2}}{\Gamma\left(\frac{3N}{2}+1\right)}\frac{3N}{2}\frac{\Delta E}{E} \tag{4.71}$$

This equation allows us to calculate entropy, $S = k\ln\Omega$, in the thermodynamic limit, that is for $N\to\infty$, $V\to\infty$ and $E\to\infty$:

[3] The general formula for the d-dimensional volume of a hypersphere reads

$$V_d(r) = r^d\frac{\pi^{d/2}}{\Gamma\left(\frac{d}{2}+1\right)} \tag{4.66}$$

For example, if $d=2$, we recover the area of a circle:

$$V_2(r) = r^2\frac{\pi}{\Gamma(2)} = \pi r^2. \tag{4.67}$$

If $d=3$, then

$$V_3(r) = r^3\frac{\pi^{3/2}}{\Gamma(5/2)} = r^3\frac{\pi^{3/2}}{\frac{3}{2}\frac{1}{2}\Gamma(1/2)} = \frac{4}{3}r^3\frac{\pi^{3/2}}{\pi^{1/2}} = \frac{4}{3}\pi r^3. \tag{4.68}$$

We also remind that

$$\Gamma(n+1) = n! \tag{4.69}$$

$$
\begin{aligned}
S(N,V,E) &= \\
&= k\ln\Omega = k\left\{N\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] - \ln N! - \ln\Gamma\left(\frac{3N}{2}+1\right) + \ln\left(\frac{3N}{2}\right) + \ln\frac{\Delta E}{E}\right\} = \\
&= k\left\{N\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] - \ln N! - \left(\frac{3N}{2}+1\right)\ln\left(\frac{3N}{2}+1\right) + \left(\frac{3N}{2}+1\right) + \ln\left(\frac{3N}{2}\right) + \ln\frac{\Delta E}{E}\right\} \simeq \\
&\simeq k\left\{N\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] - \ln N! - \left(\frac{3N}{2}\right)\ln\left(\frac{3N}{2}\right) + \left(\frac{3N}{2}\right) + \ln\left(\frac{3N}{2}\right) + \ln\frac{\Delta E}{E}\right\}
\end{aligned}
\tag{4.72}
$$

where we used that $\ln\Gamma(x) \simeq x\ln x - x$. We now apply Stirling's approximation[4] and get

$$
\begin{aligned}
S(N,V,E) &= kN\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] + \\
&+ k\left\{-N\ln N + N - \frac{3N}{2}\ln\left(\frac{3N}{2}\right) + \frac{3N}{2} + \ln\left(\frac{3N}{2}\right) + \ln\frac{\Delta E}{E}\right\} = \\
&kN\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] + k\left\{-N\ln N + N - \left(\frac{3N}{2}-1\right)\ln\left(\frac{3N}{2}\right) + \frac{3N}{2} + \ln\frac{\Delta E}{E}\right\} \simeq \\
&\simeq kN\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] + kN\left\{-\ln N + 1 - \ln\left(\frac{3N}{2}\right)^{3/2} + \frac{3}{2} + \frac{1}{N}\ln\frac{\Delta E}{E}\right\} \simeq \\
&\simeq kN\left\{\ln\left[\frac{V}{N}\frac{(2\pi mE)^{3/2}}{h^3}\right] - \ln\left(\frac{3N}{2}\right)^{3/2} + \frac{5}{2}\right\}
\end{aligned}
\tag{4.73}
$$

and merging the two logarithms, we finally get

$$
S(N,V,E) \simeq kN\left\{\ln\left[\frac{V}{N}\left(\frac{4\pi mE}{3h^2N}\right)^{3/2}\right] + \frac{5}{2}\right\}
\tag{4.74}
$$

Eq. 4.74, referred to as the "Sackur-Tetrode formula"[5], gives the absolute entropy of a classical monoatomic ideal gas. Note that both V/N and E/N in the above equation are **intensive properties**. As such, the absolute entropy can be concisely written as $S/kN = f(\text{intensive})$, where $f(\text{intensive})$ indicates that S/kN is a function of intensive properties. Nevertheless, $S \propto Nf(\text{intensive})$ is an **extensive property** as it increases with the number of particles in the system.

Boltzmann Constant

We are now in the conditions to obtain the numerical value of the constant k. To this end, we need to calculate the pressure of the ideal gas

$$
p = T\left(\frac{\partial S}{\partial V}\right)_{N,E} = TkN\frac{\partial}{\partial V}\ln V = TkN\frac{1}{V}
\tag{4.75}
$$

[4] For integer numbers, Stirling approximation reads

$$\ln N! \simeq N\ln N - N, \quad \text{with } N \gg 1$$

while for real numbers, it reads

$$\ln\Gamma(x) \simeq x\ln x - x, \quad \text{with } x \gg 1$$

[5] Otto Sackur (1880-1914) was a German physical chemist. Hugo Tetrode (1895-1931) was a Dutch theoretical physicist. Each independently uncovered this equation in 1912.

Rearranging, we obtain

$$pV = NkT \tag{4.76}$$

and because the ideal-gas law $pV = nRT$ must hold, we conclude that

$$Nk = nR \tag{4.77}$$

equivalently

$$k = \frac{n}{N}R = \frac{R}{N_A} \tag{4.78}$$

where N_A is the Avogadro's number and R the ideal-gas constant. Therefore:

$$k \simeq 1.380649 \times 10^{-23}\text{J K}^{-1} \tag{4.79}$$

This constant is known as the **Boltzmann constant**, named after the Austrian physicist Ludwig Boltzmann, who made significant contributions to the development of statistical mechanics and the understanding of the behaviour of particles in a gas.

Intensive and Extensive Properties from Entropy

Let's differentiate the Sackur-Tetrode equation with respect to E, V and N, the three key properties of the microcanonical ensemble:

$$\left(\frac{\partial S}{\partial E}\right)_{N,V} = \frac{3}{2}\frac{kN}{E} = \frac{1}{T} \tag{4.80}$$

where we have applied the equipartition theorem[6]. Additionally, differentiating S with respect to V, one obtains

$$\left(\frac{\partial S}{\partial V}\right)_{N,E} = \frac{kN}{V} = \frac{p}{T} \tag{4.81}$$

where we have applied the ideal-gas law $pV = NkT$. Finally, we can define the *chemical potential* as

$$\mu = -T\left(\frac{\partial S}{\partial N}\right)_{V,E} = -kT\ln\left[\frac{V}{N}\left(\frac{4\pi mE}{3h^2N}\right)^{3/2}\right] = -kT\ln\left[\frac{V}{N}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] \tag{4.82}$$

where we have used $E/N = 3/2kT$. We can now write down the total derivative of S with respect to N, V and E:

$$dS = \frac{1}{T}dE + \frac{p}{T}dV - \frac{\mu}{T}dN \tag{4.83}$$

or

$$dE = TdS - pdV + \mu dN \tag{4.84}$$

It is then possible to calculate the intensive variables in terms of energy change:

$$T = \left(\frac{\partial E}{\partial S}\right)_{N,V} \tag{4.85}$$

$$p = -\left(\frac{\partial E}{\partial V}\right)_{N,S} = T\left(\frac{\partial S}{\partial V}\right)_{N,E} \tag{4.86}$$

$$\mu = \left(\frac{\partial E}{\partial N}\right)_{V,S} = -T\left(\frac{\partial S}{\partial N}\right)_{E,V} \tag{4.87}$$

[6] The equipartition theorem states that each quadratic degree of freedom, on average, possess an energy equal to $kT/2$. This theorem is key in the development of the Kinetic Theory of Gases. See Appendix F for additional details.

From here, we can also obtain the heat capacities at constant volume and constant pressure

$$C_v = \left(\frac{\partial E}{\partial T}\right)_{N,V} = T\left(\frac{\partial S}{\partial T}\right)_{N,V} = \frac{3}{2}kN \tag{4.88}$$

where we have again made use of the equipartition theorem.

$$C_p = \left(\frac{\partial H}{\partial T}\right)_{N,P} = T\left(\frac{\partial S}{\partial T}\right)_{N,P} = \frac{5}{2}kN \tag{4.89}$$

where H stands for enthalpy and we have made use of the ideal-gas law.

We stress that both properties are extensive. Nevertheless, the heat capacities per mol, that is $c_v = C_v/n$ and $c_p = C_p/n$, usually referred to as specific heats, are intensive properties. We can then obtain the Mayer relation:

$$c_p - c_v = \frac{5}{2}\frac{kN}{n} - \frac{3}{2}\frac{kN}{n} = R \tag{4.90}$$

Following similar steps, we can obtain the Gibbs free energy (G), enthalpy (H) and Helmholtz free energy (A):

$$G = A + pV = \mu N = -NkT\ln\left[\frac{V}{N}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] \tag{4.91}$$

$$H = E + pV = G + TS = \frac{5}{2}NkT \tag{4.92}$$

$$A = E - TS = \mu N - pV = -NkT\left\{\ln\left[\frac{V}{N}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] + 1\right\} \tag{4.93}$$

4.5 Quantum Microcanonical Ensemble

The quantum microcanonical ensemble refers to a collection of quantum systems that share fixed total energy, number of particles, and volume. As observed for the classical NVE ensemble, also in its quantum counterpart, individual systems are considered to be isolated and in equilibrium, meaning that they do not exchange energy, particles, or volume with their surroundings. Basically, we are going to repeat the same arguments as in the previous sections, but this time for the quantum case.

In the formulation of the quantum NVE ensemble, the correction factor $N!h^s$, which was necessary in the classical expression for the number of states Ω, is no longer required. Why? In quantum statistical mechanics, we do not integrate over phase space; instead, we directly count quantum energy eigenstates. This eliminates the need for the factor h^s because (i) quantum states are inherently discrete, and (ii) there is no need to adjust units, as was necessary when defining the density of states in the classical formulation. Furthermore, the factor $N!$ is also unnecessary because quantum mechanics inherently accounts for particle indistinguishability—only valid arrangements are counted from the outset, avoiding the overcounting issue that arises in classical mechanics. In the classical formulation, it is more practical to first count all microstates in phase space and then apply corrections, rather than restricting the counting to only distinct states from the beginning.

Once again, we consider an isolated system in equilibrium. Let's remember here some relevant results of Chapter 3:

- Time-independent Schrödinger equation

$$\hat{H}\psi = E\psi \tag{4.94}$$

- Liouville-von Neumann equation

$$\frac{\partial \hat{\rho}}{\partial t} = 0 \to [\hat{\rho}, \hat{H}] = 0 \tag{4.95}$$

- There exists an orthonormal basis $\{|\phi_n\rangle\}$ that simultaneously diagonalises both operators H and ρ.

$$\hat{H}|\varphi_n\rangle = E_n|\varphi_n\rangle \qquad \hat{\rho}|\varphi_n\rangle = P_n|\varphi_n\rangle$$

$$\hat{\rho} = \sum_n P_n|\varphi_n\rangle\langle\varphi_n| \qquad \mathrm{Tr}[\hat{\rho}] = \sum_n P_n = 1$$

$$\rho_{mn} = P_n\delta_{mn}$$

Figure 4.8: Summary of relevant results on density matrix obtained in 3

What are the values of the probabilities P_n for an isolated system in equilibrium?

4.5.1 Determination of the density matrix and number of states with energy E

An isolated system with energy E is an idealisation. In reality, we can never completely eliminate interactions with the external world. Any small error in determining the energy covers a broad spectrum of microstates. According to the Heisenberg Uncertainty Principle, obtaining the exact energy would require a measurement of infinite duration.

Instead of specifying the energy, we will define an isolated system as one in which its energy levels are within the interval ΔE, that is:

$$E_n \in [E, E + \Delta E], \text{ with } \Delta E \ll E \tag{4.96}$$

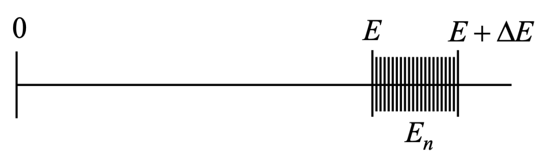


Figure 4.9: Energy levels E_n of an isolated system.

In order to deduce the value of the probabilities P_n, we again employ the postulate of equal *a priori* probability: all microstates n compatible with the macrostate of constant energy are equally probable. Therefore, if the total number of microstates compatible with $E_n \in [E, E + \Delta E]$ is Ω, then

$$P_n = \begin{cases} 1/\Omega \, \forall n & \text{if } E_n \in [E, E + \Delta E] \\ 0 \, \forall n & \text{if } E_n \notin [E, E + \Delta E] \end{cases} \tag{4.97}$$

It follows that the ensemble average of the operator $\hat{B}$ (see also Section 3.6 in Chapter 3) reads

$$\langle B \rangle = \sum_n P_n B_{nn} = \sum_n P_n \langle \phi_n|\hat{B}|\phi_n\rangle = \frac{1}{\Omega}\sum_{n^*}\langle \phi_n|\hat{B}|\phi_n\rangle \tag{4.98}$$

where n^* indicates microstates compatible with the condition $E_n \in [E, E + \Delta E]$. We can see that, within this ensemble, the expected values are averaged by assigning each of them the same statistical weight: all are equally probable. It can be demonstrated that, similar to classical formalism, entropy is connected to the number of microstates as: $S = k \ln \Omega$.

$\rho_{mn} = \begin{cases} \frac{1}{\Omega}\delta_{mn} & E_n \in [E, E+\Delta E] \\ 0 & E_n \notin [E, E+\Delta E] \end{cases}$	$\rho(p,q) = \frac{1}{h^{3N} N!} \frac{1}{\omega(E,V,N)} \delta(E-H)$ $\omega(E,V,N) = \frac{1}{h^{3N} N!} \int \delta(E - H(q,p)) dq dp$
$\Omega = \sum_{\substack{n \\ E \le E_n \le E+\Delta E}} 1$	$\phi(E,V,N) \equiv \phi(E) = \frac{1}{h^{3N} N!} \int \theta(E-H) dq dp$ $\Omega(E) = \phi(E+\Delta E) - \phi(E) \approx \frac{\partial \phi}{\partial E} \Delta E$

Figure 4.10: Essential results associated to microstates in the NVE ensemble as obtained from quantum (left column) and classical (right column) formulations.

4.5.2 Application Example 4.1: Einstein model for a monoatomic solid

The Einstein solid is a theoretical model proposed by Albert Einstein in 1907 to describe the behaviour of solids at varying temperatures[7]. It considers atoms as simple harmonic oscillators with quantised energy levels, explaining heat capacity at low temperatures by accounting for the discrete vibrational energies of atoms. More specifically, it is a set of $3N$ quantum harmonic oscillators (where N is the number of atoms), with vibration frequency ω and total energy E, isolated from the surroundings and not interacting with each other. In this example, our goal is calculating S, T, and E.

The energy of a single quantum harmonic oscillator can be expressed using the Hamiltonian operator for the harmonic oscillator. The Hamiltonian operator for a one-dimensional harmonic oscillator is given by:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{x}^2 \tag{4.99}$$

where $\hat{H}$ is the Hamiltonian operator representing the total energy, $\hat{p}$ is the momentum operator, m is the mass of the oscillator, ω is the angular frequency of the oscillator and $\hat{x}$ is the position operator. The energy eigenvalues of the harmonic oscillator system, denoted by E_i, can be found by solving the Schrödinger equation for the Hamiltonian operator:

$$\hat{H}\psi_i(x) = E_i \psi_i(x) \tag{4.100}$$

where $\psi_i(x)$ is the wavefunction corresponding to the ith energy eigenstate. The energy eigenvalues for a one-dimensional harmonic oscillator are given by:

$$E_i = \left(n_i + \frac{1}{2}\right) \hbar\omega \tag{4.101}$$

where $n_i = 0, 1, \ldots \infty$ is a non-negative integer representing the quantum number of the energy level. Each energy level n represents a quantised energy state of the harmonic oscillator. The ground state $n_i = 0$ has the lowest energy $E_0 = \frac{1}{2}\hbar\omega$, and the energy increases in discrete steps as n_i increases. In particular, the energy of each successive level is separated by $\hbar\omega$ and reads

[7]Einstein's model of a solid was an attempt to understand how heat capacity varies with temperature. At that time, classical physics couldn't explain certain observations in heat capacity at low temperatures. The classical theory predicted that heat capacity should approach zero as the temperature approaches absolute zero. However, experimental data showed that the heat capacity of solids doesn't tend to zero, but instead approaches a finite value at low temperatures.

$$E = \sum_{i=1}^{3N} E_i = \sum_{i=1}^{3N} \left(n_i + \frac{1}{2} \right) \hbar\omega \tag{4.102}$$

Let's rearrange the above equation

$$\frac{E}{\hbar\omega} = \sum_{i=1}^{3N} n_i + \sum_{i=1}^{3N} \frac{1}{2} \Longleftrightarrow \sum_{i=1}^{3N} n_i = \frac{E}{\hbar\omega} - \frac{3N}{2} \tag{4.103}$$

We now introduce the variable $M \equiv \frac{E}{\hbar\omega} - \frac{3N}{2}$, which is necessarily a positive integer number[8]. Therefore:

$$\sum_{i=1}^{3N} n_i = M \tag{4.104}$$

We have to count in how many different ways can we choose the $3N$ integers n_i so that their total sum is always M. More precisely, we need to find the multiplicity Ω compatible with a fixed macrostate (N, E), which is basically the degeneracy of the eigenstate of energy E. This is a combinatorial problem. To clarify ideas, we first consider a very simple case with $M = 6$ and $N = 1$ (only 1 atom) $\rightarrow 3N = 3$ oscillators. Let's see how many possible combinations there are, using in the table below.

Table 4.1: Number of ways of selecting 3 positive integers such that their sum equals 6. The number of possible arrangements is $\Omega = 28$.

O1	O2	O3	O1	O2	O3	O1	O2	O3	O1	O2	O3
0	0	6	1	0	5	2	1	3	3	3	0
0	1	5	1	1	4	2	2	2	4	0	2
0	2	4	1	2	3	2	3	1	4	1	1
0	3	3	1	3	2	2	4	0	4	2	0
0	4	2	1	4	1	3	0	3	5	0	1
0	5	1	1	5	0	3	1	2	5	1	0
0	6	0	2	0	4	3	2	1	6	0	0

We can consider the $3N$ oscillators as boxes where M balls are arranged, representing the value of n_i. We can count how many balls are in each box, but we don't know which balls are in each box. In other words, while boxes are distinguishable, balls are not. The number we are looking for is $\Omega(N, E)$, being the number of ways M indistinguishable balls can be arranged in $3N$ distinguishable boxes. We can imagine that contiguous boxes are separated by a wall. Since there are $3N$ boxes, there will be $3N - 1$ walls (w). Because some boxes might not contain balls (b), the b-w sequence is in general randomly distributed, such as $\{bbwwbbbw \dots wbwb\}$. In case all balls are in the first box then $\{bbbbb \dots bbbb \dots wwww \dots ww\}$. Note that both configurations are equally probable. Now, if balls and walls were **distinguishable**, then the number of ways of arranging the sum of these objects (balls plus walls) would be given by $(M + 3N - 1)!$, a permutation of $M + 3N - 1$ objects. However, the M balls are **indistinguishable** and so are the $3N - 1$ walls. Consequently the number of ways to arrange them is

$$\Omega(N, E) = \frac{(M + 3N - 1)!}{M!\,(3N - 1)!} \tag{4.105}$$

For $M = 6$ and $N = 1$, we get $\Omega = 28$, which is the number of combinations reported in Table 4.1. Substituting the value of M into the above equation, we obtain

$$\Omega(N, E) = \frac{(E/(\hbar\omega) + 3N/2 - 1)!}{(E/(\hbar\omega) - 3N/2)!\,(3N - 1)!} \tag{4.106}$$

[8] M must be a positive integer because it represents the total number of quanta in a system of N particles.

We can then calculate the entropy:

$$S(N,E) = k\ln\Omega = k\left\{\ln\left[\left(\frac{E}{\hbar\omega}+\frac{3N}{2}-1\right)!\right] - \ln\left[\left(\frac{E}{\hbar\omega}-\frac{3N}{2}\right)!\right] - \ln\left[(3N-1)!\right]\right\} \tag{4.107}$$

Applying the Stirling approximation and after a few rearrangements, we finally get

$$S(N,E) = 3Nk\left[\left(\frac{E}{3N\hbar\omega}+\frac{1}{2}\right)\ln\left(\frac{E}{3N\hbar\omega}+\frac{1}{2}\right) - \left(\frac{E}{3N\hbar\omega}-\frac{1}{2}\right)\ln\left(\frac{E}{3N\hbar\omega}-\frac{1}{2}\right)\right] \tag{4.108}$$

Note that S is an extensive property. We can now calculate the energy E. To this end, it is convenient to calculate the temperature first:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{N,V} = \frac{k}{\hbar\omega}\left[\ln\left(\frac{E}{3N\hbar\omega}+\frac{1}{2}\right) - \ln\left(\frac{E}{3N\hbar\omega}-\frac{1}{2}\right)\right] = \frac{k}{\hbar\omega}\ln\frac{\frac{E}{N\hbar\omega}+\frac{3}{2}}{\frac{E}{N\hbar\omega}-\frac{3}{2}} \tag{4.109}$$

Rearranging, we obtain

$$E = \frac{3N\hbar\omega}{2}\frac{\mathrm{e}^{\hbar\omega/kT}+1}{\mathrm{e}^{\hbar\omega/kT}-1} = \frac{3N\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2kT}\right) \tag{4.110}$$

It is now interesting to evaluate the limits of energy at very low and very large temperature, that is for $T \to 0$ and $T \to \infty$, respectively. These limits read[9]

$$\lim_{T\to\infty}\frac{3N\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2kT}\right) = 3NkT \tag{4.113}$$

$$\lim_{T\to 0}\frac{3N\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2kT}\right) = \frac{3}{2}N\hbar\omega \tag{4.114}$$

Interestingly enough, we find that in the limit of very large temperatures, we recover the classical result as $kT \gg \hbar\omega$ and the equipartion theorem is satisfied[10]. By contrast, in the limit of very low temperatures, we observe that E depends on $\hbar\omega$ only and can only take discrete values according to the number of particles in the system.

4.5.3 Application Example 4.2: Helmholtz Free Energy

In this example, our aim is determining the Helmholtz Free Energy $A(T,N)$ as a function of temperature for a system composed of N identical yet distinguishable and ideal particles, each capable of existing in 2 energy levels $\{-\epsilon, \epsilon\}$.

Let's label with n_+ and n_- the particles whose energy level is, respectively, ϵ and $-\epsilon$, such that $N = (n_+ + n_-)$ and $E = n_+\epsilon + n_-(-\epsilon) = (n_+ - n_-)\epsilon$. It follows that

[9] Note that

$$\lim_{x\to 0}\coth(x) = \lim_{x\to 0}\frac{\mathrm{e}^{x}+\mathrm{e}^{-x}}{\mathrm{e}^{x}-\mathrm{e}^{-x}} \simeq \frac{1+x+1-x}{1+x-1+x} = \frac{2}{2x} = \frac{1}{x} \tag{4.111}$$

and

$$\lim_{x\to\infty}\coth(x) = \lim_{x\to\infty}\frac{\mathrm{e}^{x}+\mathrm{e}^{-x}}{\mathrm{e}^{x}-\mathrm{e}^{-x}} = 1 \tag{4.112}$$

[10] Note that each harmonic oscillator contributes kT and not $kT/2$, because it has both kinetic and potential energy. In particular, at high temperatures, half of the energy goes to the kinetic energy term and half to the potential energy term for each harmonic oscillator, consistent with the classical equipartition theorem.

$$n_+ = \frac{1}{2}\left(N + \frac{E}{\epsilon}\right) \tag{4.115}$$

$$n_- = \frac{1}{2}\left(N - \frac{E}{\epsilon}\right) \tag{4.116}$$

It can be observed that there is a one-to-one relationship between (n_+, n_-) and (N, E). In other words, given N and E, there exists a unique compatible value of n_+ and n_-. The number of microstates compatible with a given energy E coincides with the number of distinct ways to select n_+ particles from a set of N. Mathematically:

$$\Omega(E, N) = \Omega(n_+, n_-) = \frac{N!}{n_+!(N - n_+)!} \tag{4.117}$$

We can now calculate entropy:

$$S = k \ln \Omega = k[\ln N! - \ln n_+! - \ln(N - n_+)!] \simeq k[N \ln N - n_+ \ln n_+ - n_- \ln n_-] \tag{4.118}$$

where Stirling approximation has been applied. It is convenient to define an intensive variable x as the fraction of particles in the higher energy level:

$$x = \frac{n_+}{N} = \frac{1}{2}\left(1 + \frac{E}{N\epsilon}\right) \tag{4.119}$$

$$\frac{n_-}{N} = 1 - x \tag{4.120}$$

Rearranging and substituting:

$$S = k[N \ln N - n_+ \ln n_+ - n_- \ln n_-] = kN[\ln N - \frac{n_+}{N} \ln \frac{n_+}{N} - \frac{n_+}{N} \ln N - \frac{n_-}{N} \ln \frac{n_-}{N} - \frac{n_-}{N} \ln N] \tag{4.121}$$

Finally, we obtain an expression for entropy:

$$S = -kN[x \ln x + (1 - x) \ln(1 - x)] \tag{4.122}$$

It is easy to see that, as expected, S is an extensive property. We can now calculate temperature

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{N,V} = \left(\frac{\partial S}{\partial X}\right)_N \underbrace{\left(\frac{\partial X}{\partial E}\right)_N}_{1/2N\epsilon} = \frac{k}{2\epsilon} \ln\left(\frac{1 - x}{x}\right) \tag{4.123}$$

Let's substitute x with its original value to obtain the distribution of population as a function of temperature:

$$x = \frac{n_+}{N} = \frac{1}{\mathrm{e}^{2\epsilon/kT} + 1} \tag{4.124}$$

We can finally calculate the energy E:

$$E = N\epsilon(2x - 1) = N\epsilon\left(\frac{2}{\mathrm{e}^{2\epsilon/kT} + 1} - 1\right) = -N\epsilon \tanh\left(\frac{\epsilon}{kT}\right) \tag{4.125}$$

Let's see what we get at very low and very large temperatures:

$$\lim_{T \to 0} x = 0 \tag{4.126}$$

$$\lim_{T\to 0} E = -N\epsilon \tag{4.127}$$

This result suggests that all particles are placed in the lowest energy level. The entropy is zero: there is only one state $\Omega = 1$ (there is no uncertainty).

$$\lim_{T\to\infty} x = \frac{1}{2} \tag{4.128}$$

$$\lim_{T\to\infty} E = 0 \tag{4.129}$$

In this case the populations in both energy levels distribute equally. This is the state of maximum disorder, where entropy $S = Nk\ln 2$ reaches its maximum value (null information).

We can finally calculate the Helmholtz free energy:

$$A = E - TS = -N\epsilon\tanh\left(\frac{\epsilon}{kT}\right) + NkT[x\ln x + (1-x)\ln(1-x)] \tag{4.130}$$

4.6 Conclusions

The microcanonical ensemble is of fundamental importance:

1. It is constructed in a clear way from the postulate of equal a priori probability.
2. It serves as the starting point for developing more useful equilibrium ensembles for more realistic physical situations, such as the study of thermally, mechanically, and materially coupled systems.

However, it presents significant drawbacks:

1. If applied to non-trivial problems (non-ideal systems), this ensemble is mathematically challenging to use.
2. Isolated physical systems do not exist in reality, and the study of systems interacting with their environment is often of greater interest.
3. There is no tool to experimentally measure a system's energy directly. It can only be indirectly determined from other magnitudes: P, T, etc. Furthermore, T and P can not only be measured, but also manipulated.

4.7 Problems

4.1 Calculate the entropy and the chemical potential of Ne at 300 K and 1 bar.

4.2 Consider a monoatomic ideal gas that lives at a height z above sea level, so each molecule has potential energy mgz, with g the gravitational acceleration, in addition to its kinetic energy.

(a) Find an expression for the chemical potential for the molecules at height z:

$$\mu = -kT\ln\left[\frac{V}{N}\left(\frac{4\pi mE}{3h^2N}\right)^{3/2}\right] + mgz \tag{4.131}$$

(b) Suppose you have two chunks of this gas, one at sea level and one at height z, each having the same temperature and volume. Assuming that they are in diffusive equilibrium, show that the number of molecules in the higher chunk is

$$N(z) = N(0)\mathrm{e}^{-mgz/kT}. \tag{4.132}$$

4.3 Use the microcanonical ensemble to derive the Helmholtz free energy $A = A(T, N)$ as a function of the system's temperature for a system composed of N identical and distinguishable particles. Each particle can occupy two distinct energy levels ($\pm\varepsilon$), which are non-degenerate. Additionally, examine the behaviour of entropy in the limits $T \to 0$ and $T \to \infty$.

4.4 A system is composed of N classical three-dimensional harmonic oscillators located in space. All oscillators have a frequency ω, and the system is isolated with a total energy equal to E. The Hamiltonian for this system is given by:

$$H = \sum_{i=1}^{3N} = \sum_{i=1}^{3N} \left(\frac{p_i^2}{2m} + \frac{1}{2} m\omega^2 q_i^2 \right) \tag{4.133}$$

Using the microcanonical ensemble, determine the entropy and temperature of the system. Also, discuss whether the oscillators should be treated as distinguishable or indistinguishable particles.

4.5 Consider a set of N non-interacting classical two-dimensional harmonic oscillators with frequencies ω_i and mass m_i, where i ranges from 1 to N. Calculate, using the microcanonical ensemble, the specific heat at constant volume and the chemical potential. Note that each oscillator has its own frequency.

Chapter 5

Canonical Ensemble

5.1 Introduction

In the previous chapter, we have learnt that the microcanonical (NVE) ensemble is employed to study systems with a fixed total energy and no energy exchange with the surroundings. As a result, it characterises the statistical behaviour of an isolated system, considering the multiplicity of energy states accessible to the system for a given fixed energy. We have already mentioned that the NVE ensemble presents significant drawbacks:

1. If applied to non-trivial problems (non-ideal systems), this ensemble is **mathematically challenging** to use.

2. Isolated physical systems do not exist in reality, and the study of **systems interacting with their environment is typically of greater interest**.

3. There is no instrument to directly measure or control a system's energy experimentally. It can only be indirectly determined from other quantities: pressure, temperature, etc. **Fixing temperature is much simpler as it is directly observable (with a thermometer) and easily controlled using a thermal bath**.

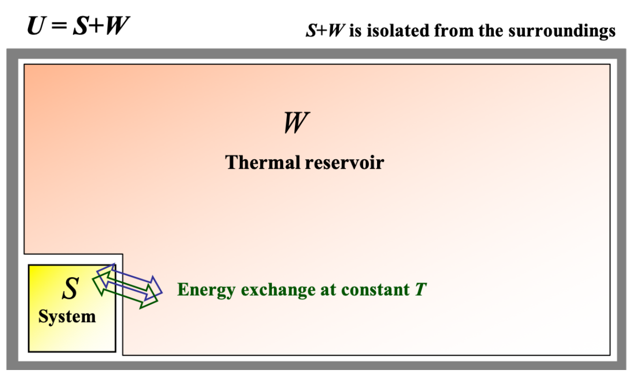


Figure 5.1: Schematic representation of a macroscopic system S in contact with a thermal reservoir W. The universe $U = S + W$ is isolated from the surroundings.

In this chapter, we introduce and describe the **canonical ensemble**, also indicated as NVT ensemble, which deals with systems that can exchange energy with a larger reservoir (bath) at a constant temperature. In the NVT ensemble, systems are allowed to interact with an external reservoir, enabling energy exchanges.

This interaction permits the system to explore various energy states within the constraints of a fixed temperature. The resulting fluctuations among energy states contribute to the statistical distribution of energy across accessible states, allowing for a probabilistic description of the system.

The members of the NVT ensemble are macroscopic systems thermally coupled through diathermic walls, allowing energy exchange between them. All systems are in equilibrium at a constant temperature (T), and the ensemble is isolated from the external surroundings as schematically illustrated in Fig. 5.1.

5.2 Design of the Canonical Ensemble

With reference to Fig. 5.1, U is an isolated *supersystem* with E_u constant. The system S is a region of U that is not strongly fluctuating and is macroscopic, but small compared to U. The complement of S in U, indicated by W, is large compared to S and acts as a thermal bath. We also know that $1 << N_s << N_w$, where N_s and N_w are, respectively, the number of particles in S and in W. Similarly, $1 << V_s << V_w$, with V_s and V_w the volume occupied, respectively, by S and W. In particular, the following relations hold:

$$N_s \to \infty;\ N_w \to \infty;\ V_s \to \infty;\ V_w \to \infty \tag{5.1}$$

and, in the thermodynamic limit, the size of the bath is infinitely larger than that of the system of interest. Mathematically, this condition is expressed as

$$\frac{N_w}{N_s} \to \infty;\ \frac{V_w}{V_s} \to \infty \tag{5.2}$$

We also assume that the density of S equals the density of W:

$$\rho = \frac{N_s}{V_s} = \frac{N_w}{V_w} \tag{5.3}$$

Finally, the internal energy of the supersystem (universe) is related to that of the system and the bath as follows:

$$E_u = E_s + E_w + E_{sw} \simeq E_s + E_w \tag{5.4}$$

where $E_s \ll E_w$ and $E_{sw} \ll E_s$ are negligible. As a consequence, given the negligible interaction between S and W, one can write the following relation:

$$E_m = E_m(N_s, V_s) \tag{5.5}$$

where E_m, which depends on the number of particles and the volume of S, denotes the energy of an energy state m in which S can be found. If the energy of S is E_m, then the energy of the thermal bath W must lie between

$$E_w = E_u - E_m \tag{5.6}$$

and

$$E_w + \Delta E. \tag{5.7}$$

This energy interval of *uncertainty* arises from the fact that energy cannot be measured exactly, as already noted in the discussion of the microcanonical ensemble in Chapter 4. From now on, to simplify the notation, we will use N and V to refer, respectively, to the number of particles and volume of S.

Partition Function

Now that the thermodynamic system of interest has been defined, the question we aim to address is: *What is the probability, P_m, that the system S is found in the energy state E_m?* To answer this question, we note that the supersystem U is isolated, with E_u, N_u, and V_u constant. This implies that U can be treated within the microcanonical ensemble framework. In particular,

$$\Omega_u = \Omega_u(E_u, N_u, V_u; \Delta E) \equiv \Omega_u(E_u, \Delta E). \tag{5.8}$$

Strictly speaking, the thermal bath W, with energy $E_u - E_m$, is not isolated, as it can exchange energy with S. Nevertheless, since $E_m \ll E_u$, we can neglect the effect of S on W and assume, to a good approximation, that W is also effectively isolated:

$$\Omega_w = \Omega_w(E_u - E_m; \Delta E). \tag{5.9}$$

We emphasize that Ω_w represents the number of configurations (states) in the universe U that are compatible with an energy E_m for S, where E_w is such that the total energy falls within the range $[E_u, E_u + \Delta E]$. Additionally, since W can be considered an isolated system (as its interactions with S can be neglected), it can be treated within the microcanonical ensemble framework, implying that all its possible configurations (states) are equally probable. In light of these preliminary considerations, let's calculate this probability:

$$P_m = \frac{\text{favourable cases}}{\text{possible cases}} = \frac{\text{configurations of } U \text{ that are compatible with } S \text{ at energy } E_m}{\text{total configurations of universe } U} \tag{5.10}$$

equivalently

$$P_m = \frac{\Omega_w(E_u - E_m; \Delta E)}{\Omega_u(E_u; \Delta E)} \tag{5.11}$$

Given that $E_m \ll E_u$, we can develop a power series of Ω_w. However, since Ω is generally a function that increases exponentially with energy, the convergence of the series will be very slow. An alternative is expanding $\ln \Omega_w$:

$$\ln \Omega_w(E_w; \Delta E) = \ln \Omega_w(E_u; \Delta E) + \underbrace{\left(\frac{\partial \ln \Omega_w}{\partial E}\right)_{E=E_u}}_{\beta} (E_w - E_u) + \cdots \simeq \ln \Omega_w(E_u; \Delta E) - \beta E_m \tag{5.12}$$

Rearranging, we obtain

$$\Omega_w(E_u - E_m; \Delta E) \simeq \Omega_w(E_u; \Delta E) \mathrm{e}^{-\beta E_m} \tag{5.13}$$

and substituting in Eq. 6.20, we get

$$P_m = \underbrace{\frac{\Omega_w(E_u; \Delta E)}{\Omega_u(E_u; \Delta E)}}_{1/Z} \mathrm{e}^{-\beta E_m} = \frac{1}{Z} \mathrm{e}^{-\beta E_m} \tag{5.14}$$

where Z is the **canonical partition function**. The best way to calculate Z is not to use Eq. 5.14, but to recall the normalisation condition of probability:

$$\sum_m P_m = \sum_m \frac{1}{Z} \mathrm{e}^{-\beta E_m} = \frac{1}{Z} \sum_m \mathrm{e}^{-\beta E_m} = 1 \tag{5.15}$$

It follows that

$$Z = \sum_m \mathrm{e}^{-\beta E_m} \tag{5.16}$$

and equivalently

$$Z = \sum_n \Omega(E_n) \mathrm{e}^{-\beta E_n} \tag{5.17}$$

where n indicates the number of energy levels. Since each energy level E_n can have multiple states associated with it, Z can be expressed either as a sum over individual states or as a sum over energy levels weighted by their degeneracy $\Omega(E_n)$, as in Eq. 5.17. The partition function Z encodes how the probabilities are **partitioned** among the different states, based on their individual energies[1]. It represents the sum over all the energy states of S compatible with the same NVT-macrostate or, equivalently, the sum over all the energy levels observed in S. The partition function is much easier to calculate than Ω and is valid for systems that are more general than the isolated one.

[1] The letter Z stands for the German word *Zustandssumme*, "sum over states"

Quantum Formulation

The quantum formulation of the canonical ensemble is represented by the density matrix ρ_{mn}, which is given by:

$$\rho_{mn} = \frac{1}{Z}\mathrm{e}^{-\beta E_m}\delta_{mn}. \tag{5.18}$$

In general, expectation values are calculated as

$$\langle B\rangle = \sum_m P_m B_m = \frac{1}{Z}\sum_m B_m \mathrm{e}^{-\beta E_m}, \tag{5.19}$$

where $\langle B\rangle = \langle\phi_m|\hat{B}|\phi_m\rangle$ is the expectation value of $\hat{B}$ in the eigenstate $|\phi_m\rangle$. One of the key advantages of the canonical ensemble is that it can be applied using any orthonormal basis $|\psi_m\rangle$, even if it does not diagonalize $\hat{H}$. This implies that solving Schrödinger's equation explicitly is not necessary. In terms of the density operator, we define:

$$\hat{\rho} = \frac{1}{Z}\mathrm{e}^{-\beta\hat{H}}. \tag{5.20}$$

It follows that

$$\langle B\rangle = \mathrm{Tr}[\hat{B}\hat{\rho}] = \frac{1}{Z}\mathrm{Tr}[\hat{B}\,\mathrm{e}^{-\beta\hat{H}}]. \tag{5.21}$$

Thus, we can use any orthonormal basis of the Hilbert space to perform numerical calculations:

$$Z = \mathrm{Tr}\left[\mathrm{e}^{-\beta\hat{H}}\right]. \tag{5.22}$$

5.3 Connection with Thermodynamics

Thermodynamics begins with a few empirical observations, which are organised and axiomatised in the form of four laws. From these laws, thermodynamics establishes relationships between several properties that characterise the thermal behaviour of a macroscopic system. However, thermodynamics presents an important drawback: **it does not allow the calculation of absolute values for any property**. For example, if we know C_V, we can determine C_P and viceversa; thermodynamics provides the relationship between them but not their specific values. Statistical Physics can resolve this inconvenience; specifically, the canonical ensemble provides a model for thermal equilibrium. More specifically, Statistical Physics is capable of obtaining thermodynamic (macroscopic) properties in terms of microscopic magnitudes of molecules. In the case of the canonical ensemble, if we know the energy spectrum $\{E_n\}$, we can deduce the thermodynamic variables. Similarly, Statistical Physics allows for the calculation of microscopic properties from macroscopic observations (for example, intermolecular interactions).

Thermodynamic variables can be classified into three groups:

- **External Parameters**, such as N (number of particles), V (volume), etc. Fixed by the experimenter or the system's environment.
- **Mechanical Properties**, such as p (pressure), U (internal energy), etc. They are averages of microscopic dynamic functions.
- **Thermal Properties**, such as T (temperature), S (entropy). They do not have a microscopic equivalent (at the particle level). They can only be defined at the macroscopic level. They are collective properties whose value is determined by the entire ensemble.

The first connection is provided by the internal energy:

$$U = \langle E\rangle = \mathrm{Tr}[\hat{H}\hat{\rho}] = \frac{1}{Z}\sum_m E_m \mathrm{e}^{-\beta E_m} \tag{5.23}$$

We note that $U = U(\beta, N, V)$, although we still have to see what β is. Now, since the system can exchange energy with the environment (in equilibrium), E is no longer fixed but can fluctuate. The internal energy is its mean value $U = \langle E \rangle$. It is possible to re-write Eq. 6.28 in a more practical fashion:

$$U = \sum_m E_m P_m = \frac{1}{Z} \sum_m E_m \mathrm{e}^{-\beta E_m} = \frac{1}{Z} \sum_m \left(-\frac{\partial}{\partial \beta} \mathrm{e}^{-\beta E_m} \right) = \frac{1}{Z} \left(-\frac{\partial}{\partial \beta} \right) \sum_m \mathrm{e}^{-\beta E_m} \tag{5.24}$$

Equivalently:

$$U = \frac{1}{Z} \left(-\frac{\partial Z}{\partial \beta} \right)_{N,V} = \left(-\frac{\partial \ln Z}{\partial \beta} \right)_{N,V} \tag{5.25}$$

To find out further connections with Thermodynamics, specifically that β is closely related to the temperature, we now consider the configuration given in Fig. 5.2.

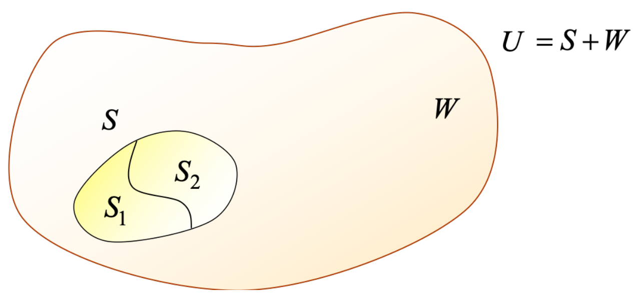


Figure 5.2: Schematic representation of a macroscopic system $S = S_1 + S_2$ in contact with a thermal reservoir W.

Systems S_1 and S_2 exchange energy between themselves and with W and the three systems are at thermal equilibrium with each other. We will assume that the sizes of S_1 and S_2 are large compared to the range of interactions between particles: the interaction energy between S_1 and S_2, although physically significant in ensuring the exchange of energy, is negligible compared to the energy of S_1 or S_2. What is the probability, $P_{n,m}$, of finding system S_1 in the energy state E_{1n} and S_2 in E_{2m}? To answer this question, we have two options:

- If we consider S as a sum of S_1 and S_2, then

$$P_{n,m} = P_n^{(1)} P_m^{(2)} = \frac{1}{Z_1} \mathrm{e}^{-\beta_1 E_{1,n}} \frac{1}{Z_2} \mathrm{e}^{-\beta_2 E_{2,m}} \tag{5.26}$$

  This is basically the product of two independent probabilities, with, in principle, $\beta_1 \neq \beta_2$.

- If we can look at S as a unique system, then

$$P_{n,m} = \frac{1}{Z} \mathrm{e}^{-\beta(E_{1,n} + E_{2,m})} \tag{5.27}$$

The probability $P_{n,m}$ must be the same regardless the option applied. This implies that

$$\frac{1}{Z_1 Z_2} \mathrm{e}^{-\beta_1 E_{1,n}} \mathrm{e}^{-\beta_2 E_{2,m}} = \frac{1}{Z} \mathrm{e}^{-\beta(E_{1,n} + E_{2,m})} \tag{5.28}$$

Since this equation is true $\forall E_{1,n}$ and $\forall E_{2,m}$, then it follows that

$$\beta_1 = \beta_2 \tag{5.29}$$

Because thermodynamics asserts that two systems in equilibrium must have the same temperature, it must be that

$$\beta = \beta(T) \tag{5.30}$$

is an **intensive** property (in particular, $\beta = 1/kT$). We also see that

$$Z = Z_1 Z_2 \tag{5.31}$$

which implies that

$$\ln Z = \ln Z_1 + \ln Z_2 \tag{5.32}$$

is an additive or **extensive** property. Moreover, we see that β **is a common property**, being the same in all systems when thermal equilibrium is achieved. By applying these two conclusions, we can define the following function

$$A \equiv -\frac{1}{\beta} \ln Z \tag{5.33}$$

which is also extensive as $A_{1+2} = A_1 + A_2$. What does A represent? To find that out, let's calculate the total derivative of βA, noting that $A = A(\beta, \{E_n\})$

$$d(\beta A) = \left.\frac{\partial(\beta A)}{\partial \beta}\right|_{\{E_n\}} d\beta + \sum_n \left.\frac{\partial(\beta A)}{\partial E_n}\right|_{\beta} dE_n \tag{5.34}$$

The first term in the right hand side of the above equation is equal to the internal energy (see also Eq. 5.25):

$$\left.\frac{\partial(\beta A)}{\partial \beta}\right|_{\{E_n\}} = -\left.\frac{\partial(\ln Z)}{\partial \beta}\right|_{\{E_n\}} = -\left.\frac{\partial(\ln Z)}{\partial \beta}\right|_{N,V} = U \tag{5.35}$$

The second term is a probability. Let's see why:

$$\left.\frac{\partial(\beta A)}{\partial E_n}\right|_{\beta} = -\left.\frac{\partial(\ln Z)}{\partial E_n}\right|_{\beta} = -\frac{1}{Z}\left.\frac{\partial Z}{\partial E_n}\right|_{\beta} = -\frac{1}{Z}\frac{\partial}{\partial E_n}\left(\sum_m \mathrm{e}^{-\beta E_m}\right)_{\beta} = \beta\,\frac{\mathrm{e}^{-\beta E_n}}{Z} = \beta P_n \tag{5.36}$$

Substituting these results in Eq. 5.34, we obtain

$$d(\beta A) = U d\beta + \beta \sum_n P_n \, dE_n \tag{5.37}$$

Since $U d\beta = d(\beta U) - \beta dU$, we can rewrite this equation as

$$d[\beta(U - A)] = \beta dU - \beta \sum_n P_n \, dE_n \tag{5.38}$$

We have now to understand what the summation in the above equation, that is $\sum_n P_n \, dE_n$, represents *thermodynamically*. To this end, we first note that dE_n is an energy change of the system following an **adiabatic volume change**[2]. Mathematically, this volume-driven energy change can be expressed as

$$E_n' = E_n + \left.\frac{\partial E_n}{\partial V}\right|_N dV \tag{5.39}$$

Therefore

$$\sum_n P_n \, dE_n = \underbrace{\sum_n P_n E_n'}_{U'} - \underbrace{\sum_n P_n E_n}_{U} = dU_{\text{adiabatic}} \tag{5.40}$$

[2]The term *adiabatic* refers to a process that occurs with no heat exchange or to a system that does not exchange heat with its surroundings

The First Law of Thermodynamics (conservation of energy) states that $dU = đQ + đW$, where $đQ$ and $đW$ represent, respectively, the amount of heat and work exchanged between the system and its surroundings[3]. Because the process we are considering is adiabatic, the First Law reduces to $dU_{\text{adiabatic}} = đW$. In particular,

$$dU_{\text{adiabatic}} = đW = \sum_n P_n \, dE_n = \sum_n P_n \left. \frac{\partial E_n}{\partial V} \right|_N dV \tag{5.43}$$

Since the work exchanged as a result of a volume change is $đW = -pdV$, we can rewrite the last equation as

$$\sum_n P_n \left. \frac{\partial E_n}{\partial V} \right|_N dV = -pdV \tag{5.44}$$

Rearranging, we obtain an expression for pressure:

$$p = -\sum_n P_n \left. \frac{\partial E_n}{\partial V} \right|_N \tag{5.45}$$

If we now substitute this result into Eq. 5.38, we get

$$d[\beta(U - A)] = \beta dU - \beta \sum_n P_n \, dE_n = \beta[dU - đW] = \beta đQ \tag{5.46}$$

It is interesting and useful to observe that, by multiplying $đQ$ by β, one can transform this inexact differential into the exact differential $d[\beta(U - A)]$. This implies that β is the *integrating factor* of $đQ$[4]. If we now apply the Second Law of Thermodynamics $đQ = TdS$, we obtain

$$d[\beta(U - A)] = \beta đQ = \beta T dS \tag{5.47}$$

This implies that the property $A = A(N, V, T) = U - TS$, defined in Eq. 5.33, is the **Helmholtz free energy**. The total differential of $A(N, V, T)$ reads

$$dA = \left. \frac{\partial A}{\partial N} \right|_{V,T} dN + \left. \frac{\partial A}{\partial V} \right|_{N,T} dV + \left. \frac{\partial A}{\partial T} \right|_{N,V} dT = -d\left(\frac{1}{\beta} \ln Z \right). \tag{5.48}$$

It follows that

$$\mu = \left. \frac{\partial A}{\partial N} \right|_{V,T} = \frac{1}{Z} \sum_m \left(\left. \frac{\partial E_m}{\partial N} \right|_{V,T} \mathrm{e}^{-\beta E_m} \right) = \left\langle \frac{\partial E}{\partial N} \right\rangle_{V,T} \tag{5.49}$$

$$p = -\left. \frac{\partial A}{\partial V} \right|_{N,T} = \frac{1}{\beta} \left. \frac{\partial \ln Z}{\partial V} \right|_{N,T} = -\frac{1}{Z} \sum_m \left(\left. \frac{\partial E_m}{\partial V} \right|_{N,T} \mathrm{e}^{-\beta E_m} \right) = -\left\langle \frac{\partial E}{\partial V} \right\rangle_{N,T} \tag{5.50}$$

$$S = -\left. \frac{\partial A}{\partial T} \right|_{N,V} = kT \left. \frac{\partial \ln Z}{\partial T} \right|_{N,V} + k \ln Z = k(\beta \langle E \rangle + \ln Z) \tag{5.51}$$

[3]By convention, we here establish that the work is taken positive if done on the system and negative if done by the system. This is just a *sign* convention and does not change anyhow the flow of our reasoning. The symbol $đ$ indicates an inexact differential, which is here employed to stress that heat and work are **path functions**: their value depends on the path followed to go from an initial state 1 to a final state 2. Mathematically, this implies that

$$\int_1^2 đW \neq W_2 - W_1 \tag{5.41}$$

and

$$\int_1^2 đQ \neq Q_2 - Q_1 \tag{5.42}$$

[4]In this context, an *integrator factor* converts an inexact differential into an exact differential. Note that $-p^{-1}$ is the integrating factor of $đW$

We already know the expression of the internal energy $U = -\left.\frac{\partial \ln Z}{\partial \beta}\right|_{N,V}$, from which the heat capacity at constant volume is obtained:

$$C_v = \left.\frac{\partial U}{\partial T}\right|_{N,V} = \left.\frac{\partial U}{\partial \beta}\frac{\partial \beta}{\partial T}\right|_{N,V} = \frac{1}{kT^2}\left.\frac{\partial^2 \ln Z}{\partial \beta^2}\right|_{N,V} = k\beta^2 \left.\frac{\partial^2 \ln Z}{\partial \beta^2}\right|_{N,V} \tag{5.52}$$

We can also express entropy in terms of probability only. Let's start with the operative definition of probability:

$$P_n = \frac{\mathrm{e}^{-\beta E_n}}{Z} \Longleftrightarrow \ln P_n = -\beta E_n - \ln Z \tag{5.53}$$

We can use P_n to determine the average value of energy:

$$\beta\langle E\rangle = \sum_n P_n \beta E_n = \sum_n P_n(-\ln P_n - \ln Z) = -\sum_n P_n \ln P_n - \ln Z \sum_n P_n = -\sum_n P_n \ln P_n - \ln Z \tag{5.54}$$

Therefore

$$S = k(\beta\langle E\rangle + \ln Z) = k\left(-\sum_n P_n \ln P_n - \cancel{\ln Z} + \cancel{\ln Z}\right) = -k\sum_n P_n \ln P_n \tag{5.55}$$

We see that entropy depends solely on the probability values: it does not possess any associated microscopic magnitude; it is a **thermal quantity**. Eq. 6.55 leads to several important conclusions:

- If the system is in the ground state (at $T = 0$ K), and this state is unique, then $P_n = 1$ for this state and zero for all other states. The value of entropy for this state is zero, which is essentially the **Third Law of Thermodynamics**.
- As the temperature increases, the number of accessible energy states increases, and the values of P_n above the ground level are no longer zero. This leads to an increase in entropy. **Higher entropy implies a higher degree of statistical disorder within the system**.

Finally, we can express the partition function in terms of states or energy levels:

$$Z = \underbrace{\sum_m}_{\text{states}} \mathrm{e}^{-\beta E_m} = \underbrace{\sum_n}_{\text{energy levels}} \sum_{[j]_n} \mathrm{e}^{-\beta E_n} = \sum_n \mathrm{e}^{-\beta E_n} \sum_{[j]_n} 1 = \sum_n \Omega(E_n)\mathrm{e}^{-\beta E_n} \tag{5.56}$$

where $[j]_n$ indicates the number of states with energy level compatible with n. We also remember that

$$P_m = \frac{1}{Z}\mathrm{e}^{-\beta E_m} \tag{5.57}$$

and

$$\langle B\rangle = \frac{1}{Z}\sum_m B_m \mathrm{e}^{-\beta E_m} \tag{5.58}$$

5.4 Partition Function. Classical and Quantum Formalism

This section aims at providing a conneciton between the canonical and microcanonical ensembles within the quantum and classical formalisms.

I. Quantum formalism

Isolated system (NVE ensemble)

$$P_m = \begin{cases} 1/\Omega & \text{if}\, E_m \in [E, E+\Delta E] \\ 0 & \text{if}\, E_m \notin [E, E+\Delta E] \end{cases}$$

$$\Omega(N,V,E) \equiv \Omega(E;\Delta E)$$

$$S = S(N,V,E) = k\ln\Omega$$

$$\langle b\rangle = \frac{1}{\Omega}\sum_n \langle\phi_n|\hat{b}|\phi_n\rangle \,\text{with}\, E_n \in [E, E+\Delta E]$$

$$\hat{H}|\phi_m\rangle = E_m|\phi_m\rangle$$

System coupled to a bath (NVT ensemble)

$$P_m = \frac{1}{Z}\mathrm{e}^{-\beta E_m} \;\; ; \;\; \hat{\rho} = \frac{1}{Z}\mathrm{e}^{-\beta\hat{H}}$$

$$Z = \sum_m \mathrm{e}^{-\beta E_m} \;\; ; \;\; Z = \mathrm{Tr}[\mathrm{e}^{-\beta\hat{H}}]$$

$$A = A(N,V,T) = -kT\ln Z$$

$$\langle b\rangle = \frac{1}{Z}\sum_n \mathrm{e}^{-\beta E_n}\langle\phi_n|\hat{b}|\phi_n\rangle]$$

$$\langle b\rangle = \mathrm{Tr}[\hat{b}\hat{\rho}] = \frac{1}{Z}\mathrm{Tr}[\hat{b}\,\mathrm{e}^{-\beta\hat{H}}]$$

The canonical partition function provides a contact point with the two ensembles:

$$Z = \underbrace{\sum_n}_{\text{energy levels}} \Omega(E_n)\mathrm{e}^{-\beta E_n} \tag{5.59}$$

II. Classical formalism

Isolated system (NVE ensemble)

$$\rho = \frac{1}{N!h^s}\frac{1}{\omega(E)}\delta(E - H(\mathbf{q},\mathbf{p}))$$

$$\omega(E) = \frac{1}{N!h^s}\int \delta(E - H(\mathbf{q},\mathbf{p}))d\mathbf{q}\,d\mathbf{p}$$

$$\Omega(E) = \omega(E)\Delta E \;\; ; \;\; S = k\ln\Omega$$

$$\langle b\rangle = \frac{1}{N!h^s\omega(E)}\int b(\mathbf{q},\mathbf{p})\delta(E - H(\mathbf{q},\mathbf{p}))d\mathbf{q}\,d\mathbf{p}$$

System coupled to a bath (NVT ensemble)

$$\rho(\mathbf{q},\mathbf{p}) = \frac{1}{N!h^s}\frac{1}{Z}\mathrm{e}^{-\beta H(\mathbf{q},\mathbf{p})}$$

$$Z = \frac{1}{N!h^s}\int \mathrm{e}^{-\beta H(\mathbf{q},\mathbf{p})}d\mathbf{q}\,d\mathbf{p}$$

$$A = -kT\ln Z \;\; ; \;\; S = -k\int \rho(\mathbf{q},\mathbf{p})\ln[N!h^s\rho(\mathbf{q},\mathbf{p})]d\mathbf{q}\,d\mathbf{p}$$

$$\langle b\rangle = \frac{1}{N!h^s}\frac{1}{Z}\int b(\mathbf{q},\mathbf{p})\mathrm{e}^{-\beta H(\mathbf{q},\mathbf{p})}d\mathbf{q}\,d\mathbf{p}$$

Again, the canonical partition function provides a contact point with the two ensembles:

$$Z = \int_0^\infty \mathrm{e}^{-\beta E}\omega(E)\,dE \tag{5.60}$$

Z is basically the Laplace transform of $\omega(E)$: $\omega(E) \to Z(\beta)$. Let's prove the above equation:

$$
\begin{aligned}
Z &= \frac{1}{N!h^s}\int \mathrm{e}^{-\beta H(\mathbf{q},\mathbf{p})} d\mathbf{q}\, d\mathbf{p} \underbrace{\int_0^\infty \delta(E - H(\mathbf{q},\mathbf{p}))\, dE}_{=1} = \\
&= \frac{1}{N!h^s}\int \mathrm{e}^{-\beta E}\, d\mathbf{q}\, d\mathbf{p} \int_0^\infty \delta(E - H(\mathbf{q},\mathbf{p}))\, dE = \\
&= \int_0^\infty \mathrm{e}^{-\beta E}\left[\frac{1}{N!h^s}\int \delta(E - H(\mathbf{q},\mathbf{p})) d\mathbf{q}\, d\mathbf{p}\right] dE = \\
&= \int_0^\infty \mathrm{e}^{-\beta E}\omega(E)\, dE
\end{aligned}
\tag{5.61}
$$

5.5 Application Example: Model for Solids.

Let's consider a set of N quantum three-dimensional harmonic oscillators, with vibration frequency ω at temperature T isolated from the surroundings, as shown in Fig. 5.3. Our aim is calculating the canonical partition function Z and, once the expression for Z is obtained, the thermodynamic properties. To this end, we make use of the Einstein solid, which consists of $3N$ distinguishable and independent oscillators of equal frequency, ω. In the Einstein crystal model, the partition function can then be derived by treating the system as a collection of $3N$ independent quantum harmonic oscillators, corresponding to the vibrational degrees of freedom of the N atoms in the crystal.

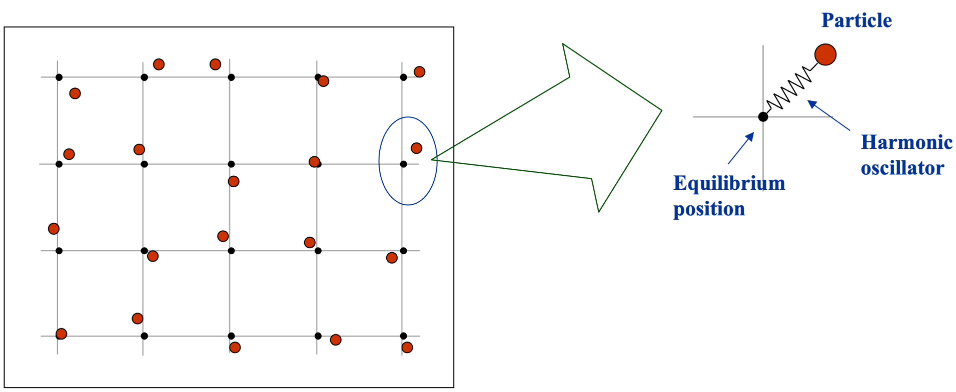


Figure 5.3: Model solid incorporating N quantum three-dimensional quantum harmonic oscillators.

To obtain Z, we should first write down the total energy of this system, which is given by

$$E = \sum_{i=1}^{3N} \epsilon_i = \sum_{i=1}^{3N} \hbar\omega\left(n_i + \frac{1}{2}\right) \tag{5.62}$$

with $n_i = 0, 1, \ldots \infty$. It follows that the partition function reads

$$Z = \sum_{\{E\}} \mathrm{e}^{-\beta E} = \sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\cdots\sum_{n_{3N}=0}^{\infty} \mathrm{e}^{-\beta\hbar\omega\sum_{i=1}^{3N}\left(n_i+\frac{1}{2}\right)} \tag{5.63}$$

which gives

$$Z = \underbrace{\sum_{n_1=0}^{\infty} \mathrm{e}^{-\beta\hbar\omega\left(n_1+\frac{1}{2}\right)}}_{Z_1}\underbrace{\sum_{n_2=0}^{\infty} \mathrm{e}^{-\beta\hbar\omega\left(n_2+\frac{1}{2}\right)}}_{Z_2}\cdots\underbrace{\sum_{n_{3N}=0}^{\infty} \mathrm{e}^{-\beta\hbar\omega\left(n_{3N}+\frac{1}{2}\right)}}_{Z_{3N}} = (Z_1)^{3N} \tag{5.64}$$

with $Z_1 = Z_2 = Z_{3N}$. In particular

$$Z_1 = \sum_{n=0}^{\infty} \mathrm{e}^{-\beta\hbar\omega\left(n+\frac{1}{2}\right)} = \mathrm{e}^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} \mathrm{e}^{-\beta\hbar\omega n} = \mathrm{e}^{-\beta\hbar\omega/2} \frac{1}{1-\mathrm{e}^{-\beta\hbar\omega}} \tag{5.65}$$

where we have made use of the property $\sum_{n=0}^{\infty} a^n = 1/(1-a)$ for $a < 1$ and $a = \mathrm{e}^{-\beta\hbar\omega}$. It follows that

$$Z_1 = \frac{1}{2\sinh(\beta\hbar\omega/2)} = \frac{1}{2\sinh(\hbar\omega/2kT)} \tag{5.66}$$

and

$$Z = \left(\frac{1}{2\sinh(\hbar\omega/2kT)}\right)^{3N} \tag{5.67}$$

The Helmholtz free energy reads

$$A = -kT\ln Z = 3NkT\ln\left[2\sinh\left(\frac{\hbar\omega}{2kT}\right)\right] \tag{5.68}$$

which is, as it is easy to check, an extensive variable. Let's now calculate entropy and internal energy U:

$$S = -\left.\frac{\partial A}{\partial T}\right|_{N,V} = -3Nk\left.\frac{\partial\{T\ln[2\sinh(\hbar\omega/2kT)]\}}{\partial T}\right|_{N,V} = -3Nk\left(\ln[2\sinh(\beta\hbar\omega/2)] - \frac{\beta\hbar\omega}{2}\coth(\beta\hbar\omega/2)\right) \tag{5.69}$$

$$U = -\left.\frac{\partial\ln Z}{\partial\beta}\right|_{N,V} = \left.\frac{\partial\{3N\ln[2\sinh(\beta\hbar\omega/2)]\}}{\partial\beta}\right|_{N,V} = 3N\frac{\cosh(\beta\hbar\omega/2)}{\sinh(\beta\hbar\omega/2)}\frac{\hbar\omega}{2} = \frac{3N\hbar\omega}{2}\coth\left(\frac{\beta\hbar\omega}{2}\right) \tag{5.70}$$

We also see that for very large temperatures, we recover the result that we obtained in the microcanonical ensemble, that is

$$\lim_{T\to\infty} U = 3NkT \tag{5.71}$$

Other thermodynamic functions can be calculated by applying the same strategy.

5.6 Application to Ideal Classical Systems: Boltzmann Statistics

What do we mean by ideal system? It's the application of Statistical Physics to a system formed by N particles, with $N \gg 1$. The resolution of the Schrödinger equation of the system, although necessary to know $E_j(N,V)$, is, in general, a prohibitive task. However, there exist a significant number of systems of great physical interest, whose Hamiltonian can be separated into independent terms:

$$H(N) = \sum_{j=1}^{N} H_j \quad \longrightarrow \quad E(N) = \sum_{j=1}^{N} \epsilon_j \tag{5.72}$$

The **ideal system** is a set where H_j involves a reduced set of degrees of freedom that do not interact with the rest, consisting of **mutually independent** particles. Consequently, the problem of N bodies can be separated into N individual one-body problems. The dynamics of each particle is not affected by the presence of other particles - some examples of this kind are included in Table 5.1.

An ideal system, as typically defined in physics, consists of non-interacting particles. If there is no interaction between particles, then no energy or momentum exchange can occur between them. This can lead to situations where the system does not evolve toward equilibrium from an arbitrary initial state, which

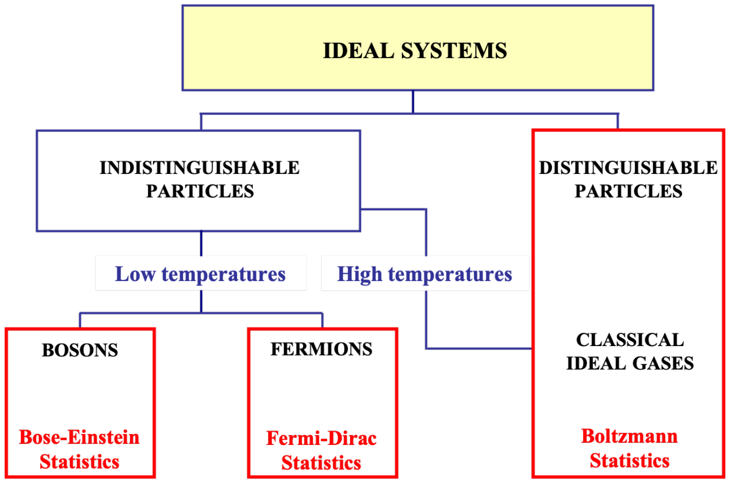


Figure 5.4: Scheme of ideal systems

Table 5.1: Examples of systems that can be treated as ideal.

System	Particle	Degrees of Freedom
Dilute gas of atoms	Atom	Translation of the center of mass Electronic-nuclear
Dilute gas of molecules	Molecule	Translation-rotation-vibration Electronic-nuclear
Solid	Atom	Vibration about the equilibrium position
Non-interacting spins	Atom	Spin

appears to contradict the Second Law of Thermodynamics[5]. Despite these limitations, this model remains very useful when weak interactions between system components exist. These interactions allow eventual equilibration, with their smallness representing minor corrections to ideal behaviour. With reference to Fig. 5.4, we will now explore the characteristics of ideal systems of distinguishable and indistinguishable particles.

5.6.1 Case 1. Ideal systems of identical and distinguishable particles

A system composed of N identical but distinguishable particles does not require consideration of the bosonic/fermionic nature of the particles. For example, atoms in a solid fit this description. They are localised at specific positions, thus, they are distinguishable. The total energy of this system can be written as:

$$E = \sum_{\alpha=1}^{N} \epsilon_{\xi}^{(\alpha)} = \epsilon_i^{(1)} + \epsilon_j^{(2)} + \epsilon_k^{(3)} + \dots \tag{5.73}$$

where $\alpha = 1, 2, \dots, N$ labels a specific particle, whose quantum state is indicated by $\xi = i, j, k, \dots$. In particular, Z can be calculated as the sum over all configurations compatible with the system total energy:

$$Z_N = \sum_{\{E\}} \mathrm{e}^{-\beta E}[\epsilon_i] \tag{5.74}$$

[5]The Second Law states that for an isolated system, entropy tends to increase over time, leading to equilibrium. However, in an ideal system, (1) if particles do not interact, their individual energies remain constant; (2) there is no mechanism for redistributing energy among the components of the system; and (3) without redistribution, a system that starts out in a non-equilibrium state (*e.g.*, with different regions having different energy distributions) will remain in that state indefinitely. Since the Second Law relies on microscopic interactions driving the system toward a more probable (higher entropy) state, a system without interactions fails to obey this principle. However, in real systems, even weak interactions restore thermalisation and entropy growth over time.

Summing for all E values is equivalent to summing over any configuration of the quantum numbers i, j, k, etc.

$$Z_N = \sum_{i,j,k,\dots} \mathrm{e}^{-\beta[\epsilon_i^{(1)}+\epsilon_j^{(2)}+\epsilon_k^{(3)}+\dots]} \tag{5.75}$$

We can solve these sums by breaking down the argument of the sum into products of exponentials and grouping:

$$Z_N = \left[\sum_i \mathrm{e}^{-\beta\epsilon_i^{(1)}}\right]\left[\sum_j \mathrm{e}^{-\beta\epsilon_j^{(2)}}\right]\left[\sum_k \mathrm{e}^{-\beta\epsilon_k^{(3)}}\right]\cdots \tag{5.76}$$

and in total, we have N factors. Since the particles are identical, the energy spectrum $\{\epsilon_i\}$ is the same for all; therefore, these N factors are identical. Let's define the partition function of a single particle:

$$Z_1 = \sum_m \mathrm{e}^{-\beta\epsilon_m} \tag{5.77}$$

where m represents the number of states. Because particles are ideal and distinguishable, then

$$Z_N = (Z_1)^N \tag{5.78}$$

if there is degeneracy[6], then Z_1 is better expressed as a sum over energy levels the form

$$Z_1 = \sum_n g_n \mathrm{e}^{-\beta\epsilon_n} \tag{5.79}$$

where g_n is the degeneracy of the energy level ϵ_n of a particle. In conclusion, for an ideal system with N distinguishable particles, **the problem of N bodies can be reduced to the problem of a single body**. If there are more species, for instance there are N_A particles of type A and N_B particles of type B, then the resulting partition function reads

$$Z_N(N_A, N_B, T, V) = (Z_1^{(A)})^{N_A}(Z_1^{(B)})^{N_B} \tag{5.80}$$

5.6.2 Case 2. Ideal systems of identical and indistinguishable particles

The expression derived in the previous section, that is $Z_N = (Z_1)^N$, is no longer valid when particles are indistinguishable (non-localised). The total energy of the ideal system results from the sum of distinct, autonomous terms.

$$E_N = \sum_{j=1}^{N} \epsilon^{(j)} \tag{5.81}$$

Ideal conditions allow for the definition of independent quantum states for each particle:

$$E_N = \epsilon_i^{(1)} + \epsilon_j^{(2)} + \epsilon_k^{(3)} + \dots \tag{5.82}$$

and the new canonical partition function reads

$$Z_N = \sum_{i,j,k,\dots} e^{-\beta\left(\epsilon_i^{(1)}+\epsilon_j^{(2)}+\epsilon_k^{(3)}+\dots\right)} \tag{5.83}$$

These sums are no longer independent of each other. The symmetry/antisymmetry of the wave function generates correlations. There is a need to distinguish between bosons and fermions.

[6]The degeneracy of a particle state is the number of different quantum states that have the same energy. It is represented mathematically by the Hamiltonian for the system having more than one linearly independent eigenstate with the same energy eigenvalue.

1. **Bosons**.

   - Example 1. There are 5 bosons in the states $\{\epsilon_a, \epsilon_b, \epsilon_c, \epsilon_d, \epsilon_e\}$. Therefore there are $W = 5!$ ways to arrange the 5 particles.
   - Example 2. There are 5 bosons in the states $\{\epsilon_a, \epsilon_a, \epsilon_a, \epsilon_b, \epsilon_b\}$. Therefore there are $W = \frac{5!}{3!2!} = 10$ ways to arrange the 5 particles.
   - Example 3. There are 5 bosons in the states $\{\epsilon_a, \epsilon_a, \epsilon_a, \epsilon_a, \epsilon_b\}$. Therefore there are $W = \frac{5!}{4!1!} = 5$ ways to arrange the 5 particles.

   In general, if there are n_1 bosons in state ϵ_1, n_2 in state ϵ_2, n_3 in state ϵ_3, ..., then $W = \frac{N!}{n_1!n_2!n_3!\cdots}$, with $N = n_1 + n_2 + n_3 + \ldots$. Consequently, for bosons, we have:

$$Z_N = \sum_i \sum_j \sum_k (\cdots) \frac{n_1!n_2!n_3!\cdots}{N!} \mathrm{e}^{-\beta\epsilon_i^{(1)}} \mathrm{e}^{-\beta\epsilon_j^{(2)}} \mathrm{e}^{-\beta\epsilon_k^{(3)}} \cdots \tag{5.84}$$

   Due to the restriction $\sum_i n_i = N$, the above summation is indeed very complicated.

2. **Fermions**.
   As far as fermions are concerned, one can apply Pauli's Exclusion Principle. It follows that either are the states empty or they are occupied by a single particle:

$$n_i = \{0, 1\} \quad \rightarrow \quad n_i! = 1 \tag{5.85}$$

   and the partition function reads

$$Z_N = \frac{1}{N!} \underbrace{\sum_i \sum_j \sum_k \cdots}_{i \neq j \neq k \neq \cdots} \mathrm{e}^{-\beta\epsilon_i^{(1)}} \mathrm{e}^{-\beta\epsilon_j^{(2)}} \mathrm{e}^{-\beta\epsilon_k^{(3)}} \cdots \tag{5.86}$$

   This is also a rather complicated summation.

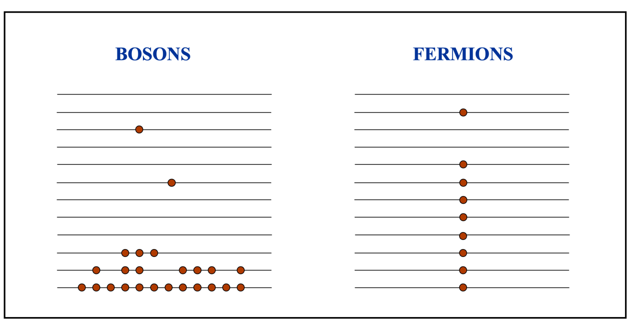


Figure 5.5: At low temperatures, the thermal energy (kT) is small, making it very difficult for a particle to get access to high energy levels. Instead, most particles remain in low-energy quantum states. Therefore, the number of accessible states is of the same order or smaller than the number of particles.

In general, the above summations for bosons and fermions are very complicated, especially so at low temperatures (see Fig. 5.5). Nevertheless, if temperature is large enough, the number of accessible states is significantly larger than the number of particles (see Fig. 5.6). This favourable condition implies that is very unlikely to find two particles in the same quantum state. Therefore, for bosons we can assume that $n_1! = n_2! \simeq 1$ and for fermions that the condition $i \neq j \neq k \neq \cdots$ does not apply. The resulting partition function for both bosons and fermions is then modified as follows

$$Z_N = \frac{1}{N!} \underbrace{\left(\sum_i e^{-\beta\epsilon_i^{(1)}}\right)\left(\sum_j e^{-\beta\epsilon_j^{(2)}}\right)\left(\sum_k e^{-\beta\epsilon_k^{(3)}}\right)\cdots}_{N \text{ summations with no restrictions}} = \frac{1}{N!}(Z_1)^N \tag{5.87}$$

where

$$Z_1 = \sum_i e^{-\beta\epsilon_i} = \sum_m g_m e^{-\beta\epsilon_m} \tag{5.88}$$

Thermodynamic systems whose partition function factorises according to these equations are known as **classical ideal systems**. Ideal systems consist of:

1. Distinguishable particles.
2. Indistinguishable particles at high temperatures (classical ideal gases).

These systems follow the so-called *Boltzmann Statistics*. A summary of their thermodynamic properties is provided in Fig. 5.7.

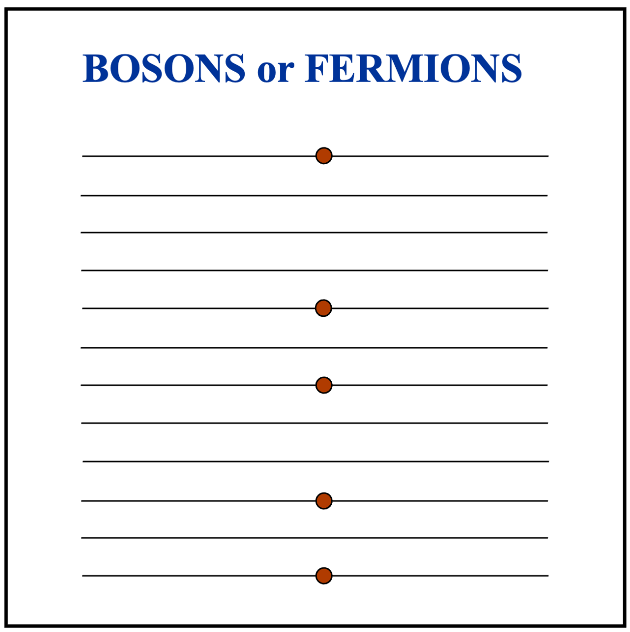


Figure 5.6: At high temperatures, the thermal energy (kT) is high, making it very easy for a particle to get access to high energy levels, distributing randomly across the entire energy spectrum. Therefore, the number of accessible states is much larger than the number of particles.

System	**Use $Z = Z_1^N$?**	**Use $Z = Z_1^N/N!$?**	**Correct Expression**
Different chemical species (e.g., O_2 and N_2 in air)	No	Yes	$Z = \frac{Z_1^{N_{O_2}}}{N_{O_2}!} \times \frac{Z_1^{N_{N_2}}}{N_{N_2}!}$
Simulation with labeled molecules	Yes	No	$Z = Z_1^N$
Particles fixed in specific positions (e.g., pinned atoms)	Yes	No	$Z = Z_1^N$
Identical gas molecules (e.g., all He atoms)	No	Yes	$Z = Z_1^N/N!$
Classical ideal gas (correct thermodynamics)	No	Yes	$Z = Z_1^N/N!$
Crystalline solids	Yes	No	$Z = Z_1^N$

Table 5.2: Partition function cases: When to use $Z = Z_1^N$ or $Z = Z_1^N/N!$

$$Z_N = \frac{1}{N!}(Z_1)^N \qquad A(T,V,N) = -kT\ln Z_N = -NkT\left[\ln\left(\frac{Z_1}{N}\right)+1\right]$$

$$U = NkT^2\left(\frac{\partial \ln Z_1}{\partial T}\right)_V$$

$$S = NkT\left(\frac{\partial \ln Z_1}{\partial T}\right)_V + kN\left[\ln\left(\frac{Z_1}{N}\right)+1\right]$$

$$P = NkT\left(\frac{\partial \ln Z_1}{\partial V}\right)_{T,N}$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = NkT^2\left(\frac{\partial^2 \ln Z_1}{\partial T^2}\right)_V + 2NkT\left(\frac{\partial \ln Z_1}{\partial T}\right)_V$$

Figure 5.7: Thermodynamic properties of a classical ideal gas, according to Boltzmann Statistics.

Justification that at large temperature the number of accessible states is much larger than the number of particles

Consider a cubic volume with side length L containing ideal particles with kinetic energy $\epsilon = \frac{p^2}{2m}$. Solving the Schrödinger equation for a single particle, one obtains that the energy of a single particle is quantised:

$$\epsilon_{n_x,n_y,n_z} = \frac{h^2}{8mL^2}\left(n_x^2 + n_y^2 + n_z^2\right) \tag{5.89}$$

where n_x, n_y, $n_z = 1, 2, 3, \ldots$ are quantum numbers. From this we can calculate the number of quantum states with energy lower or equal to ϵ_{n_x,n_y,n_z}:

$$n_x^2 + n_y^2 + n_z^2 = \frac{8mL^2}{h^2}\epsilon_{n_x,n_y,n_z} \tag{5.90}$$

which is a sphere of radius $R = \sqrt{\frac{8mL^2}{h^2}\epsilon_{n_x,n_y,n_z}}$. Calculating the number of states within this spheres is not trivial. However, it can be simplified if we assume that R is very large. In this case, R (or ϵ_{n_x,n_y,n_z}) can be taken as a continuous variable and approximate the number of states with the volume of an octant of a sphere of radius R:

$$\phi(\epsilon) = \frac{1}{8}\left(\frac{4}{3}\pi R^3\right) = \frac{\pi}{6}\left(\frac{8mL^2\epsilon}{h^2}\right)^{3/2} \tag{5.91}$$

If we now substitute reasonable values into the above equation, $m = 10^{-27}$ kg, $L = 0.1$ m, $T = 300$ K, and $\epsilon = 3kT/2$, then we get that $\phi(\epsilon) \sim 10^{27}$, which is indeed much larger than $N \sim N_A \sim 10^{23} - 10^{24}$.

5.7 Problems

5.1 Consider a system of N classical, noninteracting and distinguishable particles in contact with a thermal reservoir at temperature T. Each particle may have energies 0, $\epsilon > 0$, or 3ϵ. Obtain an expression for the canonical partition function, and calculate the internal energy per particle, $u = u(T)$. Sketch a graph of u versus T (indicate the values of u in the limits $T \to 0$ and $T \to \infty$). Calculate the entropy per particle, $s = s(T)$, and sketch a graph of s versus T.

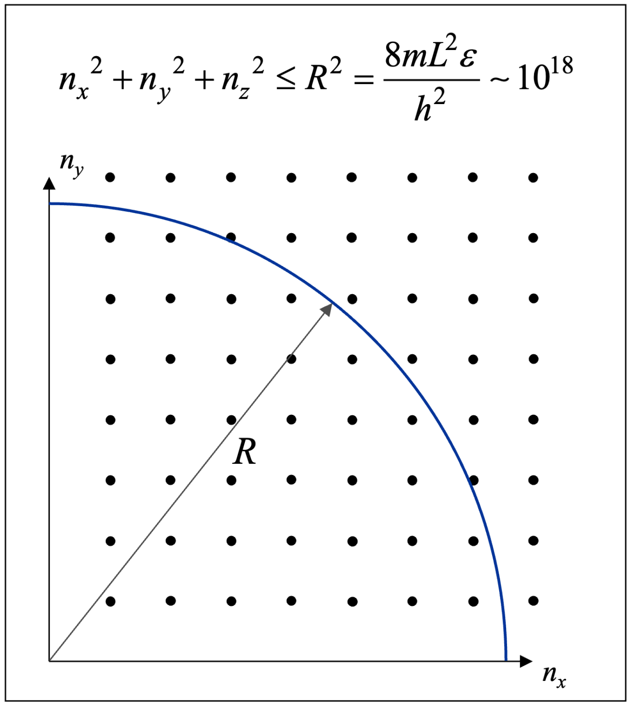


Figure 5.8: Accessible states an n-dimensional space.

5.2 Prove that the canonical partition function for an ideal gas takes the form

$$Z = \frac{1}{N!}\left(\frac{V}{\Lambda^3}\right)^N \tag{5.92}$$

where V is the volume occupied by the gas, N the number of gas molecules and $\Lambda \equiv h/\sqrt{2\pi kTm}$ the thermal wavelength.

5.3 **Exam paper - September 2008**. A classical gas at equilibrium contains N noninteracting molecules that move inside a container with cross-sectional area S and height $4L$ as shown in Fig. 5.9. A field

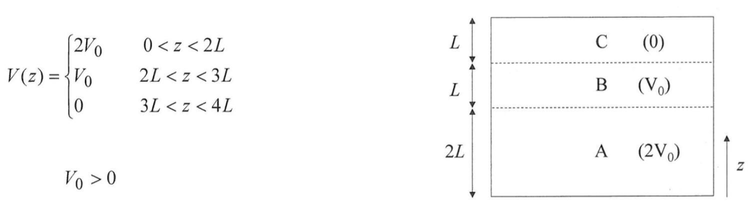


Figure 5.9: External field acting on the gas molecules (left) and geometry of the container (right).

$V = V(z)$ acts on the gas molecules as indicated in Fig. 5.9. Additionally, each molecule is characterised by internal degrees of freedom such that it can stay in two quantum energy states, 0 and ϵ, with degeneracy 1 and 3, respectively.

(a) Calculate the canonical partition function of this system.

(b) Calculate the probability that a randomly-selected molecule is in region A, B and C of the container. Analyse what value these probabilities take when the temperature is especially large and especially low.

(c) Determine the average number of molecules in region B whose energy state is ϵ.

5.4 **Exam paper - September 2009**.

A classical gas is constituted of N identical dipolar molecules that do not interact with each other, confined within a volume V at uniform temperature, and subjected to the influence of an external uniform electric field E. It is known that the Hamiltonian of a dipolar molecule is given by:

$$H_1 = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + \frac{1}{2I}\left(p_\theta^2 + \frac{p_\phi^2}{\sin^2\theta}\right) - \mu E \cos\theta \tag{5.93}$$

where m is the mass of the molecule, I is its moment of inertia, and μ is its electric dipole moment.

(a) Obtain the Helmholtz free energy and verify that it is an extensive quantity.

(b) Calculate the fraction of molecules that have an angular orientation θ within the interval $0 \leq \theta \leq \pi/4$.

5.5 A monoatomic ideal gas is contained in a centrifuge consisting of a cylinder with a radius R and length L. The cylinder rotates with an angular velocity ω around its axis of symmetry. The gas atoms have mass m and obey classical statistics.

a) Write the Hamiltonian in the reference frame rotating with the cylinder.

b) What is the partition function of the system?

c) What will be the mean number of particles per unit volume at a distance r from the axis of the cylinder?

Do not consider the effect of gravity and assume that the centrifuge has rotated long enough for the gas to reach equilibrium. (Hint: Describe the centrifugal force using a potential).

Chapter 6

Macrocanonical Ensemble

6.1 Introduction

In Chapter 4, we examined the microcanonical ensemble, focusing on isolated systems at equilibrium. In Chapter 5, we introduced the canonical ensemble, where systems in equilibrium exchange energy with their surroundings. This ensemble allows us to comprehend the behaviour of systems with a fixed energy content interacting with a thermal reservoir. Now, a key question arises: can we design an ensemble that captures the essence of macroscopic systems, ones that not only exchange energy with the external environment, but also mass? This scenario forms the basis of the macrocanonical (or grand canonical) ensemble, an ensemble tailor-made for systems achieving both energy and material equilibrium with the exterior. To this end, let's first resume the main conclusions on microcanonical and canonical ensembles.

- The **microcanonical ensemble** is mathematically challenging to employ. Moreover, the concept of isolated physical systems is more of an idealisation than a reality. In practical terms, the study of systems interacting with their environment often holds greater significance. Experimentally, there is no instrument available to directly measure or control a system's energy. Instead, it is typically inferred from other observables such as pressure, temperature, among others. By contrast, fixing the temperature proves to be more straightforward—it is directly observable through a thermometer and easily manageable with a thermal bath.

- The **canonical ensemble** proves to be a significantly more versatile ensemble, offering immense utility in addressing a multitude of physical problems within closed systems. However, the canonical ensemble exhibits significant limitations when attempting to address various other physical and chemical problems: (*1*) systems in equilibrium that exchange particles with the surroundings (open systems); (*2*) systems in equilibrium where chemical reactions take place; and (*3*) systems separated into phases in mutual coexistence (liquid-vapour, solid-vapour, etc.). These are all systems where the number of particles of each species is not constant.

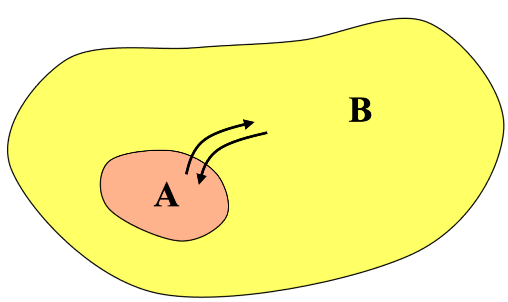


Figure 6.1: Interacting systems, A and B, exchanging energy and mass.

Let's consider an isolated system at equilibrium. This system is formed by two subsystems, A and B, that can exchange energy and mass, as shown in Fig. 6.1. If the chemical potential of A, μ_A, equals the chemical potential of B, μ_B, then there is no net mass exchange. However, if this is not the case, then there will be a mass transfer from A to B if $\mu_A > \mu_B$ or from B to A if $\mu_A < \mu_B$. We should note that this mass transfer is internal to the A+B system. In other words,

$$N_A + N_B = \text{const} \quad \rightarrow dN_A + dN_B = 0 \tag{6.1}$$

Additionally, the total entropy change of the system reads

$$dS = \left(\frac{\partial S}{\partial N_A}\right)_{N_B,U,V} dN_A + \left(\frac{\partial S}{\partial N_B}\right)_{N_A,U,V} dN_B = dN_A \left[\left(\frac{\partial S}{\partial N_A}\right) - \left(\frac{\partial S}{\partial N_B}\right)\right] \tag{6.2}$$

where the infinitesimal entropy change with mass is proportional to the chemical potential:

$$\mu = -T\left(\frac{\partial S}{\partial N}\right)_{U,V} \tag{6.3}$$

Therefore, substituting Eq. 6.3 into Eq. 6.2, we obtain

$$dS = -\frac{dN_A}{T}(\mu_A - \mu_B) \tag{6.4}$$

Note that, if $\Delta\mu \equiv \mu_A - \mu_B > 0$, then $dN_A < 0$; by contrast, if $\Delta\mu < 0$, then $dN_A > 0$. In both cases, $dS > 0$ and $dS = 0$ if no mass transfer occurs, that is if $\Delta\mu = 0$.

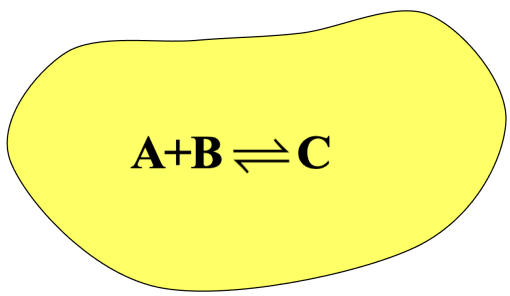


Figure 6.2: System where a chemical reaction takes place and a product forms.

Now, let's consider a system where the generic chemical reaction A + B = C occurs (see Fig. 6.2). At equilibrium, the chemical potentials of the three species are related as follows

$$\mu_A + \mu_B = \mu_C \tag{6.5}$$

Out of equilibrium, then products or reagents can form:

$$\mu_A + \mu_B > \mu_C \quad \Longrightarrow \quad \text{A + B} \rightarrow \text{C} \tag{6.6}$$

$$\mu_A + \mu_B < \mu_C \quad \Longrightarrow \quad \text{C} \rightarrow \text{A + B} \tag{6.7}$$

In Fig. 6.3, we schematically show a representative member S of the macrocanonical ensemble that exchange energy and mass with its surrounding reservoir W. The net mass exchange is zero at the thermodynamic equilibrium, when $T_s = T_w$ and $\mu_s = \mu_w$. In particular, S is confined within rigid, permeable and diathermic walls that allow mass and energy transfer, but leave temperature (T), volume (V) and chemical potential (μ) constant. To stress these conditions, the macrocanonical ensemble that we wish to design is also denoted as (μ, V, T) ensemble and the corresponding partition function as $\Xi = \Xi(\mu, V, T)$. Additionally, we will introduce the **macrocanonical potential**, also known as the grand canonical potential or the Helmholtz free energy in the macrocanonical ensemble, $J(\mu, V, T) = -kT \ln \Xi$, defined as:

$$J = U - TS - \mu N = A + \mu N \tag{6.8}$$

where U is the internal energy, S the entropy, N the number of particles and A the Helmholtz free energy. The minimisation of the macrocanonical potential at equilibrium conditions provides the most probable distribution of energy and particles for a given temperature, chemical potential, and volume.

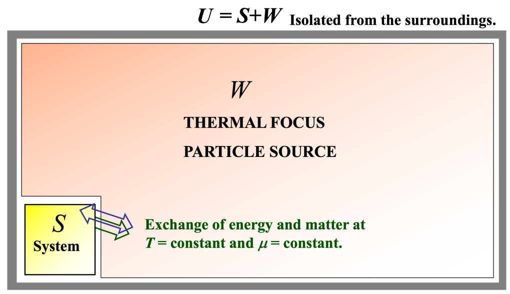


Figure 6.3: The members of the macrocanonical ensemble are macroscopic systems, replicas of the original, in equilibrium, and thermally and materially coupled to each other.

6.2 Design of the grand canonical ensemble

With reference to Fig. 6.4, the system S is very large, but much smaller than W, that is $S \ll W$, and

$$1 \ll N_s \ll N_w \tag{6.9}$$

Additionally, the numerical density is basically constant across all U:

$$\frac{N_s}{V_s} \simeq \frac{N_w}{V_w} \tag{6.10}$$

If we disregard the interaction energy between S and W, the energy of the "supersystem" U reads

$$E_u \simeq E_s + E_w, \quad \text{with } E_s \ll E_w \tag{6.11}$$

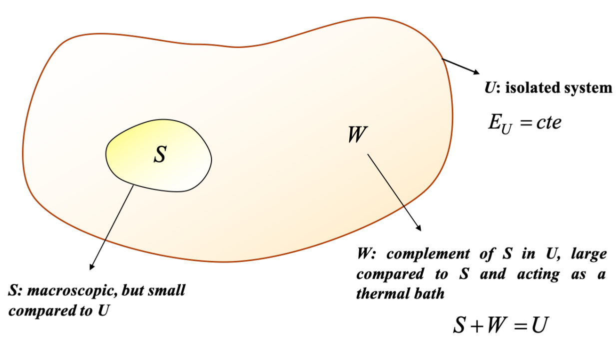


Figure 6.4: Sketch of a macroscopic system S exchanging mass and energy with its surroundings W.

Given that the Hamiltonian of the super-system U is separated into two parts $(S + W)$, the Schrödinger equation also separates:

$$\hat{H}_s|\phi_s\rangle = E_s|\phi_s\rangle \tag{6.12}$$

$$\hat{H}_w|\phi_w\rangle = E_w|\phi_w\rangle \tag{6.13}$$

where $E_s(N)$ is the energy of S, E_w is the energy of W and $|\phi_u\rangle = |\phi_s\rangle|\phi_w\rangle$. Given that S has quantum numbers that are independent on W, it makes sense to estimate the probability, P_{mN}, that S contains N particles and is in the energy state E_{mN}. Since W is much larger than S, we can also consider W as an isolated system with energy $E_w = E_u - E_{mN}$ and $N_w = N_u - N$ particles. It follows that the number of accessible states in U reads

$$\Omega_u = \Omega_u(E_u, N_u; \Delta E) \tag{6.14}$$

and the number of accessible states in U compatible with S containing N particles and having energy E_{mN}, reads

$$\Omega_w = \Omega_w(E_u - E_{mN}, N_u - N; \Delta E) \tag{6.15}$$

We can then calculate the aforementioned probability in terms of the energy and number of particles of the reservoir W:

$$P_{mN} = \frac{\text{favourable cases}}{\text{possible cases}} = \frac{\Omega_w(E_u - E_{mN}, N_u - N; \Delta E)}{\Omega_u(E_u, N_u; \Delta E)} \tag{6.16}$$

Given that $E_{mN} \ll E_u$, we can develop a power series of Ω_w. Nevertheless, since Ω_w increases exponentially with energy and converges slowly, we prefer to Taylor-expand the logarithm of Ω_w near the point (E_u, N_u):

$$\ln \Omega_w(E_w, N_w; \Delta E) \simeq \ln \Omega_w(E_u, N_u; \Delta E) + \left(\frac{\partial \ln \Omega_w}{\partial E_w}\right)_{E_w = E_u} (E_w - E_u) + \left(\frac{\partial \ln \Omega_w}{\partial N_w}\right)_{N_w = N_u} (N_w - N_u) \tag{6.17}$$

or equivalently

$$\ln \Omega_w(E_w, N_w; \Delta E) \simeq \ln \Omega_w(E_u, N_u; \Delta E) - \underbrace{\frac{\partial \ln \Omega_w}{\partial E_w}}_{\beta = 1/kT} E_{mN} - \underbrace{\frac{\partial \ln \Omega_w}{\partial N_w}}_{-\beta\mu} N \tag{6.18}$$

where we used the fact that the thermal bath W can be approximately treated as an isolated system and therefore within the microcanonical-ensemble framework. It follows that $S_w = k \ln \Omega_w$ and, from Eq. 6.3, we obtain the chemical potential in the above equation. Therefore,

$$\Omega_w(E_u - E_{mN}, N_u - N; \Delta E) \simeq \Omega_w(E_u, N_u; \Delta E) \mathrm{e}^{-\beta(E_{mN} - \mu N)} \tag{6.19}$$

and the probability reads

$$P_{mN} = \underbrace{\frac{\Omega_w(E_u, N_u; \Delta E)}{\Omega_u(E_u, N_u; \Delta E)}}_{\leq 1} \mathrm{e}^{-\beta(E_{mN} - \mu N)} \tag{6.20}$$

To calculate P_{mN}, we can impose the normalisation condition:

$$\sum_{N=0}^{\infty} \sum_m P_{mN} = 1 \tag{6.21}$$

which gives

$$\sum_{N=0}^{\infty} \sum_m \underbrace{\frac{\Omega_w(E_u, N_u; \Delta E)}{\Omega_u(E_u, N_u; \Delta E)}}_{\equiv 1/\Xi} \mathrm{e}^{-\beta(E_{mN} - \mu N)} = 1 \tag{6.22}$$

where the sum is over all possible states and number of particles of system S, without restriction. Rearranging, one obtains

$$\Xi \equiv \frac{\Omega_u(E_u, N_u; \Delta E)}{\Omega_w(E_u, N_u; \Delta E)} = \sum_{N=0}^{\infty} \sum_m \mathrm{e}^{-\beta(E_{mN} - \mu N)} \tag{6.23}$$

Substituting this result into Eq. 6.20, we obtain

$$P_{mN} = \frac{1}{\Xi} \mathrm{e}^{-\beta(E_{mN} - \mu N)} \tag{6.24}$$

where $\Xi = \Xi(\mu, T, V)$ is the **macrocanonical partition function** and P_{mN} is the **macrocanonical probability**. Note that both Ξ and P_{mN} do not depend on W.

In terms of operators:

$$\hat{\rho} = \frac{1}{\Xi} \mathrm{e}^{-\beta(\hat{H} - \mu \hat{N})} \tag{6.25}$$

$$\Xi = \mathrm{Tr}\left[\mathrm{e}^{-\beta(\hat{H} - \mu \hat{N})}\right] \tag{6.26}$$

and the average value of a generic observable reads

$$\langle b \rangle = \mathrm{Tr}\left[\hat{b}\hat{\rho}\right] = \frac{1}{\Xi} \mathrm{Tr}\left[\hat{b}\mathrm{e}^{-\beta(\hat{H} - \mu \hat{N})}\right] \tag{6.27}$$

6.3 Connection with thermodynamics

In this section, our goal is understanding what the physical meaning of μ is and how Ξ, and more generally the macrocanonical ensemble, is related to Thermodynamics. In the (μ, V, T) ensemble, both E and N are not constant as the system exchanges energy and mass with the surroundings. Nevertheless, from a Thermodynamics perspective, we can still define their average values:

$$U = \langle E \rangle = \sum_{N=0}^{\infty} \sum_m E_{mN} P_{mN} = \frac{1}{\Xi} \sum_{N=0}^{\infty} \sum_m E_{mN} \mathrm{e}^{-\beta(E_{mN} - \mu N)} \tag{6.28}$$

where E_{mN} is also a function of the volume V. Let's define the **fugacity** as $\lambda \equiv \mathrm{e}^{\beta\mu}$. Then Eq. 6.28 is modified as

$$U = \frac{1}{\Xi} \sum_{N=0}^{\infty} \sum_m E_{mN} \mathrm{e}^{-\beta E_{mN}} \lambda^N = -\frac{1}{\Xi} \sum_{N=0}^{\infty} \lambda^N \sum_m \left.\frac{\partial \mathrm{e}^{-\beta E_{mN}}}{\partial \beta}\right|_V = -\frac{1}{\Xi} \frac{\partial}{\partial \beta} \left[\sum_{N=0}^{\infty} \lambda^N \sum_m \mathrm{e}^{-\beta E_{mN}}\right]_{\lambda, V} \tag{6.29}$$

which can simply be written as

$$U = -\frac{1}{\Xi} \left(\frac{\partial \Xi}{\partial \beta}\right)_{\lambda, V} = -\left(\frac{\partial \ln \Xi}{\partial \beta}\right)_{\lambda, V} \tag{6.30}$$

We now turn our attention to the average number of particles in S, which can be estimated by using the probability P_{mN} as have done above with the internal energy:

$$\langle N \rangle = \sum_{N=0}^{\infty} \sum_m N P_{mN} = \frac{1}{\Xi} \sum_{mN} N \mathrm{e}^{-\beta E_{mN}} \mathrm{e}^{\beta \mu N} = \frac{1}{\Xi} \frac{1}{\beta} \frac{\partial}{\partial \mu} \left[\sum_{mN} \mathrm{e}^{-\beta E_{mN}} \mathrm{e}^{\beta \mu N}\right]_{\beta, V} = \frac{1}{\Xi} \frac{1}{\beta} \left.\frac{\partial \Xi}{\partial \mu}\right|_{\beta, V} \tag{6.31}$$

which can be more concisely written

$$\langle N \rangle = kT \left.\frac{\partial \ln \Xi}{\partial \mu}\right|_{T, V} \tag{6.32}$$

or, equivalently, in terms of fugacity $\lambda = e^{\beta\mu}$:

$$\langle N \rangle = kT \left.\frac{\partial \ln \Xi}{\partial \lambda}\right|_{T,V} \left.\frac{\partial \lambda}{\partial \mu}\right|_{T} = \lambda \left.\frac{\partial \ln \Xi}{\partial \lambda}\right|_{T,V} \tag{6.33}$$

Let's now consider Fig. 6.5, where we have divided system S in two subsystems, S_1 and S_2, that can exchange energy and mass with W and with each other. The probability that S_1 contains N_1 particles and is in the energy state E_{1m} and that S_2 contains N_2 particles and is in the energy state E_{2n} is given by

$$P_{(N_1,E_{1m};N_2,E_{2n})} = P_{(N_1,E_{1m})} P_{(N_2,E_{2n})} = \frac{1}{\Xi_1} e^{-\beta_1(E_{1m}-\mu_1 N_1)} \frac{1}{\Xi_2} e^{-\beta_2(E_{2n}-\mu_2 N_2)} \tag{6.34}$$

Since $S = S_1 + S_2$, the same probability can be written as

$$P_{(N_1,E_{1m};N_2,E_{2n})} = \frac{1}{\Xi} e^{-\beta(E_{1m}+E_{2n})} e^{\beta\mu(N_1+N_2)} \tag{6.35}$$

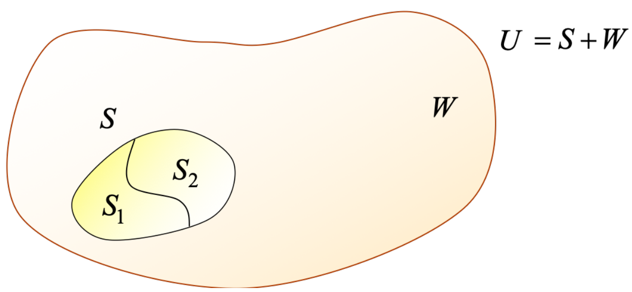


Figure 6.5: System S is divided in two subsystems, S_1 and S_2 that can exchange energy and mass with W and with each other.

It then follows that

$$\beta_1 = \beta_2 = \beta, \quad \mu_1 = \mu_2 = \mu \tag{6.36}$$

This suggests that μ does not depend on the system size or, in other words, that it is an **intensive** property. It also follows that

$$\Xi = \Xi_1 \Xi_2 \tag{6.37}$$

or

$$\ln \Xi = \ln \Xi_1 + \ln \Xi_2. \tag{6.38}$$

This result indicates that the property $\ln \Xi$ and hence also $J \equiv -kT \ln \Xi$, which we call macrocanonical potential, are **extensive** variables. In particular, the macrocanonical potential depends on temperature, chemical potential, and energy state of the system. Mathematically:

$$J = J(\beta, \mu, \{E_{mN}\}) \tag{6.39}$$

Its total differential can therefore be written as

$$d(\beta J) = \left.\frac{\partial(\beta J)}{\partial \beta}\right|_{\mu,\{E_{mN}\}} d\beta + \left.\frac{\partial(\beta J)}{\partial \mu}\right|_{\beta,\{E_{mN}\}} d\mu + \sum_{N=0}^{\infty} \sum_{m} \left.\frac{\partial(\beta J)}{\partial E_{mN}}\right|_{\beta,\mu} dE_{mN} \tag{6.40}$$

Let's calculate these three partial derivatives separately.

I

$$
\begin{aligned}
\left.\frac{\partial(\beta J)}{\partial\beta}\right|_{\mu,\{E_{mN}\}} &= -\frac{\partial \ln \Xi}{\partial \beta} = -\frac{1}{\Xi}\left.\frac{\partial \Xi}{\partial \beta}\right|_{\mu,\{E_{mN}\}} = \\
&= -\frac{1}{\Xi}\sum_{N=0}^{\infty}\sum_{m}(-E_{mN}+\mu N)\mathrm{e}^{-\beta(E_{mN}-\mu N)} = \\
&= \frac{1}{\Xi}\sum_{N=0}^{\infty}\sum_{m}E_{mN}\mathrm{e}^{-\beta E_{mN}+\beta\mu N} - \mu\frac{1}{\Xi}\sum_{N=0}^{\infty}\sum_{m}N\mathrm{e}^{-\beta E_{mN}+\beta\mu N} = \\
&= \langle E\rangle - \mu\langle N\rangle = \\
&= U - \mu\langle N\rangle
\end{aligned} \tag{6.41}
$$

II

$$
\left.\frac{\partial(\beta J)}{\partial\mu}\right|_{\beta,\{E_{mN}\}} = -\left.\frac{\partial \ln \Xi}{\partial \mu}\right|_{T,V} = -\beta\langle N\rangle \tag{6.42}
$$

III

$$
\left.\frac{\partial(\beta J)}{\partial E_{mN}}\right|_{\beta,\mu} = -\frac{1}{\Xi}\frac{\partial \Xi}{\partial E_{mN}} = -\frac{1}{\Xi}(-\beta)\mathrm{e}^{-\beta E_{mN}+\beta\mu N} = \beta P_{mN} \tag{6.43}
$$

Therefore Eq. 6.40 is re-written as

$$
d(\beta J) = (U-\mu\langle N\rangle)d\beta - \beta\langle N\rangle d\mu + \beta\sum_{Nm}P_{mN}dE_{mN} \tag{6.44}
$$

where $\sum_{Nm} \equiv \sum_{N=0}^{\infty}\sum_{m}$. We now note that:

$$
\begin{aligned}
Ud\beta &= d(U\beta) - \beta dU \\
-\mu\langle N\rangle d\beta - \beta\langle N\rangle d\mu &= -d(\beta\mu\langle N\rangle) + \beta\mu d\langle N\rangle
\end{aligned} \tag{6.45}
$$

It follows that

$$
d(\beta J) = d(U\beta) - \beta dU - d(\beta\mu\langle N\rangle) + \beta\mu d\langle N\rangle + \beta\sum_{Nm}P_{nm}dE_{mN} \tag{6.46}
$$

and rearranging, we obtain

$$
d[\beta(U-\mu\langle N\rangle - J)] = \beta dU - \beta\mu d\langle N\rangle - \beta\sum_{Nm}P_{nm}dE_{mN} \tag{6.47}
$$

Now, what is the physical meaning of the last term in the r.h.s. of the above equation? To answer this question, we first notice that we are assuming that $E_{mN} = E_{mN}(V)$ and, if the volume changes, then the energy will change accordingly. In other words, dE_{mN} is a change in the energy levels due to a volume change dV. If we impose a volume change, keeping constant all the other variables, the internal energy will also change:

$$
U = \sum_{Nm}P_{Nm}E_{Nm} \longrightarrow U' = \sum_{Nm}P_{Nm}E'_{Nm} \tag{6.48}
$$

If we now apply the First Law of Thermodynamics[1], we observe that

$$
dU = U' - U = \sum_{Nm}P_{Nm}dE_{Nm} = \sum_{Nm}P_{Nm}\left(\frac{\partial E_{Nm}}{\partial V}\right)dV = \text{đ}W = -pdV \tag{6.49}
$$

[1]The First Law of Thermodynamics, in general, is written as $dU = \text{đ}Q + \text{đ}W + \mu dN$. However, in this case, we are only changing volume, while energy and mass remain constant. It follows that the internal energy change is only due to work exchanged between the system and its surroundings.

where p stands for pressure. It becomes clear that

$$-\sum_{Nm} P_{Nm}\left(\frac{\partial E_{Nm}}{\partial V}\right) = p, \quad \text{pressure} \tag{6.50}$$

and

$$\sum_{Nm} P_{Nm} dE_{Nm} = đW, \quad \text{mechanical work.} \tag{6.51}$$

If we incorporate this result into Eq. 6.52, then we obtain

$$d[\beta(U - \mu\langle N\rangle - J)] = \beta[dU - \mu d\langle N\rangle - đW] = \beta[dU + pdV - \mu d\langle N\rangle] \tag{6.52}$$

Equivalently

$$d\left[\frac{1}{T}(U - \mu\langle N\rangle - J)\right] = \frac{1}{T}[dU + pdV - \mu d\langle N\rangle] \tag{6.53}$$

Let's now apply the Second Law of Thermodynamics and obtain the Fundamental Equation of Thermodynamics, which combines the first and the second laws:

$$đQ = TdS = dU + pdV - \mu dN \tag{6.54}$$

Rearranging, we obtain the infinitesimal change of entropy:

$$dS = \frac{1}{T}[dU + pdV - \mu dN] \tag{6.55}$$

or the infinitesimal change of internal energy:

$$dU = TdS - pdV + \mu dN \tag{6.56}$$

Comparing Eq. 6.55 with Eq. 6.53, one observes that μ is indeed the chemical potential and that the argument of the derivative in the l.h.s. of Eq. 6.53 results to be

$$U - \mu\langle N\rangle - J = TS \tag{6.57}$$

or, rearranging, that

$$J = U - TS - \mu\langle N\rangle \tag{6.58}$$

which justifies the definition of macrocanonical potential for $J(\mu, T, V) = -kT\ln\Xi$. Additionally, from Eq. 6.56, we see that

$$U = TS - pV + \mu N \tag{6.59}$$

and finally

$$J(\mu, T, V) = U - TS - \mu N = -pV = -p(T, \mu)V \tag{6.60}$$

This last equation, combined with $J = -kT\ln\Xi$, is crucial to derive all the equations of Thermodynamics. More specifically:

$$S = -\left.\frac{\partial J}{\partial T}\right|_{\mu,V} = kT\left.\frac{\partial \ln\Xi}{\partial T}\right|_{\mu,V} + k\ln\Xi, \quad \text{Entropy} \tag{6.61}$$

$$p = -\left.\frac{\partial J}{\partial V}\right|_{\mu,T} = kT\left.\frac{\partial \ln\Xi}{\partial V}\right|_{\mu,T} = \frac{kT}{V}\ln\Xi, \quad \text{Pressure} \tag{6.62}$$

$$\langle N\rangle = -\left.\frac{\partial J}{\partial \mu}\right|_{T,V} = kT\left.\frac{\partial \ln\Xi}{\partial \mu}\right|_{T,V}, \quad \text{Average number of particles} \tag{6.63}$$

$$U = -\left.\frac{\partial \ln \Xi}{\partial \beta}\right|_{\lambda,V}, \quad \text{Internal energy} \tag{6.64}$$

There is an alternative expression for entropy that relates it to the probability P_{Nm}:

$$\begin{aligned}
\frac{S}{k} &= \frac{1}{kT}(U - \mu\langle N\rangle - J) = \beta(U - \mu\langle N\rangle - J) = \\
&= \beta\left(\sum_{Nm} E_{Nm}P_{Nm}\right) - \mu\beta\sum_{Nm} NP_{Nm} + \ln\Xi = \\
&= \sum_{Nm}(\beta E_{Nm} - \mu\beta N + \ln\Xi)P_{Nm} = \\
&= \sum_{Nm}\left[-\ln\left(\frac{1}{\Xi}\mathrm{e}^{-\beta E_{Nm}+\beta\mu N}\right)\right]P_{Nm} \\
&= -\sum_{Nm} P_{Nm}\ln P_{Nm}
\end{aligned} \tag{6.65}$$

Therefore

$$S = -k\sum_{Nm} P_{Nm}\ln P_{Nm} \tag{6.66}$$

As a last step, we can connect Ξ with Z, the canonical partition function and with Ω, the microcanonical partition function. Let's first start with Z:

$$\Xi = \sum_{Nm}\mathrm{e}^{-\beta E_{Nm}}\underbrace{\mathrm{e}^{\beta\mu N}}_{\lambda^N} = \sum_{N=0}^{\infty}\lambda^N\underbrace{\sum_m \mathrm{e}^{-\beta E_{Nm}}}_{Z_N\equiv Z(N,T,V)} = \sum_{N=0}^{\infty}\lambda^N Z_N = \underbrace{Z_0}_{1} + \lambda Z_1 + \lambda^2 Z_2 + \ldots \tag{6.67}$$

This series expansion is especially useful when one studies real gases as it allows to decompose a multi-body problem into a 1-body, 2-body, … problem. This result can be applied to systems of ideal particles:

- Ideal system of distinguishable particles

$$\Xi = \sum_{N=0}^{\infty}\lambda^N Z_N = \sum_{N=0}^{\infty}(\lambda Z_1)^N = \frac{1}{1-\lambda Z_1}, \quad \text{with}\,\lambda Z_1 < 1 \tag{6.68}$$

- Ideal system of classical indistinguishable particles

$$\Xi = \sum_{N=0}^{\infty}\lambda^N Z_N = \sum_{N=0}^{\infty}\lambda^N\frac{1}{N!}Z_1^N = \sum_{N=0}^{\infty}\frac{1}{N!}(\lambda Z_1)^N = \mathrm{e}^{\lambda Z_1} \tag{6.69}$$

Now, let's see how Ξ connects with Ω:

$$\Xi(T,V,\lambda) = \sum_{N=0}^{\infty}\lambda^N\sum_n\underbrace{\Omega(E_{Nm},V,N)}_{\substack{\text{Degeneracy of the}\\ \text{energy level } E_{Nn}}}\mathrm{e}^{-\beta E_{Nn}} \tag{6.70}$$

6.4 Grand canonical partition function: quantum and classical formalism

6.4.1 Quantum formalism

- Density matrix and operator

We make use of a representation where operators $\hat{H}$ and $\hat{N}$ are diagonal, that is $\hat{H}|\Psi\rangle = E_{mN}|\Psi\rangle$ and $\hat{N}|\Psi\rangle = N|\Psi\rangle$.

$$P_{mN} = \frac{1}{\Xi}\mathrm{e}^{-\beta E_{mN}+\beta\mu N} \tag{6.71}$$

$$\rho_{(mN)(sN')} = P_{mN}\delta_{ms}\delta_{NN'} \tag{6.72}$$

$$\hat{\rho} = \frac{1}{\Xi}\mathrm{e}^{-\beta\hat{H}+\beta\mu\hat{N}} \tag{6.73}$$

- Partition function

$$\Xi = \Xi(\mu, T, V) = \sum_{N=0}^{\infty}\mathrm{e}^{\beta\mu N}\sum_{m}\mathrm{e}^{-\beta E_m} = \mathrm{Tr}\left[\mathrm{e}^{-\beta\hat{H}+\beta\mu\hat{N}}\right] \tag{6.74}$$

- Average value of a given observable b:

$$\langle b\rangle = \mathrm{Tr}[\hat{b}\hat{\rho}] = \frac{1}{\Xi}\mathrm{Tr}\left[\hat{b}\mathrm{e}^{-\beta\hat{H}+\beta\mu\hat{N}}\right] \tag{6.75}$$

6.4.2 Classical formalism

$$\rho_N(\mathbf{q},\mathbf{p}) = \frac{1}{\Xi}\frac{1}{N!h^{3N}}\mathrm{e}^{-\beta H(\mathbf{q},\mathbf{p})+\beta\mu N} \tag{6.76}$$

$$\Xi = \sum_{N=0}^{\infty}\mathrm{e}^{\beta\mu N}\frac{1}{N!h^{3N}}\int\int d\mathbf{q}\,d\mathbf{p}\,\mathrm{e}^{-\beta H(\mathbf{q},\mathbf{p})} \tag{6.77}$$

where $(\mathbf{q},\mathbf{p}) \equiv (q_1,\ldots,q_{3N},p_1,\ldots p_{3N})$.

6.5 Application example: solid-vapour coexistence

Consider a monocomponent system separated into two phases, solid and vapour, in equilibrium, as schematically represented in Fig. 6.6. The entire system is enclosed in a volume V at temperature T. Both phases can exchange particles, so $N \neq$ constant. At equilibrium, chemical potentials (or, equivalently, fugacities), pressure and temperature must be equal:

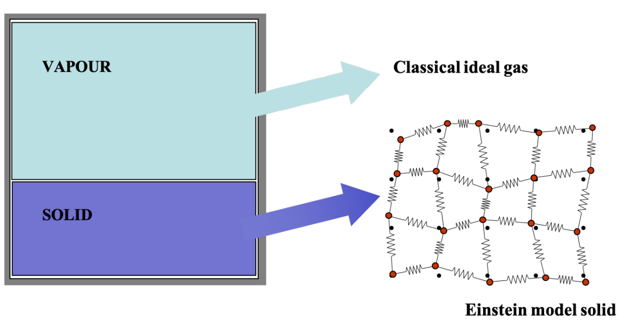


Figure 6.6: Two-phase system at equilibrium and how it can be modelled.

$$\mu_v = \mu_s, \quad \lambda_v = \lambda_s, \quad T_v = T_s, \quad p_v = p_s \tag{6.78}$$

Our interest is obtaining the vapour pressure p_v of the vapour phase as a function of the temperature T only. To this end, we assume that the system is made of a monoatomic species and apply the classical ideal-gas model. Because there is a thermodynamic equilibrium between the two phases, mass and energy can fluctuate around their equilibrium value: there is not a net mass or energy transfer, but particles can diffuse from one phase to the other. It therefore makes sense to use the macrocanonical ensemble to study this coexistence.

Let's consider the macrocanonical partition function for a classical ideal-gas system, whose particles are indistinguishable:

$$\Xi = \sum_{N=0}^{\infty} \lambda^N Z(N, V, T) = \mathrm{e}^{\lambda Z_1} \tag{6.79}$$

where Z_1 is the canonical partition function of a single gas molecule:

$$Z_1 = \frac{1}{h^3} \int d\vec{r}_1 \int d\vec{p}_1 \mathrm{e}^{-\beta H_1} = \frac{1}{h^3} \underbrace{\int dr_x\, dr_y\, dr_z}_{V} \int dp_x\, dp_y\, dp_z \mathrm{e}^{-\beta H_1} \tag{6.80}$$

and $H_1 = \frac{p_x^2+p_y^2+p_z^2}{2m}$. Substituting the Hamiltonian in the above equation, we get

$$Z_1 = \frac{1}{h^3} V \left(\int dp\, \mathrm{e}^{-\beta p^2/2m} \right)^3 = \frac{(2\pi m k T)^{3/2}}{h^3} V \tag{6.81}$$

Therefore

$$\Xi = \sum_{N=0}^{\infty} \lambda^N \frac{1}{N!} V^N \underbrace{\left(\frac{2\pi m k T}{h^2} \right)^{3N/2}}_{f(T)^N} \tag{6.82}$$

where we have defined a function of the temperature only, that is $f(T) = (2\pi m k T/h^2)^{3/2}$. In particular:

$$\Xi = \sum_{N=0}^{\infty} \frac{[\lambda V f(T)]^N}{N!} = \mathrm{e}^{\lambda V f(T)} \tag{6.83}$$

We notice that if the vapour phase was molecular, rather than monoatomic, $f(T)$ would be different, but the rest would be exactly the same. Let's now calculate the macrocanonical potential:

$$J = -kT \ln \Xi = -kT \lambda V f(T) \tag{6.84}$$

From this equation, we can obtain all the thermodynamic properties of interest:

$$pV = -J = kT\lambda V f(T) \Longleftrightarrow p = kT\lambda f(T) \tag{6.85}$$

$$\langle N \rangle = - \left. \frac{\partial J}{\partial \mu} \right|_{T,V} = - \left. \frac{\partial J}{\partial \lambda} \right|_{T,V} \left. \frac{\partial \lambda}{\partial \mu} \right|_{T,V} = \lambda V f(T) \tag{6.86}$$

from which we also get

$$\lambda = \frac{\langle N \rangle}{V} \frac{1}{f(T)} \tag{6.87}$$

This expression for fugacity can be substituted in Eq. 6.85 to obtain the ideal-gas law:

$$pV = \langle N \rangle kT \tag{6.88}$$

and to obtain the well-known expression for the internal energy of an ideal gas:

$$U = -\left.\frac{\partial \ln \Xi}{\partial \beta}\right|_{\lambda,V} = -\left.\frac{\partial \ln \Xi}{\partial T}\right|_{\lambda,V} \frac{\partial T}{\partial \beta} = kT^2 \lambda V f'(T) = kT^2 \frac{\langle N\rangle}{V} \frac{f'(T)}{f(T)} = \frac{3}{2}\langle N\rangle kT \tag{6.89}$$

We stress that this expression assumes a monatomic ideal gas. For diatomic and polyatomic gases, additional terms related to rotational and vibrational degrees of freedom may be included in the internal energy expression.

We now turn or attention to the solid phase, which can be treated quantum-mechanically. To this end, let's consider the Einstein model for solids. If the solid consists of N atoms, we place $3N$ independent and localised harmonic oscillators, each with a frequency ω. The resulting energy of the system reads

$$E = -E_0 + \sum_{i=1}^{3N} \hbar\omega \left(n_i + \frac{1}{2}\right), \quad \text{with}\, n_i = 0, 1, \ldots, \infty \tag{6.90}$$

where $E_0 = N\epsilon_0$ is the binding energy. Because these atoms are assumed distinguishable, then

$$\Xi = \sum_{N=0}^{\infty} \lambda^N Z_N = \sum_{N=0}^{\infty} (\lambda Z_1)^N \tag{6.91}$$

where Z_1, the canonical partition function of a 1-body system, reads

$$\begin{aligned} Z_1 &= \mathrm{e}^{\beta\epsilon_0} \sum_{n_x,n_y,n_z=0}^{\infty} \mathrm{e}^{-\beta\hbar\omega(n_x+1/2)} \mathrm{e}^{-\beta\hbar\omega(n_y+1/2)} \mathrm{e}^{-\beta\hbar\omega(n_z+1/2)} = \\ &= \mathrm{e}^{\beta\epsilon_0} \left(\sum_{n=0}^{\infty} \mathrm{e}^{-\beta\hbar\omega(n+1/2)}\right)^3 = \\ &= \mathrm{e}^{\beta\epsilon_0} \left[\frac{1}{2\sinh\left(\frac{\beta\hbar\omega}{2}\right)}\right]^3 \equiv g(T) \end{aligned} \tag{6.92}$$

We can substitute this result in Eq. 6.91

$$\Xi = \sum_{N=0}^{\infty} (\lambda g(T))^N = \frac{1}{1-\lambda g(T)} \tag{6.93}$$

It follows that

$$J = -kT \ln \Xi = kT \ln(1 - \lambda g(T)) \tag{6.94}$$

where $\lambda g(T) < 1$ for the above series to converge. We can now calculate pressure from the macrocanonical potential:

$$pV = -J \Longleftrightarrow p = -\frac{kT}{V} \ln(1-\lambda g(T)) \tag{6.95}$$

and the average number of atoms reads

$$\langle N\rangle = -\left.\frac{\partial J}{\partial \mu}\right|_{T,V} = -\left.\frac{\lambda}{kT}\frac{\partial J}{\partial \lambda}\right|_{T,V} = \frac{\lambda g(T)}{1-\lambda g(T)} \tag{6.96}$$

Rearranging, we obtain

$$\lambda g(T) = \frac{\langle N\rangle}{\langle N\rangle + 1} \simeq 1 \Longleftrightarrow \lambda = \frac{1}{g(T)} \tag{6.97}$$

Because vapour and solid are at equilibrium, then their fugacities must be the same :

$$\lambda_v = \lambda_s \Longleftrightarrow \frac{N_v}{V_v f(T)} = \frac{1}{g(T)} \Longleftrightarrow \frac{N_v}{V_v} = \frac{f(T)}{g(T)} \tag{6.98}$$

This expression summarises the **equilibrium condition** between the two phases. Let's incorporate this result into Eq. 6.85, where we found the pressure of the vapour phase:

$$p_v = kT\lambda_v f(T) = kT\frac{N_v}{V_v f(T)} f(T) = kT\frac{N_v}{V_v} = kT\frac{f(T)}{g(T)} \tag{6.99}$$

Note that N_v/V_v represents the vapour concentration at equilibrium. We now substitute $f(T)$ and $g(T)$ in the above equation and obtain an expression for the **vapour pressure**:

$$p_v = kT\left(\frac{2\pi mkT}{h^2}\right)^{3/2} \mathrm{e}^{-\epsilon_0/kT}\left[2\sinh\left(\frac{\hbar\omega}{2kT}\right)\right]^3 \tag{6.100}$$

The condition for the vapour-to-solid phase transition (also referred to as *deposition*) to occur is that

$$\lambda_v \geq \lambda_s \tag{6.101}$$

or, equivalently:

$$\frac{N_v}{V_v} \geq \frac{f(T)}{g(T)} \tag{6.102}$$

where $f(T)/g(T)$ is an increasing function of temperature T. If T is very large, then $f(T)/g(T) > N_v/V_v$ and the system is a pure vapour phase (no solid). There exists a critical temperature T_c at which the condition of Eq. 6.98 is satisified. In this case, the number of atoms in the vapour phase is $N_v = V_v f(T)/g(T)$ and those in the solid phase is $N_s = N - N_v$.

6.6 Conclusions

Advantages of the macrocanonical ensemble:

- Facilitates the study of systems where the number of particles is not constant. These include open systems, closed systems with chemical reactions, systems in phases of coexistence (liquid/vapor, etc.).
- Extremely useful when examining ideal systems of identical particles (bosons or fermions), where effects arising from particle indistinguishability (quantum gases) come to play.
- Not always easy to directly measure the number of particles (N) in a system. It is convenient to treat N as a variable and fix only its mean value $\langle N\rangle$.

6.7 Problems

6.1 Obtain an expression for the grand canonical partition function, macrocanonical potential, number of particles of an ideal gas.

6.2 Show that the entropy of an ideal gas as obtained by applying the grand canonical ensemble is given by the Sackur-Tetrode equation.

6.3 Show that, for a system described by the grand canonical ensemble, the following relation holds:

$$\langle NE\rangle - \langle N\rangle\langle E\rangle = \langle (N-\langle N\rangle)^2\rangle \left.\frac{\partial U}{\partial N}\right|_{T,V}$$

6.4 **Exam paper - July 2008** A closed container holds a classical ideal gas at a pressure P and temperature T. The container walls have a total of N_0 absorption sites where gas particles can be trapped in equilibrium with the gas. Each of these sites can absorb 0, 1, or 2 particles with energies 0, $-\varepsilon$, and -2ε, respectively. Exclusively using the macrocanonical ensemble,

a) Find an expression for the fugacity of the classical ideal gas, $\lambda_g = e^{\beta\mu}$, in terms of temperature and pressure (T, P).

b) Calculate the macrocanonical partition function for a single site, and deduce the mean number of absorbed particles as a function of fugacity and temperature.

c) Considering that the free gas particles and those absorbed in each site are in thermodynamic equilibrium, obtain the mean number of absorbed particles as a function of gas temperature and pressure. Physically interpret what happens in the limits $P \to \infty$ and $P \to 0$ (constant T).

6.5 A macroscopic system in equilibrium is composed of distinguishable particles of the same type, which can occupy two possible energy levels: one with zero energy and another with energy $+\varepsilon$. The upper level has a degeneracy of g, while the fundamental level is non-degenerate. Calculate the grand canonical partition function and the mean energy of the system in terms of the mean number of particles.

Chapter 7

Equivalence between ensembles

7.1 Introduction

The interpretation of ensemble equivalence, a fundamental pillar in statistical mechanics, is strictly related to the concept of *thermodynamic limit*. In finite systems, each ensemble adheres to distinct characteristic patterns, with different probability distributions that only converge when the thermodynamic system is sufficiently large. If the system is far from its thermodynamic limit, as finite systems inherently are, then it exhibits *fluctuations* that would otherwise be negligible. Consider, for example, the canonical energy distribution given by

$$P_m(E) = \frac{\mathrm{e}^{-\beta E_m}}{Z}, \tag{7.1}$$

where Z is the canonical partition function. In the thermodynamic limit, this distribution converges towards a delta function, expressed as $\delta(E - U)$. This convergence is established by examining the diminishing fluctuations, quantified by

$$\sigma^2[E] = \langle E^2 \rangle - \langle E \rangle^2 \tag{7.2}$$

which approach negligibility as $\sigma[E]/\langle E \rangle \to 0$ in the specified limit. Thermodynamic limit, fluctuations and how they connect to the concept of ensemble equivalence are the core topic of the present chapter.

7.2 Existence of the Thermodynamic Limit

The concept of ensemble equivalence asserts that different statistical ensembles become equivalent as the system size becomes sufficiently large. More specifically, as the system size increases, the statistical properties and distributions associated with these ensembles converge, leading to equivalent predictions for macroscopic observables. This convergence is a consequence of the vast number of microstates available to the system in larger configurations, smoothing out statistical fluctuations and aligning the ensembles' predictions. In practical terms, this equivalence allows one to choose the **most convenient ensemble for a particular problem**, depending on the information available or assumptions introduced. Each ensemble implies a different probability distribution of the system's microstates, which for microcanonical, canonical and macrocanonical ensembles read, respectively, $P_m = 1/\Omega$, $P_m = \mathrm{e}^{-\beta E_m}/Z$ and $P_{Nm} = \mathrm{e}^{\beta(\mu N - E_{Nm})}/\Xi$. Nevertheless, the thermodynamics of a macroscopic system at equilibrium, which can be obtained by applying the most convenient ensemble, is **unique**. It follows that ensembles must be **equivalent** to each other when the system is large enough, that is when the following conditions are satisfied

$$N \to \infty, \quad V \to \infty, \quad \rho = \frac{N}{V} = \mathrm{const} \tag{7.3}$$

Only in this limit, the different ensembles become equivalent, allowing one to choose the most convenient ensemble for each case. When taking the limit, results become independent of the system's specific shape;

there are no surface effects. Extensive variables such as A, U, S, etc., exhibit independence and phase transitions, which would not occur in finite-sized systems, can be observed[1].

In the canonical ensemble, the Helmholtz free energy is given by $A(N, V, T) = -kT \ln Z(N, V, T)$. However, *a priori*, there is no assurance that $A(N, V, T)$ exists as $N, V \to \infty$. We must demonstrate that $A(N, V, T)$ is an extensive function. Equivalently, we need to show that the following properties

$$\lim_{V \to \infty} \frac{A(\rho V, V, T)}{V} = \tilde{a}(T, \rho) \tag{7.4}$$

$$\lim_{N \to \infty} \frac{A(N, N/\rho, T)}{N} = a(T, \rho) = \frac{1}{\rho}\tilde{a}(T, \rho) \tag{7.5}$$

are finite in the thermodynamic limit and depend only on T and ρ. This is not guaranteed automatically, but depends on the form of the Hamiltonian. In other words, there may exist Hamiltonians that are pathological, so that when taking the limit, the per-particle free energy $a = A/N$ diverges. Therefore, what conditions do we need to impose on the Hamiltonian to ensure the existence of the Thermodynamic Limit? To address this question, we will assume that our system consists of N identical particles, describable using classical formalism, and that the Hamiltonian can be written as the sum of an ideal contribution, H_0, that only depends on momenta and a real contribution, H', that only depends on the positions of mutually-interacting particles. In particular:

$$H = H_0 + H' = \sum_{i=1}^{N} \frac{\vec{p}_i^{\,2}}{2m} + H'(\vec{r}_1, \vec{r}_2, \ldots, \vec{r}_N). \tag{7.6}$$

The canonical partition function of the classical ideal gas (see Eq. 6.81) reads

$$Z_0 = \frac{1}{N! h^{3N}} V^N \left\{ \int d\vec{p}\, \mathrm{e}^{-\beta H_0} \right\}^N = (2\pi m k T)^{3N/2} \frac{V^N}{N! h^{3N}} = \frac{V^N}{N!} \Lambda^{-3N} \tag{7.7}$$

where $\Lambda = h/\sqrt{2\pi m k T}$ is the thermal de Broglie wavelength. We can then re-write the ideal part of the Helmholtz free energy as follows

$$A_0(N, V, T) = -kT \ln Z_0(N, V, T) = NkT \left[\ln\left(\frac{N}{V} \Lambda^3 \right) - 1 \right] \tag{7.8}$$

where we have applied Stirling's approximation. Introducing the numerical density, $\rho \equiv N/V$, and dividing by N, one obtains the specific free energy

$$a_0(\rho, T) = \frac{A_0}{N} = kT[\ln(\rho \Lambda^3) - 1] = \text{const} \tag{7.9}$$

We conclude that, for an **ideal gas**, the free energy exists and is finite in the thermodynamic limit. Therefore, whether the thermodynamic limit exists or not depends exclusively on the nature of the interaction term between particles, H'. Let's then turn our attention to this term. To this end, let's re-write the canonical partition function in its more general form, incorporating the interaction term. It reads

$$Z(N, V, T) = \frac{1}{N! h^{3N}} \int d\vec{p}_1 \ldots d\vec{p}_N \mathrm{e}^{-\beta H_0} \int d\vec{r}_1 \ldots d\vec{r}_N \mathrm{e}^{-\beta H'} = \Lambda^{-3N} \frac{1}{N!} \underbrace{\int d\vec{r}_1 \ldots d\vec{r}_N \mathrm{e}^{-\beta H'}}_{\theta'(N,V,T)} \tag{7.10}$$

[1] Phase transitions, which are dramatic changes in the macroscopic properties of a system as it crosses certain thresholds (like melting and boiling), often arise due to the collective behaviour of a large number of particles. In infinite systems, these transitions are well-defined and can occur at specific temperatures or conditions. However, in finite systems, the behaviour can be different due to size effects and boundary conditions. For example, the finite size of the system may prevent the formation of long-range order necessary for certain phase transitions to occur. Additionally, surface effects and finite-size effects can influence the behaviour near the phase transition point. As a result, the occurrence and nature of phase transitions in finite systems may differ from those in infinite systems, and some transitions may not occur at all.

For $a(\rho, T)$ to exist, H' must meet some conditions that determine the behaviour of the above integral $\theta'(N, V, T)$. Let's assume that the real part of the Hamiltonian only depends on pair interactions, thus neglecting multi-body contributions:

$$H' = \sum_{i<j}^{N} \phi(r_{ij}) \tag{7.11}$$

We also assume that $\phi(r)$ is the van Hove potential, which represents uncharged hard-core spheres with diameter d_0 that attract each other if their centers are separated by a distance less than R_0, being the finite-range attractive tail. The van Hove potential encapsulates qualitative aspects of realistic potentials (hard core - impenetrability - and short-range attraction), such as the Lennard-Jones potential. One could also use, for the present discussion, the latter potential, but for simplicity the van Hove is preferred. It reads

$$\phi(r_{ij}) = \begin{cases} \infty & r_{ij} \leq d_0 \\ < 0 & d_0 < r_{ij} < R_0 \\ 0 & r_{ij} \geq R_0 \end{cases} \tag{7.12}$$

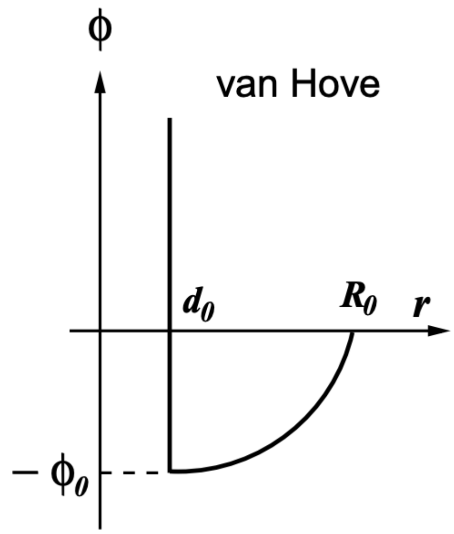


Figure 7.1: Schematic representation of the van Hove potential.

For the so-defined potential, all non-zero interactions between a general particle i and all its neighbours are established within a distance R_0 from i. Under these conditions, there exists a maximum number of particles, confined in such a volume, that can interact with i, each contributing to H' (in absolute value) less than ϕ_0[2] Mathematically:

$$|H'| \leq \underbrace{\phi_0}_{\substack{\text{contribution} \\ \text{of one particle}}} \times \underbrace{N}_{\substack{\text{total number} \\ \text{of particles}}} \times \underbrace{\alpha \frac{\frac{4\pi}{3}(R_0^3 - r_0^3)}{\frac{4\pi}{3} r_0^3}}_{\substack{\text{maximum number} \\ \text{of particles within} \\ \text{the interaction range}}} \tag{7.13}$$

where $r_0 = d_0/2$ is the particle radius and $\alpha > 0$ is a measure of the particle packing within the interaction range (spherical volume of radius R_0). Rearranging, we obtain

[2]Note that ϕ_0 is a limit value for $r_{ij} \to d_0$. This implies that a given particle-particle interaction cannot be larger than ϕ_0.

$$|H'| \leq \phi_0 N \underbrace{\left\{ \alpha \left[\left(\frac{R_0}{r_0} \right)^3 - 1 \right] \right\}}_{K} = wN \tag{7.14}$$

where $w = \phi_0 K$. The inequality $|H'| \leq wN$ or, incorporating the sign of the potential, $H' \geq -wN$ is called the **stability condition** (on the potential). The van Hove potential satisfies this condition as it has a rigid core and a lower limit, preventing the collapse of the system, i.e., the possibility of all particles falling into the attraction sphere of another (infinite particles within the interaction region). In such a case, $H' \to -\infty$ and the integral $\theta'(N, V, T)$ in Eq. 7.10 would diverge. Let's incorporate this inequality in Eq. 7.10:

$$Z(N, V, T) = \Lambda^{-3N} \frac{1}{N!} \int d\vec{r}_1 \dots d\vec{r}_N \mathrm{e}^{-\beta H'} \leq \Lambda^{-3N} \frac{1}{N!} \int d\vec{r}_1 \dots d\vec{r}_N \underbrace{\mathrm{e}^{\beta w N}}_{\text{const}} = \Lambda^{-3N} \frac{1}{N!} V^N \mathrm{e}^{\beta w N} \tag{7.15}$$

Equivalently, applying Stirling's approximation, we obtain

$$\ln Z(N, V, T) \leq -3N \ln \Lambda - N \ln N + N + N \ln V + \beta w N \tag{7.16}$$

It follows that

$$\frac{-kT \ln Z(N, V, T)}{V} = \frac{A}{V} \geq kT \frac{N}{V} (3 \ln \Lambda + \ln N - 1 - \ln V - \beta w) \tag{7.17}$$

Therefore, we get a lower bound for A/V

$$\frac{A}{V} \geq \rho k T \left[\ln(\rho \Lambda^3) - 1 - \beta w \right] \tag{7.18}$$

which is independent from N and V and therefore finite in the Thermodynamic Limit. If the stability condition is satisfied, the free energy per unit volume, A/V, has a **lower bound**. This means that A/V cannot decrease indefinitely. From a physical perspective, the stability condition implies that the interactions between particles are not attractive enough to cause the collapse of the system.

The second condition is referred to as **strong tempering condition**. Suppose that the system can be partitioned into two regions with N_1 and N_2 particles, respectively, where $N = N_1 + N_2$, separated by a given distance $R \geq d_0$ as illustrated in Fig. 7.2. We define the potential of mutual interaction between the

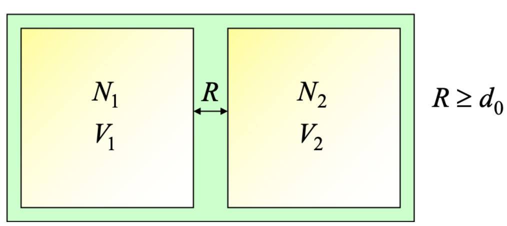


Figure 7.2: Condition of strong moderation.

particles in the two regions as:

$$\phi_{N_1 N_2}(\mathbf{r}_1, \dots, \mathbf{r}_N) \equiv H'(\mathbf{r}_1, \dots, \mathbf{r}_{N_1}, \mathbf{r}_{N_1+1}, \dots, \mathbf{r}_N) - H'_1(\mathbf{r}_1, \dots, \mathbf{r}_{N_1}) - H'_2(\mathbf{r}_{N_1+1}, \dots, \mathbf{r}_N) \tag{7.19}$$

Due to the van Hove potential, the interaction between the two groups of particles, which are separated by a distance $R \geq d_0$, is purely attractive:

$$\phi_{N_1 N_2} \leq 0, \quad \|\mathbf{r}_i - \mathbf{r}'_j\| \geq R \geq d_0, \quad \forall i, j. \tag{7.20}$$

If a potential, like the van Hove potential, allows writing such an **upper bound** for the interaction energy between two groups of particles, it is said to be a moderate (or tempered) potential. If the bound is zero, as above, it is said to be **strongly moderated**. A potential lacking this property, due to sufficiently strong long-range repulsive interactions, would cause the system to get especially unstable: the interactions are not capable of keeping the particles together, and the free energy diverges to $+\infty$.

Physically, the tempering condition ensures that repulsive interactions are not strong enough to destabilise the system. The strong tempering condition further implies that A/V decreases as V increases while keeping the density ρ constant. However, this decrease cannot continue indefinitely—thermodynamic stability requires that A/V remains finite. To establish this, we analyse how the partition function behaves when the system is divided into two subsystems as indicated in Fig. 7.2[3]. Since the interaction potential satisfies $\Phi_{N_1N_2} \leq 0$, we can derive an **upper bound** for A/V, showing that it is constrained and approaches a well-defined value in the thermodynamic limit:

$$\begin{aligned} Z(N,V,T) &\geq Z(N,V_1+V_2,T) = \\ &= \frac{1}{\Lambda^{3N}}\frac{1}{N!}\sum_{N_1}\frac{N!}{N_1!N_2!}\int_{V_1} d\vec{r}_1\ldots d\vec{r}_{N_1}\int_{V_2} d\vec{r}_{N_1+1}\ldots d\vec{r}_N \mathrm{e}^{-\beta\phi_{N_1N_2}}\mathrm{e}^{-\beta H_1'}\mathrm{e}^{-\beta H_2'} \end{aligned} \tag{7.21}$$

We now eliminate the sum (as this reinforces the inequality), incorporate the condition found above ($\phi_{N_1N_2} \leq 0$) and then obtain

$$Z(N,V,T) \geq \frac{1}{\Lambda^{3N}}\frac{1}{N_1!N_2!}\int_{V_1} d\vec{r}_1\ldots d\vec{r}_{N_1}\mathrm{e}^{-\beta H_1'}\int_{V_2} d\vec{r}_{N_1+1}\ldots d\vec{r}_N \mathrm{e}^{-\beta H_2'} = Z_1 \times Z_2 \tag{7.22}$$

It then follows that

$$A = -kT\ln Z \leq -kT\ln Z_1 - kT\ln Z_2 = A_1 + A_2 \tag{7.23}$$

Equivalently

$$\frac{A}{V} \leq \frac{A_1 + A_2}{V} \tag{7.24}$$

In conclusion, when taking the thermodynamic limit, A/V converges to a fixed value independent of V, meaning it is an intensive variable:

$$\lim_{N\to\infty, V\to\infty}\frac{A}{V} = \tilde{a}(T,\rho) \tag{7.25}$$

For neutral molecules (interacting through a Lennard-Jones potential), both conditions are satisfied: stability and strong tempering. For charged systems, Dyson and Lenard (1967-1968) provided a rigorous proof that a globally neutral system of electrically charged particles is stable, as long as the particles are fermions. However, the strong tempering condition is violated. In 1964, Fisher demonstrated that the strong tempering condition can be relaxed to the weak tempering condition:

$$\Phi_{N_1N_2} \leq \frac{N_1N_2w'}{R^{3+\epsilon}} \quad |\mathbf{r}_i - \mathbf{r}_j| \geq R \geq R_0 \tag{7.26}$$

with R_0 and w' constants. In 1969, Lebowitz and Lieb finally provided a proof of the existence of the thermodynamic limit for charged particles. The key lies in the screening of charged particles with those of opposite sign [16, 17, 18].

7.3 Fluctuations

Any deviation of a mechanical variable with respect to its mean value is called **fluctuation**. If x is a macroscopic variable, then

[3]Note that the volume of the system, V, includes the region separating the two volumes V_1 and V_2 and therefore $V \geq V_1+V_2$.

$$\delta x = x - \langle x \rangle \tag{7.27}$$

measures its deviation from its average value. At equilibrium, macroscopic variables do not exhibit a defined value, but rather fluctuate over time around their average value. These fluctuations should **not be considered as random noise** as they can represent (or be used to represent) other macroscopic properties of the same equilibrium state. The study of fluctuations is essential in Statistical Physics. The most obvious reason is to understand how deviations from the mean values will occur. If the dispersion is significant, experimentally we will observe a series of different values that will differ from each other. Nevertheless, we know that, experimentally, macroscopic properties do not fluctuate, suggesting that deviations from the average value should be small. This important property is key to establish the equivalence between ensembles. In particular, the **mean-squared fluctuation** is calculated as follows

$$\langle \delta x^2 \rangle = \langle (x - \langle x \rangle)^2 \rangle = \langle x^2 \rangle - 2\langle x \rangle\langle x \rangle + \langle x \rangle^2 = \langle x^2 \rangle - \langle x \rangle^2 \tag{7.28}$$

whereas the **relative mean-squared fluctuation** is defined as $\sqrt{\langle \delta x^2 \rangle}/\langle x \rangle$. We will see that

$$\lim_{\substack{N\to\infty \\ V\to\infty \\ N/V=\text{const}}} \frac{\sqrt{\langle \delta x^2 \rangle}}{\langle x \rangle} \to 0 \tag{7.29}$$

We shall first focus on energy fluctuations. In the canonical ensemble, energy can change between 0 and ∞, whereas in the microcanonical ensemble, it is constrained to a specific value. Therefore, how can it be possible that the two formalisms are consistent?

7.3.1 Fluctuations in Energy. Equivalence between canonical and microcanonical ensembles.

In case of energy, fluctuations increases with the system size, that is $\langle \delta E^2 \rangle \sim N$. However, what is really important is not the absolute fluctuation, but rather the relative fluctuation. For example, if $\langle \delta E \rangle \simeq 10^{20}$ and $\sqrt{\langle \delta E^2 \rangle} \simeq 10^{10}$, then the relative mean-squared fluctuation is small:

$$\frac{\sqrt{\langle \delta E^2 \rangle}}{\langle E \rangle} = \frac{\sqrt{kT^2 C_v}}{U} \sim \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}} \xrightarrow{N\to\infty} 0 \tag{7.30}$$

where C_v is the heat capacity at constant volume and, together with U, is an extensive property[4]. We notice that the ratio between energy fluctuations is negligible as compared to the energy average value, even of the former are substantial. In the thermodynamic limit, where $N \to \infty$, this ratio goes to zero. We need to prove how one can obtain the heat capacity at constant volume from the energy mean-squared fluctuations in Eq. 7.30:

$$\begin{aligned}
\langle \delta E^2 \rangle = \langle E^2 \rangle - \langle E \rangle^2 &= \frac{\sum_r E_r^2 \mathrm{e}^{-\beta E_r}}{\sum_r \mathrm{e}^{-\beta E_r}} - \left(\frac{\sum_r E_r \mathrm{e}^{-\beta E_r}}{\sum_r \mathrm{e}^{-\beta E_r}} \right)^2 = \\
&= -\frac{\partial}{\partial \beta} \left(\frac{\sum_r E_r \mathrm{e}^{-\beta E_r}}{\sum_r \mathrm{e}^{-\beta E_r}} \right)_{\{E_r\}} = \\
&= -\left.\frac{\partial U}{\partial \beta}\right|_{\{E_r\}} = -\left.\frac{\partial U}{\partial \beta}\right|_{N,V} = -\frac{\partial T}{\partial \beta} \left.\frac{\partial U}{\partial T}\right|_{N,V} = \\
&= kT^2 \left.\frac{\partial U}{\partial T}\right|_{N,V} = kT^2 C_v
\end{aligned} \tag{7.31}$$

This formula, which unveils the equivalence between energy variance[5] and heat capacity at constant volume, is especially relevant and beautiful, as it relates the fluctuations of a microscopic property (E) with

[4] Note that extensive properties depend on the system size. It follows that they are proportional to N.
[5] Note that $\langle \delta E^2 \rangle = \langle E^2 \rangle - \langle E \rangle^2$ is the energy variance, also indicated as σ_E^2.

a macroscopic property (C_v) that measures the ability of a system to exchange heat with its surroundings and can be easily obtained experimentally. It is now clearer that relevance of Eq. 7.30, which highlights that (i) relative energy fluctuations are completely negligible and (ii) the energy of a system in the canonical ensemble is equal to the internal energy[6]. This result explains why, in the thermodynamic limit, **canonical and microcanonical ensembles are equivalent**.

We can deepen our discussion a bit further. From the above conclusions, it follows the probability of observing a member of the canonical ensemble with $\langle E\rangle \neq U$ is practically zero and the energy probability distribution is, therefore, a Gaussian function that, in the limit, tends to a Dirac δ distribution. This probability distribution, $P(E)$, in the canonical ensemble reads

$$P(E) = \frac{1}{Z}\Omega(E)\mathrm{e}^{-\beta E} \tag{7.32}$$

where Z, the canonical partition function, depends on N, V and T, but not on E; $\Omega(E)$, the number of microstates with energy E, increases exponentially with E; and $\mathrm{e}^{-\beta E}$, the Boltzmann factor, decreases with E. We therefore expect that $\Omega(E)\mathrm{e}^{-\beta E}$ should exhibit an extremum point at a given value of energy E^*, as indicated in Fig. 7.3 for details.

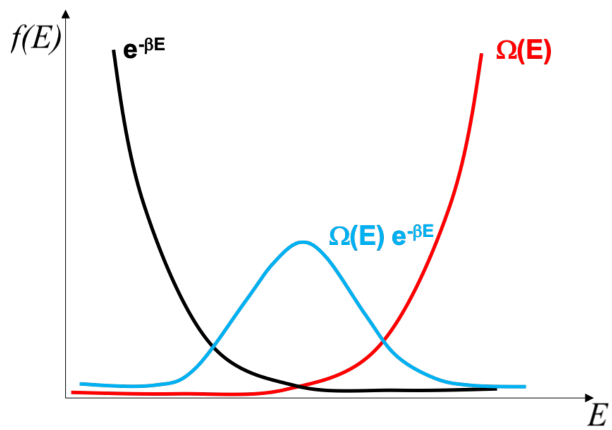


Figure 7.3: Probability distributions.

Let's find out whether this is a maximum or a minimum and where exactly E^* is located:

$$\begin{aligned}\frac{\partial}{\partial E}\left[\mathrm{e}^{-\beta E}\Omega(E)\right]_{E=E^*} = 0 \Longleftrightarrow \frac{\partial}{\partial E}\ln\left[\mathrm{e}^{-\beta E}\Omega(E)\right]_{E=E^*} = 0 \Longleftrightarrow \\ \Longleftrightarrow \frac{\partial}{\partial E}\left[-\beta E + \ln\Omega(E)\right]_{E=E^*} = 0 \Longleftrightarrow \\ \Longleftrightarrow \left.\frac{\partial \ln\Omega(E)}{\partial E}\right|_{E=E^*} = \beta\end{aligned} \tag{7.33}$$

It should be noticed that, because $S = k\ln\Omega$, the above equation can be rewritten as

$$\left.\frac{\partial S}{\partial E}\right|_{E=E^*} = \frac{1}{T} \tag{7.34}$$

If we now compare this result with the entropy change with internal energy from Thermodynamics, that is

$$\left.\frac{\partial S}{\partial U}\right|_{N,V} = \frac{1}{T} \tag{7.35}$$

[6]The ensemble-average energy $\langle E\rangle$ strictly equals the internal energy U in the thermodynamic limit where Eq. 7.30 is valid. Otherwise, we can say that $\langle E\rangle$ is almost equal to U.

we conclude that $E^* = U$. It follows that the most probable configuration is one where the energy of the system is exactly U. Now, is this point a maximum?

$$\begin{aligned}
\left.\frac{\partial^2 \ln P(E)}{\partial E^2}\right|_{E=E^*} &= \left.\frac{\partial^2(-\beta E + \ln \Omega)}{\partial E^2}\right|_{E=E^*} = \left.\frac{\partial^2 \ln \Omega}{\partial E^2}\right|_{E=U} = \\
&= \frac{1}{k}\left.\frac{\partial^2 S}{\partial U^2}\right|_{N,V} = \frac{1}{k}\frac{\partial}{\partial U}\left(\frac{\partial S}{\partial U}\right)_{N,V} = \frac{1}{k}\frac{\partial}{\partial U}\left(\frac{1}{T}\right) = \\
&= -\frac{1}{kT^2}\left(\frac{\partial T}{\partial U}\right)_{N,V} = -\frac{1}{kT^2 C_v} < 0
\end{aligned} \tag{7.36}$$

The last inequality is due to the fact that $T > 0$ and $C_v > 0$. We therefore conclude that the function $f(E) = \Omega(E)\mathrm{e}^{-\beta E}$ has a maximum in $E = U$. Now, to find the form of the probability distribution around $E = U$, we can expand it in power series:

$$\begin{aligned}
\ln P(E) &= \ln P(U) + \underbrace{\left.\frac{\partial \ln P(E)}{\partial E}\right|_{E=U}}_{0}(E-U) + \frac{1}{2}\left.\frac{\partial^2 \ln P(E)}{\partial E^2}\right|_{E=U}(E-U)^2 + \dots \\
&= \ln P(U) - \frac{(E-U)^2}{2kT^2C_v} + \dots = \ln\left(\frac{P(U)}{\mathrm{e}^{(E-U)^2/2kT^2C_v}}\right) + \dots
\end{aligned} \tag{7.37}$$

We therefore obtain

$$P(E) \simeq P(U)\mathrm{e}^{-(E-U)^2/2kT^2C_v} \tag{7.38}$$

One can conclude that, close to the maximum, $P(E)$ behaves as a **Gaussian distribution**[7] with variance kT^2C_v. If we use the reduced variable E/U, then it becomes a Gaussian distribution centred in 1 and variance

$$\sigma^2 = \frac{kT^2C_v}{U^2} \sim \frac{N}{N^2} = \frac{1}{N} \longrightarrow 0 \tag{7.40}$$

Basically, the width of the distribution tends to zero and at its peak reads

$$P(E/U = 1) = \frac{1}{\sqrt{2\pi\sigma^2}} = \frac{U}{\sqrt{2\pi kT^2C_v}} \sim \frac{N}{\sqrt{N}} = \sqrt{N} \longrightarrow \infty \tag{7.41}$$

Therefore, in the canonical ensemble, the energy fluctuates around a mean value (which corresponds to a microcanonical ensemble with a temperature function equal to the temperature of the canonical thermal bath) in a Gaussian manner, with a width of the order of $1/\sqrt{N}$. In the thermodynamic limit, the distribution of energy per particle in the canonical ensemble can be approximated by a Dirac delta function. If the heat capacity diverges (i.e., grows rapidly with some power of N), the entire argument fails. This happens at critical points of phase transitions. However, the equivalence of ensembles remains true at critical points (although it needs to be demonstrated in a more elaborate manner).

Given the probability distribution $P(E)$, the canonical partition function can be estimated as follows

[7] The general form of the Gaussian probability density distribution is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \tag{7.39}$$

where μ is the mean and σ the standard deviation, while the variance of the distribution is σ^2. In the above case, U is the mean, $\sqrt{kT^2C_v}$ the standard deviation and kT^2C_v the variance.

$$
\begin{aligned}
\underbrace{\int_{-\infty}^{+\infty} P(E)dE}_{1} \simeq & \int_{-\infty}^{+\infty} P(U)\mathrm{e}^{-(E-U)^2/2kT^2C_v}dE = \\
&= \underbrace{\frac{\mathrm{e}^{\beta kT\ln Z}}{Z}}_{1}\int_{-\infty}^{+\infty} \mathrm{e}^{-(E-U)^2/2kT^2C_v}dE = \\
&= \frac{\mathrm{e}^{-\beta A}}{Z}\int_{-\infty}^{+\infty} \mathrm{e}^{-(E-U)^2/2kT^2C_v}dE = \\
&= \frac{\mathrm{e}^{-\beta(U-TS)}}{Z}\int_{-\infty}^{+\infty} \mathrm{e}^{-(E-U)^2/2kT^2C_v}dE = \\
&= \frac{\mathrm{e}^{-\beta(U-TS)}}{Z}\sqrt{2kT^2C_v}\int_{-\infty}^{+\infty} \mathrm{e}^{-x^2}dx = \\
&= \frac{\mathrm{e}^{-\beta(U-TS)}}{Z}\sqrt{2kT^2\pi C_v} \Longleftrightarrow \\
\Longleftrightarrow Z(N,V,T) = & \,\mathrm{e}^{-\beta(U-TS)}\sqrt{2kT^2\pi C_v}
\end{aligned}
\tag{7.42}
$$

where we have set $x = (E-U)/\sqrt{2kT^2C_v}$. It follows that the Helmholtz free energy reads

$$
A \equiv -kT\ln Z \simeq U - TS - \frac{1}{2}kT\ln(2\pi kT^2C_v) \simeq U - TS \tag{7.43}
$$

In the thermodynamic limit, $N \gg \ln(N)$ and the last term in the r.h.s. can be neglected. As such, $A = U-TS$ in a macroscopic system, confirming that there are no differences between the results in the two ensembles in the thermodynamic limit.

7.3.2 Fluctuations in the Number of Particles. Equivalence between canonical and macrocanonical ensembles.

In the canonical ensemble, we know that N is fixed, while it can fluctuate between 0 and $+\infty$ in the macrocanonical ensemble. Nevertheless, we will show here that, in practice, the relative fluctuation of N around its average value $\langle N\rangle$ is negligible. Let's calculate the variance σ_N^2:

$$
\sigma_N^2 = \langle(\delta N)^2\rangle = \langle N^2\rangle - \langle N\rangle^2 = \frac{\sum_{Nm} N^2\mathrm{e}^{\beta(\mu N - E_{mN})}}{\sum_{Nm}\mathrm{e}^{\beta(\mu N - E_{mN})}} - \left[\frac{\sum_{Nm} N\mathrm{e}^{\beta(\mu N - E_{mN})}}{\sum_{Nm}\mathrm{e}^{\beta(\mu N - E_{mN})}}\right]^2 \tag{7.44}
$$

where we have introduced a short form for the double summation: $\sum_{Nm} = \sum_{N=0}^{\infty}\sum_m$, with m the number of quantum states. We also see that

$$
\frac{1}{\beta}\left.\frac{\partial\langle N\rangle}{\partial\mu}\right|_{T,V} = \frac{1}{\beta}\frac{\partial}{\partial\mu}\left(\frac{\sum_{Nm} N\mathrm{e}^{\beta(\mu N - E_{mN})}}{\sum_{Nm}\mathrm{e}^{\beta(\mu N - E_{mN})}}\right)_{\beta,V} = \underbrace{\frac{\sum_{Nm} N^2\mathrm{e}^{\beta(\mu N - E_{mN})}}{\sum_{Nm}\mathrm{e}^{\beta(\mu N - E_{mN})}}}_{\langle N^2\rangle} - \underbrace{\left[\frac{\sum_{Nm} N\mathrm{e}^{\beta(\mu N - E_{mN})}}{\sum_{Nm}\mathrm{e}^{\beta(\mu N - E_{mN})}}\right]^2}_{\langle N\rangle^2} \tag{7.45}
$$

So, we find that

$$
\frac{1}{\beta}\left.\frac{\partial\langle N\rangle}{\partial\mu}\right|_{T,V} = \langle N^2\rangle - \langle N\rangle^2 \tag{7.46}
$$

Because μ and T are intensive properties, the l.h.s. of the above equation increases with N. Consequently, it follows that $\langle N^2\rangle - \langle N\rangle^2 \sim N$. Let's now see the mean-squared relative fluctuation:

$$\frac{\sqrt{\langle \delta N^2 \rangle}}{\langle N \rangle} = \frac{\sqrt{\langle N^2 \rangle - \langle N \rangle^2}}{\langle N \rangle} = \frac{1}{\langle N \rangle} \sqrt{\frac{1}{\beta} \left(\frac{\partial \langle N \rangle}{\partial \mu} \right)_{T,V}} \sim \frac{\sqrt{N}}{N} \sim \frac{1}{\sqrt{N}} \xrightarrow{N \to \infty} 0 \tag{7.47}$$

One can conclude that the relative fluctuations are insignificant in the thermodynamic limit, where $N \to \infty$, suggesting the equivalence between canonical and macrocanonical ensembles. Now, what is the physical meaning of the derivative in Eq. 7.46? To answer this question, we make use of Thermodynamics and, precisely, of the macrocanonical potential as obtained in Eq. 6.60:

$$J(\mu, T, V) = -pV \tag{7.48}$$

Additionally, applying the First and Second Law of Thermodynamics, we know that

$$dJ = d(U - TS - \mu\langle N \rangle) = \underbrace{đQ - đW + \mu d\langle N \rangle}_{dU} - TdS - SdT - \mu d\langle N \rangle - \langle N \rangle d\mu = -pdV - SdT - \langle N \rangle d\mu \tag{7.49}$$

It follows that

$$dJ = -d(pV) = -pdV - Vdp = -pdV - SdT - \langle N \rangle d\mu \Longleftrightarrow d\mu = \frac{V}{\langle N \rangle} dp - \frac{S}{\langle N \rangle} dT \tag{7.50}$$

and the change of chemical potential with $\langle N \rangle$ at constant T and V reads

$$\left. \frac{\partial \mu}{\partial \langle N \rangle} \right|_{T,V} = \frac{V}{\langle N \rangle} \left. \frac{\partial p}{\partial \langle N \rangle} \right|_{T,V} \tag{7.51}$$

To find a more suitable expression, one can calculate how pressure changes with $\langle N \rangle$ and V:

$$\left. \frac{\partial p}{\partial \langle N \rangle} \right|_{T,V} = \left. \frac{\partial p}{\partial \rho} \right|_{T} \left. \frac{\partial \rho}{\partial \langle N \rangle} \right|_{V} = \left. \frac{\partial p}{\partial \rho} \right|_{T} \frac{1}{V} \Longleftrightarrow \left. \frac{\partial p}{\partial \rho} \right|_{T} = V \left. \frac{\partial p}{\partial \langle N \rangle} \right|_{T,V} \tag{7.52}$$

and

$$\left. \frac{\partial p}{\partial V} \right|_{T,N} = \left. \frac{\partial p}{\partial \rho} \right|_{T} \left. \frac{\partial \rho}{\partial V} \right|_{N} = \left. \frac{\partial p}{\partial \rho} \right|_{T} \left(-\frac{\langle N \rangle}{V^2} \right) \Longleftrightarrow \left. \frac{\partial p}{\partial \rho} \right|_{T} = \left(-\frac{V^2}{\langle N \rangle} \right) \left. \frac{\partial p}{\partial V} \right|_{T,N} \tag{7.53}$$

It follows that

$$V \left. \frac{\partial p}{\partial \langle N \rangle} \right|_{T,V} = \left(-\frac{V^2}{\langle N \rangle} \right) \left. \frac{\partial p}{\partial V} \right|_{T,N} \tag{7.54}$$

and substituting this result into Eq. 7.51, one obtains:

$$\left. \frac{\partial \mu}{\partial \langle N \rangle} \right|_{T,V} = -\frac{V^2}{\langle N \rangle^2} \left. \frac{\partial p}{\partial V} \right|_{T,N} \tag{7.55}$$

We can now go back to Eq. 7.46 and calculate the fluctuations in the number of particles:

$$
\begin{aligned}
\langle N^2\rangle - \langle N\rangle^2 =& \\
&= kT\,\frac{\partial \langle N\rangle}{\partial \mu}\bigg|_{T,V} = \\
&= kT\frac{1}{\frac{\partial \mu}{\partial\langle N\rangle}\Big|_{T,V}} = \\
&= kT\frac{1}{-\frac{V^2}{\langle N\rangle^2}\,\frac{\partial p}{\partial V}\Big|_{T,N}} = \\
&= kT\frac{1}{-\frac{V^2}{\langle N\rangle^2}\,\frac{1}{\frac{\partial V}{\partial p}\big|_{T,N}}} = \\
&= kT\frac{\langle N\rangle^2}{V}\underbrace{\left[-\frac{1}{V}\,\frac{\partial V}{\partial p}\bigg|_{T,N}\right]}_{\kappa_T}
\end{aligned}
\tag{7.56}
$$

where κ_T is the **isothermal compressibility**, an intensive property that quantifies how much a substance will change its volume when subjected to a change in pressure at constant temperature. Let's rewrite this equation as it proves the equivalence between the canonical and macrocanonical ensembles in the thermodybamic limit:

$$
\langle N^2\rangle - \langle N\rangle^2 = kT\frac{\langle N\rangle^2}{V}\kappa_T \tag{7.57}
$$

and in the limit

$$
\frac{\sqrt{\langle N^2\rangle - \langle N\rangle^2}}{\langle N\rangle} \sim \frac{\sqrt{\langle N^2\rangle / V}}{\langle N\rangle} \sim \frac{1}{\sqrt{V}} \xrightarrow{V\to\infty} 0 \tag{7.58}
$$

Compressible systems (such as a diluted gas) are more prone to local density fluctuations than incompressible systems (such as solids). For stable thermodynamic systems, it holds true that $\frac{\partial V}{\partial p} < 0$, thereby making κ_T and the fluctuation of the particle number positive. The only exception occurs precisely at the **critical point** of the liquid-gas transition (see Fig. 7.4), where:

$$
\left(\frac{\partial p}{\partial V}\right)_T = 0 \quad \Rightarrow \quad \kappa_T \to \infty \tag{7.59}
$$

At the critical point, the system becomes infinitely compressible. This leads to the appearance of density fluctuations that tend to become extremely large. Macroscopically, this phenomenon is observable and is referred to as **critical opalescence**. In this situation, the macrocanonical ensemble yields results that are not identical to those obtained with the canonical ensemble. Here, the use of the canonical ensemble leads to a misinterpretation of the results. We can only apply the macrocanonical ensemble.

7.4 Conclusions

In conclusion, when taking the thermodynamic limit, all equilibrium ensembles are equivalent: we can use whichever suits us best. For instance,

$$
Z(N,V,T) = \sum_E \Omega(N,V,E)\mathrm{e}^{-E/kT} \tag{7.60}
$$

Nevertheless, we have seen that $P(E) = \frac{1}{Z}\Omega(E)\mathrm{e}^{-e/kT}$ is a Gaussian function especially narrow around $\langle E\rangle$. In the thermodynamic limit, where $\langle N\rangle \to \infty$, only a single value of E is significant, that is $E = \langle E\rangle = U$. Therefore, in the summation of the above equation, the only term contributing to the sum is

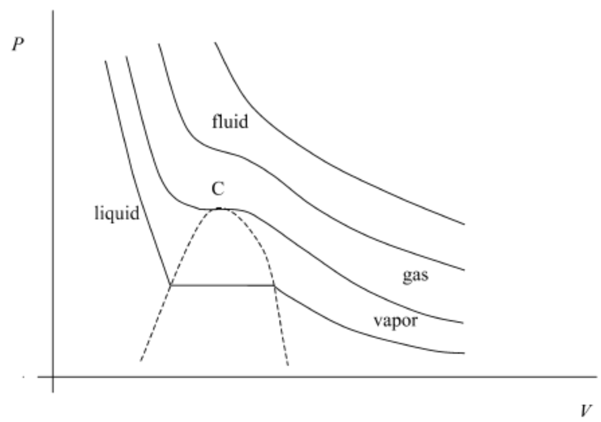


Figure 7.4: Pressure *vs* volume phase diagram with four exemplary isotherms. Point C represents the critical point, where $(\partial p/\partial V)_T = 0$ and thermal compressibility and fluctuations diverge.

$$Z(N, V, T) = \sum_E \Omega(N, V, U)\mathrm{e}^{-U/kT} \tag{7.61}$$

Even if systems in the canonical ensemble have all different values of energy, what really happens is that the members of the ensemble distribute uniformly and it is practically impossible to find a member of the ensemble with a particular value of $E \neq \langle E \rangle = U$ or $N \neq \langle N \rangle$. In the thermodynamic limit, the distribution becomes uniform with fixed values of N and E. It follows that:

$$A = -kT \ln Z = U - kT \ln \Omega(N, V, U) = U - TS \tag{7.62}$$

and the microcanonical partition function

$$\Xi(N, V, E) = \sum_N \mathrm{e}^{\beta\mu N} \sum_E \Omega(N, V, E)\mathrm{e}^{-\beta E} \simeq \Omega(N, V, U)\mathrm{e}^{-\beta U + \beta\mu N} \tag{7.63}$$

with

$$J = -kT \ln \Xi \sim U - TS - \mu N. \tag{7.64}$$

7.5 Problems

7.1 Show that, in the macrocanonical ensemble, the energy fluctuation reads

$$\langle \delta E^2 \rangle = kT^2 C_v + kT \frac{\langle N \rangle^2}{V} \kappa_T \left(\frac{\partial U}{\partial N} \right)^2_{T,V} \tag{7.65}$$

7.2 Get the grand canonical partition function of a classical ideal gas without internal structure, in terms of V, T, and μ. From it, calculate the entropy. Verify that, by expressing the entropy in terms of (T, V, N), the same expression is obtained as that obtained using canonical and microcanonical ensembles.

7.3 **Exam paper - February 2009**. Consider an ideal gas that consists of identical magnetic particles of mass m. This gas, which is at temperature T and occupies a volume V, is under the action of a uniform magnetic field $\vec{B}$. The energy of each particle is the sum of two contributions: one is due to translational motion, which can be treated within the classical formalism, and the other to the interaction between the intrinsic dipolar moment and the applied field:

$$\epsilon = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + \Delta_B \tag{7.66}$$

where Δ_B is the magnetic energy that can assume three different states depending on the orientation of the particle dipolar moment, $\vec{\mu_d}$, with respect to $\vec{B}$:

$$\Delta_B = \begin{cases} -\mu_d B & \text{if } \vec{\mu_d} \text{ and } \vec{B} \text{ are parallel} \\ 0 & \text{if } \vec{\mu_d} \text{ is not oriented} \\ +\mu_d B & \text{if } \vec{\mu_d} \text{ and } \vec{B} \text{ are antiparallel} \end{cases} \tag{7.67}$$

Under these conditions, determine:

(a) The system canonical partition function Z as a function of N, V and T;

(b) The fraction of particles that are found in each of the three above-mentioned magnetic states;

(c) The system internal energy U as a function of N, V and T;

(d) The macrocanonical partition function, Ξ as a function of T, V, and $\lambda = \exp(\beta\mu)$, with μ the chemical potential.

(e) The internal energy, this time within the macrocanonical formalism, as a function of N, V and T. Show that the result is the same as that obtained *via* the canonical formalism.

7.4 Consider a mixture of two classical ideal gases with N particles ($i = 1, 2$). The only difference between the gases is the mass of the particles (m_1 and m_2, respectively). The equilibrium mixture is at pressure P and temperature T, occupying volume V.

(a) Calculate the canonical partition function.

(b) Compare the result with the case where the two gases are in separate containers at the same pressure and temperature.

(c) Calculate the macrocanonical partition function for the mixture.

(d) Find the expression for the entropy of the mixture (defined as the difference between the two situations described above) in terms of the molar fractions $x_i = N_i/N$.

7.5 Consider a system of identical but distinguishable particles, each of which has two states, with energies ϵ and $-\epsilon$ available to it. Use the microcanonical, canonical and grand canonical ensembles to calculate the mean entropy per particle as a function of the mean energy per particle in the limit of a very large system. Verify that all three ensembles yield identical results in this limit[8].

[8] Problem taken from *Problems on Statistical Mechanics* by Dalvit, Frastai and Lawrie.

Appendices

Appendix A

A Brief Introduction to the Theory of Probability[1]

A.1 Introduction

First, we will introduce some basic concepts of probability theory. A set S containing all possible outcomes of a random experiment is called the *sample space*. If the sample space has a finite number of points, it is called a *finite sample space*. If it has as many points as the natural numbers, it is called a *countably infinite sample space*, and if it belongs to the real line, for example, it is called an *uncountably infinite sample space*. In general, a finite or countably infinite sample space is referred to as a discrete sample space, while an uncountably infinite one is called a non-discrete sample space.

An event is a subset A of the sample space S, that is, a set of possible outcomes. As specific events, we have the entire sample space S, which is called the certain event, and the empty set $\oslash$, which is called the impossible event since an element of $\oslash$ cannot occur.

By using set operations on events in S, we can obtain new events. For example, let A and B be events, then:

1. $A \cup B$ is the event "A or B," which is called the union.

2. $A \cap B$ is the event "A and B," which is called the intersection.

3. A' is the event "not A," which is called the complement or negation of A.

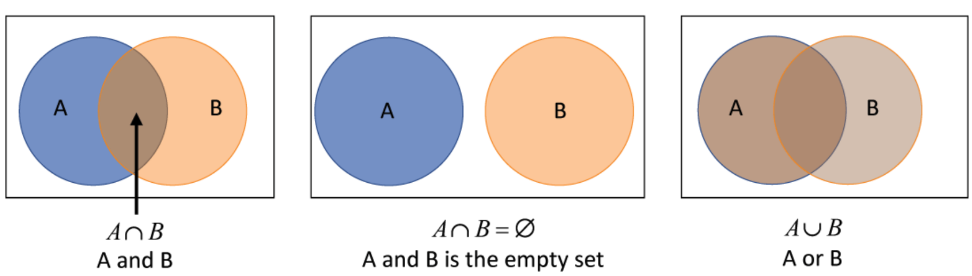


Figure A.1: Sample space and subsets.

[1]The content of this appendix has been adapted from the notes kindly shared by Dr Pablo Villegas.

A.2 The Concept of Probability

The first concept we need to introduce is that of probability, which we can define as a quantitative measure of the likelihood or chance that a certain event will occur. We will explore how one can formalise these concepts into a mathematical theory. To this end, there are two important procedures by which we can calculate the probability of an event, and they are based on the following two definitions.

A.2.1 Definitions

Definition 2.1. Classical: If an event can occur in h different ways out of a total of n possible ways, all of which are equally likely, then the probability of the event is h/n.

Definition 2.2. Frequentist: If after n repetitions of an experiment, where n is very large, the event occurs h times, then the probability of that event is h/n. This is also referred to as the empirical probability of an event.

Both approaches have drawbacks; the first one because "equally likely" is a vague definition, and the second one because a "large number" is also an ambiguous concept. Due to these difficulties, the axiomatic approach to probability is proposed.

A.2.2 Kolmogorov's Axioms

Axiom 1. The probability of the entire sample space S is non-negative and less than or equal to 1. Mathematically, this can be expressed as

$$0 \leq p(S) \leq 1 \tag{A.1}$$

Axiom 2. The probability of the certain event S is equal to 1, denoted symbolically by

$$P(S) = 1 \tag{A.2}$$

Axiom 3. If $A_1, A_2, \ldots$ are mutually exclusive (pairwise disjoint) events, then:

$$P(A_1 \cup A_2 \cup \ldots) = \sum_i P(A_i) \tag{A.3}$$

And for the particular case of two events:

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) \tag{A.4}$$

A.2.3 Important Theorems on Probability

Theorem 2.1

For any event A, the following inequality holds:

$$0 \leq P(A) \leq 1 \tag{A.5}$$

This means that the probability $P(A)$ is bounded between 0 and 1.

Theorem 2.2

The impossible event has a probability of zero, $P(\emptyset) = 0$.

Theorem 2.3

If A' is the complement of A, then

$$P(A') = 1 - P(A) \tag{A.6}$$

Theorem 2.4

If A and B are two arbitrary events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \tag{A.7}$$

Theorem 2.5

For any events A and B,

$$P(A) = P(A \cap B) + P(A \cap B') \tag{A.8}$$

A.2.4 Conditional Probability

Consider two events, A and B, where $P(A) > 0$. Let's delve into $P(A|B)$, which is the probability of event A occurring, given that event B has already occurred. We denote this as $P(A|B)$, and it's read as "the probability of A given B." It's crucial to note that there doesn't have to be a direct cause-and-effect or time-related connection between A and B. A can happen before, after, or simultaneously with B. Additionally, A can influence B, B can influence A, or there might be no causal connection at all. Causality and temporal relations fall outside the scope of probability. Whether they matter or not depends on how we interpret the events. Let's illustrate this with a classic example: tossing a coin followed by rolling a die. What's the probability of getting a 6 on the die, given that a head has already appeared on the coin? This probability is denoted as $P(6|C)$.

In a nutshell, within a probability framework and with two events A and B (or outcomes) where $P(B) > 0$, the conditional probability of A given B is defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{A.9}$$

Alternatively, we can express it as:

$$P(A \cap B) = P(B) \cdot P(A|B) = P(A) \cdot P(B|A) \tag{A.10}$$

A.2.5 Combinatorics

In many cases, the number of sample points in a sample space is not very large, making it feasible to directly enumerate or count the sample points needed to calculate probabilities. However, when it becomes impossible to perform this process, combinatorial analysis comes into play.

Example 2.1

Let's consider tossing a coin three times. It's immediately apparent how different outcomes of heads and tails can occur when repeating the experiment.

Combinatorics deals with the number of different ways to consider sets formed from elements of a given set, while adhering to specific rules such as size, order, repetition, or partition. Therefore, a combinatorial problem typically involves establishing a rule about how the groupings should be and determining how many of them meet that rule.

Permutations

A significant type of arrangement is known as permutations. Let's assume we have n distinct objects, and we want to arrange these objects in a sequence. Since there are n ways to choose the first element, and once that's chosen, we have $(n-1)$ ways to choose the second element, and so on, when we reach the k-th element, we have only $[n-(k-1)]$ possible elements to choose from. This leads us to the fact that we have $n(n-1)(n-2)\ldots 2\cdot 1$ ways to arrange the set. Therefore, given a finite set A of n elements, the number of all its permutations is equal to $n!$, where:

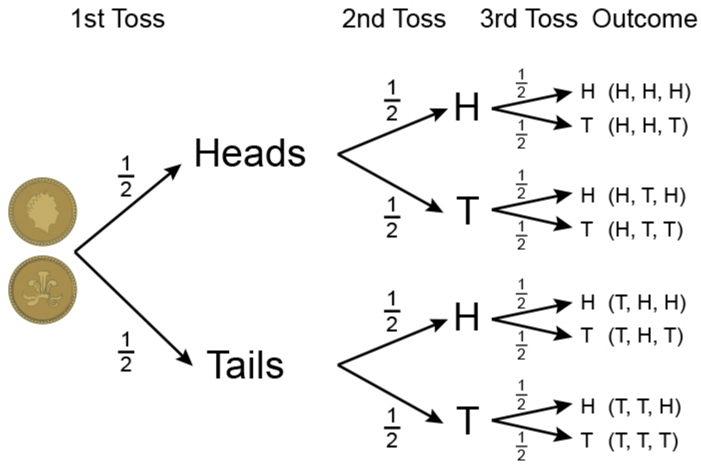


Figure A.2: Tree diagram illustrating the possibilities of getting 'head' and 'tail' successively when tossing a coin.

$$n! = n(n-1)(n-2)\dots 1 \tag{A.11}$$

Now, let's consider that we have n distinct objects, and we want to arrange k of these objects in a sequence. Since there are n ways to choose the first object, $(n-1)$ ways to choose the second, and $(n-k+1)$ ways to choose the k-th object, it's easy to deduce that the number of different permutations we can create is given by:

$$n(n-1)(n-2)\dots(n-k+1) = \frac{n!}{(n-k)!} \tag{A.12}$$

Combinations

In a permutation, the order of objects is significant. For example, "abc" is a different permutation from "bca". However, in many situations, we are only concerned with selecting objects without regard to their order. These selections are known as combinations. For instance, "abc" and "bca" represent the same combination. We have already introduced these concepts in Chapter 1. The number of ways to select k elements from a set of n, or combinations of n taken k at a time, is denoted as $C_{(n,k)}$, C_k^n, or $\binom{n}{k}$ and is calculated as follows:

$$\binom{n}{k} = C_{(n,k)} = \frac{n!}{k!(n-k)!} \tag{A.13}$$

These concepts find applications in various areas of physics, including statistical mechanics, quantum mechanics, and particle physics, where the arrangement and selection of particles or states are fundamental to understanding physical phenomena. In summary, permutations and combinations play a crucial role in solving problems related to arranging or selecting objects, making them invaluable tools for physicists and researchers alike.

A.3 Random Variables and Probability Distribution

In the realm of statistical physics, let's explore the concept of random variables. Imagine that we assign a number to each point in the sample space. We now have a function defined across this sample space, and this function is known as a random variable (or stochastic variable). Typically, we represent it using an uppercase letter, such as X or Y.

In general, a random variable holds physical, geometric, or other meaningful significance within a given context. It serves as a bridge between the theoretical framework and real-world observations. A random variable that can assume either a finite set or a countably infinite set of values is termed a *discrete random variable*. On the other hand, a random variable that can take an uncountably infinite number of values is referred to as a *continuous random variable*. The distinction between discrete and continuous random variables is fundamental in statistical physics, as it reflects the underlying nature of the phenomena being studied. Understanding these variables is essential for modelling and analysing complex systems in the field of physics.

In summary, random variables, denoted by symbols like X or Y, are central to statistical physics. They act as mathematical representations of physical or geometric properties, helping us make sense of the often intricate and unpredictable behaviour of physical systems.

A.3.1 Discrete Distributions

Consider a discrete random variable X. Let's assume that the possible values it can take are given as $x_1, x_2, x_3, \ldots$ in a specific order. Additionally, let's assume that these values are associated with probabilities given by:

$$P(X = x_k) = f(x_k), \quad k = 1, 2, \ldots \tag{A.14}$$

In this context, $f(x)$ is defined as the probability function or probability distribution. It's expressed as:

$$P(X = x) = f(x) \tag{A.15}$$

In general, we designate $f(x)$ as a probability function if it satisfies two fundamental conditions:

1. $f(x) \geq 0$ for all values of x. This ensures that the probability of any outcome is non-negative.

2. The sum of the probabilities for all possible values of X must equal 1, which is expressed as:

$$\sum_k f(x_k) = 1 \tag{A.16}$$

This condition ensures that the total probability across all possible outcomes of the random variable X is equal to 1.

In statistical physics and probability theory, probability functions like $f(x)$ play a crucial role in describing the likelihood of different events or states. They are essential tools for analysing and understanding the behaviour of discrete random variables in various physical systems.

Example 3.1

Calculate the probability function corresponding to the random variable, the number of heads (X), that we can obtain when tossing a coin two times.
Assuming that getting heads (H) and tails (T) are equiprobable, we have:

$$\begin{aligned} P(HH) &= \frac{1}{4} \\ P(HT) &= \frac{1}{4} \\ P(TH) &= \frac{1}{4} \\ P(TT) &= \frac{1}{4} \end{aligned} \tag{A.17}$$

So, the probability distribution is as follows:

$$
\begin{aligned}
P(X=0) &= P(HH) = \frac{1}{4} \\
P(X=1) &= P(HT \cup TH) = P(HT) + P(TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \\
P(X=2) &= P(TT) = \frac{1}{4}
\end{aligned}
\tag{A.18}
$$

Hence, the probability distribution is given in the following table:

x	$f(x)$
0	1/4
1	1/2
2	1/4

(A.19)

Cumulative Distribution Function

The probability distribution should not be confused with the cumulative distribution function or distribution function, which is defined for a random variable X as:

$$F(x) = P(X \le x) \tag{A.20}$$

where x can be any real number, $x \in (-\infty, \infty)$. The cumulative distribution function $F(x)$ has the following properties:

1. $F(x)$ is increasing.
2. $\lim_{x\to-\infty} F(x) = 0$; $\lim_{x\to\infty} F(x) = 1$
3. $F(x)$ is continuous from the right, i.e., $\lim_{h\to 0^+} F(x+h) = F(x)$ for all x.

Discrete Random Variables

In the case of a discrete random variable, the cumulative distribution function for a random variable X can be obtained from its probability function as follows:

$$F(x) = \sum_{u \le x} f(u) \tag{A.21}$$

where the summation includes all values u taken by X such that $u \le x$.

Example 3.2

Consider the discrete random variable from the previous example (the number of heads (X) obtained when tossing a coin two times). The cumulative distribution function is defined as follows:

$$F(x) = \begin{cases} 0 & \text{if } -\infty < x < 0 \\ 1/4 & \text{if } 0 \le x < 1 \\ 3/4 & \text{if } 1 \le x < 2 \\ 1 & \text{if } 2 \le x < \infty \end{cases} \tag{A.22}$$

And its graph is as follows:

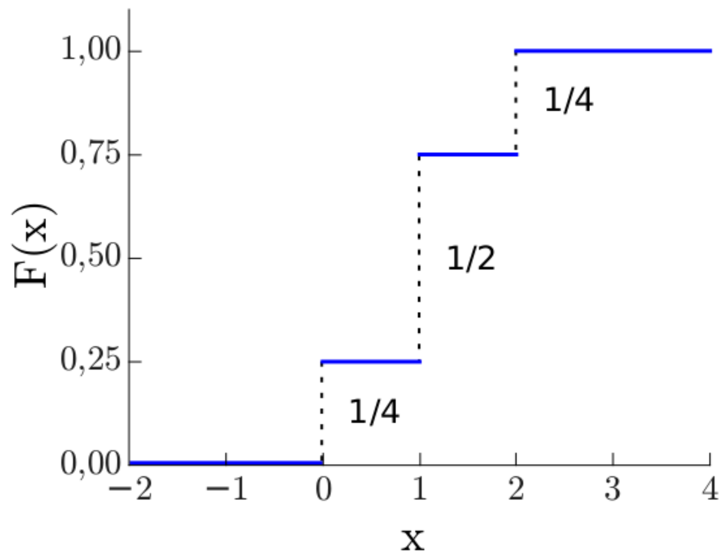


Figure A.3: Cumulative Distribution Function for X.

A.3.2 Continuous Distributions

In the realm of probability and statistics, we encounter a distinct concept when dealing with continuous random variables, denoted as X. Unlike discrete variables, it doesn't make sense to define a probability function $P(X = x)$ for every specific value of x, as the probability of a precise value is infinitesimal, $P(X = x) = 0$, for any x. However, we introduce an essential concept known as the probability distribution function, or more commonly, the cumulative distribution function (CDF), denoted as $F(x)$. The primary difference here is that this function is continuous, unlike the step-like functions we see in discrete cases. Therefore, we say that a non-discrete random variable X is purely continuous if its cumulative distribution function can be represented as:

$$f(x) = \frac{dF}{dx} \tag{A.23}$$

This means that $f(x)$ is the derivative of $F(x)$ with respect to x. The cumulative distribution function $F(x)$ signifies the probability that the random variable X takes on a value less than or equal to x:

$$F(x) = P(X \leq x) \tag{A.24}$$

A continuous random variable X must adhere to the following properties:

1. Non-negativity: $f(x) \geq 0$ for all x.
2. Total Probability: The integral of $f(x)$ over the entire real line is equal to 1:

$$\int_{-\infty}^{\infty} f(x)dx = 1 \tag{A.25}$$

3. Cumulative Distribution: The probability that X is less than x is given by the integral of $f(u)$ from $-\infty$ to x:

$$\int_{-\infty}^{x} f(u)du = F(x) \tag{A.26}$$

4. Probability within an Interval: The probability that X lies within the interval $(a < X < b)$ is calculated as:

$$P(a < X < b) = \int_a^b f(x)dx \tag{A.27}$$

These properties collectively define continuous random variables and enable us to work with probability distributions in a continuous setting, which is crucial in various areas of statistics and physics.

Measures of Central Tendency

We can utilise the probability distribution function to compute certain measures of location and dispersion for a variable. Firstly, let's define the mode as the most probable value or the one that occurs with the highest frequency in our probability distribution. In cases where multiple values have equal probabilities, a distribution can be bimodal, trimodal, or multimodal. On the other hand, we define the median of a continuous variable X as the point $x_{0.5}$ such that $F(x_{0.5}) = 0.5$. In other words, it's the value x for which we reach 50% of the cumulative probability. More generally, we can define the percentile as the point x_p for which $F(x_p) = p$. In particular, the first quartile is $Q_1 = x_{0.25}$, and the third quartile is $Q_3 = x_{0.75}$. What about discrete variables?

In the case of discrete variables, this definition of the median isn't appropriate because it's possible that there's no value x such that $F(x) = 0.5$. Therefore, we employ the following definition for any percentile, which should satisfy:

$$P(X \leq x) \geq p \tag{A.28}$$

$$P(X \geq x) \geq p \tag{A.29}$$

And particularly for $p = 0.5$:

$$P(X \leq x) \geq 0.5 \tag{A.30}$$

$$P(X \geq x) \geq 0.5 \tag{A.31}$$

These definitions allow us to calculate central tendency measures for both continuous and discrete variables.

Graphical Interpretation

If $f(x)$ is the probability density function for a random variable X, we can represent it graphically. Since $f(x) \geq 0$, the curve cannot fall below the x-axis. Furthermore, the entire area bounded by the curve must be equal to 1, ensuring that the probability distribution is correctly normalized. It's easy to see that the probability of X lying between a and b is represented by the integral under the curve. On the other hand, the function $F(x)$ is monotonically increasing and grows from 0 to 1. To graphically represent this concept, consider the following:

$$\int_a^b p(x)dx \tag{A.32}$$

This integral represents the probability of X falling within the interval $[a, b]$, and it corresponds to the area under the probability density function curve between a and b. Additionally, the cumulative distribution function $F(x)$ satisfies the following:

$$0 \leq F(x) \leq 1, \tag{A.33}$$

where $F(x)$ starts at 0 and monotonically increases to 1, reflecting the cumulative probability of X being less than or equal to x.To visually understand these concepts, you can plot the probability density function $p(x)$ and cumulative distribution function $F(x)$ on a graph as shown in Fig. A.4

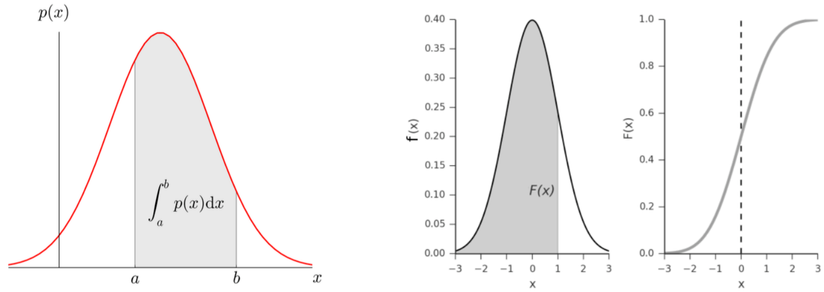


Figure A.4: Probability in an Interval $[a, b]$ (left); Calculation of the Cumulative Distribution Function (right).

A.3.3 Change of Variables

Given the probability distributions of one or more random variables, it can be of interest to find the distributions of other random variables that depend on them.

Discrete Variables

Theorem 3.1

Let X be a discrete random variable with a probability function $f(x)$. Suppose the discrete random variable U is defined in terms of X as $U = \phi(X)$, where for each value of X, there corresponds one and only one value of U, and vice versa, such that $X = \psi(U)$. Then, the probability function of U is given by:

$$g(u) = f[\psi(u)] \tag{A.34}$$

Continuous Variables

Theorem 3.2

Let X be a continuous random variable with a probability density function $f(x)$. Let's define $U = \phi(X)$, where $X = \psi(U)$. Then, the probability density function of U is given by $g(u)$ where:

$$g(u)|du| = f(x)|dx| \tag{A.35}$$

Or equivalently,

$$g(u) = f(x)\frac{|dx|}{|du|} = f[\psi(u)]|\psi'(u)| \tag{A.36}$$

Theorem 3.3 - Transformation of Joint Probability Distributions

Let X and Y be continuous random variables with a joint probability density function *f(x, y)*. Define $U = \phi_1(X, Y)$ and $V = \phi_2(X, Y)$, where $X = \psi_1(U, V)$ and $Y = \psi_2(U, V)$. Then, the joint probability density function of U and V is given by *g(u, v)*, where:

$$g(u, v)|dudv| = f(x, y)|dxdy| \tag{A.37}$$

Or equivalently,

$$g(u, v) = f(x, y)\left|\frac{\partial(x, y)}{\partial(u, v)}\right| = f[\psi_1(u, v), \psi_2(u, v)]|J| \tag{A.38}$$

Where J is the Jacobian determinant [2] and is given by:

[2] Change of variables in probability distributions is significant, and in the case of continuous variables, it involves the Jacobian of the transformation.

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial v} & \frac{\partial y}{\partial u} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \tag{A.39}$$

A.4 Moments and Cumulants

Once probability distributions have been introduced, the next step is to characterise them. So far, we have seen measures of centrality for a variable (mode and median). Another measure of centrality is the mean.

Let's consider generating a sample from a discrete variable with a probability function $P(\cdot)$. In this case, the sample mean is given by:

$$\bar{x} = \sum_i x_i P(X = x_i) \tag{A.40}$$

where $P(X = x_i)$ is the probability that our stochastic variable takes a certain value x_i. It's easy to see that for continuous distributions, we simply replace the sum with an integral:

$$\bar{x} = \int x f(x) dx \tag{A.41}$$

The mean is commonly denoted as $E[X]$ or μ. In the same way that we defined the mean, we can define moments of higher order. Formally, for $k \in \mathbb{N}$, the k-th moment is $E[X^k]$ [3]. Analogously to the previous definition, it is calculated as follows:

$$\langle x^k \rangle = \sum_i x_i^k P(X = x_i) \tag{A.42}$$

for the discrete case. By contrast, for the continuous case, it reads

$$\langle x^k \rangle = \int x^k f(x) dx \tag{A.43}$$

Some moments have special names. The first moment is the mean, the second moment is related to variance σ^2, the third moment is related to skewness (indicating asymmetry), and the fourth moment is related to kurtosis (analyzing the concentration of values around the mean). Finally, the k-th central moment is defined as $E[(X - \mu)^k]$, where μ is the mean.

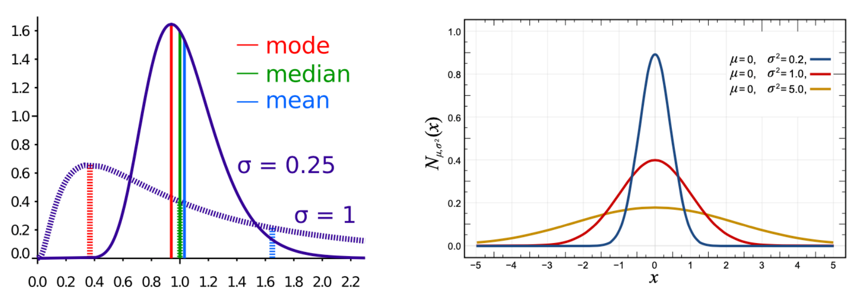


Figure A.5: Comparison of mean, median, and mode for two log-normal distributions with different variance (left). Normal distribution with different variance values (right).

[3]Indistinctly, from now on, we will use both E and $\langle\cdot\rangle$ to refer to any moment of a probability distribution. During the course, we will adopt $\langle\cdot\rangle$ to avoid confusion with the energy of a system.

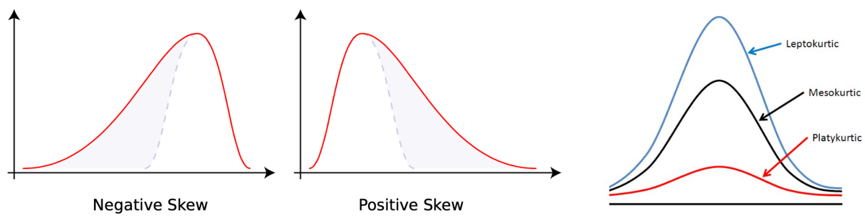


Figure A.6: Negative and Positive Skewness (left). Kurtosis (right).

A.4.1 Moment Generating Function

The moment generating function of a random variable X is defined as:

$$M_X(t) := \mathbb{E}[e^{tX}], \quad t \in \mathbb{R} \tag{A.44}$$

provided that this expectation exists. Here, $\mathbb{E}$ represents the expectation operator, and t is a real-valued parameter. This function provides a unique way to capture the moments (average values) of a probability distribution by using the exponential function. In the discrete case, when X is a discrete random variable taking values x_i with corresponding probabilities $P(X = x_i)$, the moment generating function can be expressed as

$$M_X(t) = \sum_{i=1}^{\infty} e^{tx_i} P(X = x_i) \tag{A.45}$$

This equation sums the contributions of each possible value x_i of X weighted by their respective probabilities. In the continuous case, when X is a continuous random variable with probability density function $f(x)$, the moment generating function is given by the following integral:

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x)dx \tag{A.46}$$

The moment generating function is a powerful tool in statistics, as it allows us to calculate moments of any order and provides valuable insights into the shape and characteristics of a probability distribution. The term "moment-generating function" derives its name from its remarkable property. If it exists in a neighborhood of $t = 0$, it enables the generation of moments of a probability distribution:

$$\mathbb{E}(X^n) = M_X^{(n)}(0) = \left.\frac{d^n M_X}{dt^n}\right|_{t=0} \tag{A.47}$$

Remarkably, if the moment generating function is defined in such an interval, it uniquely determines the probability distribution associated with it. This property makes the moment-generating function a powerful tool for statistical analysis.

A.4.2 Cumulant Generating Function

A key issue with moment generating functions is that moments and the moment generating function itself do not always exist because the defining integrals may not always converge. In contrast, the characteristic function always exists and can be used as an alternative. The characteristic function of a random variable or its probability distribution is a real-valued function that takes complex values. It enables the application of analytical methods (i.e., functional analysis) in probability studies.

For a continuous random variable X:

$$\phi_X : \mathbb{R} \to \mathbb{C}; \quad \phi_X(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} f_X(x)dx \tag{A.48}$$

Similarly, for a discrete random variable:

$$\phi_X(t) = \sum_{i=1}^{\infty} e^{itx_i} P(X = x_i) \tag{A.49}$$

When moments of a random variable exist, they can be calculated using derivatives of the characteristic function:

$$\phi_X^{(n)}(0) = i^n \mathbb{E}[X^n] \tag{A.50}$$

Cumulants (κ_n) can be defined based on the natural logarithm of the characteristic function:

$$H(t) = \log \mathbb{E}\left[e^{itX}\right] = \sum_{n=1}^{\infty} \kappa_n \frac{(it)^n}{n!} = \mu it - \sigma^2 \frac{t^2}{2} + ... \tag{A.51}$$

where $H(t)$ is referred to as the second characteristic function. In statistical physics, many extensive quantities (proportional to the size or volume of the system) are related to cumulants of a random variable. As you'll learn during the course, the canonical partition function is determined by $Z(\beta) = e^{-\beta E}$, where E represents the system's energy, and $\beta = \frac{1}{kT}$ with k being Boltzmann's constant and T the temperature. The Helmholtz free energy is given by $F = -\beta^{-1} \log Z$, and it is closely related to the cumulant generating function for energy. From this expression, various thermodynamic properties of the system, such as internal energy, entropy, or specific heat, can be derived.

A.5 Special Distributions

A.5.1 Binomial Distribution

Let's imagine we're conducting an experiment, such as repeatedly flipping a coin, rolling a die, or drawing marbles from an urn (where we can look for success in a specific event, *e.g.*, getting heads or drawing a 4). Let p be the probability of success in an experiment of this kind. Then $q = 1 - p$ is the probability of failure. The probability that the event occurs exactly x times in n trials (i.e., n successes and $n - x$ failures) is given by the probability mass function:

$$P(X = x) = \binom{n}{x} p^x q^{n-x} = \frac{n!}{x!(n-x)!} p^x q^{n-x} \tag{A.52}$$

where the random variable X denotes the number of successes in n trials, and $x = 0, 1, \ldots, n$.

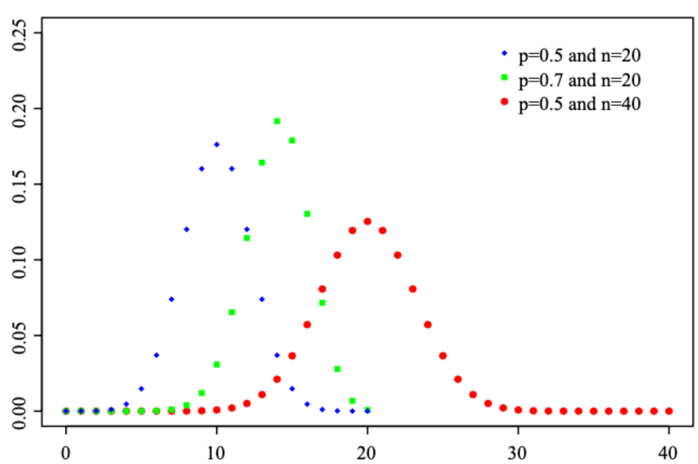


Figure A.7: Probability distribution function for different values of p and n.

Property	Value
Mean (μ)	np
Variance (σ^2)	npq
Skewness	$\frac{q-p}{\sqrt{npq}}$
Kurtosis	$\frac{1-6pq}{npq}$
Moment Generating Function	$q + pe^t n$
Characteristic Function	$q + pe^{it} n$

Table A.1: Properties of the binomial distribution.

A.5.2 Poisson Distribution

Let X be a discrete random variable that can take values 0, 1, 2, ... such that the probability function of X is given by

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!} \quad \text{for } x = 0, 1, 2, \ldots \tag{A.53}$$

where λ is a positive constant. This distribution is called the Poisson distribution.

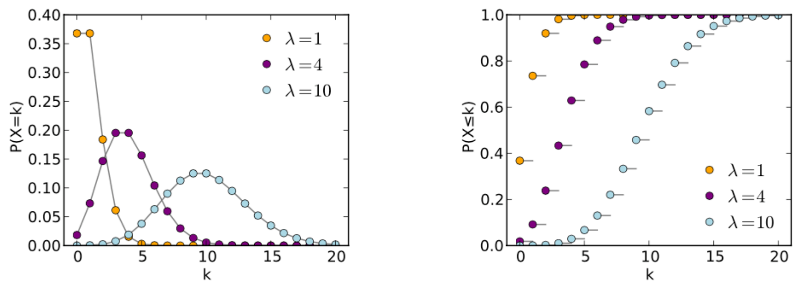


Figure A.8: Poisson distribution function for different values of λ (left) and cumulative distribution function (right).

Some important properties of this distribution are presented in Table 2.2. If we consider a binomial distribution with a large n and a very small occurrence probability p, such that $q = 1 - p$ approaches 1, the event is termed rare. For such cases, the binomial distribution closely approximates the Poisson distribution with $\lambda = np$.

Property	Value
Mean (μ)	λ
Variance (σ^2)	λ
Skewness	$\sqrt{\frac{1}{\lambda}}$
Kurtosis	$3 + \frac{1}{\lambda}$
Moment Generating Function	$M(t) = e^{\lambda(e^t - 1)}$
Characteristic Function	$\phi(t) = e^{\lambda(e^{it} - 1)}$

Table A.2: Properties of the Poisson distribution.

A.5.3 Normal Distribution

One of the most important examples of a continuous probability distribution is the normal or Gaussian distribution. The probability density function for this distribution is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{A.54}$$

where μ is the mean, and σ^2 the variance. Some properties of the normal distribution are presented in the following table:

Property	**Formula**
Mean (μ)	μ
Variance (σ^2)	σ^2
Skewness	0
Kurtosis	0
Moment Generating Function	$M(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$
Characteristic Function	$\phi(t) = e^{i\mu t - \frac{1}{2}\sigma^2 t^2}$

Table A.3: Properties of the Gaussian distribution.

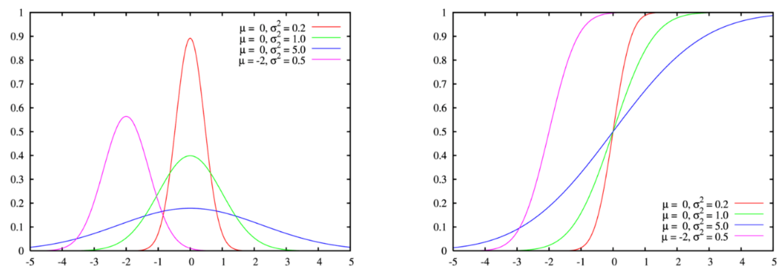


Figure A.9: Gaussian distribution function for different values of λ (left) and cumulative distribution function (right)

A.5.4 Law of Large Numbers

The strong law of large numbers states that if $X_1, X_2, X_3, \ldots$ is an infinite sequence of independent random variables, each with a finite expected value μ, then the sample average

$$\overline{X}_n = \frac{X_1 + X_2 + \ldots + X_n}{n} \tag{A.55}$$

converges in probability to μ. In other words,

$$P\left(\lim_{n\to\infty} \overline{X}_n = \mu\right) = 1 \tag{A.56}$$

A.5.5 Central Limit Theorem

Central Limit Theorem

Let $S_n := X_1 + X_2 + \ldots + X_n$, and assume that $X_1, X_2, \ldots$ is a sequence of independent and identically distributed random variables with mean μ and finite variance $\sigma^2 < \infty$. When the condition $n \to \infty$ is met, the standardized random variable $\frac{\sqrt{n}(S_n - n\mu)}{\sqrt{\sigma^2}}$ converges to a standard normal distribution $N(0,1)$:

$$\lim_{n\to\infty}\left(\frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n\sigma^2}}\right) \to N(0,1). \tag{A.57}$$

Proof. Consider a sequence of identically distributed random variables $X_1, X_2, \ldots, X_n$, each with mean μ and variance σ^2. It's immediate that the sum $X_1 + X_2 + \ldots + X_n$ has a mean of $n\mu$ and a variance of $n\sigma^2$. Let's define the following standardized variable:

$$Z_n = \frac{\sum_{i=1}^{n}(X_i - \mu)}{\sqrt{n\sigma^2}} = \sum_{i=1}^{n} \frac{X_i - \mu}{\sqrt{n\sigma^2}} = \sum_{i=1}^{n} \frac{1}{\sqrt{n}} Y_i, \tag{A.58}$$

where we've introduced the variable $Y_i = \frac{X_i - \mu}{\sigma}$, which has a mean of 0 and a variance of 1. The characteristic function of Y_n is given by:

$$\varphi_{Y_n}(t) = E[e^{itY_n}] = \prod_{i=1}^{n} \varphi_{Y_i}\left(\frac{t}{\sqrt{n}}\right). \tag{A.59}$$

Since Y_i's are independent and identical, we can simplify this to:

$$\varphi_{Y_n}(t) = \left[\varphi_{Y_1}\left(\frac{t}{\sqrt{n}}\right)\right]^n. \tag{A.60}$$

Now, using the Taylor expansion $e^x = \lim_{n\to\infty}(1 + x/n)^n$ and taking the limit:

$$\begin{aligned} \varphi_{Y_1}(t) &= \lim_{n\to\infty}\left(1 + \frac{it}{\sqrt{n}} - \frac{t^2}{2n}\right)^n \\ &= \lim_{n\to\infty}\left(1 + \frac{it}{\sqrt{n}} - \frac{t^2}{2n}\right)^n e^{-t^2/2} \\ &= e^{-t^2/2}, \end{aligned}$$

where we have used the fact that $1/n \to 0$ as $n \to \infty$. Thus, $\varphi_{Y_n}(t) \to e^{-t^2/2}$, which is the characteristic function of the standard normal distribution $N(0,1)$. Therefore, as n approaches infinity, the distribution of Z_n converges to a standard normal distribution $N(0,1)$. Consequently, the sum $\sum_{i=1}^{n} X_i$ converges to a normal distribution $N(n\mu, n\sigma^2)$.

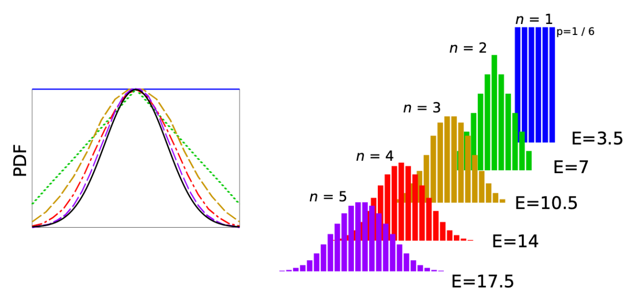


Figure A.10: The evolution of the probability distribution of the sum of dice is depicted, illustrating its convergence to a normal distribution, in accordance with the Central Limit Theorem. In the left figure, a comparison with a normal distribution (black curve) is shown.

Appendix B

Derivation of the Euler-Lagrange equation

We begin with the Principle of Least Action, which states that the path taken by a physical system between two points in configuration space is the one for which the action functional is minimized. The action S is defined as the integral of the Lagrangian L over time:

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt \tag{B.1}$$

where q represents the generalized coordinates, $\dot{q}$ represents their time derivatives, and t is time itself. The endpoints of this integral correspond to the initial time t_1 and the final time t_2.

To find the path that minimizes the action, we use the principle of stationary action, which states that the true path of the system is the one for which the action is stationary, i.e., its variation is zero:

$$\delta S = 0 \tag{B.2}$$

To compute this variation, we introduce small perturbations δq around the actual path q:

$$q(t) \to q(t) + \delta q(t) \tag{B.3}$$

where $\delta q(t_1) = \delta q(t_2) = 0$ since we fix the endpoints. The perturbed action becomes:

$$\delta S \approx \int_{t_1}^{t_2} \left[L(q + \delta q, \dot{q} + \delta \dot{q}, t) - L(q, \dot{q}, t) \right] dt \tag{B.4}$$

Expanding this expression to first order in δq and $\delta \dot{q}$, we have:

$$\delta S \approx \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right] dt \tag{B.5}$$

$$\delta S \approx \left[\frac{\partial L}{\partial q} \delta q \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q\, dt \tag{B.6}$$

Since $\delta q(t_1) = \delta q(t_2) = 0$, the boundary terms vanish:

$$\delta S \approx - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q\, dt \tag{B.7}$$

For the action to be stationary, δS must be zero for arbitrary variations δq. This leads to the Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0 \tag{B.8}$$

To obtain Hamilton's equations, we introduce the generalized momenta p_i, which are defined as:

$$p_i = \frac{\partial L}{\partial \dot{q}_i} \tag{B.9}$$

Now, we can rewrite the Euler-Lagrange equations in terms of q_i, p_i, and the Hamiltonian H:

$$\frac{d}{dt}p_i - \frac{\partial L}{\partial q_i} = 0 \tag{B.10}$$

$$\dot{q}_i \frac{d}{dt}p_i - \dot{q}_i \frac{\partial L}{\partial q_i} = 0 \tag{B.11}$$

$$\frac{dH}{dt} = \sum_i \left(\dot{q}_i \frac{dp_i}{dt} + p_i \frac{d\dot{q}_i}{dt} \right) - \frac{dL}{dt} \tag{B.12}$$

Using the Euler-Lagrange equations ($\frac{d}{dt}p_i - \frac{\partial L}{\partial q_i} = 0$) and the fact that $L = L(q_i, \dot{q}_i, t)$, we can simplify this expression to:

$$\frac{dH}{dt} = -\frac{dL}{dt} \tag{B.13}$$

Now, we express $\frac{dL}{dt}$ using the chain rule:

$$\frac{dL}{dt} = \sum_i \left(\frac{\partial L}{\partial q_i}\frac{dq_i}{dt} + \frac{\partial L}{\partial \dot{q}_i}\frac{d\dot{q}_i}{dt} + \frac{\partial L}{\partial t} \right) \tag{B.14}$$

Since the Euler-Lagrange equations imply that $\frac{d}{dt}p_i - \frac{\partial L}{\partial q_i} = 0$, we have $\frac{\partial L}{\partial q_i} = \frac{d}{dt}p_i$. Therefore, the sum in the expression for $\frac{dL}{dt}$ becomes:

$$\sum_i \left(\frac{d}{dt}p_i \frac{dq_i}{dt} + p_i \frac{d\dot{q}_i}{dt} + \frac{\partial L}{\partial t} \right) \tag{B.15}$$

Using the definition of momenta $p_i = \frac{\partial L}{\partial \dot{q}_i}$ and rearranging terms, we obtain:

$$\frac{dH}{dt} + \frac{\partial L}{\partial t} = 0 \tag{B.16}$$

If the Lagrangian L does not explicitly depend on time ($\frac{\partial L}{\partial t} = 0$), then we obtain Hamilton's equations:

$$\frac{dH}{dt} = 0 \quad (1) \tag{B.17}$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad (2) \tag{B.18}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} \quad (3) \tag{B.19}$$

These are Hamilton's equations, which describe the evolution of the generalized coordinates q_i and momenta p_i over time in terms of the Hamiltonian H.

Appendix C

Time reversibility of the Hamiltonian equations of motion

To verify the time reversibility of the Hamiltonian equations of motion, let's consider the standard Hamiltonian equations:

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i} \quad \text{and} \quad \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i} \tag{C.1}$$

Now, let's reverse the sign of all momenta (p_i) and the sign of time (t) in these equations:

$$\frac{dq_i}{dt} = -\frac{\partial H}{\partial(-p_i)} \quad \text{and} \quad \frac{d(-p_i)}{dt} = \frac{\partial H}{\partial q_i} \tag{C.2}$$

Simplifying, we obtain:

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i} \quad \text{and} \quad \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i} \tag{C.3}$$

These are the same original Hamiltonian equations. Therefore, we have demonstrated that by reversing the sign of all momenta and the sign of time in the Hamiltonian equations, the resulting equations are identical to the original ones. This confirms the time reversibility of the Hamiltonian equations of motion.

Appendix D

Inner product between states

In quantum mechanics, the inner product (also known as the scalar product or dot product) between two quantum states $|\psi_1\rangle$ and $|\psi_2\rangle$ is a fundamental mathematical operation. It is denoted by $\langle\psi_1|\psi_2\rangle$, where:

- $\langle$ represents the bra (conjugate transpose) of $|\psi_1\rangle$[1].
- $|\psi_1\rangle$ and $|\psi_2\rangle$ are the quantum states.

The inner product is defined as follows:

$$\langle\psi_1|\psi_2\rangle = \int \psi_1^*(x)\psi_2(x)\, dx \tag{D.1}$$

Here's what each component means:

- $\psi_1^*(x)$ is the complex conjugate of the wave function $\psi_1(x)$ associated with $|\psi_1\rangle$.
- $\psi_2(x)$ is the wave function associated with $|\psi_2\rangle$.

The inner product yields a complex number, and it provides important information in quantum mechanics:

1. Probability: The squared magnitude of the inner product $|\langle\psi_1|\psi_2\rangle|^2$ gives the probability that, if the system is in state $|\psi_2\rangle$, it will be found in state $|\psi_1\rangle$ upon measurement.
2. Orthogonality: If $\langle\psi_1|\psi_2\rangle = 0$, it indicates that the two states $|\psi_1\rangle$ and $|\psi_2\rangle$ are orthogonal (perpendicular) to each other. In this case, they are mutually exclusive in terms of certain properties.

[1] In quantum mechanics, the "bra" is the conjugate transpose (also known as the Hermitian conjugate or adjoint) of a quantum state represented as a ket. Let's break down what each term means:

1. **Ket ($|\psi\rangle$)**: A ket is a notation used to represent a quantum state in Dirac notation. It typically appears as $|\psi\rangle$, where $|\psi\rangle$ is the state vector. It represents the state of a quantum system.
2. **Bra ($\langle\psi|$)**: The bra is essentially the conjugate transpose of the ket. It is written as $\langle\psi|$, where $\langle\psi|$ is the conjugate transpose of $|\psi\rangle$. It represents the dual or "adjoint" state associated with the ket.

The bra is obtained by taking the complex conjugate (changing the sign of the imaginary part) of the elements in the ket and then transposing the resulting row vector into a column vector. Mathematically, if $|\psi\rangle$ is represented as a column vector:

$$|\psi\rangle = \begin{bmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_n \end{bmatrix}$$

Then the bra $\langle\psi|$ is obtained as:

$$\langle\psi| = \begin{bmatrix} \psi_1^* & \psi_2^* & \dots & \psi_n^* \end{bmatrix}$$

Here, ψ_i represents the components of the ket $|\psi\rangle$, and ψ_i^* represents the complex conjugates of those components. The bra and ket together are used to compute inner products (scalar products) in quantum mechanics, which are essential for calculating probabilities, expectation values, and other quantum properties. The notation $\langle\psi|\psi\rangle$ represents the inner product of a state with itself, yielding a real, non-negative value.

3. Expectation Values: The inner product is used to calculate the expectation values of physical observables. For instance, the expectation value of an operator A in state $|\psi_1\rangle$ is given by $\langle\psi_1|A|\psi_1\rangle$.

4. Completeness: In the context of a complete set of quantum states, the inner product can be used to expand any state as a linear combination of basis states.

The inner product is a crucial mathematical tool in quantum mechanics, providing a way to quantify relationships between quantum states, calculate probabilities, and make predictions about measurement outcomes.

Appendix E

Proof of Liouville's Theorem

Let's consider a point in the phase space Γ at a given instant of time. This point corresponds to a generic microstate $(q_1, \ldots, q_s, p_1, \ldots, p_s)$. At a time $t + dt$, this point will evolve to a new microstate as coordinates and momenta change over time. To this end, we define a *velocity* vector $\vec{v}$ in Γ as follows

$$\vec{v} = (\dot{q}_1, \dot{q}_2, \ldots, \dot{q}_s, \dot{p}_1, \dot{p}_2, \ldots, \dot{p}_s) \tag{E.1}$$

Let's now take a given region of the phase space, which we call Γ_1 whose surface is Σ, as indicated in the left frame of Fig. E.1.

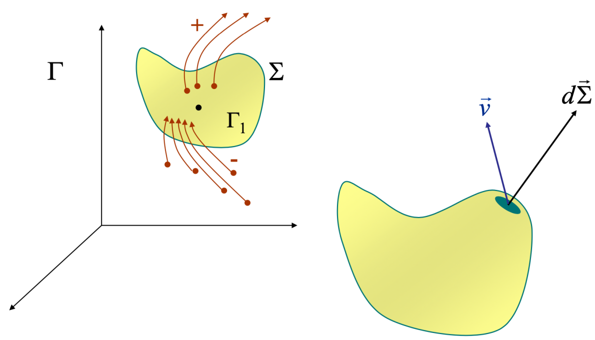


Figure E.1: ently long time.

The probability $P(t)$ that the system is found in Γ_1 at time t is given by

$$P(t) = \int_{\Gamma_1} \rho(q_1, \ldots, q_s, p_1, \ldots, p_s; t) dq_1, \ldots, dp_s, \tag{E.2}$$

which can be simplified to

$$P(t) = \int_{\Gamma_1} \rho(\mathbf{q}, \mathbf{p}; t)\, d\mathbf{q}\, d\mathbf{p} \tag{E.3}$$

At time $t + dt$, this probability will change as $\rho(\mathbf{q}, \mathbf{p}; t)$ changes over time. Because the points in the phase space are actually system's microstates that cannot appear or disappear, the only way $P(t)$ can change is that the microstates enter or exit region Γ_1 through its surface Σ. In other words, $P(t)$ changes over time because there is a **flux of probabilities** through Σ. The net flux is given by

$$\int_{\Sigma} \rho(\mathbf{q}, \mathbf{p}; t)\, \vec{v} \cdot d\vec{\Sigma} \tag{E.4}$$

If the flux is positive, microstates leaves Γ_1, if negative then they enter Γ_1 (see Fig. E.1). The conservation law (*generation=(output-input)+accumulation*) or Reynolds Transport Theorem applied to the number of microstates gives

$$\frac{D(\text{microstates})}{Dt} = \frac{d}{dt}\iiint_{\text{CV}} \rho(\mathbf{q},\mathbf{p};t)\,d\mathbf{q}\,d\mathbf{p} + \oint_{\text{CS}} \rho(\mathbf{q},\mathbf{p};t)(\vec{v}\cdot\hat{n})d\Sigma = 0 \tag{E.5}$$

where CV and CS refer, respectively, to Control Volume (Γ_1) and control surface (Σ). A couple of considerations: (1) because the number of microstates is conserved, the generation term is zero; and (2) the closed integral includes inward and outward contribution. In words, the number of microstates can change because it increases or decreases within CV and because more microstates can enter or exit through the CS. More concisely:

$$\frac{d}{dt}\int_{\Gamma_1} \rho(\mathbf{q},\mathbf{p};t)\,d\mathbf{q}\,d\mathbf{p} = -\int_{\Sigma} \rho(\mathbf{q},\mathbf{p};t)\,\vec{v}\cdot d\vec{\Sigma} \tag{E.6}$$

Let's now apply the Gauss theorem:

$$\int_{\Sigma} \rho(\mathbf{q},\mathbf{p};t)\,\vec{v}\cdot d\vec{\Sigma} = \int_{\Gamma_1} \nabla\cdot(\rho\vec{v})\,d\mathbf{q}\,d\mathbf{p} \tag{E.7}$$

In addition, we can also rewrite the left hand side of Eq. E.6 as follows

$$\frac{d}{dt}\int_{\Gamma_1} \rho(\mathbf{q},\mathbf{p};t)\,d\mathbf{q}\,d\mathbf{p} = \int_{\Gamma_1} \frac{\partial\rho(\mathbf{q},\mathbf{p};t)}{\partial t}\,d\mathbf{q}\,d\mathbf{p} \tag{E.8}$$

Therefore

$$\int_{\Gamma_1} \left[\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\vec{v})\right] d\mathbf{q}\,d\mathbf{p} = 0 \tag{E.9}$$

If we now take into account that Γ_1 is an arbitrary volume (more precisely, hypervolume), then the above equation implies that

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\vec{v}) = 0 \tag{E.10}$$

This result, which resembles the continuity equation, apart from the fact that ρ represents a probability density and space is $2s$-dimensional, is known as *Liouville Equation*. Let's obtain alternative forms of Eq. E.10:

$$\nabla\cdot(\rho\vec{v}) = \nabla\cdot(\rho\dot{q}_1,\ldots,\rho\dot{q}_s,\rho\dot{p}_1,\ldots,\rho\dot{p}_s) = \frac{\partial}{\partial q_1}(\rho\dot{q}_1)+\cdots+\frac{\partial}{\partial q_s}(\rho\dot{q}_s)+\frac{\partial}{\partial p_1}(\rho\dot{p}_1)+\cdots+\frac{\partial}{\partial p_s}(\rho\dot{p}_s) \tag{E.11}$$

More concisely:

$$\nabla\cdot(\rho\vec{v}) = \sum_{i=1}^{s}\left[\frac{\partial}{\partial q_i}(\rho\dot{q}_i)+\frac{\partial}{\partial p_i}(\rho\dot{p}_i)\right] = \sum_{i=1}^{s}\left[\frac{\partial\rho}{\partial q_i}\dot{q}_i+\frac{\partial\rho}{\partial p_i}\dot{p}_i\right] + \rho\sum_{i=1}^{s}\left[\frac{\partial\dot{q}_i}{\partial q_i}+\frac{\partial\dot{p}_i}{\partial p_i}\right] \tag{E.12}$$

We notice that the last term of the right hand side equals zero, because $\dot{q}_i = \partial H/\partial p_i$ and $\dot{p}_i = -\partial H/\partial q_i$. Therefore, the above equation takes the form

$$\nabla\cdot(\rho\vec{v}) = \sum_{i=1}^{s}\left(\frac{\partial\rho}{\partial q_i}\dot{q}_i+\frac{\partial\rho}{\partial p_i}\dot{p}_i\right) \tag{E.13}$$

Substituting this result into Eq. E.10, we get

$$\frac{\partial\rho}{\partial t} + \sum_{i=1}^{s}\left(\frac{\partial\rho}{\partial q_i}\dot{q}_i+\frac{\partial\rho}{\partial p_i}\dot{p}_i\right) = 0 \tag{E.14}$$

This is the total derivative of ρ with respect to t as we can easily see from

$$\begin{aligned} d\rho &= \frac{\partial \rho}{\partial t}dt + \sum_{i=1}^{s}\left(\frac{\partial \rho}{\partial q_i}dq_i + \frac{\partial \rho}{\partial p_i}dp_i\right) \Longleftrightarrow \\ \frac{d\rho}{dt} &= \frac{\partial \rho}{\partial t} + \sum_{i=1}^{s}\left(\frac{\partial \rho}{\partial q_i}\frac{dq_i}{dt} + \frac{\partial \rho}{\partial p_i}\frac{dp_i}{dt}\right) = \\ &= \frac{\partial \rho}{\partial t} + \sum_{i=1}^{s}\left(\frac{\partial \rho}{\partial q_i}\dot{q}_i + \frac{\partial \rho}{\partial p_i}\dot{p}_i\right) = \\ &= 0 \end{aligned} \tag{E.15}$$

We then obtain an alternative form of the Liouville equation, in terms of total derivative of the probability density:

$$\frac{d\rho}{dt} = 0 \tag{E.16}$$

From this result, it follows that the flux of probabilities is *incompressible*. A given hypervolume Γ_1 can change shape, but its volume does not. A microstate can move, but its probability density does not change over time. There are other two forms of the Liouville Equation. Let's derive both of them. To this end, let's pick an element of volume of the phase space, $dq\,dp$ at time t and let's make it evolve over time to $dq_\tau\,dp_\tau$. All microstates that were included in $dq\,dp$ at time t will be found in $dq_\tau\,dp_\tau$ at time $t+\tau$. Mathematically:

$$\rho(\mathbf{q}, \mathbf{p}; t)d\mathbf{q}\,d\mathbf{p} = \rho(\mathbf{q}_\tau, \mathbf{p}_\tau; t+\tau)d\mathbf{q}_\tau\,d\mathbf{p}_\tau \tag{E.17}$$

Because the probability density does not change over time (see Eq. E.16), then it follows that

$$d\mathbf{q}\,d\mathbf{p} = d\mathbf{q}_\tau\,d\mathbf{p}_\tau \tag{E.18}$$

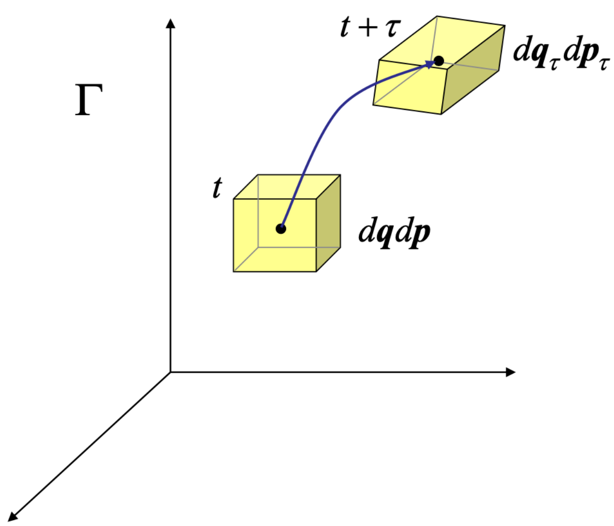


Figure E.2: The hypervolume of phase space remains invariant during the natural evolution of the system: $d\mathbf{q}d\mathbf{p} = d\mathbf{q}_\tau d\mathbf{p}_\tau$.

This is a new form of Liouville equation, which indicates that the volume in the phase space does not change over time. In other words the Jacobian of the transformation from $\mathbf{q}\,\mathbf{p}$ to $\mathbf{q}_\tau\,\mathbf{p}_\tau$ is equal to 1. See also Fig. E.2. Finally, let's obtain the last available form of Liouville Equation, employing the results we already know. From Eq. E.14, we have

$$\frac{\partial \rho}{\partial t} + \sum_{i=1}^{n} \left[\frac{\partial \rho}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \rho}{\partial p_i} \frac{\partial H}{\partial q_i} \right] = 0 \tag{E.19}$$

We can now make use of the Poisson brackets:

$$\frac{\partial \rho}{\partial t} + \{\rho, H\} = 0 \tag{E.20}$$

Equivalently:

$$\frac{\partial \rho}{\partial t} = -\{\rho, H\} = \{H, \rho\} \tag{E.21}$$

Appendix F

A brief introduction to the Kinetic Theory of Gases

In what follows, we will introduce few key concepts that are crucial to understand the *Kinetic Theory of Gases*. To this end, we imagine to have a system containing many gas molecules. The volume occupied by this system is very large compared to the size of the molecules. The internal energy of this system consists of several contributions: the kinetic energy, E_K, associated with the motion of the center of mass of the gas molecules; the potential energy associated with the intermolecular interactions, $E_{P,inter}$; and the potential energy associated with the intramolecular interactions, $E_{P,intra}$. Each gas molecule can translate and rotate in space. Additionally, they might vibrate as a result of intramolecular forces acting between atoms forming the molecule. All these kinds of motion are generally referred to as *degrees of freedom*. For instance, a single atom can have three *translational* degrees of freedom if it freely moves in the three spatial directions. Additional degrees of freedom, such as *rotational* and *vibrational*, are usually related to the internal atomic structure. According to the simplest theory of gases, rotational, translational and vibrational degrees of freedom are equally probable at all temperatures. In reality, they are not as their contribution to the internal energy of the system is actually different. Nevertheless, the Kinetic Theory of Gases assumes that the internal energy distributes itself equally among all the possible degrees of freedom of a gas molecule. Equivalently, each independent degree of freedom has an equal amount of energy. This assumption is the core of the so-called **equipartition theorem**.

The equipartition theorem states that each quadratic degree of freedom[1] will, on average, possess an energy equal to $k_BT/2$, where $k_B = 1.38064852 \times 10^{-23}$ J K^{-1} is the Boltzmann's constant and T the absolute temperature. This result is rather intuitive as a system generally prefers to spread out its energy evenly amongst all the accessible modes of motion. In particular, the kinetic and potential energies associated with translational, rotational and vibrational energy are:

- $E_{K,t} = \frac{1}{2}mv^2$, for each translational degree of freedom
- $E_{K,r} = \frac{1}{2}I\omega^2$, for each rotational degree of freedom
- $E_{K,v} + E_{P,v} = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$, for each vibrational degrees of freedom

These degrees of freedom have a quadratic dependence on velocity (or angular velocity in the case of rotation) and therefore all follow the equipartition theorem. The total internal energy of an ideal gas containing N molecules is then

$$U = N_{df}N\frac{1}{2}k_BT \tag{F.1}$$

where N_{df} is the total number of degrees of freedom. For example, a diatomic molecule has 3 translational, 2 rotational and 1 vibrational degrees of freedom, and therefore its internal energy is $U = 3Nk_BT$. Nevertheless, the vibrational degree of freedom is only considered at high temperatures. At room temperature,

[1] A *quadratic degree of freedom* is one for which the energy depends on the square of some property.

the internal energy is only due to the three translational and two rotational degrees of freedom, that is $U = 5Nk_BT/2$. An ideal monatomic gas atom has no rotational degrees of freedom or internal degrees of freedom, but only three translational degrees of freedom, and its internal energy is $U = 3N_Ak_BT/2$. To build a simple gas model, the Kinetic Theory of Gases makes few crucial assumptions:

1. **The number of molecules in the gas is large, and the average distance between them is large compared with their dimensions**. In other words, the molecules are imagined to be point-like and the volume occupied by them is negligible. This implies that the above mentioned intramolecular interactions can be neglected.

2. **The molecules obey Newton's laws of motion, but as a whole they move randomly**.

3. **The molecules interact only by short-range forces during elastic collisions**. Basically, the above mentioned intermolecular contributions to the potential energy can be neglected: molecules behave as hard-core particles. They cannot overlap, but will not repel or attract eachother.

4. **The molecules make elastic collisions with the walls and time of collision is negligible compared with time between collisions**. This means that there are not energy dissipations and the momentum is conserved.

5. **All molecules are identical**. In other words, the gas is a pure substance.

Pressure of an Ideal Gas

Consider an ideal gas consisting of N molecules of mass m inside a container of volume V and pressure P. The gas molecules collide elastically with each other and the walls of the container. The pressure that the gas exerts on the container is due to the elastic collisions of the molecules with the walls of the container. Let's consider a molecule that moves with velocity $\vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z$ and collide with a wall of the container, as shown in Fig. F.1

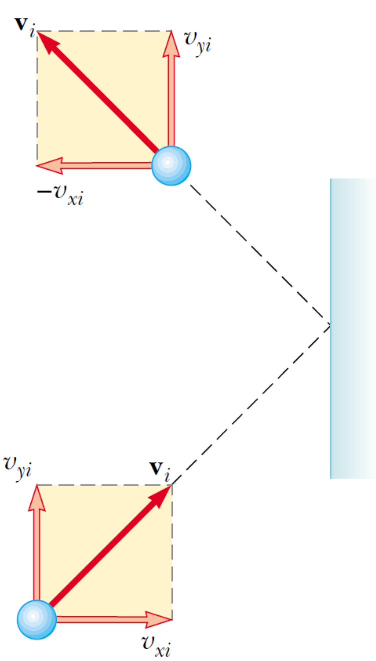


Figure F.1: A molecule makes an elastic collision with the wall of the container.

Since the collisions are elastic, the momentum is conserved and the x-component of the velocity changes direction:

$$|\Delta\vec{p}| = |m\vec{v}_2 - m\vec{v}_1| = m(-v_x + v_y + vz - v_x - v_y - v_z) = -2mv_x \tag{F.2}$$

From the *impulse-momentum theorem*, we know that the change in momentum of an object equals the impulse applied to it:

$$\vec{F}\Delta t_{col} = \Delta\vec{p} = -2m\vec{v}_x, \tag{F.3}$$

where Δt_{col} is the (very short) time of collision. For the same molecule to make another collision with the same wall, it must travel a distance equal to twice the box length ($2d$) in the x direction. The time interval between these two collisions is $\Delta t = 2d/v_x$. Note that $\Delta t_{col} << \Delta t$. How many molecules will collide within a given area A of the wall normal to x in the time interval Δt? The total mass of N molecules can be expressed as:

$$m_{tot} = \rho V = \rho A d = \rho A \frac{v_x \Delta \mathrm{t}}{2} \tag{F.4}$$

The total momentum exerted by all these molecules in the time interval Δt is

$$F_{tot}\Delta t = \Delta p_{tot} = 2m_{tot}v_x = 2\rho A \frac{v_x \Delta t}{2} v_x \tag{F.5}$$

We observe that the force of collision with the molecules and the wall is independent of the time interval and is given by

$$F_{tot} = \rho A v_x^2 \Leftrightarrow \frac{F_{tot}}{A} = P = \rho v_x^2 = \frac{m_{tot}}{V} v_x^2 = \frac{mN}{V} v_x^2 \tag{F.6}$$

Now, we note that $v_x^2 = v_y^2 = v_z^2 = \bar{v}^2/3$, where $\bar{v}$ is the average velocity. Therefore:

$$PV = N\frac{m\bar{v}^2}{3} = nN_A\frac{m\bar{v}^2}{3} \tag{F.7}$$

where $N_A = 6.022 \times 10^{23}$ mol^{-1} is the Avogadro number. We note that $\bar{v}^2$ represents three translational degrees of freedom, each of which has an energy $k_B T/2$, so that

$$PV = nN_A\frac{3k_BT}{3} = nN_Ak_BT \tag{F.8}$$

Finally, we note that $N_A k_T$ is the universal gas constant $R = 8.314$ J K^{-1} mol^{-1}:

$$PV = nRT \tag{F.9}$$

This equation is known as the ideal gas equation of state also known as the **Ideal Gas Law**. Although we started with atomistic description of the collisions of individual gas molecules satisfying the principles of conservation of energy and momentum, we ended up with a relationship between the macroscopic variables pressure, volume, and temperature that are measurable properties of the system.

Appendix G

Solutions of Problems

G.1 Problem Chapter 1

Problem 1.1

Let k be the number of Y molecules, and thus, the number of Z molecules is $n-k$. According to the problem, the ratio of Y to Z is $2:1$, so we have $k = \frac{2n}{3}$. The total number of ways the reaction can occur is the number of ways to choose k Y molecules out of n, which is given by the binomial coefficient $\binom{n}{k}$. Therefore, the solution is:

$$\text{Number of ways} = \binom{n}{k} = \binom{n}{\frac{2n}{3}} \tag{G.1}$$

This expression represents the number of ways the reaction can occur with the specified ratio of Y to Z molecules.

Problem 1.2

When rolling three distinct dice, each die can have values from 1 to 6. We want to find the probability of obtaining a sum of six points.

1. Possible combinations for a sum of six:
   - (1, 1, 4)
   - (1, 2, 3)
   - (1, 3, 2)
   - (1, 4, 1)
   - (2, 1, 3)
   - (2, 2, 2)
   - (2, 3, 1)
   - (3, 1, 2)
   - (3, 2, 1)
   - (4, 1, 1)

   There are a total of 10 favorable outcomes.

2. Total possible outcomes for rolling three dice: Since each die has 6 faces, there are $6^3 = 216$ possible outcomes when rolling three dice.

Now, since all of the configurations are equally probable, the probability (P) of obtaining a sum of six is given by:

$$P = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}} = \frac{10}{216} \tag{G.2}$$

Problem 1.3

Let's define with n_1, n_2 and n_3 the number of particles in each compartment, with $n_1 + n_2 + n_3 = N$. To find the probability of the most likely macrostate, that is the macrostate with the most probable number of particles in each compartment, we need to calculate the maximum value of the probability. Mathematically:

$$P_N = \frac{\text{favourable configurations}}{\text{total configurations}} = \frac{W_N(n)}{N_T} = \frac{N!}{n_1!n_2!(N-n_1-n_2)!3^N} \tag{G.3}$$

In particular:

$$\begin{aligned}
\left.\frac{\partial P_N}{\partial n_1}\right|_{n_2,n_3} = 0 \Leftrightarrow \left.\frac{\partial \ln(P_N)}{\partial n_1}\right|_{n_2,n_3} = 0 \\
&\Leftrightarrow \frac{\partial}{\partial n_1}\left[\ln N! - \ln n_1! - \ln n_2! - \ln(N-n_1-n_2)! - N\ln 3\right]_{n_2,n_3} = 0 \\
&\Leftrightarrow \frac{\partial}{\partial n_1}\{(N\ln N - N) - (n_1 \ln n_1 - n_1) - (n_2 \ln n_2 - n_2) + \\
&- [(N-n_1-n_2)\ln(N-n_1-n_2) - (N-n_1-n_2)] - N\ln 3\}_{n_2,n_3} = 0 \\
&\Leftrightarrow -\ln n_1 - 1 + 1 + \ln(N-n_1-n_2) + \frac{N-n_1-n_2}{N-n_1-n_2} - 1 = 0 \\
&\Leftrightarrow \ln\frac{N-n_1-n_2}{n_1} = 0 \\
&\Leftrightarrow \frac{N-n_1-n_2}{n_1} = 1 \\
&\Leftrightarrow N = 2n_1 + n_2
\end{aligned} \tag{G.4}$$

Similarly, if one differentiates P_N with respect to n_2, obtains

$$\left.\frac{\partial P_N}{\partial n_2}\right|_{n_1,n_3} = 0 \Leftrightarrow N = 2n_2 + n_1 \tag{G.5}$$

It follows that

$$n_1 = n_2 = \frac{1}{3}N \quad \rightarrow \quad n_3 = N - n_1 - n_2 = \frac{1}{3}N \tag{G.6}$$

Problem 1.4

The mean value $\langle x \rangle$ is given by the integral:

$$\langle x \rangle = \int_0^\infty x p(x)\,dx = \int_0^\infty x e^{-x}\,dx = \underbrace{-xe^{-x} + \int_0^\infty e^{-x}\,dx}_{\text{integration by parts}} = 1 \tag{G.7}$$

This represents the average value of the random variable x. For independent variables, the expected sum is twice the mean, and the expected product is the square of the mean:

$$\langle x_1 + x_2 \rangle = \langle x_1 \rangle + \langle x_2 \rangle = 2\langle x \rangle = 2 \tag{G.8}$$

$$\langle x_1 x_2 \rangle = \langle x_1 \rangle \langle x_2 \rangle = \langle x \rangle^2 = 1. \tag{G.9}$$

These values represent the expected sum and product of two independent random variables.

Problem 1.5

It is easy to see that the probability $P(n)$ is normalised:

$$\sum_{n=0}^{\infty} P(n) = \mathrm{e}^{-\lambda} \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} = \mathrm{e}^{-\lambda}\mathrm{e}^{\lambda} = 1; \tag{G.10}$$

To calculate the mean value of n, we see that

$$\langle n \rangle = \sum_{n=0}^{\infty} nP(n) = \mathrm{e}^{-\lambda} \sum_{n=0}^{\infty} n\frac{\lambda^n}{n!} = \mathrm{e}^{-\lambda} \left(\lambda \frac{\partial}{\partial \lambda} \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} \right) = \mathrm{e}^{-\lambda}\lambda\mathrm{e}^{\lambda} = \lambda \tag{G.11}$$

Finally, to calculate $\langle (\delta n)^2 \rangle$, we note that

$$\begin{aligned}
\langle n^2 \rangle = \langle n(n-1) + n \rangle &= \langle n(n-1) \rangle + \underbrace{\langle n \rangle}_{\lambda} = \\
&= \sum_{n=0}^{\infty} n(n-1)P(n) + \lambda = \\
&= \sum_{n=0}^{\infty} n(n-1)\frac{\lambda^n \mathrm{e}^{-\lambda}}{n!} + \lambda = \\
&= \mathrm{e}^{-\lambda} \sum_{n=2}^{\infty} \frac{\lambda^n}{(n-2)!} + \lambda = \\
&= \mathrm{e}^{-\lambda}\lambda^2 \sum_{n=2}^{\infty} \frac{\lambda^{n-2}}{(n-2)!} + \lambda = \\
&= \mathrm{e}^{-\lambda}\lambda^2\mathrm{e}^{\lambda} + \lambda = \\
&= \lambda^2 + \lambda
\end{aligned} \tag{G.12}$$

Therefore

$$\langle (\delta n)^2 \rangle = \langle (n - \langle n \rangle)^2 \rangle = \langle n^2 \rangle - \langle n \rangle^2 = \lambda^2 + \lambda - \lambda^2 = \lambda \tag{G.13}$$

Problem 1.6

Let's start with the mean:

$$\langle x \rangle = \int_{-\infty}^{\infty} xp(x)\, dx = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x \exp\left[-\frac{(x-a)^2}{2\sigma^2} \right] dx \tag{G.14}$$

Let $\eta = x - a$; then $d\eta = dx$

$$\langle x \rangle = a \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{\eta^2}{2\sigma^2} \right] d\eta}_{1} + \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \eta \exp\left[-\frac{\eta^2}{2\sigma^2} \right] d\eta}_{0} = a \tag{G.15}$$

Now the variance:

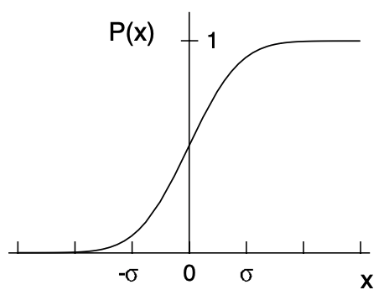
$$
\begin{aligned}
\langle x^2 \rangle &= \int_{-\infty}^{\infty} x^2 p(x)\, dx = \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} (\eta^2 - 2\eta a + a^2) \exp\left[-\frac{\eta^2}{2\sigma^2}\right] d\eta = \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \underbrace{\int_{-\infty}^{\infty} \eta^2 \exp\left[-\frac{\eta^2}{2\sigma^2}\right] d\eta}_{\sqrt{2\pi}\sigma^3} + 0 + a^2 = \\
&= \sigma^2 + a^2
\end{aligned}
\tag{G.16}
$$

It follows that:

$$
\mathrm{Var}(x) = \langle x^2 \rangle - \langle x \rangle^2 = \sigma^2 \tag{G.17}
$$

The standard deviation is the square root of the variance, that is σ. Note that the Gaussian density is completely specified by two parameters, its mean and its variance. In the case where $a = 0$ the cumulative function is

$$
P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{x} \exp[-\frac{\zeta^2}{2\sigma^2}]\, d\zeta
$$



The integral can not be expressed as a closed form expression using simple functions. It can be found tabulated as the "error function".

Problem 1.7

Let's check the normalisation first:

$$
\begin{aligned}
\int_{-\infty}^{\infty} p(y) dy &= \int_{-\infty}^{\infty} \sum_{n=0}^{\infty} p(n)\delta(y-n) dy = \\
&= \sum_{n=0}^{\infty} p(n) \underbrace{\int_{-\infty}^{\infty} \delta(y-n) dy}_{1} = \sum_{n=0}^{\infty} p(n) = \\
&= \mathrm{e}^{-rL} \sum_{n=0}^{\infty} \frac{1}{n!}(rL)^n = \mathrm{e}^{-rL}\mathrm{e}^{rL} = \\
&= 1
\end{aligned}
\tag{G.18}
$$

Now, let's calculate the mean:

$$\begin{aligned}\langle n\rangle &= \int_{-\infty}^{\infty} yp(y)dy = \sum_{n=0}^{\infty} p(n) \underbrace{\int_{-\infty}^{\infty} y\delta(y-n)dy}_{n} = \\ &= \sum_{n=0}^{\infty} np(n) = \mathrm{e}^{-rL} \sum_{n=0}^{\infty} \frac{n}{n!}(rL)^n\end{aligned} \tag{G.19}$$

This sum can be done by differentiating the previous sum used to check the normalisation.

$$\frac{d}{dr}\left[\sum_{n=0}^{\infty} \frac{1}{n!}(rL)^n\right] = \sum_{n=0}^{\infty} \frac{n}{n!} r^{n-1} L^n = \frac{1}{r}\sum_{n=0}^{\infty} \frac{n}{n!}(rL)^n \tag{G.20}$$

Alternatively

$$\frac{d}{dr}\left[\sum_{n=0}^{\infty} \frac{1}{n!}(rL)^n\right] = \frac{d}{dr}(\mathrm{e}^{rL}) = L\mathrm{e}^{rL} \tag{G.21}$$

Equating the two results gives the required sum:

$$\sum_{n=0}^{\infty} \frac{n}{n!}(rL)^n = rL\mathrm{e}^{rL} \tag{G.22}$$

Finally,

$$\langle n\rangle = \mathrm{e}^{-rL} rL\mathrm{e}^{rL} = rL = \text{rate} \times \text{interval} \tag{G.23}$$

To calculate the variance, we first need to find the mean square:

$$\langle n^2\rangle = \sum_{n=0}^{\infty} n^2 p(n) = \mathrm{e}^{-rL}\sum_{n=0}^{\infty} \frac{n^2}{n!}(rL)^n \tag{G.24}$$

Again, we proceed by differentiating the previous sum, the one worked out for the mean.

$$\frac{d}{dr}\left[\sum_{n=0}^{\infty} \frac{n}{n!}(rL)^n\right] = \sum_{n=0}^{\infty} \frac{n^2}{n!} r^{n-1} L^n = \frac{1}{r}\sum_{n=0}^{\infty} \frac{n^2}{n!}(rL)^n \tag{G.25}$$

Alternatively

$$\frac{d}{dr}\left[\sum_{n=0}^{\infty} \frac{n}{n!}(rL)^n\right] = \frac{d}{dr}(rL\mathrm{e}^{rL}) = L\mathrm{e}^{rL} + rL^2\mathrm{e}^{rL} \tag{G.26}$$

Again, equating the two results gives the required sum:

$$\sum_{n=0}^{\infty} \frac{n^2}{n!}(rL)^n = rL\mathrm{e}^{rL} + (rL)^2\mathrm{e}^{rL} \tag{G.27}$$

Concluding the calculation, we obtain

$$\langle n^2\rangle = \mathrm{e}^{-rL}[rL + (rL)^2]\mathrm{e}^{rL} = rL + (rL)^2 = \langle n\rangle + \langle n\rangle^2 \tag{G.28}$$

It follows that the variance reads

$$\mathrm{Var(n)} = \langle n^2\rangle - \langle n\rangle^2 = \langle n\rangle = rL \tag{G.29}$$

The Poisson probability is most conveniently expressed in terms of its mean:

$$p(n) = \frac{1}{n!}\langle n\rangle^n \mathrm{e}^{-\langle n\rangle}, \quad \text{where}\langle n\rangle = rL. \tag{G.30}$$

G.2 Problem Chapter 2

Problem 2.1

The probability P is defined in the range $x = y = [1, 10]$. Let's first check whether, in this range, P is normalised. If not, we need to normalise it.

$$\begin{aligned}
P(x, y, 0) &= \int_1^{10}\int_1^{10} |4e^{-(x+y)}|^2\, dy\, dx \\
&= 16\int_1^{10} e^{-2x}\left(\int_1^{10} e^{-2y}\, dy\right) dx \\
&= 16\int_1^{10} e^{-2x}\left(-\frac{1}{2}e^{-2y}\Big|_1^{10}\right) dx \\
&= 8\int_1^{10} e^{-x2}\left(-e^{-20}+e^{-2}\right) dx \\
&= 8\int_1^{10} (e^{-2}-e^{-20})e^{-2x}\, dx \\
&= 8\left((e^{-2}-e^{-20})\int_1^{10} e^{-2x}\, dx\right) \\
&= 8\left((e^{-2}-e^{-20})\left(-\frac{1}{2}e^{-2x}\Big|_1^{10}\right)\right) \\
&= 4\left((e^{-2}-e^{-20})(-e^{-20}+e^{-2})\right) \\
&= 4\left((e^{-2}-e^{-20})^2\right) \simeq 0.07326
\end{aligned} \tag{G.31}$$

The probability of finding a particle in the range $\Delta x = [2, 4]$ and $\Delta y = [3, 6]$ at time $t = 0$ is given by

$$P(\Delta x, \Delta y; 0) = \frac{1}{0.07326}\int_2^4 dx \int_3^6 dy|\Psi(x, y; 0)|^2 \tag{G.32}$$

In particular,

$$\begin{aligned}
P(\Delta x, \Delta y; 0) &= \frac{1}{0.07326}\int_2^4 dx\int_3^6 dy\, |4e^{-(x+y)}|^2 = \int_2^4 dx\int_3^6 dy\, 16e^{-2(x+y)} = \\
&= \frac{16}{0.07326}\int_2^4 dx\, e^{-2x}\int_3^6 dy\, e^{-2y} = \\
&= \frac{16}{0.07326}\left[-\frac{e^{-2x}}{2}\right]_2^4\left[-\frac{e^{-2y}}{2}\right]_3^6 = \ldots
\end{aligned} \tag{G.33}$$

Problem 2.2

To determine the total number of microstates compatible with fixed values of n_+ and n_- (denoted as $W(n_+, n_-)$), you can use the following expression:

$$W(n_+, n_-) = \frac{N!}{n_+!n_-!} \tag{G.34}$$

This equation provides the number of microstates for a given configuration of particles with n_+ in the higher energy state and n_- in the lower energy state. Regarding temperature effects:

- At very low temperatures, you might expect the system to predominantly occupy the state with the lowest energy (n_- dominates) due to the Boltzmann factor.

- At very high temperatures, the distribution of particles among energy states becomes more uniform as thermal energy overcomes energy differences (n_+ and n_- approach each other).

Keep in mind that these are simplified expectations, and real systems may exhibit more complex behaviours at extreme temperatures.

Problem 2.3

Let's consider the three cases separately:

1. For bosons, each particle can occupy the same energy level, and they are indistinguishable. We have two energy levels with respective degeneracies: ϵ with $g_1 = 2$ and 2ϵ with $g_2 = 3$. The macrostate we seek is characterised by $E = 3\epsilon$ and $N = 2$. Using the technique of Fig. 2.10, we obtain that $\Omega(2\epsilon) = 3$, $\Omega(3\epsilon) = 6$ and $\Omega(4\epsilon) = 6$. Finally, the total number of microstates (at any energy) is 15. More efficiently, we can get the same result by applying Eq. 2.68. Check that this is the case.

2. For fermions, the Pauli Exclusion Principle states that no two fermions can occupy the same quantum state simultaneously. In this case, we can't have two fermions in the same energy level. Following the technique reported in Fig. 2.11, we obtain $\Omega(2\epsilon) = 1$, $\Omega(3\epsilon) = 6$ and $\Omega(4\epsilon) = 3$. The total number of microstates (at any energy) is therefore 10. More efficiently, we can get the same result by applying Eq. 2.69. Check that this is the case.

3. Classical particles. In this case, the particles are distinguishable from each other, so we don't have to consider indistinguishability or the Pauli Exclusion Principle. We have at least two options to solve this problem:

   - We use the above result for bosons and notice that the 2 classical particles are inter-changeable because distinguishable. As such, we just multiply by 2 the number of microstates compatible with $E = 3\epsilon$ found in the system of bosons and obtain $\Omega(3\epsilon) = 12$.
   - We can calculate the probability of observing one of the two particles, say A, in the energy level ϵ_1, which is $g_1/(g_1 + g_2) = 2/5$, and multiply it by the probability of observing the other particle, say B, in the energy level ϵ_2, which is $g_2/(g_1 + g_2) = 3/5$. We then get $2/5 \times 3/5 = 6/25$, which must be multiplied by 2 as particles A and B can exchange their positions. So, we get 12/25.

   We note that both options are not of general applications to all available macrostates. The best way to proceed is applying Eq. 2.70:

   $$W_c(3\epsilon) = N!\,\Pi_i \frac{g_i^{n_i}}{n_i!} = 2!\frac{2^1}{1!}\frac{3^1}{1!} = 2 \times 2 \times 3 = 12 \tag{G.35}$$

   It also follows that the total number of microstates (at any energy) is 25.

Problem 2.4

Let's write the total energy of the particle:

$$E = \frac{p^2}{2m} + Ax^2 + Bx^4 \tag{G.36}$$

To obtain the trajectories of the phase space, we just need to calculate the particle momentum p:

$$p(x) = \pm[2m(E - Ax^2 - Bx^4)]^{1/2} \tag{G.37}$$

where $A = -1$, $B = 1$ and $m = 1/2$. Substituting these values, the above equation reads

$$p(x) = \pm(E + x^2 - x^4)^{1/2} \tag{G.38}$$

Let's first plot the potential $V(x) = -x^2 + x^4$ in Fig. G.1.

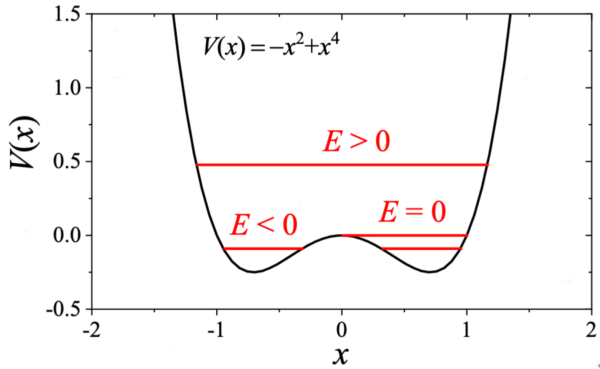


Figure G.1: Particle potential. The red lines indicate the regions where E is positive, negative or zero.

We see that $V(x)$ exhibits two minima at

$$\frac{dV(x)}{dx} = 0 \Longleftrightarrow -2x + 4x^3 = 0 \Longleftrightarrow x = \begin{cases} -1/\sqrt{2} \simeq -0.707 = -x_{min} \\ 0 = x_{max} \\ 1/\sqrt{2} \simeq 0.707 = x_{min} \end{cases} \tag{G.39}$$

In particular, $V(x_{max}) = 0$ and $V(x_{min}) = V(-x_{min}) = -0.25$. Considering Eq. G.38, we conclude that

- The region of the phase space where $E < -0.25$ is forbidden.
- If $-0.25 < E < 0$, the particle is confined in one of the two wells indicated in Fig. G.1 and cannot explore other regions.
- If $E > 0$, then the particle has access to the full region above the wells.

Let's now calculate the inversion points, that is the points for which $p = 0$:

$$p = 0 \Longleftrightarrow E + x_0^2 - x_0^4 = 0 \Longleftrightarrow E + y_0 - y_0^2 = 0 \tag{G.40}$$

where $y_0 = x_0^2$. The general solution of this equation is:

$$y_0 = \frac{1 \pm \sqrt{1+4E}}{2} \Longleftrightarrow x_0 = \pm\sqrt{\frac{1 \pm \sqrt{1+4E}}{2}} \tag{G.41}$$

Let's analyse the specific solutions separately:

- For $E = 0$, $x_0 = \{-1, 0, 1\}$;
- If $E > 0$, $x_0 = \{-\sqrt{\frac{1+\sqrt{1+4E}}{2}}, \sqrt{\frac{1+\sqrt{1+4E}}{2}}\}$;
- If $-0.25 < E < 0$, $x_0 = \{\underbrace{-\sqrt{\frac{1+\sqrt{1-4|E|}}{2}}}_{a}, \underbrace{-\sqrt{\frac{1-\sqrt{1-4|E|}}{2}}}_{b}, \underbrace{\sqrt{\frac{1-\sqrt{1-4|E|}}{2}}}_{c}, \underbrace{\sqrt{\frac{1+\sqrt{1-4|E|}}{2}}}_{d}\}$

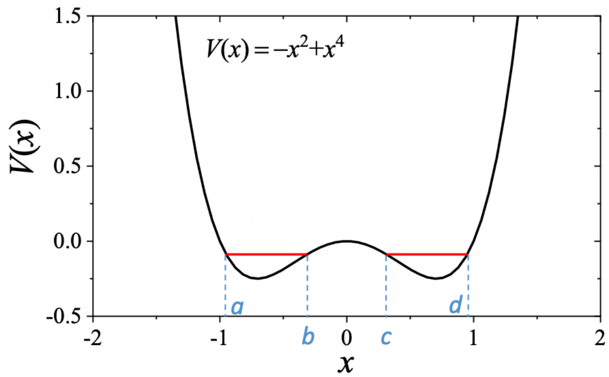


Figure G.2: Inversion points for $-0.25 < E < 0$.

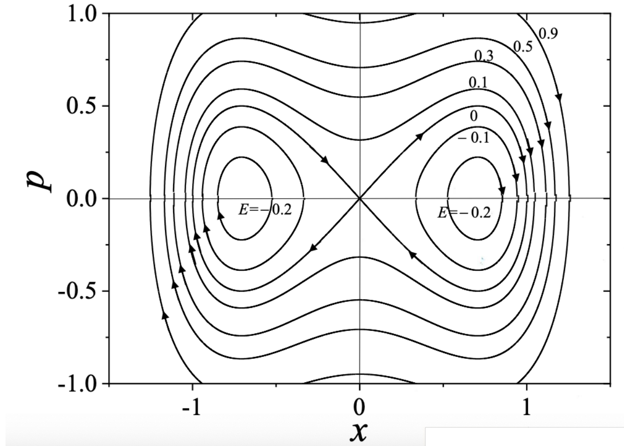


Figure G.3: Trajectories in the phase space.

Problem 2.5

(a) We first notice that the identity matrix reads

$$\hat{\mathbf{I}} = \sum_{k=1}^{\infty} |\phi_k\rangle\langle\phi_k| \tag{G.42}$$

Therefore

$$\begin{aligned}\langle \mathbf{r}' \left| \mathrm{e}^{-\beta \hat{H}} \right| \mathbf{r} \rangle &= \langle \mathbf{r}' | \mathrm{e}^{-\beta \hat{H}} \sum_{k=1}^{\infty} |\phi_k\rangle\langle\phi_k| \, \mathbf{r}\rangle = \\ &= \langle \mathbf{r}' | \sum_{k=1}^{\infty} \mathrm{e}^{-\beta \hat{H}} \, |\phi_k\rangle\langle\phi_k| \, \mathbf{r}\rangle = \\ &= \sum_{k=1}^{\infty} \mathrm{e}^{-\beta \hat{H}} \langle \mathbf{r}' | \phi_k\rangle\langle\phi_k | \mathbf{r}\rangle = \\ &= \sum_{k=1}^{\infty} \mathrm{e}^{-\beta \hat{H}} \phi_k(\mathbf{r}')\phi_k^*(\mathbf{r})\end{aligned} \tag{G.43}$$

(b) Now, we wish to calculate the above quantity for the specific case of a free particle confined in a cubic box of side L. The Hamiltonian operator reads

$$\hat{H} = \frac{p^2}{2m} = -\frac{\hbar^2}{2m}\nabla^2 = -\frac{\hbar^2}{2m}\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right] \tag{G.44}$$

with

$$\hat{H}\phi(x, y, z) = E\phi(x, y, z) \tag{G.45}$$

where $E = \epsilon_x + \epsilon_y + \epsilon_z$. It follows that

$$\hat{H}\phi(x, y, z) = -\frac{\hbar^2}{2m}\left[\frac{\partial^2 \phi_x}{\partial x^2} + \frac{\partial^2 \phi_y}{\partial y^2} + \frac{\partial^2 \phi_z}{\partial z^2}\right] = \epsilon_x\phi_x + \epsilon_y\phi_y + \epsilon_z\phi_z \tag{G.46}$$

In particular, for the x direction

$$\frac{\partial^2 \phi_x}{\partial x^2} = -\underbrace{\frac{2m\epsilon_x}{\hbar^2}}_{k_x^2}\phi_x \tag{G.47}$$

This is a second-order differential equation with general solution

$$\phi_x = A\cos(k_x x) + B\sin(k_x x) \tag{G.48}$$

To set the boundary conditions, we note that, at each face of the cubic box, the wave function ϕ_x must vanish (go to zero). This condition ensures that the particle does not penetrate the walls of the box. In particular, $\phi_x(0) = A = 0$ and $\phi_x(L) = B\sin(k_x L) = 0$. The latter condition implies that either $B = 0$ (and therefore $\phi_x = 0$) or $k_x L = n_x\pi$, with n_x positive integer (negative values of n_x only imply a change in the sign of $\sin(k_x)$ and thus do not represent new quantum states). For this case, it follows that

$$k_x^2 = \frac{\pi^2 n_x^2}{L^2} = \frac{2m\epsilon_x}{\hbar^2} \Longleftrightarrow \epsilon_x = \frac{\hbar^2\pi^2}{2mL^2}n_x^2 = \frac{h^2 n_x^2}{8mL^2} \tag{G.49}$$

We finally obtain

$$\phi_x = B\sin\left(\frac{\pi n_x x}{L}\right) \tag{G.50}$$

To find B, which represents the amplitude of the wave function (expected positive), let's impose normalisation:

$$\begin{aligned} 1 = \int_0^L \left|\phi_x^2\right| &= |B|^2 \int_0^L \sin^2\left(\frac{\pi n_x x}{L}\right) dx = \\ &= |B|^2 \int_0^L \frac{1}{2}\left[1 - \cos\left(\frac{2\pi n_x x}{L}\right)\right] dx = \\ &= |B|^2 \left[\frac{L}{2} - \frac{L}{2\pi n_x}\sin\left(\frac{2\pi n_x x}{L}\Bigg|_0^L\right)\right] = \\ &= |B|^2 \frac{L}{2} \Longleftrightarrow \\ &\Longleftrightarrow B = \sqrt{\frac{2}{L}} \end{aligned} \tag{G.51}$$

Therefore

$$\phi_x = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi n_x x}{L}\right) \tag{G.52}$$

For the three dimensions:

$$\phi = \sqrt{\frac{8}{L^3}} \sin\left(\frac{\pi n_x x}{L}\right) \sin\left(\frac{\pi n_y y}{L}\right) \sin\left(\frac{\pi n_z z}{L}\right) \tag{G.53}$$

and

$$E = \epsilon_x + \epsilon_y + \epsilon_z = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2) \tag{G.54}$$

We can now calculate the quantity from Eq. G.43:

$$\begin{aligned}
\langle \mathbf{r}' \left| \mathrm{e}^{-\beta \hat{H}} \right| \mathbf{r} \rangle &= \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} \mathrm{e}^{-\beta \frac{h^2}{8mL^2}(n_x^2+n_y^2+n_z^2)} \phi(x', y', z') \phi^*(x, y, z) = \\
&= \left[\sum_{n_x=1}^{\infty} \mathrm{e}^{-\beta \frac{h^2}{8mL^2} n_x^2} \frac{2}{L} \sin\left(\frac{\pi n_x x}{L}\right) \sin\left(\frac{\pi n_x x'}{L}\right) \right] \cdot \\
&\cdot \left[\sum_{n_y=1}^{\infty} \mathrm{e}^{-\beta \frac{h^2}{8mL^2} n_y^2} \frac{2}{L} \sin\left(\frac{\pi n_y y}{L}\right) \sin\left(\frac{\pi n_y y'}{L}\right) \right] \cdot \\
&\cdot \left[\sum_{n_z=1}^{\infty} \mathrm{e}^{-\beta \frac{h^2}{8mL^2} n_z^2} \frac{2}{L} \sin\left(\frac{\pi n_z z}{L}\right) \sin\left(\frac{\pi n_z z'}{L}\right) \right] = \\
&= \frac{2}{L} \int_0^{\infty} \mathrm{e}^{-\beta \frac{h^2}{8mL^2} n_x^2} \sin\left(\frac{\pi n_x x}{L}\right) \sin\left(\frac{\pi n_x x'}{L}\right) dn_x \cdot \\
&\cdot \frac{2}{L} \int_0^{\infty} \mathrm{e}^{-\beta \frac{h^2}{8mL^2} n_y^2} \sin\left(\frac{\pi n_y y}{L}\right) \sin\left(\frac{\pi n_y y'}{L}\right) dn_y \cdot \\
&\cdot \frac{2}{L} \int_0^{\infty} \mathrm{e}^{-\beta \frac{h^2}{8mL^2} n_z^2} \sin\left(\frac{\pi n_z z}{L}\right) \sin\left(\frac{\pi n_z z'}{L}\right) dn_z
\end{aligned} \tag{G.55}$$

To solve these integrals, we note that each of them (for instance, the first one) can be rewritten as

$$\begin{aligned}
I(n_x) &= \int_0^{\infty} \mathrm{e}^{-a n_x^2} \sin(b n_x) \sin(c n_x)\, dn_x = \\
&= \int_0^{\infty} \mathrm{e}^{-a n_x^2} \frac{1}{2} \left[\cos(\underbrace{(b-c)}_{A} n_x) - \cos(\underbrace{(b+c)}_{B} n_x) \right] dn_x = \\
&= \frac{1}{2} \int_0^{\infty} \mathrm{e}^{-a n_x^2} \cos(A n_x)\, dn_x - \frac{1}{2} \int_0^{\infty} \mathrm{e}^{-a n_x^2} \cos(B n_x)\, dn_x = \\
&= \frac{1}{2} \left(\frac{1}{2} \sqrt{\frac{\pi}{a}} \mathrm{e}^{-\frac{A^2}{4a}} \right) - \frac{1}{2} \left(\frac{1}{2} \sqrt{\frac{\pi}{a}} \mathrm{e}^{-\frac{B^2}{4a}} \right) = \\
&= \frac{1}{4} \sqrt{\frac{\pi}{a}} \left(\mathrm{e}^{-\frac{A^2}{4a}} - \mathrm{e}^{-\frac{B^2}{4a}} \right)
\end{aligned} \tag{G.56}$$

where we introduced the constants $a \equiv \beta h^2/8mL^2$, $b \equiv \pi x/L$ and $c \equiv \pi x'/L$. Therefore, we obtain:

$$
\begin{aligned}
\langle \mathbf{r}' \left| \mathrm{e}^{-\beta H} \right| \mathbf{r} \rangle = \left(\frac{1}{h} \sqrt{\frac{2m\pi}{\beta}} \right)^3 &\cdot \left(\mathrm{e}^{-\frac{2m\pi^2 (x-x')^2}{\beta h^2}} - \mathrm{e}^{-\frac{2m\pi^2 (x+x')^2}{\beta h^2}} \right) \cdot \\
&\cdot \left(\mathrm{e}^{-\frac{2m\pi^2 (y-y')^2}{\beta h^2}} - \mathrm{e}^{-\frac{2m\pi^2 (y+y')^2}{\beta h^2}} \right) \cdot \\
&\cdot \left(\mathrm{e}^{-\frac{2m\pi^2 (z-z')^2}{\beta h^2}} - \mathrm{e}^{-\frac{2m\pi^2 (z+z')^2}{\beta h^2}} \right)
\end{aligned}
\tag{G.57}
$$

(c) Using the result from part (b), we now want to obtain the partition function, defined as

$$
Z = \mathrm{Tr} \left[\mathrm{e}^{-\beta \hat{H}} \right] = \int \langle \mathbf{r} \left| \mathrm{e}^{-\beta \hat{H}} \right| \mathbf{r} \rangle \, d\mathbf{r}
\tag{G.58}
$$

In particular (note that the three integrals - over x, y, and z - are identical as space is isotropic):

$$
\begin{aligned}
Z &= \left(\frac{1}{h} \sqrt{\frac{2m\pi}{\beta}} \right)^3 \left[\int_0^L \left(\mathrm{e}^{-\frac{2m\pi^2 (x-x)^2}{\beta h^2}} - \mathrm{e}^{-\frac{2m\pi^2 (x+x)^2}{\beta h^2}} \right) dx \right]^3 = \\
&= \left(\frac{1}{h} \sqrt{\frac{2m\pi}{\beta}} \right)^3 \left[\int_0^L \left(1 - \mathrm{e}^{-\frac{2m\pi^2 (2x)^2}{\beta h^2}} \right) dx \right]^3 = \\
&= \left(\frac{1}{h} \sqrt{\frac{2m\pi}{\beta}} \right)^3 \left[\int_0^L \left(1 - \mathrm{e}^{-\frac{8m\pi^2 x^2}{\beta h^2}} \right) dx \right]^3 = \\
&= \left(\frac{1}{h} \sqrt{\frac{2m\pi}{\beta}} \right)^3 \left[L - \underbrace{\frac{1}{2} \sqrt{\frac{\pi \beta h^2}{8m\pi^2}}}_{<<L} \right]^3 \simeq \\
&\simeq \left(\frac{2m\pi}{\beta h^2} \right)^{3/2} L^3 = \\
&= \left(\frac{2m\pi}{\beta h^2} \right)^{3/2} V = \\
&= \Lambda^{-3} V
\end{aligned}
\tag{G.59}
$$

where Λ is the thermal de Broglie wavelength.

G.3 Problem Chapter 3

Problem 3.1

A free particle of mass m is constrained to move in a two-dimension phase space, with a Hamiltonian given by $H = p^2/2 + q^2/2$. Let's verify that the area of the rectangular enclosure shown in Fig. G.4 remains invariant during the natural evolution of the system.

In order to demonstrate that the area of the rectangular enclosure remains invariant during the natural evolution of the system, let's begin by defining the rectangular enclosure in the phase space and then make use of Liouville's theorem. The phase space, in this case, is two-dimensional, represented by the coordinates q and momenta p. At time $t = 0$, the rectangular enclosure can be defined in terms of its four corners: (q_0, p_0), $(2q_0, p_0)$, $(q_0, 2p_0)$ and $(2q_0, 2p_0)$, where q_0, $2q_0$, p_0, and $2p_0$ are given values. The total area is then $A(t = 0) = A_0 = q_0 p_0$. Let's write the equations of motion:

$$
\dot{p} = -\frac{\partial H}{\partial q} = -\frac{\partial (p^2/2 + q^2/2)}{\partial q} = -q
\tag{G.60}
$$

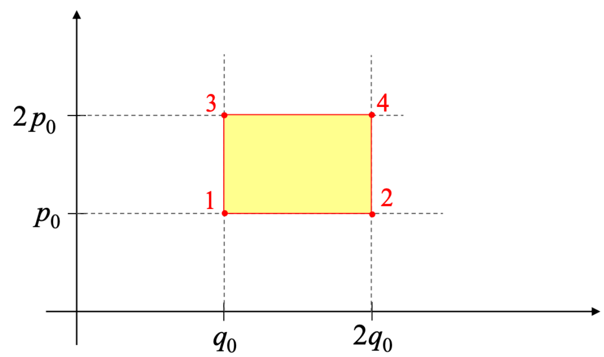


Figure G.4: Region of the phase space where a free particle of mass m is constrained to move.

and

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{\partial (p^2/2 + q^2/2)}{\partial p} = p \tag{G.61}$$

Differentiating with respect to time, we obtain:

$$\ddot{q} = \dot{p} = -q \Longleftrightarrow \ddot{q} + q = 0 \tag{G.62}$$

The general solution of this second-order differential equation is

$$q(t) = c_1 \cos(t) + c_2 \sin(t) \tag{G.63}$$

and

$$p(t) = \dot{q} = -c_1 \sin(t) + c_2 \cos(t) \tag{G.64}$$

At $t = 0$, we get:

$$q_0 = c_1, \quad p_0 = c_2 \tag{G.65}$$

It follows that

$$q(t) = q_0 \cos(t) + p_0 \sin(t), \quad p(t) = -q_0 \sin(t) + p_0 \cos(t) \tag{G.66}$$

The new coordinates are obtained with a linear transformation of the old coordinates. In particular, we notice that

$$q^2 + p^2 = q_0^2 + p_0^2 \tag{G.67}$$

Basically, each point of the rectangle, over time, follows a circular trajectory of radius $R = \sqrt{q_0^2 + p_0^2}$, maintaining its relative distance to all the other points of the rectangle, as indicated in Fig. G.5. This implies that the straight lines of the rectangular enclosure remain straight over time.

More specifically, the locations of the four points at time t are

1. $a_1 = (q_0, p_0) \longrightarrow a_1' = (q_0 \cos(t) + p_0 \sin(t), -q_0 \sin(t) + p_0 \cos(t))$
2. $a_2 = (2q_0, p_0) \longrightarrow a_2' = (2q_0 \cos(t) + p_0 \sin(t), -2q_0 \sin(t) + p_0 \cos(t))$
3. $a_3 = (q_0, 2p_0) \longrightarrow a_3' = (q_0 \cos(t) + 2p_0 \sin(t), -q_0 \sin(t) + 2p_0 \cos(t))$
4. $a_4 = (2q_0, 2p_0) \longrightarrow a_4' = (2q_0 \cos(t) + 2p_0 \sin(t), -2q_0 \sin(t) + 2p_0 \cos(t))$

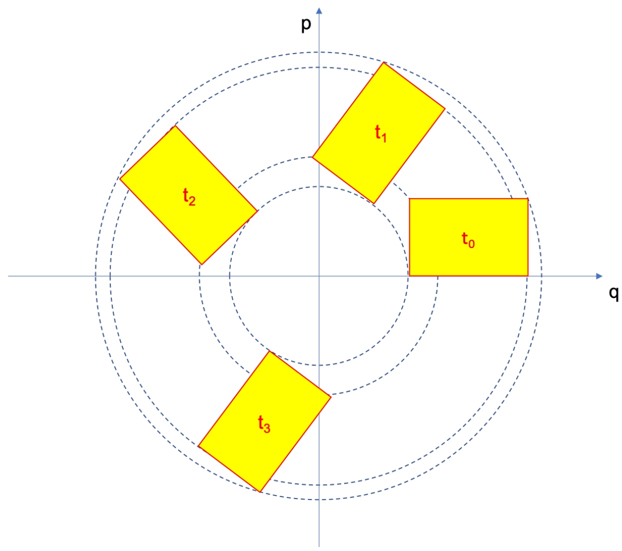


Figure G.5: Time evolution of the phase space where a free particle of mass m is constrained to move.

Problem 3.2

(a) Given that $H = E = \text{const}$, the trajectories satisfy the equation $E = (q^2 + p^2)/2$, from which we obtain

$$q^2 + p^2 = 2E. \tag{G.68}$$

It follows that the trajectories in the phase space are circumferences of radius $R = \sqrt{2E}$. We should now figure out their direction: clockwise or counterclockwise. Let's take a convenient point at $t = 0$, that is $(q_0, 0)$. At a longer time, this point will be at:

$$\begin{aligned} q(t) &= q_0 \cos t \\ p(t) &= -q_0 \sin t \end{aligned} \tag{G.69}$$

If t is sufficiently small, then $q(t) \simeq q_0$ and $p(t) \simeq -q_0\omega t$. The negative sign indicates that the direction of the trajectories is clockwise (see Fig. G.6).

(b) The probability density function must satisfy the condition

$$\int_\Gamma \rho(q, p, t) dqdp = 1 \tag{G.70}$$

This integral assumes a value only in the region Γ_0 of the phase space where ρ is non-zero, which is for $q_0 > 0$, $p_0 < 0$ and within the circumference $q_0^2 + p_0^2 < 1$, corresponding to the fourth quadrant of the circumference of Fig. G.6 with $R = 1$. In particular:

$$\int_{\Gamma_0} \rho(q_0, p_0, 0) dqdp = \int_{\Gamma_0} \text{const}\, dqdp = \text{const} \underbrace{\int_{\Gamma_0} dqdp}_{\pi R^2/4} = 1 \tag{G.71}$$

It follows that the value of the constant is $4/\pi$.

(c) From Liouville's theorem, we know that the density of an ensemble of systems remains constant as they evolve according to Hamilton's equations of motion:

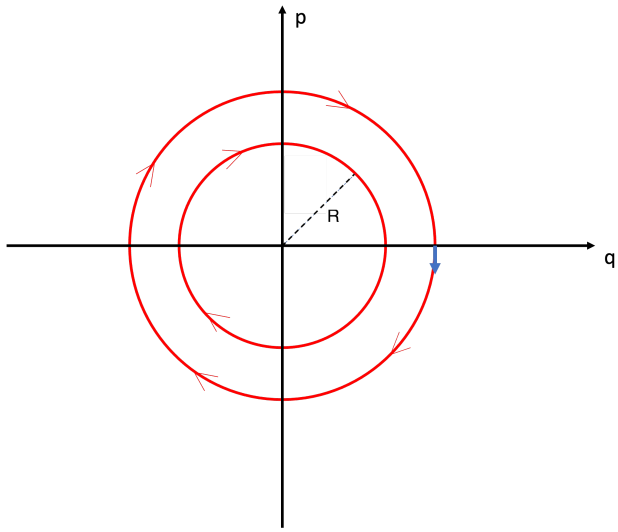


Figure G.6: Trajectories in the phase space. Arrows indicate the direction of motion along circumferences of radius $R = \sqrt{2E}$.

$$\frac{d\rho}{dt} = 0 \tag{G.72}$$

With reference to Fig. G.7, one can write that

$$\rho(q_0, p_0, t = 0) = \rho(q(t), p(t), t) \tag{G.73}$$

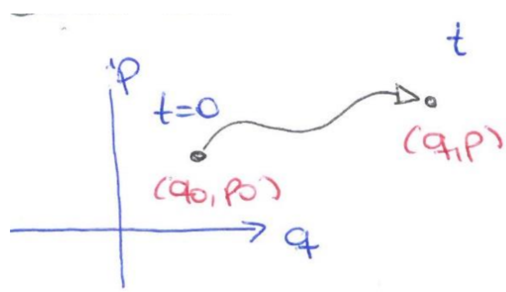


Figure G.7: Time evolution of a point in the phase space.

where

$$\begin{aligned} q(t = \pi/2) &= q_0 \cos\left(\frac{\pi}{2}\right) + p_0 \sin\left(\frac{\pi}{2}\right) = p_0 \\ p(t = \pi/2) &= -q_0 \sin\left(\frac{\pi}{2}\right) + p_0 \cos\left(\frac{\pi}{2}\right) = -q_0 \end{aligned} \tag{G.74}$$

It follows that $q_0 = -p$ and $p_0 = q$. Substituting this result into the expression of $\rho(q_0, p_0, t = 0)$, we obtain

$$\rho(q_0, p_0, t=0) = \rho(q, p, t=\pi/2\omega) = \\ = \begin{cases} 4/\pi & -p > 0, \quad q < 0, \quad q^2 + (-p)^2 < 1 \\ 0 & \text{elsewhere} \end{cases} = \\ = \begin{cases} 4/\pi & q < 0, \quad p < 0, \quad q^2 + p^2 < 1 \\ 0 & \text{elsewhere} \end{cases} \tag{G.75}$$

Problem 3.3

(a) Let's differentiate the Hamiltonian to obtain the equations of motion:

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{p}{m} \\ \dot{p} = -\frac{\partial H}{\partial q} = -m\omega^2 q \tag{G.76}$$

from which, we get

$$\ddot{q} = \frac{\dot{p}}{m} = -\omega^2 q \Longleftrightarrow \ddot{q} + \omega^2 q = 0 \tag{G.77}$$

The general solution of this homogeneous second order differential equation reads

$$q(t) = A\cos(\omega t) + B\sin(\omega t) \tag{G.78}$$

and

$$p(t) = m\dot{q}(t) = -Am\omega\sin(\omega t) + Bm\omega\cos(\omega t) \tag{G.79}$$

At $t = 0$, we obtain

$$q_0 = A \\ p_0 = Bm\omega \tag{G.80}$$

We can use this result to rewrite $q(t)$ and $p(t)$ as functions of q_0, p_0 and time t:

$$q(t) = q_0\cos(\omega t) + \frac{p_0}{m\omega}\sin(\omega t) \\ p(t) = -m\omega q_0\sin(\omega t) + p_0\cos(\omega t) \tag{G.81}$$

(b) The Jacobian of the transformation $\mathcal{T}$ reads

$$J = \frac{\partial(q_t, p_t)}{\partial(q_0, p_0)} = \begin{vmatrix} \frac{\partial q_t}{\partial q_0} & \frac{\partial q_t}{\partial p_0} \\ \frac{\partial p_t}{\partial q_0} & \frac{\partial p_t}{\partial p_0} \end{vmatrix} = \begin{vmatrix} \cos(\omega t) & \frac{1}{m\omega}\sin(\omega t) \\ -m\omega\sin(\omega t) & \cos(\omega t) \end{vmatrix} = \cos^2(\omega t) + \sin^2(\omega t) = 1 \tag{G.82}$$

We conclude that the Jacobian is indeed equal to 1 and consequently the transformation $\mathcal{T}$ satisfies Liouville's theorem.

(c) To calculate this probability, we will make use of the ergodic hypothesis, stating that time average and ensemble average are equivalently under specific conditions:

$$dP(q) = \rho(q)dq = \lim_{T\to\infty} \frac{\Delta t}{T} \tag{G.83}$$

where Δt is the time interval over which the particle is in $[q, q + dq]$ and T is the total time of measurement. This equation can be further simplified if we take into account that the movement is periodic:

$$\lim_{T\to\infty} \frac{\Delta t}{T} = \frac{2dt}{\tau} \tag{G.84}$$

where $2dt$ is the time spent in $[q, q + dq]$ over the time period τ. Let's calculate dt first:

$$E = \frac{1}{2m}p^2 + \frac{1}{2}m\omega^2 q^2 = \frac{1}{2}m\dot{q}^2 + \frac{1}{2}m\omega^2 q^2 = \frac{1}{2}m\left(\frac{dq}{dt}\right)^2 + \frac{1}{2}m\omega^2 q^2 \tag{G.85}$$

Separating variables, we obtain

$$\frac{dq}{dt} = \sqrt{\frac{2}{m}\left(E - \frac{1}{2}m\omega^2 q^2\right)} \Longleftrightarrow dt = \frac{dq}{\sqrt{\frac{2E}{m} - \omega^2 q^2}} \tag{G.86}$$

Let's now calculate the period τ:

$$\tau = 2\int_{-q_{\max}}^{q_{\max}} \frac{dq}{\sqrt{\frac{2E}{m} - \omega^2 q^2}} \tag{G.87}$$

To find $q_{\max}$, we need to calculate the inversion points of the trajectory by imposing $dq/dt = \dot{q} = 0$. Since $p = m\dot{q}$, it follows that

$$p = 0 \Longleftrightarrow E = \frac{1}{2}m\omega^2 q_{\max}^2 \Longleftrightarrow q_{\max} = \sqrt{\frac{2E}{m\omega^2}} \equiv a \tag{G.88}$$

It follows that

$$\tau = 2\int_{-a}^{a} \frac{dq}{\sqrt{\frac{2E}{m} - \omega^2 q^2}} = \frac{2}{\omega}\int_{-a}^{a} \frac{dq}{\sqrt{\frac{2E}{m\omega^2} - q^2}} = \frac{2}{\omega}\int_{-a}^{a} \frac{dq}{\sqrt{a^2 - q^2}} \tag{G.89}$$

Solving this integral, we obtain

$$\tau = \left[\frac{2}{\omega}\arcsin\left(\frac{q}{a}\right)\right]_{-a}^{a} = \frac{2}{\omega}[\arcsin(1) - \arcsin(-1)] = \frac{2}{\omega}\left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right] = \frac{2\pi}{\omega} \tag{G.90}$$

Finally:

$$dP(q) = \rho dq = \frac{2dt}{\tau} = 2\frac{\omega}{2\pi}\frac{dq}{\sqrt{\frac{2E}{m} - \omega^2 q^2}} = \frac{1}{\pi}\frac{dq}{\sqrt{\frac{2E}{m\omega^2} - q^2}} \tag{G.91}$$

Problem 3.4

The ground state for each group of particles corresponds to the lowest energy level and it is represented in Fig. G.8. Distinguishable particles and bosons exhibit the same ground state with energy 0 and degeneracy 1, whereas fermions exhibit a ground-state energy equal to 6ϵ and degeneracy 2. The number of quantum states is obtained in Fig. G.9. Four distinguishable particles can be arranged across $\Omega_{\text{tot}} = 24$ quantum states, four bosons across $\Omega_{\text{tot}} = 4$ quantum states and no options exist to organise four fermions within a total energy of 3ϵ.

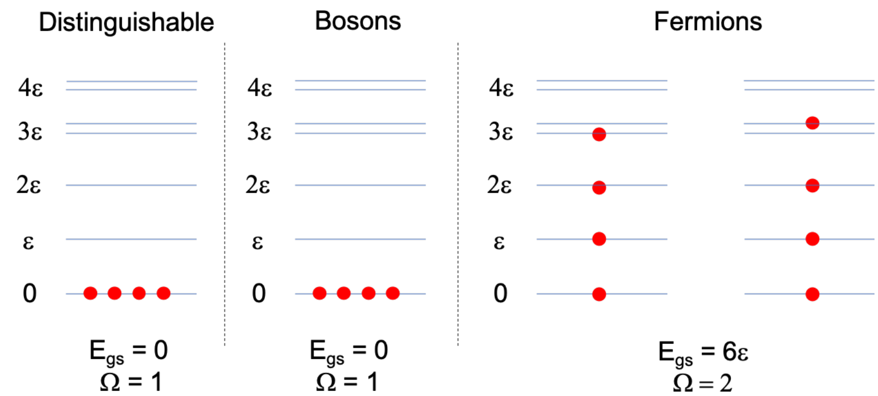


Figure G.8: Ground states for distinguishable and undistinguishable (bosons and fermions) particles.

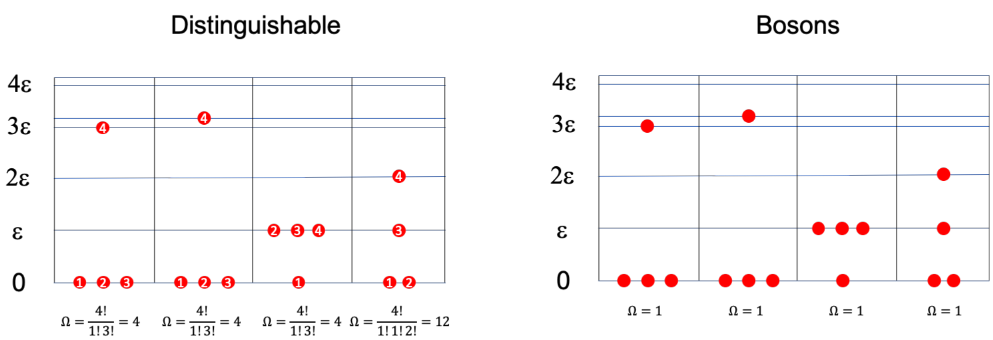


Figure G.9: Quantum states for $N = 4$ distinguishable and undistinguishable (bosons) particles associated to a total energy $E = 3\epsilon$.

Problem 3.5

The density operator can be written as follows

$$\hat{\rho} = \sum_i P_i |\psi_i\rangle\langle\psi_i| \tag{G.92}$$

Case (a). The source only produces photons polarised in the x direction. Therefore $P_1 = 1$ and $P_2 = 0$. It follows that

$$\hat{\rho}_a = \sum_i P_i |\psi_i\rangle\langle\psi_i| = |1\rangle\langle 1| \tag{G.93}$$

and the density matrix takes the form:

$$\rho_a = \begin{pmatrix} \langle 1|\hat{\rho}|1\rangle & \langle 1|\hat{\rho}|2\rangle \\ \langle 2|\hat{\rho}|1\rangle & \langle 2|\hat{\rho}|2\rangle \end{pmatrix} = \begin{pmatrix} \langle 1|1\rangle\langle 1|1\rangle & \langle 1|1\rangle\langle 1|2\rangle \\ \langle 2|1\rangle\langle 1|1\rangle & \langle 2|2\rangle\langle 2|2\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \tag{G.94}$$

Case (b). The source is incoherent and produces photons with both polarisations randomly. Therefore $P_1 = 0.5$ and $P_2 = 0.5$. It follows that

$$\hat{\rho}_b = \sum_i P_i |\psi_i\rangle\langle\psi_i| = \frac{1}{2}|1\rangle\langle 1| + \frac{1}{2}|2\rangle\langle 2| \tag{G.95}$$

and the density matrix takes the form:

$$\rho_b = \begin{pmatrix} 0.5\langle 1|1\rangle\langle 1|1\rangle + 0.5\langle 1|2\rangle\langle 2|1\rangle & 0.5\langle 1|1\rangle\langle 1|2\rangle + 0.5\langle 1|2\rangle\langle 2|2\rangle \\ 0.5\langle 2|1\rangle\langle 1|1\rangle + 0.5\langle 2|2\rangle\langle 2|1\rangle & 0.5\langle 2|1\rangle\langle 1|2\rangle + 0.5\langle 2|2\rangle\langle 2|2\rangle \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} \tag{G.96}$$

Case (c). The source produces photons coherently with a phase difference α.

$$\hat{\rho}_c = |\phi\rangle\langle\phi| = \frac{1}{2}\left[|1\rangle + \mathrm{e}^{i\alpha}|2\rangle\right]\left[|1\rangle + \mathrm{e}^{-i\alpha}|2\rangle\right] \tag{G.97}$$

Let's calculate each element of the density matrix separately:

$$\begin{aligned}
\langle 1|\hat{\rho}_c|1\rangle &= \frac{1}{2}\left[\langle 1|1\rangle + \cancel{\mathrm{e}^{i\alpha}\langle 1|2\rangle}\right]\left[\langle 1|1\rangle + \cancel{\mathrm{e}^{-i\alpha}\langle 2|1\rangle}\right] = \frac{1}{2} \\
\langle 1|\hat{\rho}_c|2\rangle &= \frac{1}{2}\left[\langle 1|1\rangle + \cancel{\mathrm{e}^{i\alpha}\langle 1|2\rangle}\right]\left[\cancel{\langle 1|2\rangle} + \mathrm{e}^{-i\alpha}\langle 2|2\rangle\right] = \frac{1}{2}\mathrm{e}^{-i\alpha} \\
\langle 2|\hat{\rho}_c|1\rangle &= \frac{1}{2}\left[\cancel{\langle 2|1\rangle} + \mathrm{e}^{i\alpha}\langle 2|2\rangle\right]\left[\langle 1|1\rangle + \cancel{\mathrm{e}^{-i\alpha}\langle 2|1\rangle}\right] = \frac{1}{2}\mathrm{e}^{i\alpha} \\
\langle 2|\hat{\rho}_c|2\rangle &= \frac{1}{2}\left[\cancel{\langle 2|1\rangle} + \mathrm{e}^{i\alpha}\langle 2|2\rangle\right]\left[\cancel{\langle 1|2\rangle} + \mathrm{e}^{-i\alpha}\langle 2|2\rangle\right] = \frac{1}{2}
\end{aligned} \tag{G.98}$$

More concisely

$$\rho_c = \begin{pmatrix} 0.5 & 0.5\mathrm{e}^{-i\alpha} \\ 0.5\mathrm{e}^{i\alpha} & 0.5 \end{pmatrix} \tag{G.99}$$

Verification:

1. **Density matrix is self-adjoint:**

$$\rho_a^\dagger = \rho_a, \quad \rho_b^\dagger = \rho_b, \quad \rho_c^\dagger = \rho_c$$

We can show this for ρ_c:

$$\rho_c = \begin{pmatrix} 0.5 & 0.5\mathrm{e}^{-i\alpha} \\ 0.5\mathrm{e}^{i\alpha} & 0.5 \end{pmatrix} \to \rho_c^T = \begin{pmatrix} 0.5 & 0.5\mathrm{e}^{i\alpha} \\ 0.5\mathrm{e}^{-i\alpha} & 0.5 \end{pmatrix} \to \rho_c^\dagger = \begin{pmatrix} 0.5 & 0.5\mathrm{e}^{-i\alpha} \\ 0.5\mathrm{e}^{i\alpha} & 0.5 \end{pmatrix} \tag{G.100}$$

2. **The eigenvalues of the three operators are real numbers:**

$$\lambda_{1,a} = 0; \quad \lambda_{2,a} = 1$$

$$\lambda_{1,b} = 0.5; \quad \lambda_{2,b} = 0.5$$

$$\lambda_{1,c} = 0; \quad \lambda_{2,c} = 1$$

3. **The eigenvectors corresponding to distinct eigenvalues of the density matrix are orthogonal to each other**. For instance for matrix ρ_a:

$$\vec{v}_{1,a} = (0,1); \vec{v}_{2,a} = (1,0) \to \langle \vec{v}_{1,a}|\vec{v}_{1,b}\rangle = 0$$

G.4 Problem Chapter 4

Problem 4.1

We can safely assume that Ne is ideal under the conditions of large temperature (much larger than the critical temperature, which is 44.45 K) and low pressure (much lower than the critical pressure, which is 26.6 bar). We know that the entropy of an ideal monoatomic gas is provided by the Sackur-Tetrode equation:

$$S(N,V,E) = kN\left\{\ln\left[\frac{V}{N}\left(\frac{4\pi mE}{3h^2N}\right)^{3/2}\right] + \frac{5}{2}\right\}, \tag{G.101}$$

which can also be written as

$$S(N,V,E) = kN \ln \left[\left(\frac{2\pi mkT}{h^2} \right)^{3/2} \frac{V\mathrm{e}^{5/2}}{N} \right] \tag{G.102}$$

At this point, to find a numerical value, we need some data. Specifically, $m = 3.351^{-26}$ kg, $k = 1.3807 \times 10^{-23}$ JK^{-1}, $h = 6.6262 \times 10^{-34}$ Js and $p = 10^5$ Pa. Substituting these values in the above equations, we get

$$\frac{V}{N} = \frac{kT}{p} = 4.142 \times 10^{-26}\,\mathrm{m}^3 \tag{G.103}$$

and

$$\left(\frac{2\pi mkT}{h^2} \right)^{3/2} = 8.853 \times 10^{31}\,\mathrm{m}^{-3} \tag{G.104}$$

Therefore

$$s = \frac{S}{N} = 2.432 \times 10^{-22}\,\mathrm{J\ K}^{-1} \tag{G.105}$$

To calculate the chemical potential, from thermodynamics we know that

$$\mu = -T \left. \frac{\partial S}{\partial N} \right|_{E,V} \tag{G.106}$$

We therefore need to differentiate the Sackur-Tetrode equation with respect to N:

$$\mu = -kT \left. \frac{\partial \left\{ N \left\{ \ln \left[\frac{V}{N} \left(\frac{4\pi mE}{3h^2 N} \right)^{3/2} \right] + \frac{5}{2} \right\} \right\}}{\partial N} \right|_{E,V} = -kT \ln \left[\frac{V}{N} \left(\frac{4\pi mE}{3h^2 N} \right)^{3/2} \right] \tag{G.107}$$

We can now substitute $E = \frac{3}{2}NkT$ and obtain

$$\mu = -kT \ln \left[\frac{V}{N} \left(\frac{2\pi mkT}{h^2} \right)^{3/2} \right] = -6.261 \times 10^{-20}\,\mathrm{J} \tag{G.108}$$

Problem 4.2

(a) Let's start from the Sackur-Tetrode equation:

$$S(N,V,E) = kN \left\{ \ln \left[\frac{V}{N} \left(\frac{4\pi mE}{3h^2 N} \right)^{3/2} \right] + \frac{5}{2} \right\}, \tag{G.109}$$

The energy being held constant in the system is the mechanical energy, not just the kinetic energy that we see in the above equation. Remember that the kinetic energy is all we get if the gas is ideal. Therefore, in the Sackur-Tetrode equation, we substitute the kinetic energy E using the total (mechanical) energy and the potential energy given by gravity and depending on z:

$$U = E + Nmgz \tag{G.110}$$

Note that mgz is the potential energy acting on a single particle. We need to multiply by N to get the total potential energy. It follows that the above Sackur-Tetrode equation can be re-written as

$$S(N,V,E) = kN \left\{ \ln \left[\frac{V}{N} \left(\frac{4\pi m(U - Nmgz)}{3h^2 N} \right)^{3/2} \right] + \frac{5}{2} \right\}, \tag{G.111}$$

Now, simplifying we get

$$S(N,V,E) = kN\left\{\ln\left[C(U-Nmgz)^{3/2}\right] - \ln N^{5/2} + \frac{5}{2}\right\}, \tag{G.112}$$

where

$$C = V\left(\frac{4\pi m}{3h^2}\right)^{3/2} \tag{G.113}$$

We can now calculate the chemical potential from

$$\mu = -T\left.\frac{\partial S}{\partial N}\right|_{E,V} \tag{G.114}$$

In particular

$$\begin{aligned}
\mu =& -kT\left[\ln\left(C(U-Nmgz)^{3/2}\right) - \ln N^{5/2} + \frac{5}{2}\right] + \\
& - NkT\left[\frac{\frac{3}{2}C(U-Nmgz)^{1/2}(-mgz)}{C(U-Nmgz)^{3/2}} + \frac{1}{N^{5/2}}\left(\frac{5}{2}N^{3/2}\right)\right] = \\
& = -kT\left[\ln\left(C(U-Nmgz)^{3/2}\right) - \ln N^{5/2} + \frac{5}{2}\right] + \\
& + \frac{3}{2}NkT\frac{mgz}{(U-Nmgz)} + \frac{5kT}{2} = \\
& = -kT\left[\ln\left(C(U-Nmgz)^{3/2}\right) - \ln N^{5/2}\right] + \\
& + \frac{3}{2}NkT\frac{mgz}{(U-Nmgz)}
\end{aligned} \tag{G.115}$$

Now, we see that $U - Nmgz = E = \frac{3}{2}NkT$. Therefore:

$$\begin{aligned}
\mu &= -kT\left[\ln\left(C(\frac{3}{2}NkT)^{3/2}\right) - \ln N^{5/2}\right] + \frac{3}{2}NkT\frac{mgz}{\frac{3}{2}NkT} = \\
&= -kT\left[\ln\left(C(\frac{3}{2}NkT)^{3/2}\right) - \ln N^{5/2}\right] + mgz
\end{aligned} \tag{G.116}$$

and substituting the constant C, we finally obtain the expected result.

(b) If the two volumes of gas are at thermodynamic equilibrium, then

$$\mu(z) = \mu(0) \tag{G.117}$$

It follows that

$$\begin{aligned}
& -kT\ln\left[\frac{V}{N(z)}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] + mgz = -kT\ln\left[\frac{V}{N(0)}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] \Longleftrightarrow \\
& \Longleftrightarrow \ln\left[\frac{V}{N(z)}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] - \frac{mgz}{kT} = \ln\left[\frac{V}{N(0)}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] \Longleftrightarrow \\
& \Longleftrightarrow \frac{V}{N(z)}\left(\frac{2\pi mkT}{h^2}\right)^{3/2} \mathrm{e}^{-mgz/kT} = \frac{V}{N(0)}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}
\end{aligned} \tag{G.118}$$

Finally, we obtain

$$N(z) = N(0)\mathrm{e}^{-mgz/kT} \tag{G.119}$$

Problem 4.3

The number of particles with energy $-\epsilon$ is $n_- = N - n_+$. In particular:

$$n_- + n_+ = N; \quad (n_+ - n_-)\epsilon = E \quad \Longrightarrow \quad n_+ = \frac{1}{2}\left(N + \frac{E}{\epsilon}\right) \quad n_- = \frac{1}{2}\left(N - \frac{E}{\epsilon}\right) \tag{G.120}$$

The number of microstates with energy E is equal to the number of ways to choose n_+ particles from a set of N.

$$\Omega(E, N) = \binom{N}{n_+} = \frac{N!}{n_+!(N - n_+)!} \tag{G.121}$$

We can safely assume that $n_+ \to \infty$, $n_- \to \infty$ and $N \to \infty$, apply Stirling's approximation and hence calculate the system's entropy:

$$S = k \ln \Omega = k \left[N \ln N - N - n_+ \ln n_+ + n_+ - (N - n_+)\ln(N - n_+) + (N - n_+)\right] \tag{G.122}$$

Let's now introduce the variable $x = \frac{n_+}{N} = \frac{1}{2}\left(1 + \frac{E}{N\epsilon}\right)$. The above equation can be re-written as

$$\begin{aligned} S = kN\left[\ln N - 1 - x \ln x - x \ln N + x - (1-x)\ln(1-x) - (1-x)\ln N + 1 - x\right] \simeq \\ \simeq -kN\left[x \ln x + (1-x)\ln(1-x)\right] \end{aligned} \tag{G.123}$$

which is clearly an extensive property. The temperature can be calculated as

$$\frac{1}{T} = \left.\frac{\partial S}{\partial E}\right|_N = \left.\frac{\partial S}{\partial x}\right|_N \left.\frac{\partial x}{\partial E}\right|_N = \frac{1}{2N\epsilon}\left.\frac{\partial S}{\partial x}\right|_N = \frac{k}{2\epsilon}\ln\left(\frac{1-x}{x}\right) \tag{G.124}$$

from which we get

$$x = \frac{n_+}{N} = \frac{1}{\mathrm{e}^{2\epsilon/kT} + 1}; \quad E = (n_+ - n_-)\epsilon = -N\epsilon \tanh\left(\frac{\epsilon}{kT}\right) \tag{G.125}$$

We can now calculate the Helmholtz free energy:

$$A(T, N) = E(T, N) - TS(T, N) = -N\epsilon \tanh\left(\frac{\epsilon}{kT}\right) + kTN\left[x \ln x (1-x)\ln(1-x)\right] \tag{G.126}$$

If $T \to 0$, then $\tanh(\epsilon/kT) \to 1$ and $E = -N\epsilon$, and also $x \to 0$ and $S \to 0$. By contrast, if $T \to \infty$, then $\tanh(\epsilon/kT) \to 0$ and $E = 0$, and also $x \to 1/2$ and $S \to Nk \ln 2$.

Problem 4.4

We can calculate the entropy from the equation:

$$S = k \ln \Omega \tag{G.127}$$

where

$$\Omega = \frac{\partial \phi}{\partial E}\Delta E \tag{G.128}$$

and ϕ is the number of microstates with energy in the interval $[0, E]$. In particular

$$\phi = \frac{1}{h^{3N}}\int \theta(E - H)\, d\mathbf{q}\, d\mathbf{p} = \frac{1}{h^{3N}}\int \theta(E - H)\, dq_1 \ldots dq_{3N} dp_1 \ldots dp_{3N} \tag{G.129}$$

where θ is the Heaviside step function. Note the factor h^{3N} in the denominator, which turns the number of states into a dimensionless observable. Additionally, the harmonic oscillators can only vibrate around their position, suggesting that the system studied is a solid and thus that particles can be assumed distinguishable. This factor represents the Boltzmann correction in the case of *distinguishable* particles. More specifically

$$\phi = \frac{1}{h^{3N}} \int \theta \left[E - \sum_{i=1}^{3N} \left(\frac{p_i^2}{2m} + \frac{1}{2} m\omega^2 q_i^2 \right) \right] dq_1 \ldots dq_{3N} dp_1 \ldots dp_{3N} \tag{G.130}$$

We now perform the following variable substitution for the first $3N$ degrees of freedom:

$$x_i = \frac{p_i}{\sqrt{2m}} \quad i = 1, \ldots, 3N \tag{G.131}$$

and for the remaining 3N degrees of freedom:

$$x_{3N+i} = \sqrt{\frac{m\omega^2}{2}} q_i \quad i = 1, \ldots, 3N \tag{G.132}$$

The integral takes the form

$$\phi = \left(\frac{2}{\hbar\omega} \right)^{3N} \int \theta \left[E - \sum_{i=1}^{6N} x_i^2 \right] dx_1 \ldots dx_{6N} \tag{G.133}$$

where the integral represents the volume of a 6N-hypersphere of radius $\sqrt{E}$. Therefore:

$$\phi = \left(\frac{2}{\hbar\omega} \right)^{3N} \frac{\pi^{3N}}{\Gamma(3N+1)} E^{3N} = \left(\frac{2}{\hbar\omega} \right)^{3N} \frac{\pi^{3N}}{3N\Gamma(3N)} E^{3N} \tag{G.134}$$

We finally find that

$$\Omega = \left(\frac{2\pi E}{\hbar\omega} \right)^{3N} \frac{1}{(3N-1)!} \frac{\Delta E}{E} \simeq \left(\frac{E}{\hbar\omega} \right)^{3N} \frac{1}{(3N)!} \frac{\Delta E}{E} \tag{G.135}$$

The entropy reads

$$S = k \ln \Omega \simeq 3Nk \left[\ln \left(\frac{E}{3\hbar\omega N} \right) + 1 \right] \tag{G.136}$$

and the temperature:

$$\frac{1}{T} = \left. \frac{\partial S}{\partial E} \right|_N = \frac{3Nk}{E} \tag{G.137}$$

from which $E = 3NkT$.

Problem 4.5

The expression we will use for the microcanonical ensemble is as follows:

$$0 \leq E \leq H \tag{G.138}$$

$$\Omega(E) = \frac{1}{h^f} \int_{E_0 \leq H \leq E} dq \, dp \tag{G.139}$$

where $f = 2N$ and $E_0 = 0$. First, we will calculate $\Omega(E)$, which is related to the system's entropy, using the equation:

$$S = k \ln \Omega(E) \tag{G.140}$$

On the other hand, we know that the potential of the two-dimensional harmonic oscillator is given by the expression:

$$V(x,y) = \frac{1}{2}m\omega^2(x^2+y^2) \tag{G.141}$$

So, the Hamiltonian would be in the following form:

$$H = \sum_{i=1}^{N}\left(\frac{p_i^2}{2m_i} + \frac{1}{2}m_i\omega_i^2(x_i^2+y_i^2)\right) = E \tag{G.142}$$

We have that the oscillators in our problem are two-dimensional, and therefore, each particle has two degrees of freedom, so the total number of degrees of freedom for the system will be $2N$. Before calculating $\Omega(E)$, let's manipulate the above expression to simplify the integral when solving it:

$$1 = \sum_{i=1}^{N}\left(\frac{p_i^2}{m_iE} + \frac{1}{2E}m_i(\omega_i^2x_i^2+\omega_i^2y_i^2)\right) \tag{G.143}$$

Now, let's perform a change of variables:

$$X_i^2 = \omega_i\sqrt{\frac{m_i}{2}}x_i$$
$$Y_i^2 = \omega_i\sqrt{\frac{m_i}{2}}y_i$$
$$v_i^2 = \frac{p_i}{\sqrt{2m_i}}$$

This transformation simplifies the expression to:

$$1 = \frac{1}{\sqrt{E}}\sum_{i=1}^{N}\left(v_i^2+X_i^2+Y_i^2\right) \tag{G.144}$$

Taking into account the change of variables, $\Omega(E)$ is now given by the following expression:

$$\Omega = \frac{1}{\Pi_{i=1}^{N}\omega_i^2}\left(\frac{2}{h}\right)^{2N}\underbrace{\int_{0\leq H\leq E} d^2v_1\ldots d^2v_N\, d^2X_1\ldots d^2X_N\, d^2Y_1\ldots d^2Y_N}_{\text{volume of a hypersphere}} \tag{G.145}$$

This expression represents the volume of a hypersphere of radius E and dimension $4N$, which can be calculated using the formula for the volume of a hypersphere:

$$V_{4N}(E) = \frac{\pi^{\frac{4N}{2}}}{\Gamma\left(\frac{4N}{2}+1\right)}E^{2N} = \frac{(E\pi)^{2N}}{(2N)!} \tag{G.146}$$

Therefore, Ω can be expressed as:

$$\Omega = \frac{1}{(2N)!}\frac{1}{\Pi_{i=1}^{N}\omega_i^2}\left(\frac{2\pi E}{h}\right)^{2N} = \frac{1}{(2N)!}\frac{1}{\Pi_{i=1}^{N}\omega_i^2}\left(\frac{E}{\hbar}\right)^{2N} \tag{G.147}$$

and then we can calculate the entropy:

$$\begin{aligned} S = k\ln\Omega &= k\ln\left[\frac{1}{(2N)!}\frac{1}{\Pi_{i=1}^{N}\omega_i^2}\left(\frac{E}{\hbar}\right)^{2N}\right] \\ &= 2k\left[N\left(\ln\left(\frac{E}{2N\hbar}\right)+1\right) - \sum_{i=1}^{N}\ln\omega_i\right] \end{aligned} \tag{G.148}$$

The specific heat at constant volume reads

$$c_V = \frac{C_V}{N} = \frac{1}{N}\left.\frac{\partial E}{\partial T}\right|_V = \frac{2Nk}{N} = 2k \tag{G.149}$$

Finally, the chemical potential reads

$$-\frac{\mu}{T} = \left.\frac{\partial S}{\partial N}\right|_{E,V} = 2k \ln\frac{E}{2N\hbar} \Longleftrightarrow \mu = -2kT\ln\frac{E}{2N\hbar} \tag{G.150}$$

G.5 Problem Chapter 5

Problem 5.1

The canonical partition function is given by

$$Z = Z_1^N = \left[\sum_{j=1}^{3}\exp(-\beta\epsilon_j)\right]^N = [1+\exp(-\beta\epsilon)+\exp(-3\beta\epsilon)]^N \tag{G.151}$$

The internal energy can be calculated as

$$U = -\left.\frac{\partial \ln Z}{\partial\beta}\right|_{N,V} = -\left.\frac{N\partial\ln[1+\exp(-\beta\epsilon)+\exp(-3\beta\epsilon)]}{\partial\beta}\right|_{N,V} = \frac{N\exp(-\beta\epsilon)\epsilon[1+3\exp(-2\beta\epsilon)]}{1+\exp(-\beta\epsilon)+\exp(-3\beta\epsilon)} \tag{G.152}$$

At $T\to\infty$, we have:

$$U|_{T\to\infty} = N\frac{4\epsilon}{3} \tag{G.153}$$

and the specific internal energy reads

$$u|_{T\to\infty} = \frac{4\epsilon}{3} \tag{G.154}$$

By contrast, for $T\to 0$, the specific internal energy tends to zero:

$$u|_{T\to 0} \sim \epsilon\exp(-\beta\epsilon) \sim 0 \tag{G.155}$$

Let's now consider the specific entropy:

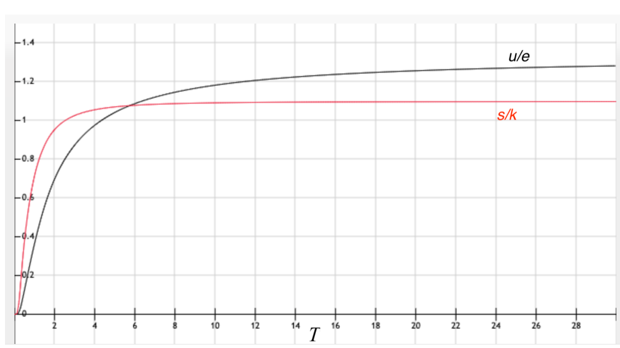


Figure G.10: Specific internal energy and entropy.

$$
\begin{aligned}
S &= \\
&= kT\left.\frac{\partial \ln Z}{\partial T}\right|_{N,V} + k\ln Z = \\
&= kT\left.\frac{\partial \ln Z}{\partial \beta}\right|_{N,V}\frac{\partial \beta}{\partial T} + k\ln Z = \\
&= k\left(-\frac{1}{T}\left.\frac{\partial \ln Z}{\partial \beta}\right|_{N,V} + k\ln Z\right) = \\
&= k\left[\frac{1}{T}\frac{N\exp(-\beta\epsilon)\epsilon[1+3\exp(-2\beta\epsilon)]}{1+\exp(-\beta\epsilon)+\exp(-3\beta\epsilon)} + N\ln[1+\exp(-\beta\epsilon)+\exp(-3\beta\epsilon)]\right]
\end{aligned} \tag{G.156}
$$

For $T \to \infty$, this expression reduces to

$$s|_{T\to\infty} \sim k\left(\frac{\epsilon}{T} + \ln 3\right) = k\ln 3 \tag{G.157}$$

and for $T \to 0$, we obtain

$$s|_{T\to 0} \sim k\frac{\epsilon\exp(-\beta\epsilon)}{T} \sim 0 \tag{G.158}$$

Problem 5.2

The canonical partition function reads

$$Z_N = Z(N,V,T) = \frac{1}{N!h^s}\int \mathrm{e}^{-\beta H} d^s q\, d^s p \tag{G.159}$$

A bit more explicitly:

$$Z_N = \frac{1}{N!h^{3N}}\int \mathrm{e}^{-\beta\sum_{i=1}^N p_i^2/2m} d^{3N}q\, d^{3N}p = \tag{G.160}$$

This can be further simplified to

$$Z_N = \frac{V^N}{N!h^{3N}}\left(\int_{-\infty}^{+\infty}\mathrm{e}^{-\beta p^2/2m}dp\right)^{3N} = \frac{V^N}{N!h^{3N}}\left(\int_{-\infty}^{+\infty}\sqrt{2kTm}\mathrm{e}^{-x^2}dx\right)^{3N} \tag{G.161}$$

where we have used the following variable substitution:

$$x = \frac{p}{\sqrt{2kTm}}; \quad dx = \frac{1}{\sqrt{2kTm}}dp \tag{G.162}$$

Then we have:

$$Z_N = \frac{V^N}{N!}\left(\frac{\sqrt{2\pi kTm}}{h}\right)^{3N} \tag{G.163}$$

where we have used

$$\int_{-\infty}^{+\infty}\mathrm{e}^{-x^2}dx = \sqrt{\pi} \tag{G.164}$$

Finally, we obtain the canonical partition function for an ideal gas:

$$Z_N = \frac{1}{N!}\left(\frac{V}{\Lambda^3}\right)^N. \tag{G.165}$$

Problem 5.3

Because molecules are not interacting with each other, we can assume that the gas is ideal. The canonical partition function of N noninteracting molecules can be written as a function of the canonical partition function of a single molecule:

$$Z_N = \frac{1}{N!} Z_1^N \tag{G.166}$$

where

$$Z_1 = \frac{1}{h^3} \int \mathrm{e}^{-\beta H_1(\mathbf{q},\mathbf{p}) dq^3\, dp^3} \tag{G.167}$$

The Hamiltonian of a generic molecule reads

$$H_i = \frac{p^2}{2m} + V(z) + \varepsilon_i \tag{G.168}$$

where ε_{i} is the energy associated to the internal degrees of freedom of a generic molecule i. Since the Hamiltonian of a molecule is the sum of different terms, the canonical partition function of a molecule also factors into a classical part Z_{cl} and a quantum part related to the internal degree of freedom, Z_{in}, i.e.

$$Z_1 = Z_{cl} \times Z_{in} \tag{G.169}$$

In particular

$$Z_{in} = \sum_{\varepsilon=0,\epsilon} \Omega(\varepsilon) \mathrm{e}^{-\beta\varepsilon} = 1 + 3\mathrm{e}^{-\beta\epsilon} \tag{G.170}$$

and

$$\begin{aligned}
Z_{cl} &= \frac{1}{h^3} \int_S dq_x\, dq_y \int_0^{4L} dq_z \int dp_x\, dp_y\, dp_z \mathrm{e}^{-\beta\left(\frac{p^2}{2m}+V(z)\right)} = \\
&= \frac{1}{h^3} S \left[\int_0^{2L} \mathrm{e}^{-2\beta V_0} dq_z + \int_{2L}^{3L} \mathrm{e}^{-\beta V_0} dq_z + \int_{3L}^{4L} dq_z \right] \int dp_x\, dp_y\, dp_z \mathrm{e}^{-\beta\frac{p^2}{2m}} = \\
&= \frac{1}{h^3} SL(2\mathrm{e}^{-2\beta V_0} + \mathrm{e}^{-\beta V0} + 1) \int dp_x\, dp_y\, dp_z \mathrm{e}^{-\beta\frac{p^2}{2m}} = \\
&= \frac{V'}{h^3} (2\mathrm{e}^{-2\beta V_0} + \mathrm{e}^{-\beta V0} + 1) \int dp_x\, dp_y\, dp_z \mathrm{e}^{-\beta\frac{p^2}{2m}}
\end{aligned} \tag{G.171}$$

where $V' = SL$ is a reference volume, equal to $1/4$ of the volume of the container. Therefore

$$Z_1 = Z_{cl} \times Z_{in} = \frac{V'}{h^3}(1 + 3\mathrm{e}^{-\beta\epsilon})(2\mathrm{e}^{-2\beta V_0} + \mathrm{e}^{-\beta V0} + 1) \int dp_x\, dp_y\, dp_z \mathrm{e}^{-\beta\frac{p^2}{2m}} \tag{G.172}$$

To solve the last integral, we apply a variable substitution for each p_i, with $i = x, y, z$:

$$w_i = \frac{p_i}{\sqrt{2kTm}}; \quad dw_i = \frac{1}{\sqrt{2kTm}} dp_i \tag{G.173}$$

It follows that

$$
\begin{aligned}
Z_1 &= \frac{V'}{h^3}(1+3\mathrm{e}^{-\beta\epsilon})(2\mathrm{e}^{-2\beta V_0}+\mathrm{e}^{-\beta V0}+1)(2kTm)^{\frac{3}{2}}\left(\underbrace{\int \mathrm{e}^{-w^2}dw}_{\sqrt{\pi}}\right)^3 = \\
&= \left(\frac{\sqrt{(2\pi kTm)}}{h^2}\right)^3 V'(1+\mathrm{e}^{-3\beta\epsilon})(2\mathrm{e}^{-2\beta V_0}+\mathrm{e}^{-\beta V0}+1) = \\
&= \frac{V'}{\Lambda^3}(1+3\mathrm{e}^{-\beta\epsilon})(2\mathrm{e}^{-2\beta V_0}+\mathrm{e}^{-\beta V0}+1)
\end{aligned}
\tag{G.174}
$$

where Λ is the de Broglie thermal wave length. Let's now calculate the probability that a randomly-selected molecule is in region A, B and C of the container. To this end, we just need to realise that the only difference between the three regions is given to the potential energy $V(z)$. As such, we have

I. $P_A = \frac{\int \mathrm{e}^{-\beta H_1} d\mathbf{q}\, d\mathbf{p}}{Z_1} = \frac{2\mathrm{e}^{-2\beta V_0}}{2\mathrm{e}^{-2\beta V_0}+\mathrm{e}^{-\beta V0}+1}$

II. $P_B = \frac{\mathrm{e}^{-\beta V_0}}{2\mathrm{e}^{-2\beta V_0}+\mathrm{e}^{-\beta V0}+1}$

III. $P_C = \frac{1}{2\mathrm{e}^{-2\beta V_0}+\mathrm{e}^{-\beta V0}+1}$

If $T\to\infty$, then $P_A = 1/2$ and $P_B = P_C = 1/4$. By contrast, if $T\to 0$, then $P_A = P_B = 0$ and $P_C = 1$.

Finally, we want to determine the average number of molecules in region B with energy state ϵ. To this end, we first calculate the probability to find a molecule in region B with energy state ϵ, which reads:

$$
P(\epsilon|B) = P_B \times P_\epsilon = \frac{\mathrm{e}^{-\beta V_0}}{2\mathrm{e}^{-2\beta V_0}+\mathrm{e}^{-\beta V0}+1} \times \frac{3\mathrm{e}^{-\beta\epsilon}}{1+3\mathrm{e}^{-\beta\epsilon}}
\tag{G.175}
$$

It follows that

$$
\langle N\rangle_{B,\epsilon} = N\frac{\mathrm{e}^{-\beta V_0}}{2\mathrm{e}^{-2\beta V_0}+\mathrm{e}^{-\beta V0}+1} \times \frac{3\mathrm{e}^{-\beta\epsilon}}{1+3\mathrm{e}^{-\beta\epsilon}}
\tag{G.176}
$$

Problem 5.4

(a) To obtain the Helmholtz free energy, we need to calculate the partition function. Since this gas can be assumed as an ideal one, the following equation holds

$$
Z_N = \frac{1}{N!}Z_1^N
\tag{G.177}
$$

where the partition function of a single dipolar molecule, Z_1, depends on five different degrees of freedom $(q_x, q_y, q_z, \theta, \phi)$ and reads

$$
\begin{aligned}
Z_1 &= \frac{1}{h^5}\int dq_x\, dq_y\, dq_z\, d\theta\, d\phi\, dp_x\, dp_y\, dp_z\, dp_\theta\, dp_\phi\, \mathrm{e}^{-\beta H_1} \\
&= \frac{1}{h^5}\int_{-\infty}^{+\infty} dp_x\, \mathrm{e}^{-\beta p_x^2/2m}\int_{-\infty}^{+\infty} dp_y\, \mathrm{e}^{-\beta p_y^2/2m}\int_{-\infty}^{+\infty} dp_z\, \mathrm{e}^{-\beta p_z^2/2m} \\
&\int_{-\infty}^{+\infty} dp_\phi\, \mathrm{e}^{-\beta p_\phi^2/2I\sin^2\theta}\int_{-\infty}^{+\infty} dp_\theta\, \mathrm{e}^{-\beta p_\theta^2/2I}\int_V dq_x\, dq_y\, dq_z \int_0^{2\pi} d\phi \int_0^{\pi} d\theta\, \mathrm{e}^{-\beta(-\mu E\cos\theta)}
\end{aligned}
\tag{G.178}
$$

To solve the first three integrals, we apply a variable substitution for each p_i, with $i = x, y, z$:

$$w_i = \frac{p_i}{\sqrt{2kTm}}; \quad dw_i = \frac{1}{\sqrt{2kTm}} dp_i \tag{G.179}$$

Similarly, for the fourth integral, the variable substitution is

$$w_\phi = \frac{p_\phi}{\sqrt{2kTI \sin^2 \theta}}; \quad dw_\phi = \frac{1}{\sqrt{2kTI \sin^2 \theta}} dp_\phi \tag{G.180}$$

and for the fifth integral

$$w_\theta = \frac{p_\theta}{\sqrt{2kTI}}; \quad dw_\theta = \frac{1}{\sqrt{2kTI}} dp_\theta \tag{G.181}$$

Therefore, we get

$$\begin{aligned} Z_1 &= \frac{1}{h^5} (2\pi mkT)^{\frac{3}{2}} (2\pi IkT)^{\frac{1}{2}} V 2\pi \int_0^\pi d\theta \, (2\pi IkT \sin^2 \theta)^{\frac{1}{2}} \mathrm{e}^{\beta \mu E \cos\theta} = \\ &= \frac{4\pi^2 VIkT (2\pi mkT)^{\frac{3}{2}}}{h^5} \int_0^\pi d\theta \, \sin\theta \, \mathrm{e}^{\beta \mu E \cos\theta} \end{aligned} \tag{G.182}$$

In the last integral, we apply the following variable substitution:

$$x = \cos\theta; \quad dx = -\sin\theta \, d\theta \tag{G.183}$$

It follows that

$$\begin{aligned} Z_1 &= \frac{4\pi^2 VIkT (2\pi mkT)^{\frac{3}{2}}}{h^5} \int_{-1}^1 dx \, \mathrm{e}^{\beta \mu E x} = \\ &= \frac{4\pi^2 VIkT (2\pi mkT)^{\frac{3}{2}}}{h^5} \frac{1}{\beta \mu E} \left(\mathrm{e}^{\beta \mu E} - \mathrm{e}^{-\beta \mu E} \right) = \\ &= \frac{4\pi^2 VIkT (2\pi mkT)^{\frac{3}{2}}}{h^5} \frac{2kT}{\mu E} \sinh \left(\frac{\mu E}{kT} \right) = \\ &= \frac{8(\pi kT)^2 VI (2\pi mkT)^{\frac{3}{2}}}{h^5 \mu E} \sinh \left(\frac{\mu E}{kT} \right) \end{aligned} \tag{G.184}$$

The Helmholtz free energy of an ideal gas containing N dipolar molecules reads

$$\begin{aligned} A &= -kT \ln Z_N = -kT \ln \left(\frac{1}{N!} Z_1^N \right) = -kT \left(-\ln N! + N \ln Z_1 \right) \simeq \\ &\simeq -kT \left(-N \ln N + N + N \ln Z_1 \right) = -NkT \left(\ln \frac{Z_1}{N} + 1 \right) = \\ &= -NkT \left\{ \ln \left[\frac{8(\pi kT)^2 VI (2\pi mkT)^{\frac{3}{2}}}{h^5 N \mu E} \sinh \left(\frac{\mu E}{kT} \right) \right] + 1 \right\} \end{aligned} \tag{G.185}$$

(b) The fraction of molecules that have an angular orientation θ within the interval $0 \leq \theta \leq \pi/4$ is given by the following probability:

$$\frac{N_\theta}{N} = P \left(\theta \in \left[0, \frac{\pi}{4} \right] \right) \tag{G.186}$$

where the probability P is obtained using the above calculated partition function:

$$P\left(\theta \in \left[0, \frac{\pi}{4}\right]\right) = \frac{1}{Z_1 h^5} \int_V dq_x\, dq_y\, dq_z \int_{-\infty}^{+\infty} dp_x\, dp_y\, dp_z \int_{-\infty}^{+\infty} dp_\theta\, dp_\phi \int_0^{2\pi} d\phi \int_0^{\pi/4} d\theta\, \mathrm{e}^{-\beta H_1} \tag{G.187}$$

Note that the difference between the two is just in the last integral. Therefore:

$$P\left(\theta \in \left[0, \frac{\pi}{4}\right]\right) = \frac{\int_0^{\pi/4} d\theta\, \sin\theta \mathrm{e}^{\beta\mu E\cos\theta}}{\int_0^{\pi} d\theta\, \sin\theta \mathrm{e}^{\beta\mu E\cos\theta}} \tag{G.188}$$

We apply the following variable substitution:

$$x = \cos\theta; \quad dx = -\sin\theta\, d\theta \tag{G.189}$$

and obtain

$$P\left(\theta \in \left[0, \frac{\pi}{4}\right]\right) = \frac{-\int_1^{\sqrt{2}/2} dx\, \mathrm{e}^{\beta\mu E x}}{-\int_1^{-1} dx\, \mathrm{e}^{\beta\mu E x}} = \frac{\mathrm{e}^{\beta\mu E\frac{\sqrt{2}}{2}} - \mathrm{e}^{\beta\mu E}}{\mathrm{e}^{-\beta\mu E} - \mathrm{e}^{\beta\mu E}} \tag{G.190}$$

Problem 5.5

(a) The Hamiltonian is the sum of two contributions:

$$H = T + V \tag{G.191}$$

where

$$T = \sum_{i=1}^{N} \frac{p_i^2}{2m} \tag{G.192}$$

and V cane be obtained from the corresponding centrifugal force:

$$F_{\mathrm{centr}} = -\frac{dV}{dr} = \frac{mv^2}{r} = m\omega^2 r \tag{G.193}$$

Therefore

$$V(r) = -\int m\omega^2 r\, dr = -\frac{m\omega^2 r^2}{2} \tag{G.194}$$

It follows that the Hamiltonian reads

$$H = \frac{1}{2}\sum_{i=1}^{N}\left(\frac{p_i^2}{m} - m\omega^2 r_i^2\right) \tag{G.195}$$

(b) To calculate the canonical partition function of an ideal gas, we notice that it can be expressed as

$$Z = \frac{Z_1^N}{N!} \tag{G.196}$$

where Z_1 is the partition function of a single gas atom. In particular

$$Z_1 = \frac{1}{h^3}\int \mathrm{e}^{-\frac{\beta}{2}\left(\frac{p^2}{2m} - m\omega^2 r^2\right)}\, d^3\mathbf{r}\, d^3\mathbf{p} \tag{G.197}$$

Let's solve these two integrals separately and in cylindrical coordinates:

$$I_p = \int_{-\infty}^{+\infty} \mathrm{e}^{-\frac{\beta p^2}{2m}} \underbrace{d^3\mathbf{p}}_{4\pi p^2 dp} = 4\pi \int_0^{+\infty} p^2 \mathrm{e}^{-\frac{\beta p^2}{2m}}\, dp \tag{G.198}$$

We now apply the following variable substitution:

$$t = \frac{\beta p^2}{2m}; \quad p = \sqrt{\frac{2m}{\beta}} t^{1/2}; \quad dt = \frac{\beta p}{m} dp \tag{G.199}$$

Therefore

$$\begin{aligned} I_p &= \frac{4\pi m}{\beta} \int_0^{+\infty} \sqrt{\frac{2m}{\beta}}\; t^{\frac{1}{2}} \mathrm{e}^{-t}\, dt = \\ &= 2\pi \left(\frac{2m}{\beta}\right)^{3/2} \int_0^{+\infty} t^{\frac{1}{2}} \mathrm{e}^{-t}\, dt \\ &= 2\pi \left(\frac{2m}{\beta}\right)^{3/2} \underbrace{\int_0^{+\infty} t^{\left(\frac{3}{2}-1\right)} \mathrm{e}^{-t}\, dt}_{\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}} = \\ &= \left(\frac{2m\pi}{\beta}\right)^{\frac{3}{2}} \end{aligned} \tag{G.200}$$

Let's now solve the second integral of the above equation:

$$I_r = \int_{-\infty}^{+\infty} \mathrm{e}^{\frac{\beta}{2} m\omega^2 r^2} \underbrace{d^3\mathbf{r}}_{r\, dr\, d\theta\, dz} = \int_{-\frac{L}{2}}^{\frac{L}{2}} dz \int_0^{2\pi} d\theta \int_0^R r \mathrm{e}^{\frac{\beta}{2} m\omega^2 r^2} dr = L 2\pi \int_0^R r \mathrm{e}^{\frac{\beta}{2} m\omega^2 r^2} dr \tag{G.201}$$

We apply the following variable substitution:

$$t = \frac{\beta m\omega^2 r^2}{2}; \quad r = \sqrt{\frac{2t}{\beta m\omega^2}}; \quad dt = \beta m\omega^2 r dr \tag{G.202}$$

Therefore

$$I_r = \frac{2\pi L}{\beta m\omega^2} \int_0^{\frac{\beta}{2} m\omega^2 R^2} \mathrm{e}^t dt = \frac{2\pi L}{\beta m\omega^2} \left(\mathrm{e}^{\frac{\beta}{2} m\omega^2 R^2} - 1\right) \tag{G.203}$$

We can finally obtain the partition function of a single atom:

$$Z_1 = \frac{1}{h^3} I_p I_r = \frac{1}{h^3} \left(\frac{2\pi}{\beta}\right)^{\frac{5}{2}} \frac{\sqrt{m} L}{\omega^2} \left(\mathrm{e}^{\frac{\beta}{2} m\omega^2 R^2} - 1\right) \tag{G.204}$$

and the total partition function reads

$$Z = \frac{Z_1^N}{N!} = \frac{1}{h^3 N!} \left[\left(\frac{2\pi}{\beta}\right)^{\frac{5}{2}} \frac{\sqrt{m} L}{\omega^2} \left(\mathrm{e}^{\frac{\beta}{2} m\omega^2 R^2} - 1\right)\right]^N \tag{G.205}$$

(c) Finally, the mean number of atoms per unit volume at a distance r from the axis of the cylinder can be calculated as follows

$$N(r) = C \mathrm{e}^{-\beta V(r)} \tag{G.206}$$

where C is a constant that can be found by imposing normalisation on $N(r)$:

$$\int N(r)\, d^3\mathbf{r} = N \tag{G.207}$$

In particular, using a similar variable substitution a that employed to solve integral I_r, we obtain

$$\int C\mathrm{e}^{-\beta V(r)} = C\int_{-\frac{L}{2}}^{\frac{L}{2}} dz \int_0^{2\pi} d\theta \int_0^R \mathrm{e}^{\frac{\beta}{2}m\omega^2 r^2} dr = C\frac{2\pi L}{\beta m\omega^2}\left(\mathrm{e}^{\frac{\beta}{2}m\omega^2 R^2} - 1\right) = N \tag{G.208}$$

We find that

$$C = \frac{N\beta m\omega^2}{2\pi L\left(\mathrm{e}^{\frac{\beta}{2}m\omega^2 R^2} - 1\right)} \tag{G.209}$$

and the number of atoms per unit volume reads

$$n(r) = \frac{N(r)}{V} = \frac{n\beta m\omega^2 \mathrm{e}^{\frac{\beta}{2}m\omega^2 r^2}}{2\pi L\left(\mathrm{e}^{\frac{\beta}{2}m\omega^2 R^2} - 1\right)} \tag{G.210}$$

G.6 Problem Chapter 6

Problem 6.1

The macrocanonical partition function can be expressed in terms of the canonical partition function:

$$\Xi(\mu, V, T) = \sum_{N=0}^{\infty} \lambda^N Z_N \tag{G.211}$$

where $\lambda = \mathrm{e}^{\beta\mu}$ is the gas fugacity. In addition, from Eq. G.165, we also know the expression of the canonical partition function Z_N. Therefore

$$\Xi(\mu, V, T) = \sum_{N=0}^{\infty} \mathrm{e}^{\beta\mu N}\frac{1}{N!}\left(\frac{V}{\Lambda^3}\right)^N = \sum_{N=0}^{\infty}\frac{1}{N!}\left(\mathrm{e}^{\beta\mu}\frac{V}{\Lambda^3}\right)^N = \sum_{N=0}^{\infty}\frac{1}{N!}x^N \tag{G.212}$$

where $x = \mathrm{e}^{\beta\mu}\frac{V}{\Lambda^3}$. The last term is the exponential function in series form. We then obtain:

$$\Xi(\mu, V, T) = \sum_{N=0}^{\infty}\frac{1}{N!}x^N = \exp\left(\mathrm{e}^{\beta\mu}\frac{V}{\Lambda^3}\right) \tag{G.213}$$

The macrocanonical potential can be easily obtained from the partition function:

$$J(\mu, V, T) = -kT\ln\Xi = -kT\mathrm{e}^{\beta\mu}\frac{V}{\Lambda^3} \tag{G.214}$$

Finally, the average particle number reads

$$\langle N(\mu, V, T)\rangle = -\left.\frac{\partial J}{\partial\mu}\right|_{T,V} = \mathrm{e}^{\beta\mu}\frac{V}{\Lambda^3} \tag{G.215}$$

Problem 6.2

To calculate the ideal-gas entropy, we can make use of the macrocanonical potential:

$$S(\mu, V, T) = -\left.\frac{\partial J}{\partial T}\right|_{\mu,V} = -\frac{\partial}{\partial T}\left[-kT\mathrm{e}^{\mu/kT}\frac{V}{\Lambda^3}\right] \tag{G.216}$$

Note that the thermal wavelength depends on T:

$$S(\mu, V, T) = -\frac{\partial}{\partial T}\left[-kT\mathrm{e}^{\mu/kT}\frac{V}{\Lambda^3}\right] = k\frac{\partial}{\partial T}\left[T\mathrm{e}^{\mu/kT}\frac{V(2\pi mkT)^{\frac{3}{2}}}{h^3}\right] = \frac{kV}{h^3}(2\pi mk)^{\frac{3}{2}}\frac{\partial}{\partial T}\left[T^{\frac{5}{2}}\mathrm{e}^{\mu/kT}\right] \tag{G.217}$$

Differentiating with respect to T, we obtain

$$S(\mu, V, T) = \frac{kV}{h^3}(2\pi mkT)^{\frac{3}{2}}\mathrm{e}^{\mu/kT}\left(\frac{5}{2} - \frac{\mu}{kT}\right) = k\underbrace{\frac{V}{\Lambda^3}\mathrm{e}^{\mu/kT}}_{=N \text{ (see 6.1)}}\left(\frac{5}{2} - \frac{\mu}{kT}\right) = kN\left(\frac{5}{2} - \frac{\mu}{kT}\right) \tag{G.218}$$

Inverting N for μ/kT, we obtain the usual form of the Sackur-Tetrode equation:

$$S(\mu, V, T) = kN\left(\frac{5}{2} + \ln\frac{V}{N\Lambda^3}\right) \tag{G.219}$$

Problem 6.3

In the grand canonical ensemble, the internal energy reads

$$U = U(T, V, \mu) = \langle E\rangle = \frac{1}{\Xi}\sum_{N=0}^{\infty}\mathrm{e}^{\beta\mu N}\sum_m E_{Nm}\mathrm{e}^{-\beta E_{Nm}} \tag{G.220}$$

The total differential of U is

$$dU = \left.\frac{\partial U}{\partial T}\right|_{V,\mu}dT + \left.\frac{\partial U}{\partial V}\right|_{T,\mu}dV + \left.\frac{\partial U}{\partial \mu}\right|_{T,V}d\mu \tag{G.221}$$

The change of U with N can be written as

$$\left.\frac{\partial U}{\partial N}\right|_{T,V} = \left.\frac{\partial U}{\partial \mu}\right|_{T,V}\left.\frac{\partial \mu}{\partial N}\right|_{T,V} \tag{G.222}$$

The change of the chemical potential with respecto to the average number of particles N can be obtained from the following equation

$$\langle N^2\rangle - \langle N\rangle^2 = \frac{1}{\beta}\left.\frac{\partial\langle N\rangle}{\partial \mu}\right|_{T,V} \tag{G.223}$$

We can then plug this result into Eq. G.222 and obtain:

$$\left.\frac{\partial U}{\partial N}\right|_{T,V} = \left.\frac{\partial U}{\partial \mu}\right|_{T,V}\frac{1}{\beta(\langle N^2\rangle - \langle N\rangle^2)} \tag{G.224}$$

In this equation, we can further expand the change of U with μ as follows

$$
\begin{aligned}
\left.\frac{\partial U}{\partial \mu}\right|_{T,V} &= \frac{\partial}{\partial \mu}\left[\frac{1}{\Xi}\sum_{Nm} \mathrm{e}^{\beta\mu N} E_{Nm}\mathrm{e}^{-\beta E_{Nm}}\right]_{T,V} = \\
&= -\left.\frac{\partial \Xi}{\partial \mu}\right|_{T,V}\frac{1}{\Xi^2}\left(\sum_{Nm}\mathrm{e}^{\beta\mu N}E_{Nm}\mathrm{e}^{-\beta E_{Nm}}\right) + \\
&+ \frac{1}{\Xi}\sum_{Nm}\beta N E_{Nm}\mathrm{e}^{\beta\mu N}\mathrm{e}^{-\beta E_{Nm}}
\end{aligned}
\tag{G.225}
$$

Now, from Eq. 6.63, we know that

$$
\left.\frac{\partial \ln \Xi}{\partial \mu}\right|_{T,V} = \beta\langle N\rangle \Longleftrightarrow \left.\frac{\partial \ln \Xi}{\partial \Xi}\right|_{T,V}\left.\frac{\partial \Xi}{\partial \mu}\right|_{T,V} = \frac{1}{\Xi}\left.\frac{\partial \Xi}{\partial \mu}\right|_{T,V} = \beta\langle N\rangle \tag{G.226}
$$

Substituting this result in the above equations, we get

$$
\left.\frac{\partial U}{\partial \mu}\right|_{T,V} = -\beta\langle N\rangle\underbrace{\frac{1}{\Xi}\left(\sum_{Nm}\mathrm{e}^{\beta\mu N}E_{Nm}\mathrm{e}^{-\beta E_{Nm}}\right)}_{\langle E\rangle} + \beta\langle NE\rangle = -\beta\langle N\rangle\langle E\rangle + \langle NE\rangle \tag{G.227}
$$

From which we obtain the desired relation.

Problem 6.4

(a) To find an expression for the fugacity, λ_g, of the classical ideal gas, we note that

$$
\lambda_g = e^{\beta\mu} \Longleftrightarrow \mu = kT\ln\lambda_g = \left.\frac{\partial A}{\partial N}\right|_{V,T} \tag{G.228}
$$

where A is the Helmholtz free energy that can be obtained from the canonical partition function Z_N:

$$
\begin{aligned}
A = -kT\ln Z_N = -kT\ln\left(\frac{1}{N!}Z_1^N\right) = -kT[-\ln N! + N\ln Z_1] \simeq \\
\simeq kT[N\ln N - N - N\ln Z_1] = -NkT\left(\ln\frac{Z_1}{N} + 1\right)
\end{aligned}
\tag{G.229}
$$

It follows that

$$
\begin{aligned}
\mu = \left.\frac{\partial A}{\partial N}\right|_{V,T} &= \frac{\partial}{\partial N}\left[-NkT\left(\ln\frac{Z_1}{N}+1\right)\right] = \\
&= -kT\frac{\partial}{\partial N}\left[N\left(\ln\frac{Z_1}{N}+1\right)\right] = -kT\ln\frac{Z_1}{N}
\end{aligned}
\tag{G.230}
$$

Note that Z_1 is independent from N and can be written as

$$
Z_1 = \frac{1}{h^3}\int_V dq_x\,dq_y\,dq_z \int_{-\infty}^{+\infty} dp_x \int_{-\infty}^{+\infty} dp_y \int_{-\infty}^{+\infty} dp_z \mathrm{e}^{-\beta H_1} \tag{G.231}
$$

where $H_1 = \mathrm{e}^{p^2/2m}$. Solving the integral by variable substitution, we obtain

$$
Z_1 = \frac{V}{h^3}(2m\pi kT)^{\frac{3}{2}} \tag{G.232}
$$

We can now obtain the expression for fugacity:

$$\lambda_g = \frac{N}{Z_1} = \frac{Nh^3}{V(2m\pi kT)^{\frac{3}{2}}} \tag{G.233}$$

Because the gas is ideal, the above ratio N/V can be rewritten by using the ideal-gas law:

$$pV = NkT \Longleftrightarrow \frac{N}{V} = \frac{p}{kT}. \tag{G.234}$$

Therefore

$$\lambda_g = \frac{p}{kT}\frac{h^3}{(2m\pi kT)^{\frac{3}{2}}} \tag{G.235}$$

An alternative form to solve this problem is calculating the macrocanonical partition function, which, for an ideal system of classical undistinguishable particles, reads

$$\Xi = \sum_{N=0}^{\infty} \lambda_g Z_N = \mathrm{e}^{\lambda_g Z_1} \tag{G.236}$$

Hence, we obtain the macrocanonical potential:

$$J = -kT \ln \Xi = -kT\lambda_g Z_1 \tag{G.237}$$

Now, we differentiate J with respect to V as follows

$$p = -\left.\frac{\partial J}{\partial V}\right|_{\mu,T} = \frac{\partial}{\partial V}(kT\lambda_g Z_1) = kT\lambda_g \frac{\partial Z_1}{\partial V} = kT\lambda_g \frac{(2\pi mkT)^{3/2}}{h^3} \tag{G.238}$$

It follows that

$$\lambda_g = \frac{p}{kT}\frac{h^3}{(2\pi mkT)^{\frac{3}{2}}} \tag{G.239}$$

which is the same results as that obtained above.

(b) To solve this problem, we can use the property obtained in Eq. 6.37, which, for N_0 absorption sites, reads

$$\Xi = \Pi_{i=1}^{N_0} \Xi_i \tag{G.240}$$

and because all sites are equivalent, their partition function is the same:

$$\Xi = \Pi_{i=1}^{N_0} \Xi_i = \Xi_1^{N_0} \tag{G.241}$$

In particular

$$\Xi_1 = \sum_{N=0}^{2} \lambda_{g,abs} Z_N = Z_0 + \lambda_{g,abs} Z_1 + \lambda_{g,abs}^2 Z_2 = 1 + \lambda_{g,abs}\mathrm{e}^{-\beta\epsilon} + \lambda_{g,abs}^2 \mathrm{e}^{-2\beta\epsilon} \tag{G.242}$$

It follows that

$$\Xi = \Xi_1^{N_0} = \left(1 + \lambda_{g,abs}\mathrm{e}^{-\beta\epsilon} + \lambda_{g,abs}^2 \mathrm{e}^{-2\beta\epsilon}\right)^{N_0} \tag{G.243}$$

From this, we obtain the macrocanonical potential:

$$J = -kT \ln \Xi = -kTN_0 \ln \left(1 + \lambda_{g,abs} e^{-\beta\epsilon} + \lambda_{g,abs}^2 e^{-2\beta\epsilon}\right) \tag{G.244}$$

and the average number of particles

$$\begin{aligned}\langle N \rangle = - \left.\frac{\partial J}{\partial \mu}\right|_{T,V} &= \frac{\partial J}{\partial \lambda_{g,abs}} \frac{\partial \lambda_{g,abs}}{\partial \mu} = -\beta\lambda_{g,abs} \frac{\partial J}{\partial \lambda_{g,abs}} = \\ &= \lambda_{g,abs} N_0 \frac{e^{-\beta\epsilon} + 2\lambda_{g,abs} e^{-2\beta\epsilon}}{1 + \lambda_{g,abs} e^{-\beta\epsilon} + \lambda_{g,abs}^2 e^{-2\beta\epsilon}}\end{aligned} \tag{G.245}$$

(c) To obtain the mean number of absorbed particles at equilibrium, we notice that

$$\mu_{g,abs} = \mu_g \Longleftrightarrow \lambda_{g,abs} = \lambda_g = \frac{p}{kT} \frac{h^3}{(2\pi mkT)^{\frac{3}{2}}} \tag{G.246}$$

We can just substitute this result in the equation of $\langle N \rangle$ found above. In particular, if $P \to \infty$ then also $\lambda_{g,abs} \to \infty$ and $\langle N \rangle = 2N_0$. By contrast, if $P \to 0$, then $\lambda_{g,abs} \to 0$ and $\langle N \rangle = 0$.

Problem 6.5

We have N distinguishable non-interacting particles, so the canonical partition function factorises as $Z_N = Z_1^N$. If $\lambda = e^{\beta\mu}$ is the fugacity for chemical potential μ, then the grand canonical partition function will be

$$\Xi(T, \lambda) = \sum_{N=0}^{\infty} \lambda^N Z_1^N = \frac{1}{1 - \lambda Z_1} \tag{G.247}$$

where we have assumed that $\lambda Z_1 < 1$ for the series to converge. The canonical partition function for a single particle in this particular case is

$$Z_1 = 1 + g e^{-\beta\epsilon} \tag{G.248}$$

from which we obtain

$$\Xi(T, \lambda) = \frac{1}{1 - \lambda(1 + g e^{-\beta\epsilon})} \tag{G.249}$$

Now, we calculate the grand canonical potential:

$$J = kT \ln \Xi = -kT \ln(1 - \lambda Z_1) = -kT \ln \left[1 - \lambda(1 + g e^{-\beta\epsilon})\right] \tag{G.250}$$

From which we obtain the mean number of particles in the system:

$$\langle N \rangle = \left.\frac{\partial J}{\partial \mu}\right|_{T,V} = \left.\frac{\partial J}{\partial \lambda}\right|_{T,V} \frac{\partial \lambda}{\partial \mu} = \beta\lambda \left.\frac{\partial J}{\partial \lambda}\right|_{T,V} = \beta\lambda kT \frac{Z_1}{1 - \lambda Z_1} = \frac{\lambda Z_1}{1 - \lambda Z_1} \tag{G.251}$$

To calculate the mean energy of the system, we apply Eq. 6.30:

$$\langle E \rangle = -\left.\frac{\partial \ln \Xi}{\partial \beta}\right|_{\lambda,V} \tag{G.252}$$

It is important to note that this derivative is performed while keeping the fugacity λ (or the product $\beta\mu$) constant, not just the chemical potential μ. Starting from the expression for Ξ and taking the derivative, we obtain

$$\langle E \rangle = g\epsilon \frac{\lambda e^{-\beta\epsilon}}{1 - \lambda(1 + g e^{-\beta\epsilon})} = \epsilon \langle N \rangle \frac{g e^{-\beta\epsilon}}{1 + g e^{-\beta\epsilon}} \tag{G.253}$$

where in the last expression, we have used $\langle N \rangle / Z_1 = \frac{\lambda}{1 - \lambda Z_1}$, as calculated earlier. This way, we obtain the same result as we would have derived using the canonical ensemble. The quantity

$$\langle n_+ \rangle \equiv \langle N \rangle \frac{g e^{-\beta\epsilon}}{1 + g e^{-\beta\epsilon}} \tag{G.254}$$

is the mean number of particles in the excited state.

G.7 Problem Chapter 7

Problem 7.1

Within the frame of the macrocanonical ensemble, we can express the energy fluctuations as follows

$$\sigma_E^2 = \langle(\delta E)^2\rangle = \langle E^2\rangle - \langle E\rangle^2 = \frac{\sum_{Nm} E_m^2 \mathrm{e}^{\beta\mu N - \beta E_m}}{\sum_{Nm} \mathrm{e}^{\beta\mu N - \beta E_m}} - \left[\frac{\sum_{Nm} E_m \mathrm{e}^{\beta\mu N - \beta E_m}}{\sum_{Nm} \mathrm{e}^{\beta\mu N - \beta E_m}}\right]^2 \tag{G.255}$$

Now we introduce the fugacity, $\lambda = \mathrm{e}^{\beta\mu}$:

$$\langle(\delta E)^2\rangle = \frac{\sum_{Nm} E_m^2 \lambda^N \mathrm{e}^{-\beta E_m}}{\sum_{Nm} \lambda^N \mathrm{e}^{-\beta E_m}} - \left[\frac{\sum_{Nm} E_m \lambda^N \mathrm{e}^{-\beta E_m}}{\sum_{Nm} \lambda^N \mathrm{e}^{-\beta E_m}}\right]^2 = -\frac{\partial}{\partial\beta}\left(\frac{\sum_{Nm} E_m \lambda^N \mathrm{e}^{-\beta E_m}}{\sum_{Nm} \lambda^N \mathrm{e}^{-\beta E_m}}\right)_{\lambda,V} \tag{G.256}$$

The last term in the r.h.s. of the above equation is the internal energy U. Therefore

$$\langle(\delta E)^2\rangle = -\left.\frac{\partial U}{\partial\beta}\right|_{\lambda,V} = kT^2 \left.\frac{\partial U}{\partial T}\right|_{\lambda,V} \tag{G.257}$$

It is important to notice that the internal energy change with temperature at constant λ and V is not the heat capacity at constant volume,

$$\left.\frac{\partial U}{\partial T}\right|_{\lambda,V} \neq C_v \tag{G.258}$$

because N is not constant. This derivative has a rather intriguing interpretation that is worth understanding. Let's first apply the chain rule to expand it:

$$\left.\frac{\partial U}{\partial T}\right|_{\lambda,V} = \underbrace{\left.\frac{\partial U}{\partial T}\right|_{N,V}}_{C_v} + \left.\frac{\partial U}{\partial N}\right|_{T,V} \left.\frac{\partial N}{\partial T}\right|_{\lambda,V} \tag{G.259}$$

The first factor of the second term can be further expanded:

$$\left.\frac{\partial U}{\partial N}\right|_{T,V} = T\left.\frac{\partial S}{\partial N}\right|_{T,V} + \mu \tag{G.260}$$

where we have applied $dU = TdS - pdV + \mu dN$ and because V in this case is constant it reduces to $dU = TdS + \mu dN$. In addition, from $dA = -SdT - pdV + \mu dN$, we obtain

$$\left.\frac{\partial\mu}{\partial T}\right|_{N,V} = \frac{\partial^2 A}{\partial T\partial N} = \frac{\partial^2 A}{\partial N\partial T} = -\left.\frac{\partial S}{\partial N}\right|_{T,V} \tag{G.261}$$

Therefore

$$\left.\frac{\partial U}{\partial N}\right|_{T,V} = -T\left.\frac{\partial\mu}{\partial T}\right|_{N,V} + \mu \tag{G.262}$$

Let's go back to Eq. G.259 and consider the last term, to which we apply again the chain rule:

$$\left.\frac{\partial N}{\partial T}\right|_{\lambda,V} = \left.\frac{\partial N}{\partial T}\right|_{\mu,V} + \left.\frac{\partial N}{\partial\mu}\right|_{T,V}\left.\frac{\partial\mu}{\partial T}\right|_{\lambda,V} \tag{G.263}$$

We now note that

$$\left.\frac{\partial\mu}{\partial T}\right|_{\lambda,V} = \left.\frac{\partial(kT\ln\lambda)}{\partial T}\right|_{\lambda,V} = k\ln\lambda = \frac{\mu}{T} \tag{G.264}$$

Substituting this result into Eq. G.263, we obtain

$$\left.\frac{\partial N}{\partial T}\right|_{\lambda,V} = \left.\frac{\partial N}{\partial T}\right|_{\mu,V} + \left.\frac{\partial N}{\partial \mu}\right|_{T,V} \frac{\mu}{T} \tag{G.265}$$

We now notice that[1]

$$\left.\frac{\partial N}{\partial T}\right|_{\mu,V} \left.\frac{\partial T}{\partial \mu}\right|_{N,V} \left.\frac{\partial \mu}{\partial N}\right|_{T,V} = -1 \Longleftrightarrow \left.\frac{\partial N}{\partial T}\right|_{\mu,V} = -\left.\frac{\partial \mu}{\partial T}\right|_{N,V} \left.\frac{\partial N}{\partial \mu}\right|_{T,V} \tag{G.266}$$

Therefore:

$$\left.\frac{\partial N}{\partial T}\right|_{\lambda,V} = \left.\frac{\partial N}{\partial T}\right|_{\mu,V} + \left.\frac{\partial N}{\partial \mu}\right|_{T,V} \frac{\mu}{T} = -\left.\frac{\partial N}{\partial \mu}\right|_{T,V} \left.\frac{\partial \mu}{\partial T}\right|_{N,V} + \left.\frac{\partial N}{\partial \mu}\right|_{T,V} \frac{\mu}{T} = \frac{1}{T}\left.\frac{\partial N}{\partial \mu}\right|_{T,V} \left[\mu - T\left.\frac{\partial \mu}{\partial T}\right|_{N,V}\right] \tag{G.267}$$

We can finally go back to Eq. G.259 and substitute all the results we have obtained:

$$\left.\frac{\partial U}{\partial T}\right|_{\lambda,V} = C_v + \left[\mu - T\left.\frac{\partial \mu}{\partial T}\right|_{N,V}\right] \frac{1}{T}\left.\frac{\partial N}{\partial \mu}\right|_{T,V} \left[\mu - T\left.\frac{\partial \mu}{\partial T}\right|_{N,V}\right] = C_v + \frac{1}{T}\left.\frac{\partial N}{\partial \mu}\right|_{T,V} \left(\left.\frac{\partial U}{\partial N}\right|_{T,V}\right)^2 \tag{G.268}$$

The energy fluctuations take thus the following form:

$$\langle(\delta E)^2\rangle = kT^2 C_v + kT\left.\frac{\partial N}{\partial \mu}\right|_{T,V} \left(\left.\frac{\partial U}{\partial N}\right|_{T,V}\right)^2 = kT^2 C_v + kT\frac{N^2}{V}\kappa_T \left(\left.\frac{\partial U}{\partial N}\right|_{T,V}\right)^2 \tag{G.269}$$

which is the result we were seeking.

Problem 7.2

We have a classical ideal gas of indistinguishable particles. It follows that the canonical partition function factorises in this case as

$$Z_N(T,V) = \frac{Z_1^N}{N!}, \tag{G.270}$$

where

[1]To prove the given expression:

$$\left.\frac{\partial x}{\partial y}\right|_z \left.\frac{\partial y}{\partial z}\right|_x \left.\frac{\partial z}{\partial x}\right|_y = -1$$

we can use the chain rule for partial derivatives. Let's start by expressing the derivatives in terms of total differentials:

$$\left.\frac{\partial x}{\partial y}\right|_z = \frac{dx}{dy} - \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial x}{\partial z}\right)_y$$

$$\left.\frac{\partial y}{\partial z}\right|_x = \frac{dy}{dz} - \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial y}{\partial x}\right)_z$$

$$\left.\frac{\partial z}{\partial x}\right|_y = \frac{dz}{dx} - \left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial z}{\partial y}\right)_x$$

Now, multiply these expressions together:

$$\begin{aligned}
&\left(\frac{dx}{dy} - \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial x}{\partial z}\right)_y\right) \cdot \left(\frac{dy}{dz} - \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial y}{\partial x}\right)_z\right) \cdot \left(\frac{dz}{dx} - \left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial z}{\partial y}\right)_x\right) \\
&= -\left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial x}{\partial z}\right)_y \cdot \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial y}{\partial x}\right)_z \cdot \left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial z}{\partial y}\right)_x \\
&= -1
\end{aligned}$$

So, the given expression is indeed equal to -1.

$$Z_1 = \frac{V}{\Lambda(T)^3} \tag{G.271}$$

is the canonical partition function of a classical free particle and we have defined the thermal de Broglie wavelength

$$\Lambda(T) = \frac{h}{2\pi m k_B T}. \tag{G.272}$$

Introducing the fugacity $\lambda = e^{\beta\mu}$, the grand canonical partition function is written as

$$\Xi(T, V, \lambda) = \sum_{N=0}^{\infty} \lambda^N Z_N(T, V) = \sum_{N=0}^{\infty} \lambda^N \frac{Z_1^N}{N!} = e^{\lambda Z_1} = e^{\frac{\lambda V}{\Lambda(T)^3}} \tag{G.273}$$

Therefore, the grand canonical potential is

$$J = k_B T \ln \Xi = k_B T \lambda Z_1 = k_B T e^{\beta\mu} \frac{V}{\Lambda(T)^3}. \tag{G.274}$$

From J, we can now calculate the entropy of the system.

$$S = \left.\frac{\partial J}{\partial T}\right|_{\mu,V} = \frac{V}{h^3}(2\pi m k_B T)^{3/2} k_B \frac{\partial}{\partial T}\left(T^{5/2} \mathrm{e}^{\beta\mu}\right) = \mathrm{e}^{\beta\mu} \frac{V}{\Lambda(T)^3}\left(\frac{5}{2}k_B - \frac{\mu}{T}\right) \tag{G.275}$$

And the average number of particles

$$\langle N \rangle = \left.\frac{\partial J}{\partial \mu}\right|_{T,V} = \frac{\partial}{\partial \mu}\left(k_B T \mathrm{e}^{\beta\mu} \frac{V}{\Lambda(T)^3}\right) = \mathrm{e}^{\beta\mu} \frac{V}{\Lambda(T)^3} \tag{G.276}$$

or equivalently,

$$\beta\mu = \ln\left(\frac{\langle N \rangle}{V} \Lambda(T)^3\right) = \ln\left(\langle n \rangle \Lambda^3\right) \tag{G.277}$$

where we have defined the average density $\langle n \rangle = \langle N \rangle / V$. We can now use the last two expressions in the formula for entropy, obtaining

$$S = \langle N \rangle k_B \left(\frac{5}{2} - \beta\mu\right) = \langle N \rangle k_B \left[\frac{5}{2} - \ln(\langle n \rangle \Lambda^3)\right] \tag{G.278}$$

which is nothing but the Sackur-Tetrode equation for the entropy of the ideal gas that we also derived from the canonical ensemble. Let's show that, starting from Eq. 5.51:

$$S = k_B(\beta\langle E \rangle + \ln Z_N) \tag{G.279}$$

Taking into account the equipartition theorem, we know that the internal energy of the ideal gas is $\langle E \rangle = \frac{3}{2}\langle N \rangle k_B T$. We can use this expression to eliminate the temperature dependence in the previous entropy expression in favour of energy, obtaining

$$\begin{aligned} S &= k_B(\beta \frac{3}{2}\langle N \rangle k_B T + \ln Z_N) = k_B(\frac{3}{2}\langle N \rangle + \ln Z_N) = \\ &= \frac{3}{2}\langle N \rangle k_B + k_B \ln\left(\frac{V^N}{N!\Lambda^{3N}}\right) = \\ &= \langle N \rangle k_B \left[\frac{5}{2} - \ln(\langle n \rangle)\Lambda^3)\right] \end{aligned} \tag{G.280}$$

where we have applied Stirling's approximation and $\langle n \rangle \equiv \langle N \rangle / V$ is the numerical density. Using again the equipartition theorem, we can rewrite Eq. G.278 as

$$S = Nk_B\left(\frac{5}{2} + \ln\left[\frac{V}{\langle N\rangle}\left(\frac{4\pi m}{3h^2}\frac{\langle E\rangle}{\langle N\rangle}\right)^{3/2}\right]\right) \tag{G.281}$$

which, once again, is the entropy of the ideal gas that we had also calculated from the microcanonical ensemble (see Eq. 4.74). This allows us to observe the equivalence of the results obtained in all three ensembles.

Problem 7.3

(a) The system canonical partition function reads

$$Z = \frac{1}{N!}Z_1^N \tag{G.282}$$

where Z_1, the partition function of a single particle, can be obtained as follows

$$Z_1 = \frac{1}{h^3}\underbrace{\int_V dq_x\, dq_y\, dq_z}_{V}\int_{-\infty}^{+\infty} dp_x \int_{-\infty}^{+\infty} dp_y \int_{-\infty}^{+\infty} dp_z \mathrm{e}^{-\beta H_1} \tag{G.283}$$

where $H_1 = p^2/2m + \Delta_B$. The above equation can then be re-written as follows

$$Z_1 = \frac{V}{h^3}\int_{-\infty}^{+\infty} dp_x \int_{-\infty}^{+\infty} dp_y \int_{-\infty}^{+\infty} dp_z \mathrm{e}^{-\beta p^2/2m}\mathrm{e}^{-\beta\Delta_B} \tag{G.284}$$

The last energetic contribution is a constant, which depends on mutual orientation between the dipolar moment and applied field. To solve the integrals, we apply a variable substitution for each p_i, with $i = x, y, z$:

$$w_i = \frac{p_i}{\sqrt{2kTm}}; \quad dw_i = \frac{1}{\sqrt{2kTm}}dp_i \tag{G.285}$$

Therefore, we get

$$Z_1 = \frac{(2\pi mkT)^{\frac{3}{2}}V}{h^3}\mathrm{e}^{-\beta\Delta_B} \tag{G.286}$$

Depending on the orientation of the particle dipole moment, Δ_B assumes different values. Therefore

$$Z = \frac{1}{N!}Z_1^N = \frac{1}{N!}\left[\frac{(2\pi mkT)^{\frac{3}{2}}V}{h^3}\right]^N \left(\mathrm{e}^{\beta\mu_d B} + 1 + \mathrm{e}^{-\beta\mu_d B}\right)^N \tag{G.287}$$

(b) If we define with n_{par}, n_{antipar} and n_{rand}, the number of particles whose dipolar moment is, respectively, parallel, antiparallel and randomly oriented with $\vec{B}$, the fraction of particles that are found in each of the three magnetic states is

$$\begin{aligned} P_{\text{par}} &= \frac{n_{\text{par}}}{N} = \frac{\mathrm{e}^{\beta\mu_d B}}{\mathrm{e}^{\beta\mu_d B} + 1 + \mathrm{e}^{-\beta\mu_d B}} \\ P_{\text{antipar}} &= \frac{n_{\text{antipar}}}{N} = \frac{\mathrm{e}^{-\beta\mu_d B}}{\mathrm{e}^{\beta\mu_d B} + 1 + \mathrm{e}^{-\beta\mu_d B}} \\ P_{\text{rand}} &= \frac{n_{\text{rand}}}{N} = \frac{1}{\mathrm{e}^{\beta\mu_d B} + 1 + \mathrm{e}^{-\beta\mu_d B}} \end{aligned} \tag{G.288}$$

(c) Internal energy as a function of N, V and T:

$$U = kT^2 \frac{\partial \ln Z}{\partial T} = -\frac{\partial \ln Z}{\partial \beta} = -N \frac{\partial \ln Z_1}{\partial \beta} \tag{G.289}$$

Note that

$$Z_1 = \frac{(2\pi m)^{\frac{3}{2}} V}{h^3 \beta^{\frac{3}{2}}} \left(\mathrm{e}^{\beta \mu_d B} + 1 + \mathrm{e}^{-\beta \mu_d B} \right) \tag{G.290}$$

Therefore

$$\begin{aligned} U &= -N \frac{\partial}{\partial \beta} \left[-\ln \beta^{\frac{3}{2}} + \ln \left(\mathrm{e}^{\beta \mu_d B} + 1 + \mathrm{e}^{-\beta \mu_d B} \right) \right] = \\ &= \frac{3}{2} NkT - N \mu_d B \frac{\mathrm{e}^{\beta \mu_d B} - \mathrm{e}^{-\beta \mu_d B}}{\mathrm{e}^{\beta \mu_d B} + 1 + \mathrm{e}^{-\beta \mu_d B}} \end{aligned} \tag{G.291}$$

(d) Macrocanonical partition function:

$$\Xi = \sum_{N=0}^{\infty} \frac{1}{N!} (\lambda Z_1)^N = \mathrm{e}^{\lambda Z_1} \tag{G.292}$$

with $\lambda = \mathrm{e}^{\beta \mu}$ and μ the chemical potential.

(e) we now employ the macrocanonical formalism to obtain the internal energy:

$$U = -\left. \frac{\partial \ln \Xi}{\partial \beta} \right|_{\lambda, V} = -\frac{\partial}{\partial \beta} (\lambda Z_1) = -\lambda \frac{\partial Z_1}{\partial \beta} = \lambda Z_1 \left(-\frac{\partial \ln Z_1}{\partial \beta} \right) \tag{G.293}$$

From Eq. G.289, we know that $-\partial \ln Z_1 / \partial \beta = U/N$. Therefore:

$$U = \lambda Z_1 \left[\frac{3}{2} kT - \mu_d B \frac{\mathrm{e}^{\beta \mu_d B} - \mathrm{e}^{-\beta \mu_d B}}{\mathrm{e}^{\beta \mu_d B} + 1 + \mathrm{e}^{-\beta \mu_d B}} \right] \tag{G.294}$$

Let's now find what λZ_1 is. Using the grand potential we have:

$$J = -kT \ln \Xi = -kT \lambda Z_1 \tag{G.295}$$

Additionally, we know that

$$\left. \frac{\partial J}{\partial \mu} \right|_{T,V} = -\langle N \rangle \tag{G.296}$$

alternatively:

$$\left. \frac{\partial J}{\partial \mu} \right|_{T,V} = \left. \frac{\partial J}{\partial \lambda} \right|_{T,V} \left. \frac{\partial \lambda}{\partial \mu} \right|_{T,V} = \beta \lambda \left. \frac{\partial J}{\partial \lambda} \right|_{T,V} = \frac{1}{kT} \lambda (-kT) Z_1 = -\lambda Z_1 \tag{G.297}$$

It follows that

$$\langle N \rangle = \lambda Z_1 \tag{G.298}$$

and therefore

$$U = N \left[\frac{3}{2} kT - \mu_d B \frac{\mathrm{e}^{\beta \mu_d B} - \mathrm{e}^{-\beta \mu_d B}}{\mathrm{e}^{\beta \mu_d B} + 1 + \mathrm{e}^{-\beta \mu_d B}} \right] \tag{G.299}$$

which is the same expression for the internal energy as that obtained above.

Problem 7.4

(a) The system consists of a binary mixture of two ideal gases, that we call A and B. The global partition function, following Eq. 5.32, can be written as

$$Z_N = Z_A Z_B \tag{G.300}$$

where $N = N_A + N_B$ is the sum of the particles of gas A and gas B. Each partition function can be obtained using the partition function of a single particle:

$$Z_A = \frac{Z_{A,1}^{N_A}}{N_A!}; \quad Z_B = \frac{Z_{B,1}^{N_B}}{N_B!} \tag{G.301}$$

In particular

$$Z_{A,1} = \frac{1}{h^3}\int_{-\infty}^{\infty} d^3\mathbf{p}\, d^3\mathbf{q}\, \mathrm{e}^{-\beta H_{A,1}} \tag{G.302}$$

where $H_{A,1} = \frac{\mathbf{p}^2}{2m_A}$ is the Hamiltonian of gas A. It follows that

$$Z_{A,1} = \frac{V}{h^3}\left[\int_{-\infty}^{\infty} dp\, \mathrm{e}^{-\beta p^2/2m_A}\right]^3 \tag{G.303}$$

This integral can be solved with the usual variable substitution:

$$t = \frac{p}{\sqrt{2kTm_A}}; \quad dt = \frac{dp}{\sqrt{2kTm_A}} \tag{G.304}$$

Therefore

$$Z_{A,1} = \frac{V}{h^3}\left[\sqrt{2kTm_A}\underbrace{\int_{-\infty}^{\infty} dt\, \mathrm{e}^{-t^2}}_{\sqrt{\pi}}\right]^3 = \frac{(2\pi kTm_A)^{\frac{3}{2}}}{h^3}V = \frac{V}{\Lambda_A^3} \tag{G.305}$$

where Λ_A is the de Broglie thermal wavelength for gas A. Similarly, for gas B, one obtains:

$$Z_{B,1} = \frac{V}{\Lambda_B^3} \tag{G.306}$$

and the total canonical partition function reads

$$Z_N = Z_A Z_B = \frac{Z_{A,1}^{N_A}}{N_A!}\frac{Z_{B,1}^{N_B}}{N_B!} = \frac{V^N}{(\Lambda_A\Lambda_B)^3 N_A! N_B!} \tag{G.307}$$

(b) Now the two gases are in separate containers, say of volume V_A and V_B, at the same pressure and temperature. Container with volume V_A contains N_A particles of gas A and container with volume V_B contains N_B particles of gas B. The resulting partition function reads:

$$Z_N = \frac{Z_{A,1}^{N_A}}{N_A!}\frac{Z_{B,1}^{N_B}}{N_B!} = \frac{V_A^{N_A}}{\Lambda_A^3 N_A!}\frac{V_B^{N_B}}{\Lambda_B^3 N_B!} \tag{G.308}$$

Note that since particles of type A are still distinguishable from particles of type B, one has to divide by $N_A!$ and $N_B!$ rather than by $N!$. We also observe that this partition function differ only in the volume from that calculated in (a).

(c) Let's now calculate the grand canonical partition function of the mixture. To this end, let's remember, from Eq. 6.37, that

$$\Xi_{AB} = \Xi_A \Xi_B \tag{G.309}$$

Therefore, we can calculate the total partition function as the product of the partition functions of the two ideal gases separately. In particular:

$$\Xi_{AB} = \Xi_A \Xi_B = \left(\sum_{N=0}^{\infty} \lambda_A^N Z_A \right) \left(\sum_{N=0}^{\infty} \lambda_B^N Z_B \right) \tag{G.310}$$

Since the two gases are at equilibrium

$$\mu_A = \mu_B \Longleftrightarrow \lambda_A = \lambda_B = \lambda \tag{G.311}$$

Consequently,

$$\begin{aligned}
\Xi_{AB} &= \left(\sum_{N=0}^{\infty} \lambda^N Z_A \right) \left(\sum_{N=0}^{\infty} \lambda^N Z_B \right) = \\
&= \underbrace{\left[\sum_{N=0}^{\infty} \frac{1}{N!} \left(\lambda Z_{A,1}\right)^N \right]}_{\mathrm{e}^{\lambda Z_{A,1}}} \underbrace{\left[\sum_{N=0}^{\infty} \frac{1}{N!} \left(\lambda Z_{B,1}\right)^N \right]}_{\mathrm{e}^{\lambda Z_{B,1}}} = \\
&= \exp\left[\lambda \left(Z_{A,1} + Z_{B,1}\right)\right]
\end{aligned} \tag{G.312}$$

Now we need to consider that the two gases are mixed together in the same container of volume V. Therefore

$$\Xi_{AB} = \exp\left[\lambda \left(\frac{V}{\Lambda_A^3} + \frac{V}{\Lambda_B^3} \right)\right] = \exp\left(\lambda V \frac{\Lambda_A^3 + \Lambda_B^3}{(\Lambda_A \Lambda_B)^3} \right) \tag{G.313}$$

(d) The entropy change between the mixed state and the separated-gas state can be calculated from the change in free energy with temperature:

$$\Delta S = - \left. \frac{\partial \Delta A}{\partial T} \right|_{N_A, N_B, T} = - \frac{\partial}{\partial T} \left(A_{A+B} - (A_A + A_B) \right) \tag{G.314}$$

Let's calculate this free-energy change:

$$
\begin{aligned}
\Delta A &= -kT\ln Z_{AB} + kT\ln(Z_A Z_B) = kT\ln\left(\frac{Z_A Z_B}{Z_{AB}}\right) = \\
&= kT\ln\left(\underbrace{\frac{(\Lambda_A\Lambda_B)^3 N_A! N_B!}{V^N}}_{Z_{AB}^{-1}} \underbrace{\frac{V_A^{N_A}}{\Lambda_A^3 N_A!}}_{Z_A} \underbrace{\frac{V_B^{N_B}}{\Lambda_B^3 N_B!}}_{Z_B}\right) = \\
&= kT\ln\left(\frac{V_A^{N_A} V_B^{N_B}}{V^N}\right) = \\
&= kT\left(N_A\ln V_A + N_B\ln V_B - N\ln V\right) = \\
&= kT\left(N_A\ln V_A - N_A\ln V + N_B\ln V_B - N_B\ln V\right) = \\
&= kT\left(N_A\ln\frac{V_A}{V} + N_B\ln\frac{V_B}{V}\right) = \\
&= NkT\left(x_A\ln\frac{V_A}{V} + x_B\ln\frac{V_B}{V}\right)
\end{aligned}
\tag{G.315}
$$

where $x_A \equiv N_A/N$ and $x_B \equiv N_B/N$. Therefore:

$$
\Delta S = -\frac{\partial \Delta A}{\partial T} = -Nk\left(x_A\ln\frac{V_A}{V} + x_B\ln\frac{V_B}{V}\right) \tag{G.316}
$$

Problem 7.5

(a) The microcanonical ensemble describes an isolated system with a fixed number N of particles and a fixed total energy E. The numbers of particles n_+ and n_- with energies ε and $-\varepsilon$ respectively are clearly constrained by the two relations $E = n_+\varepsilon - n_-\varepsilon$ and $N = n_+ + n_-$. Defining the quantity $x = E/N\varepsilon$ (which is proportional to the energy per particle), we easily find that $n_+ = N(1+x)/2$ and $n_- = N(1-x)/2$. The partition function is just the number of microstates consistent with these constraints and is equal to the number of ways of choosing, say, n_+ particles from N, namely

$$
\Omega(E,N) = \binom{N}{n_+} = \frac{N!}{n_+! n_-!} \tag{G.317}
$$

The thermodynamic interpretation of this ensemble is through the definition of the entropy $S_M(E,N) = k\ln[\Omega(E,N)]$. In the limit when N is very large, we may use Stirling's approximation $\ln N! \simeq N\ln N - N$ to obtain the entropy per particle as

$$
s = \lim_{N\to\infty}\left(\frac{S_M}{N}\right) = k\left[\ln 2 - \frac{1}{2}(1+x)\ln(1+x) - \frac{1}{2}(1-x)\ln(1-x)\right] \tag{G.318}
$$

(b) The canonical ensemble describes a system with a fixed number of particles in contact with a heat bath at a fixed temperature. The partition function is

$$
\begin{aligned}
Z(T,N) &= \underbrace{\sum_n \Omega e^{-\beta E_n}}_{\text{sum over microstates}} = \sum_{n_+=0}^{N}\binom{N}{n_+} e^{-\beta\varepsilon n_+} e^{-\beta\varepsilon(N-n_+)} = \\
&= \underbrace{\sum_m e^{-\beta E_m}}_{\text{sum over energy levels}} = Z_1^N = \left(e^{-\beta\varepsilon} + e^{\beta\varepsilon}\right)^N
\end{aligned}
\tag{G.319}
$$

where Z_1 is the partition function of a single particle (which has two different states) and $Z = Z_1^N$ applies as particles are identical and distinguishable. The thermodynamic interpretation is through the Helmholtz free energy $A(T, N) = -kT \ln[Z(T, N)]$. The entropy in this ensemble is obtained from the thermodynamic relation $A = U - TS$, where the internal energy is identified as the mean value.

$$U_C = \langle E \rangle_C = -\left.\frac{\partial(\ln Z)}{\partial \beta}\right|_N = -N\varepsilon \tanh(\beta\varepsilon) \tag{G.320}$$

The mean energy per particle is now identified as U_c/N, and the parameter x introduced above is $x = U_c/N\varepsilon = -\tanh(\beta\varepsilon)$. In order to obtain the entropy as a function of x, we must first solve this relation to obtain the temperature in terms of x:

$$\beta\varepsilon = \frac{1}{2}\ln\left(\frac{1-x}{1+x}\right) = \frac{1}{2}[\ln(1-x) - \ln(1+x)] \tag{G.321}$$

With this result ni hand, we find that

$$\begin{aligned} U_C &= N\varepsilon x \\ A_C &= -kT\ln Z = -NkT\left[\ln 2 - \frac{1}{2}\ln(1+x) - \frac{1}{2}\ln(1-x)\right] \\ S_C &= \frac{U_C - A_C}{T} = Nk\left[\ln 2 - \frac{1}{2}(1+x)\ln(1+x) - \frac{1}{2}(1-x)\ln(1-x)\right] \end{aligned} \tag{G.322}$$

and the entropy per particle $s = S_C/N$ reproduces the microcanonical result.

(c) The grand canonical ensemble describes a system in equilibrium with a reservoir, with which ti can exchange both energy and particles, so both the energy and the number of particles fluctuate. The reservoir is characterised by a definite temperature $T = 1/k\beta$ and chemical potential μ or fugacity $\lambda = \mathrm{e}^{\beta\mu}$. The partition function is

$$\Xi(T, \mu) = \sum_{N=0}^{\infty} \lambda^N Z(T, N) = \frac{1}{1 - \lambda\left(\mathrm{e}^{-\beta\varepsilon} + \mathrm{e}^{\beta\varepsilon}\right)} \tag{G.323}$$

and the thermodynamic interpretation is through the grand potential $J(T, \mu) = kT \ln \Xi(T, \mu)$. The entropy will be obtained through the thermodynamic relation $J = TS - U + \mu N$, where the particle number and internal energy are identified as the mean values:

$$\begin{aligned} N_G &= \langle N \rangle_G = \lambda \left.\frac{\partial \ln \Xi}{\partial \lambda}\right|_\beta = \frac{\lambda(\mathrm{e}^{-\beta\varepsilon} + \mathrm{e}^{\beta\varepsilon})}{1 - \lambda(\mathrm{e}^{-\beta\varepsilon} + \mathrm{e}^{\beta\varepsilon})} \\ U_G &= \langle E \rangle_G = \lambda \left.\frac{\partial \ln \Xi}{\partial \beta}\right|_\lambda = -\varepsilon\frac{\lambda(\mathrm{e}^{-\beta\varepsilon} + \mathrm{e}^{\beta\varepsilon})}{1 - \lambda(\mathrm{e}^{-\beta\varepsilon} + \mathrm{e}^{\beta\varepsilon})} = -N_G\varepsilon \tanh(\beta\varepsilon) \end{aligned} \tag{G.324}$$

As above, we would like to express thermodynamic functions in terms of the parameter $x = U_G/N_G\varepsilon = -\tanh(\beta\varepsilon)$ and N_G, rather than T and μ. So, we first solve the above equations to obtain

$$\begin{aligned} \beta\varepsilon &= \frac{1}{2}[\ln(1-x) - \ln(1+x)] \\ \beta\mu &= \ln\left(\frac{N_G}{N_G + 1}\right) - \ln 2 + \frac{1}{2}[\ln(1+x) + \ln(1-x)] \end{aligned} \tag{G.325}$$

With these results in hand, we can calculate

$$\begin{aligned}
U_G &= N_G \varepsilon x \\
J &= kT \ln \Xi = kT \ln(N_G + 1) \\
S_G &= \frac{1}{T}(J + U_G - \mu N_G) = \\
&= kN_G \left[\frac{\ln(N_G+1)}{N_G} + \ln\left(1 + \frac{1}{N_G}\right) + \ln 2 - \frac{1}{2}(1+x)\ln(1+x) - \frac{1}{2}(1-x)\ln(1-x)) \right]
\end{aligned} \tag{G.326}$$

On taking the limit that N_G is very large, we find for the entropy per particle that

$$s = \lim_{N_G \to \infty} \left(\frac{S_G}{N_G} \right) = k \left[\ln 2 - \frac{1}{2}(1+x)\ln(1+x) - \frac{1}{2}(1-x)\ln(1-x) \right] \tag{G.327}$$

which agrees with the canonical and microcanonical versions.

Bibliography

[1] P. Pathria, *Statistical Mechanics* (2nd edition), Butterworth-Heinemann, Oxford (1996).

[2] R. Balescu, *Equilibrium and Non-equilibrium Statistical Mechanics*, Wiley & Sons, New York (1975).

[3] D. A. McQuarrie, *Statistical Mechanics*, Harper & Row, New York (1976).

[4] M. Le Bellac, F. Mortessagne and G. G. Batrouni, *Equilibrium and non-equilibrium statistical thermodynamics*, Cambridge University Press, Cambridge (2004).

[5] J. J. Brey Abalo, J. de la Rubia Pacheco and J. de la Rubia Sánchez, *Mecánica Estadística*, UNED, Madrid (2001).

[6] D. Chandler, *Introduction to Modern Statistical Mechanics*, Oxford University Press, Oxford (1987).

[7] D. A. R. Dalvit, J. Frastai and I. D. Lawrie, *Problems on Statistical Mechanics*. Institute of Physics Publishing, Bristol, 1999.

[8] L. Yung-Kuo, *Problems and Solutions on Thermodynamics and Statistical Mechanics*. World Scientific Publishing, Singapore, 1990.

[9] C. Fernández Tejero and J. M. Rodríguez Parrondo, *100 problemas de Física Estadística*. Alianza, Madrid, 1996.

[10] R. Kubo, *Statistical Mechanics: an Advanced Course with Problems and Solutions*. 2nd edition, North-Holland, Amsterdam, 1999.

[11] D. J. Amit and Y. Verbin, *Statistical Physics. An Introductory Course*. World Scientific Publishing, Singapore, 1999.

[12] M. R. Spiegel, J. Schiller and R. A. Srinivasan (2003). *Probabilidad y Estadística*. Editorial Schaum. McGraw-Hill.

[13] J. J. Brey and J. de la Rubia Pacheco (2001). *Mecánica estadística*. Universidad Nacional de Educación a Distancia, UNED.

[14] L. Pauling and E. Bright Wilson (2012), *Introduction to Quantum Mechanics with Applications to Chemistry*, USA: Dover Publications.

[15] E. U. Condon and P. M. C. Morse (1963), *Quantum Mechanics*, McGraw-Hill Book Company.

[16] F.J. Dyson, A. Lenard, J. Math. Phys. 8, 423, 1967.

[17] M. Fisher, Arch. Rat. Mech. Anal. 17, 377, 1964.

[18] J. Lebowitz, E. Lieb, Phys. Rev. Lett. 22, 631, 1969